


# ATS3085Sx Datasheet

## Actions® ATS3085Sx™ QFN68

**Low Power Solution for**
**Dual mode Bluetooth smart watch**

**CPU + DSP Dual-core Single-chip**
**Bluetooth V5.3**

*Version: V1.0*

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# Key Features

## System

- 192MHz(Max) ARM MSTAR processor Core
- 192MHz(Max) CEVA TL420 DSP Core
- Memory Protect Unit(MPU)
- Internal 1168KB RAM for data and program
- Build in 32KB Execute-in-place(Xip) Cache to QSPI NOR flash and 32KB cache to OPI PSRAM
- Support QSPI STR/DTR serial Flash for custom defined software
- Support 32MHz OSC and 32.768KHz OSC with on-chip PLL
- Fully configurable PEQ, up to 20 segments
- Support RTC and Build-in timers
- Support Hi-res when A2DP
- Support echo cancellation
- Support ACTIONS single microphone noise reduction
- Support ACTIONS dual microphone noise reduction
- Support sensor hub

## Secure Subsystem

- Support hardware AES-128/192/256 Encrypt & Decrypt in ECB/CTR/CBC/CBC-CTS mode
- Support hardware TRNG used to collects random bits from the ring oscillator, 64 random bits per time.
- Support SHA-1/SHA-224/SHA-256.
- Support hardware CRC-32/CRC-16
- Support 1024bits efuse
- Support 64bit CHIP unique ID
- Support RSA 2048bits signature
- Support secure boot
- Support firmware encryption
- Support On-the-fly decryption of external memories (QSPI NorFlash/Octo-SPI PSRAM), part of which can be encrypted by AES-128 in counter mode

## Bluetooth

- Support Bluetooth V5.3
- Support LE Audio(LC3 Codec)
- Supports basic rate and enhanced data rate
- Supports Secure Simple Pairing
- Supports Low Power Mode
- Bluetooth Dual Mode support: Simultaneous LE and BR / EDR
- Supports multiple Low Energy states
- Fast AGC control to improve receiving dynamic range
- Supports AFH to dynamically detect channel quality to improve transmission quality
- Supports GFSK, π/4 DQPSK and 8DPSK modulation
- Supports Power / Enhanced Power Control
- LE Data Packet Length Extension
- LE 2M PHY
- Support long range BLE
- Support AOA/AOD
- Channel Selection Algorithm #2
- Support PTA (Packet Traffic Arbiter) when co-existing with WI-FI
- Support WCI2 when co-existing with 4G

## Bluetooth Performance

- Max transmitting output power: BR 13dBm, BLE 14dBm
- Bluetooth receiving sensitivity: -96.5dBm@GFSK, -95dBm@π/4 DQPSK, -89.5dBm@8DPSK modulation
- Bluetooth receiving sensitivity (BLE mode): -99.5dBm@1Mbps, -96dBm@2Mbps

## Audio

- Build in stereo 24-bit input sigma-delta ADC, SNR<=98dB, THD+N>=-88dB(A-weighting)
- ADC supports sample rate 8k/12k/11.025k/16k/22.05k/24k/32k/44.1k/48k/88.2k/96kHz
- Supports stereo single-ended input or differential analog microphone
- Supports stereo digital microphones.
- Build in single channel 24-bit input sigma-delta DAC, SNR<=113dB, THD+N>=-98dB, DR<=111dB, Noise>=2.7uVrms(A-weighting)
- DAC supports sample rate 8k/12k/11.025k/16k/22.05k/24k/32k/44.1k/48

k/88.2k/96kHz
- Support 1*I2S/PCM TX with master and slave mode, and another I2S TX with Master mode Only, both interfaces sampling rate from 8KHz to 192KHz
- Support 1*I2S/PCM RX with master and slave mode, sampling rate from 8KHz to 192KHz

## Storage

- Support 2*QSPI interface for SPINOR or S PINAND, 1*OPI for PSRAM.
- Support 2*SD/MMC card interface
- SIP 64Mbit/32Mbit Octo-SPI PSRAM for gorgeous UI performance.

## Graphics/Multimedia

- 192MHz 2.5D GPU
  Support any curves and any font rendering by HW tessellation, NO CPU overhead.
  2.5D perspective transformation.
  Any angle primitive rotation, flip and mirror.
  Anti-aliasing: 16x Anti-aliasing without extra performance drop and memory overhead.
  Solid, linear gradient paint and pattern paint.
  Texture mapping; Point sampling, linear and bilinear filtering.
  Scissoring and viewport clipping.
  Porter Duff Alpha Blending plus additional blend modes.
  8/16/24/32 bits source image and frame color formats.
  Index 1/2/4/8 format source image supported.
  Command buffer based DMA to minimize CPU overhead.
  Lossy compression to save system memory.

- 2D GPU
  RGB565/666/888/ARGB/ARGB8565/ARGB6666/ Ax/ABMP/XRGB/CLUT source data format
  Support 256 Level Alpha
  Support three Image Layer
  Support any angle rotate with RGB565/ARGB/BGRA source
  The Layer1/2 Support Up/down scaling

- JPEG Baseline Decoder

## Display

- Support RGB888/RGB666/RGB565 Source Data Format
- Support 8bit active (TFT) LCD panels with digital CPU interface
- Support (TFT) LCD panels with 3wire/4wire SPI option I/II interface
- Support 2lane SPI LCD interface
- Support SDR/DDR QSPI LCD interface
- Support TR-J/TR-S parallel HSYNC/VSYNC LCD interface

## Power Management

- Operation voltage: I/O 1.8V or 3.3V, Core 0.6V~1.3V
- Supports Li-Ion battery and 5V power supply
- Supports 5V power supply plugged in reset
- Integrated linear battery charger with power path management up to 400mA charging current, which supports CC/CV mode
- Integrated 12bit SAR A/D converters for battery voltage monitor, temperature monitor and wire-controller
- Integrated 2*DC-DC buck converters, which can be switch to LDO mode
- Integrated 6*Load switch, which can be used as power supply for peripheral.
- Supports DC5V insertion detection and pull out detection
- Supports adaptive voltage scaling technology for power optimization
- Low Power Consumption:
  - Typical BR Sniff Current: <= 90μA@500ms, Txpower = 8dbm
  - Typical BLE Sniff Current: <= 65μA@500ms, Txpower = 8dbm
  - Typical Simultaneous BR and BLE Sniff Current: <= 110μA@500ms, Txpower = 8dbm
  - Standby mode: <= 20uA@Vbat = 3.8V
  - Shutdown mode: 2μA@Vbat = 3.8V
  - Run mode: <= 22 μA/MHz @BAT 3.8 V @CPU 192Mhz
  - Run mode: <= 12 μA/MHz @BAT 3.8 V @CPU 64Mhz

## Flexible I/O

- Support 45 GPIOs at most
- Support 6 LRADC at most
- Support 4 PWM for lamp controller at most
- Support USB2.0 full speed
- Serial Interface:
  - 3*UART, Maximum Rate: 4Mbaud
  - 4*I2C, support for standard mode (100 kHz), Fast mode (400 kHz), and Fast mode+ (1 MHz)
  - 2*SPIMT and 2*I2CMT for sensor device
  - In addition to NOR, and QSPI_LCD, there

are 3*SPI and 1*QSPI interfaces for users. The Maximum Rate of all the SPI/QPI interfaces is 96MHz, Maximum Rate of PSRAM interface is 140Mhz and Maximum Rate of NorFlash interface is 106Mhz.

## Applications

- Dual mode Bluetooth smart watch

## Package

- QFN-68 (7*7*0.75mm, Pitch 0.35m)

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# Revision History

| Date | Revision | Description |
|---|---|---|
| 2023-07-19 | V1.0 | Release version |
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# 1 Introduction

## 1.1Overview

The ATS3085Sx is a highly integrated single-chip Bluetooth audio solution targeting Bluetooth smart watch markets with high performance, low power consumption, and low cost.

ATS3085Sx adopts Dual-core architecture. An ARM MSTAR core is used as application processor, and a companion CEVA TL420 DSP core is used to efficiently handle audio processing. The on-chip RAM and ROM are integrated for various Bluetooth audio processing applications.

ATS3085Sx integrates a Bluetooth controller that is fully compliant to V5.3 standard and is 100% backward compatible to V5.0/4.2/2.1 Bluetooth specifications. ATS3085Sx supports dual mode (BR/EDR + Low Energy Controllers) operations. Both links in BR/EDR and LE can be active simultaneously. Additionally, ATS3085Sx's embedded BLE Audio features bring new audio listening experiences and wonderful call experiences.

Well suited for wearable applications, ATS3085Sx integrates a 2.5D GPU, a 2D GPU and a built-in LCD controller providing excellent display effects while maintaining extremely low power consumptions. It supports LCD panels with 3-line SPI, 4-line SPI, Dual SPI, QSPI, DDR QSPI, TR-I/TR-S parallel HSYNC/VSYNC, or MCU interfaces. The SoC also integrates a sensor hub and a complete set of advanced power management circuits, which further enables ATS3085Sx to provide extremely low power consumption while maintaining high degrees of usability and achieving high marks in performance.

ATS3085Sx has a flexible memory configuration and a rich set of interfaces such as SD/SPI/USB2.0 FS/UART/I2C/PWM/I2S/PCM TX&RX. The ATS3085Sx architecture is fully programmable and can easily be adapted to work with application. It also has the minimum package & boasts the most compact BOM in industry, making it an ideal SoC candidate for all-in-one communication & wearable application devices.

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## 1.2Features table

| Features | ATS3085S | ATS3085S4 |
|---|---|---|
| Package | QFN68 | |
| Body Size | 7x7x0.75 | |
| Application | Smart Watch | |
| CPU | ARM MSTAR | |
| DSP | CEVA TL420 | |
| 2D GPU | √ | |
| 2.5D GPU | √ | |
| RAM(KB) | 1168 | |
| Nor flash | External, 4xIO | |
| PSRAM | Built-in 64Mbit, 8xIO, DDR | Built-in 32Mbit, 8xIO, DDR |
| Bluetooth | 5.3 | |
| Power | Li-ion | |
| Charger | 400mA(Max) | |
| DCDC | 2 | |
| Shipping Mode | 2uA | |
| Standby | 20uA | |
| GPIO | 45 | |
| WIO | 4(at most) | |
| SARADC channel | 6 | |
| LCD | QSPI/QSPI DDR/SPI/2lane SPI/CPU8080/TR-J/TR-S parallel HSYNC/VSYNC | |
| SDC | 2 | |
| Audio DAC | Differential/Single-End | |
| I2S/PCM | 1*I2S IN+2*I2S OUT | |
| Microphones | 2 channel Single-End/1 channel Differential/2DMIC | |
| USB 2.0 | FS | |
| UART | 3 | |
| QSPI | 3 | |
| SPI(Include QSPI) | 4 | |
| SPIMT | 2 | |
| I2C | 4 | |
| I2CMT | 2 | |
| PWM | 4 | |
| LOSC | √ | |

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## 1.3Application Diagram

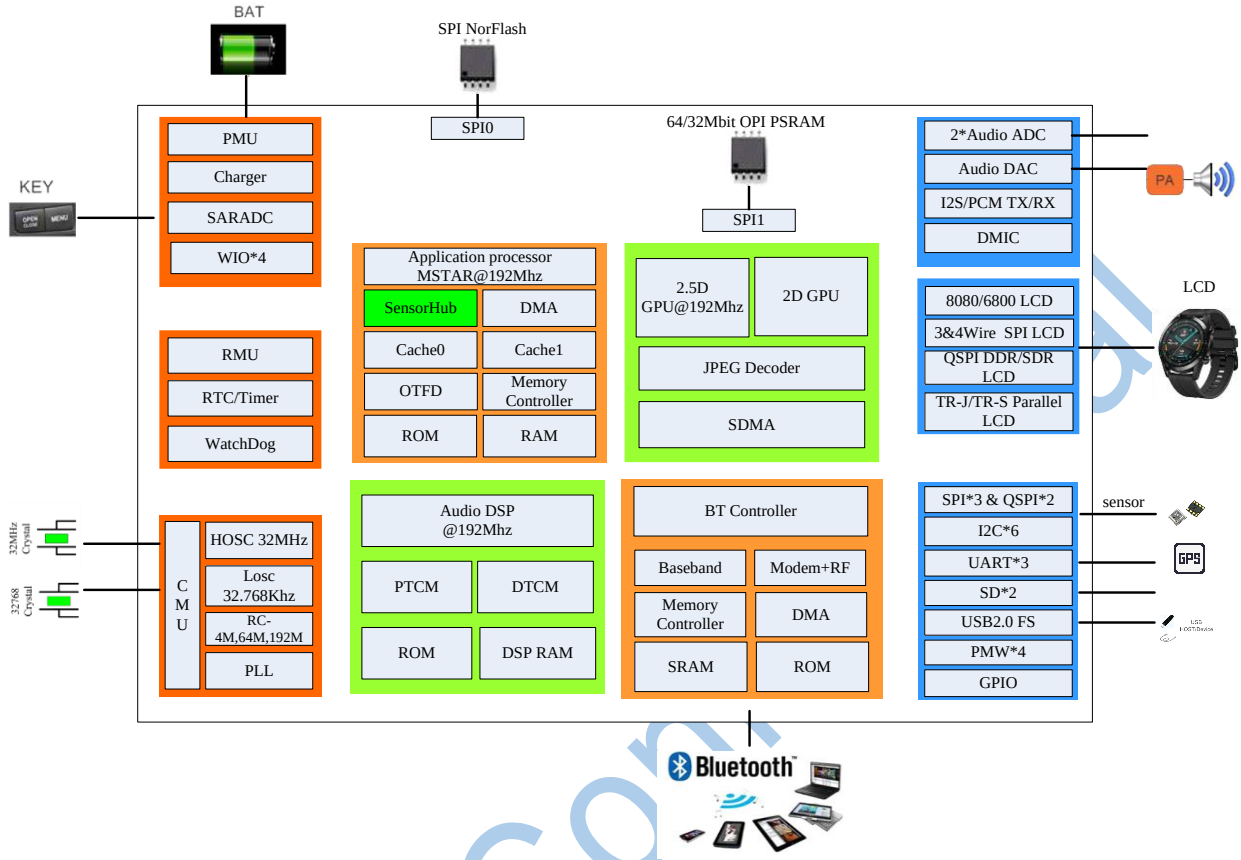


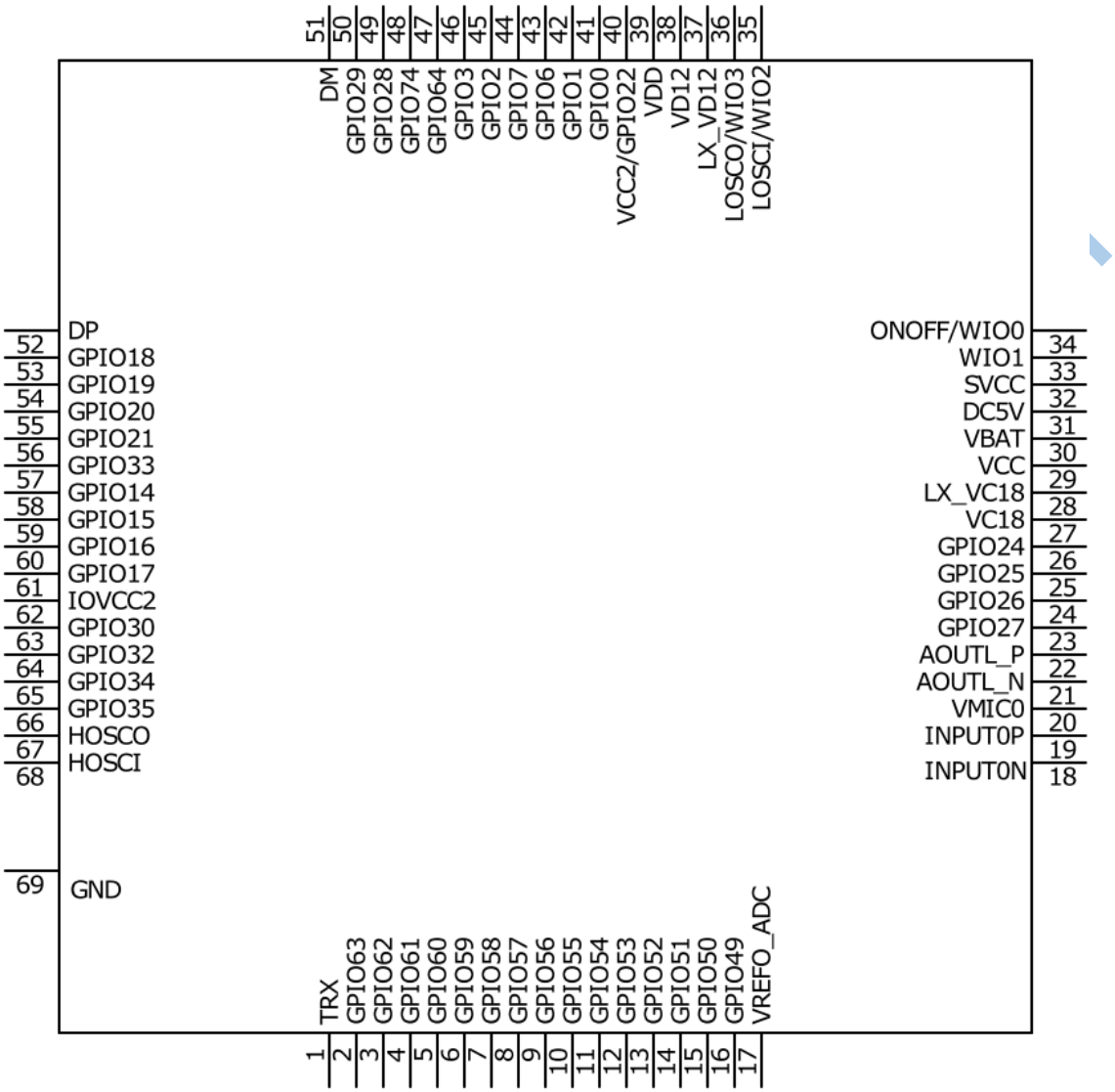

*Figure 1-1 ATS3085Sx Application Diagram*

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# 1.4Pin Assignment and Descriptions

## 1.4.1 Pin Assignment




## 1.4.2 Pin Description

***Table 1-1 ATS3085Sx Pin Description***

| Pin No. | Pin Name | Pin Type | Pad drive level | GPIO Initial state | MFP Description (5bits) | | Power domain |
|---|---|---|---|---|---|---|---|
| | | | | | AD_SELECT=0 | AD_SELECT=1 | |
| 1 | TRX | RF | | | Bluetooth antenna IO | | |
| 2 | GPIO63 | I/O | Level 1/2/3/4 | Z | Bit63 of General purpose I/O port<br>GPIO<br>SPI2_MISO<br>SPI3_MISO<br>UART0_TX | | VC18 |

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| | | | | | | |
|---|---|---|---|---|---|---|
| | | | | | UART1_TX<br>UART2_TX<br>SD0_CMD<br>PPI_TRIGGER6<br>IISG0_MCLK<br>IISG1_MCLK<br>I2C3_DAT<br>IISTX1_DOUT<br>SPIMT1_SS2<br>LRADC3<br>HOSCOUT<br>LOSCOUT<br>IOVCC4_OUT(1.8v) | |
| 3 | GPIO62 | I/O | Level 1/2/3/4 | Z | Bit62 of General purpose I/O port<br>GPIO<br>SPI3_IO3<br>UART0_RX<br>UART1_RX<br>UART2_RX<br>SD0_DAT1<br>PPI_TRIGGER5<br>IISTX0_DOUT<br>IISRX0_DIN<br>I2C2_DAT<br>SPIMT1_SS1<br>LRADC4 | VC18 |
| 4 | GPIO61 | I/O | Level 1/2/3/4 | Z | Bit61 of General purpose I/O port<br>GPIO<br>SPI3_IO2<br>SD0_DAT0<br>PPI_TRIGGER4<br>IISG0_LRCLK<br>IISG1_LRCLK<br>I2C2_CLK<br>SPIMT1_SS3<br>LRADC5<br>HOSCOUT<br>LOSCOUT | VC18 |
| 5 | GPIO60 | I/O | Level 1/2/3/4 | Z | Bit60 of General purpose I/O port<br>GPIO<br>SPI2_MOSI<br>I2C1_DAT<br>SD0_CLK0<br>PPI_TRIGGER11<br>IISG0_BCLK<br>IISG1_BCLK<br>I2C3_CLK<br>I2CMT1_DAT<br>HOSCOUT<br>LOSCOUT | VC18 |
| 6 | GPIO59 | I/O | Level 1/2/3/4 | Z | Bit59 of General purpose I/O port<br>GPIO<br>SPI2_MISO<br>I2C1_CLK | VC18 |

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| | | | | | | |
|---|---|---|---|---|---|---|
| | | | | | SD0_CLK1<br>PPI_TRIGGER10<br>IISG0_MCLK<br>IISG1_MCLK<br>I2C3_DAT<br>SPIMT0_SS1<br>I2CMT1_CLK<br>WCI_RX | |
| 7 | GPIO58 | I/O | Level 1/2/3/4 | Z | Bit58 of General purpose I/O port<br>GPIO<br>SPI2_CLK<br>I2C0_DAT<br>SD0_DAT3<br>PPI_TRIGGER9<br>IISTX0_DOUT<br>I2C3_CLK<br>SPIMT0_SS3<br>I2CMT0_DAT<br>WCI_TX | VC18 |
| 8 | GPIO57 | I/O | Level 1/2/3/4 | Z | Bit57 of General purpose I/O port<br>GPIO<br>SPI2_SS<br>I2C0_CLK<br>SD0_DAT2<br>PPI_TRIGGER8<br>I2C2_DAT<br>PWM1<br>IISTX1_DOUT<br>SPIMT0_SS2<br>I2CMT0_CLK<br>HOSCOUT<br>LOSCOUT | VC18 |
| 9 | GPIO56 | I/O | Level 1/2/3/4 | Z | Bit56 of General purpose I/O port<br>GPIO<br>SPI2_MOSI<br>SPI3_MOSI<br>UART2_RTS<br>SD0_DAT7<br>PPI_TRIGGER7<br>IISTX0_DOUT<br>IISRX0_DIN<br>I2C2_CLK<br>I2C3_DAT<br>PWM3<br>SPIMT1_MOSI<br>I2CMT1_DAT | VC18 |
| 10 | GPIO55 | I/O | Level 1/2/3/4 | Z | Bit55 of General purpose I/O port<br>GPIO<br>SPI2_MISO<br>SPI3_MISO<br>UART2_CTS<br>I2C1_DAT<br>SD0_DAT6<br>PPI_TRIGGER6 | VC18 |

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| | | | | | | |
|---|---|---|---|---|---|---|
| | | | | | IISG1_LRCLK<br>I2C3_CLK<br>PWM2<br>SD1_CLK<br>SPIMT1_MISO<br>I2CMT1_CLK | |
| 11 | GPIO54 | I/O | Level 1/2/3/4 | Z | Bit54 of General purpose I/O port<br>GPIO<br>SPI2_CLK<br>SPI3_CLK<br>UART2_RX<br>I2C1_CLK<br>SD0_DAT5<br>PPI_TRIGGER5<br>IISG1_BCLK<br>I2C2_DAT<br>PWM1<br>SD1_DAT3<br>SPIMT1_CLK<br>I2CMT0_DAT | VC18 |
| 12 | GPIO53 | I/O | Level 1/2/3/4 | Z | Bit53 of General purpose I/O port<br>GPIO<br>SPI2_SS<br>SPI3_SS<br>UART2_TX<br>I2C0_DAT<br>SD0_DAT4<br>PPI_TRIGGER4<br>IISG1_MCLK<br>I2C2_CLK<br>PWM0<br>SD1_DAT2<br>SPIMT1_SS0<br>I2CMT0_CLK<br>HOSCOUT<br>LOSCOUT | VC18 |
| 13 | GPIO52 | I/O | Level 1/2/3/4 | Z | Bit52 of General purpose I/O port<br>GPIO<br>SPI2_MOSI<br>UART1_RTS<br>I2C0_CLK<br>I2C1_DAT<br>PPI_TRIGGER3<br>IISTX0_DOUT<br>I2C3_DAT<br>PWM3<br>SD1_DAT0<br>IISTX1_DOUT<br>SPIMT0_MOSI<br>I2CMT1_DAT | VC18 |
| 14 | GPIO51 | I/O | Level 1/2/3/4 | Z | Bit51 of General purpose I/O port<br>GPIO<br>SPI2_MISO<br>UART1_CTS | VC18 |

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| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | I2C1_CLK<br>PPI_TRIGGER2<br>I2C3_CLK<br>PWM2<br>SD1_DAT1<br>PTA_GRANT<br>SPIMT0_MISO<br>I2CMT1_CLK | | |
| 15 | GPIO50 | I/O | Level 1/2/3/4 | Z | Bit50 of General purpose I/O port<br>GPIO<br>SPI2_CLK<br>UART1_RX<br>I2C0_DAT<br>PPI_TRIGGER1<br>I2C2_DAT<br>PWM1<br>SD1_CLK<br>BT_ACCESS<br>SPIMT0_CLK<br>I2CMT0_DAT | | VC18 |
| 16 | GPIO49 | I/O | Level 1/2/3/4 | Z | Bit49 of General purpose I/O port<br>GPIO<br>SPI2_SS<br>UART1_TX<br>I2C0_CLK<br>PPI_TRIGGER0<br>I2C2_CLK<br>PWM0<br>SD1_CMD<br>BT_REQ<br>SPIMT0_SS0<br>I2CMT0_CLK | | VC18 |
| 17 | VREF0_ADC | AO | | | Voltage reference of audio ADC | | |
| 18 | INPUT0N | I/O | Level 1/2/3/4 | L | Bit91 of GPIO | INPUT0N/INPUT1 | VC18 |
| 19 | INPUT0P | I/O | Level 1/2/3/4 | L | Bit90 of GPIO | INPUT0P/INPUT0 | VC18 |
| 20 | VMIC0 | PWR | | | Power supply of MIC | | VC18 |
| 21 | AOUTN | I/O | Level 1/2/3/4 | L | Bit95 of GPIO | Audio DAC differential output(AOUTN); | VC18 |

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| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 22 | AOUTP | I/O | Level 1/2/3/4 | L | Bit94 of GPIO | Audio DAC differential output(AOUTP);<br><br>Audio DAC single end output(AOUT) | VC18 |
| 23 | GPIO27 | I/O | Level 1/2/3/4 | X | Bit27 of General purpose I/O port<br>GPIO<br>SPI3_IO3<br>UART2_RX<br>I2C1_DAT<br>SD0_DAT5<br>PPI_TRIGGER1<br>IISTX0_DOUT<br>I2C3_CLK<br>LCD_D7<br>SD1_CLK<br>SPIMT1_MOSI<br>I2CMT1_DAT<br>LRADC6<br>VCC_OUT(3.3v) | | VCC |
| 24 | GPIO26 | I/O | Level 1/2/3/4 | Z | Bit26 of General purpose I/O port<br>GPIO<br>SPI3_SS<br>UART2_TX<br>I2C1_CLK<br>SD0_DAT4<br>PPI_TRIGGER0<br>IISG0_LRCLK<br>IISG1_LRCLK<br>I2C2_DAT<br>LCD_D6<br>SD1_CMD<br>SPIMT1_MISO<br>I2CMT1_CLK<br>VCC_OUT(3.3v) | | VCC |
| 25 | GPIO25 | I/O | Level 1/2/3/4 | Z | Bit25 of General purpose I/O port<br>GPIO<br>SPI3_IO2<br>UART1_RX<br>I2C0_DAT<br>SD0_CMD<br>PPI_TRIGGER11<br>IISG0_BCLK<br>IISG1_BCLK<br>I2C2_CLK<br>I2C3_DAT<br>LCD_D5<br>SPIMT1_CLK<br>I2CMT0_DAT<br>LRADC5<br>HOSCOUT | | VCC |

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| | | | | | | |
|---|---|---|---|---|---|---|
| | | | | | LOSCOUT<br>WCI_TX | |
| 26 | GPIO24 | I/O | Level 1/2/3/4 | Z | Bit24 of General purpose I/O port<br>GPIO<br>SPI3_CLK<br>UART1_TX<br>I2C0_CLK<br>I2C1_DAT<br>SD0_CLK0<br>PPI_TRIGGER10<br>IISG0_MCLK<br>IISG1_MCLK<br>I2C3_CLK<br>LCD_D4<br>SPIMT0_SS1<br>SPIMT1_SS0<br>I2CMT0_CLK<br>WCI_RX | VCC |
| 27 | VC18 | PWR | | | Output of buck VC18(1.8V) | VC18 |
| 28 | LX_VC18 | PWR | | | LX of buck VC18 | VC18 |
| 29 | VCC | PWR | | | Output of regulator VCC | VCC |
| 30 | VBAT | PWR | | | Battery Voltage input | VBAT |
| 31 | DC5V | PWR | | | 5.0V Voltage | DC5V |
| 32 | SVCC | | | | Standby power supply regulator | SVCC |
| 33 | WIO1 | I/O | Level 1/2/3/4 | Z | Bit1 of wake up I/O<br>LRADC1<br>I2C1_DAT<br>LOSCOUT | SVCC |
| 34 | WIO0 | I/O | Level 1/2/3/4 | Z | Bit0 of wake up I/O<br>ONOFF<br>I2C1_CLK<br>LRADC1 | SVCC |
| 35 | LOSCI | AI | Level 1/2/3/4 | Z | Bit2 of wake up I/O<br>I2C0_CLK<br>Input of LOSC(32768Hz) | SVCC |
| 36 | LOSCO | AO | Level 1/2/3/4 | Z | Bit3 of wake up I/O<br>I2C0_DAT<br>LOSCOUT<br>Output of LOSC(32768Hz) | SVCC |
| 37 | LX_VD12 | PWR | | | LX of buck VD12 | VD12 |
| 38 | VD12 | PWR | | | Output of buck VD12(0.6~1.4V) | VD12 |
| 39 | VDD | PWR | | | Output of regulator VDD(0.6~1.3V) | VDD |
| 40 | VCC2 | PWR | | | Output of regulator VCC2 | VCC2 |
| | GPIO22 | I/O | Level 1/2/3/4 | Z | Bit22 of General purpose I/O port<br>GPIO<br>SPI3_MISO<br>UART2_CTS<br>I2C0_DAT<br>SD0_DAT6 | VCC |

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|---|---|---|---|---|---|---|
| | | | | | PPI_TRIGGER8<br>I2C2_CLK<br>PWM1<br>SD1_DAT0<br>IISTX1_DOUT<br>SPIMT0_SS3<br>SPIMT1_MISO<br>I2CMT0_CLK<br>HOSCOUT<br>LOSCOUT | |
| 41 | GPIO0 | I/O | Level 1/2/3/4 | X | Bit0 of General purpose I/O port<br>GPIO<br>SPI0_SS<br>SD0_CMD<br>SPI0_MOSI | VCC |
| 42 | GPIO1 | I/O | Level 1/2/3/4 | X | Bit1 of General purpose I/O port<br>GPIO<br>SPI0_MISO<br>SD0_CLK0<br>SPI0_CLK | VCC |
| 43 | GPIO6 | I/O | Level 1/2/3/4 | X | Bit6 of General purpose I/O port<br>GPIO<br>SPI0_D2<br>UART1_CTS<br>I2C1_CLK<br>SD0_DAT2<br>DMIC01_CLK<br>PWM2 | VCC |
| 44 | GPIO7 | I/O | Level 1/2/3/4 | X | Bit7 of General purpose I/O port<br>GPIO<br>SPI0_D3<br>UART1_RTS<br>I2C1_DAT<br>SD0_DAT3<br>DMIC01_DAT<br>PWM3 | VCC |
| 45 | GPIO2 | I/O | Level 1/2/3/4 | X | Bit2 of General purpose I/O port<br>GPIO<br>SPI0_CLK<br>SD0_DAT0<br>SPI0_SS | VCC |
| 46 | GPIO3 | I/O | Level 1/2/3/4 | X | Bit3 of General purpose I/O port<br>GPIO<br>SPI0_MOSI<br>SD0_DAT1<br>PWM0<br>SPI0_MISO | VCC |
| 47 | GPIO64 | I/O | Level 1/2/3/4 | H | Bit64 of General purpose I/O port<br>GPIO<br>SPI2_MISO<br>SPI3_MISO<br>UART0_TX<br>UART1_TX<br>UART2_TX | VCC |

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|---|---|---|---|---|---|---|
| | | | | | SD0_DAT7<br>PPI_TRIGGER11<br>IISG0_MCLK<br>IISG1_MCLK<br>I2C2_CLK<br>I2C3_DAT<br>PWM0<br>SD1_DAT1<br>IISTX1_DOUT<br>SPIMT0_SS0<br>I2CMT1_DAT<br>HOSCOUT<br>LOSCOUT<br>IOVCC1_OUT(3.3v) | |
| 48 | GPIO74 | I/O | Level 1/2/3/4 | Z | Bit74 of General purpose I/O port<br>GPIO<br>SPI2_MOSI<br>SPI3_MOSI<br>UART0_RX<br>UART2_RX<br>I2C0_CLK<br>I2C1_DAT<br>SD0_DAT6<br>PPI_TRIGGER4<br>IISG0_BCLK<br>IISG1_BCLK<br>I2C2_DAT<br>I2C3_CLK<br>DMIC01_CLK<br>PWM1<br>SD1_DAT0<br>SPIMT0_CLK<br>I2CMT1_CLK<br>LRADC6<br>HOSCOUT<br>LOSCOUT<br>WCI_RX | VCC |
| 49 | GPIO28 | I/O | Level 1/2/3/4 | X | Bit28 of General purpose I/O port<br>GPIO<br>SPI3_IO2<br>UART0_TX<br>I2C0_CLK<br>SD0_DAT6<br>PPI_TRIGGER2<br>IISTX0_DOUT<br>I2C2_CLK<br>SD1_DAT1<br>BT_UART_TX<br>SPIMT1_SS0<br>I2CMT0_CLK | VCC |
| 50 | GPIO29 | I/O | Level 1/2/3/4 | X | Bit29 of General purpose I/O port<br>GPIO<br>GPIO29<br>SPI2_SS | VCC |

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|---|---|---|---|---|---|---|
| | | | | | SPI3_IO3<br>UART0_RX<br>I2C0_DAT<br>SD0_DAT7<br>PPI_TRIGGER3<br>IISRX0_DIN<br>I2C2_DAT<br>SD1_DAT0<br>IISTX1_DOUT<br>BT_UART_RX<br>SPIMT1_SS1<br>I2CMT0_DAT | |
| 51 | GPIO76 | I/O | Level 1/2/3/4 | Firmware upgrade (L),<br><br>Firmware boot(L) | Bit76 of General purpose I/O port<br>GPIO<br>SPI2_MOSI<br>SPI3_MOSI<br>UART0_RX<br>UART2_TX<br>IISG0_LRCLK<br>IISG1_LRCLK<br>I2C2_DAT<br>USBDM<br>DMIC01_DAT<br>PWM3<br>SD1_CLK<br>BT_UART_RX<br>SPIMT1_SS2<br>I2CMT0_DAT<br>I2CMT1_DAT<br>WCI_RX | VCC |
| 52 | GPIO75 | I/O | Level 1/2/3/4 | Firmware upgrade (H),<br><br>Firmware boot(L) | Bit75 of General purpose I/O port<br>GPIO<br>SPI2_MISO<br>SPI3_MISO<br>UART0_TX<br>UART2_RX<br>IISG0_BCLK<br>IISG1_BCLK<br>I2C2_CLK<br>USBDP<br>DMIC01_CLK<br>PWM2<br>SD1_DAT3<br>BT_UART_TX<br>SPIMT1_SS3<br>I2CMT0_CLK<br>I2CMT1_CLK<br>WCI_TX | VCC |
| 53 | GPIO18 | I/O | Level 1/2/3/4 | Z | Bit18 of General purpose I/O port<br>GPIO<br>SPI2_MISO<br>SPI3_MISO<br>UART1_CTS<br>I2C0_CLK | VCC |

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| | | | | | | |
|---|---|---|---|---|---|---|
| | | | | | SD0_DAT0<br>PPI_TRIGGER4<br>IISTX0_DOUT<br>IISRX0_DIN<br>PWM1<br>LCD_D4<br>SD1_DAT2<br>IISTX1_DOUT<br>PTA_GRANT<br>SPIMT0_MISO<br>SPIMT1_MISO<br>I2CMT0_CLK<br>I2CMT1_CLK | |
| 54 | GPIO19 | I/O | Level 1/2/3/4 | Z | Bit19 of General purpose I/O port<br>GPIO<br>SPI2_MOSI<br>SPI3_MOSI<br>UART1_RTS<br>I2C0_DAT<br>SD0_DAT1<br>PPI_TRIGGER5<br>IISG0_MCLK<br>IISG1_MCLK<br>PWM2<br>LCD_D5<br>SD1_DAT3<br>SPIMT0_MOSI<br>SPIMT1_MOSI<br>I2CMT0_DAT<br>I2CMT1_DAT<br>HOSCOUT<br>LOSCOUT | VCC |
| 55 | GPIO20 | I/O | Level 1/2/3/4 | Z | Bit20 of General purpose I/O port<br>GPIO<br>SPI2_SS<br>SPI3_SS<br>UART1_TX<br>UART2_TX<br>I2C1_CLK<br>SD0_DAT2<br>PPI_TRIGGER6<br>IISG0_BCLK<br>IISG1_BCLK<br>PWM3<br>LCD_D6<br>SPIMT0_SS0<br>SPIMT1_SS0<br>I2CMT1_CLK<br>LRADC2 | VCC |
| 56 | GPIO21 | I/O | Level 1/2/3/4 | Z | Bit21 of General purpose I/O port<br>GPIO<br>SPI2_CLK<br>SPI3_CLK<br>UART1_RX | VCC |

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| | | | | | | |
|---|---|---|---|---|---|---|
| | | | | | UART2_RX<br>I2C0_CLK<br>I2C1_DAT<br>SD0_DAT3<br>PPI_TRIGGER7<br>IISG0_LRCLK<br>IISG1_LRCLK<br>PWM0<br>LCD_D7<br>SPIMT0_CLK<br>SPIMT1_CLK<br>I2CMT1_DAT<br>LRADC3 | |
| 57 | GPIO33 | I/O | Level 1/2/3/4 | Z | Bit33 of General purpose I/O port<br>GPIO<br>SPI2_MOSI<br>I2C0_DAT<br>PPI_TRIGGER2<br>LCD_RD#<br>SPIMT0_MOSI<br>I2CMT0_DAT<br>IOVCC2_OUT(1.8/3.3v) | IOVCC2 |
| 58 | GPIO14 | I/O | Level 1/2/3/4 | Z | Bit14 of General purpose I/O port<br>GPIO<br>UART0_TX<br>I2C0_CLK<br>PPI_TRIGGER0<br>IISTX0_DOUT<br>PWM0<br>LCD_D0<br>BT_UART_TX<br>SPIMT0_SS3<br>SPIMT1_MISO<br>I2CMT0_CLK | IOVCC2 |
| 59 | GPIO15 | I/O | Level 1/2/3/4 | Z | Bit15 of General purpose I/O port<br>GPIO<br>UART0_RX<br>I2C0_DAT<br>PPI_TRIGGER1<br>IISRX0_DIN<br>PWM1<br>LCD_D1<br>BT_UART_RX<br>SPIMT0_SS2<br>SPIMT1_MOSI<br>I2CMT0_DAT | IOVCC2 |
| 60 | GPIO16 | I/O | Level 1/2/3/4 | Z | Bit16 of General purpose I/O port<br>GPIO<br>SPI2_SS<br>UART1_TX<br>I2C1_DAT<br>PPI_TRIGGER2<br>IISG0_MCLK<br>IISG1_MCLK | IOVCC2 |

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| | | | | | | |
|---|---|---|---|---|---|---|
| | | | | | PWM2<br>LCD_D2<br>BT_REQ<br>SPIMT1_SS1<br>I2CMT1_DAT | |
| 61 | GPIO17 | I/O | Level 1/2/3/4 | Z | Bit17 of General purpose I/O port<br>GPIO<br>SPI2_CLK<br>UART1_RX<br>I2C1_CLK<br>PPI_TRIGGER3<br>IISG0_BCLK<br>IISG1_BCLK<br>I2C3_CLK<br>PWM3<br>LCD_D3<br>BT_ACCESS<br>SPIMT1_SS0<br>I2CMT1_CLK | IOVCC2 |
| 62 | IOVCC2 | PWR | | | Power supply for GPIO Group2(1.8/3.1V) | IOVCC2 |
| 63 | GPIO30 | I/O | Level 1/2/3/4 | Z | Bit30 of General purpose I/O port<br>GPIO<br>SPI2_SS<br>PPI_TRIGGER0<br>IISG0_LRCLK<br>IISG1_LRCLK<br>I2C2_DAT<br>I2C3_DAT<br>DMIC01_CLK<br>LCD_CS0<br>SPIMT1_CLK | IOVCC2 |
| 64 | GPIO32 | I/O | Level 1/2/3/4 | Z | Bit32 of General purpose I/O port<br>GPIO<br>SPI2_MISO<br>I2C0_CLK<br>PPI_TRIGGER1<br>I2C2_CLK<br>LCD_RS_CD<br>IISTX1_DOUT<br>SPIMT0_MISO<br>I2CMT0_CLK<br>HOSCOUT<br>LOSCOUT | IOVCC2 |
| 65 | GPIO34 | I/O | Level 1/2/3/4 | Z | Bit34 of General purpose I/O port<br>GPIO<br>SPI2_CLK<br>I2C1_CLK<br>PPI_TRIGGER3<br>DMIC01_CLK<br>LCD_WR#<br>BT_UART_TX<br>SPIMT0_CLK<br>I2CMT1_CLK | IOVCC2 |

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| | | | | | | |
|---|---|---|---|---|---|---|
| 66 | GPIO35 | I/O | Level 1/2/3/4 | Z | Bit35 of General purpose I/O port<br>GPIO<br>SPI2_MISO<br>I2C1_DAT<br>PPI_TRIGGER4<br>I2C3_CLK<br>DMIC01_DAT<br>LCD_TE<br>BT_UART_RX<br>SPIMT0_SS1<br>I2CMT0_CLK<br>I2CMT1_DAT<br>LRADC4 | IOVCC2 |
| 67 | HOSCO | AO | | | Output of HOSC(32MHz) | |
| 68 | HOSCI | AI | | | Input of HOSC(32MHz) | |
| 69 | GND | | | | The GND of the chip | |

Note: H: high level; L:low level; Z: high resistance; X: May be change in power on;
PWR: power supply PIN; I/O: GPIO; AI: analog input PIN; AO: analog output PIN;
GND: Ground pin;

### 1.4.3 Multi-Function of GPIO Pins

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| Voltage | Power Domain | Pin Name | Analog Function | Pin Multi-Function | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 |
| 3.3 | IOVCC5=VCC | GPIO0 | | GPIO | SPI0_SS | | | | | | | | | SD0_CMD | | | | | | | | | | | | SPI0_MOSI | | | | | | | | |
| 3.3 | IOVCC5=VCC | GPIO1 | | GPIO | SPI0_MISO | | | | | | | | | SD0_CLK0 | | | | | | | | | | | | SPI0_CLK | | | | | | | | |
| 3.3 | IOVCC5=VCC | GPIO6 | | GPIO | SPI0_D2 | | | | | UART1_CTS | | | I2C1_CLK | SD0_DAT2 | | | | | | | DMIC01_CLK | PWM2 | | | | | | | | | | | |
| 3.3 | IOVCC5=VCC | GPIO7 | | GPIO | SPI0_D3 | | | | | UART1_RTS | | | I2C1_DAT | SD0_DAT3 | | | | | | | DMIC01_DAT | PWM3 | | | | | | | | | | | |
| 3.3 | IOVCC5=VCC | GPIO2 | | GPIO | SPI0_CLK | | | | | | | | | SD0_DAT0 | | | | | | | | | | | | SPI0_SS | | | | | | | | |
| 3.3 | IOVCC5=VCC | GPIO3 | | GPIO | SPI0_MOSI | | | | | | | | | SD0_DAT1 | | | | | | | | PWM0 | | | | SPI0_MISO | | | | | | | | |
| 3.3 | IOVCC1=VCC | GPIO64 | | GPIO | | | SPI2_MISO | SPI3_MISO | UART0_TX | UART1_TX | UART2_TX | | | SD0_DAT7 | PPI_Trigger11 | IISG0_MCLK | IISG1_MCLK | I2C2_CLK | I2C3_DAT | | | PWM0 | | SD1_DAT1 | IISTX1_DOUT | | | SPIMT0_SS0 | | | I2CMT1_DAT | | HOSCOUT | LOSCOUT | IOVCC1_OUT |
| 3.3 | IOVCC1=VCC | GPIO74 | | GPIO | | | SPI2_MOSI | SPI3_MOSI | UART0_RX | | UART2_RX | I2C0_CLK | I2C1_DAT | SD0_DAT6 | PPI_Trigger4 | IISG0_BCLK | IISG1_BCLK | I2C2_DAT | I2C3_CLK | | DMIC01_CLK | PWM1 | | SD1_DAT0 | | | | SPIMT0_CLK | | | I2CMT1_CLK | LRADC6 | HOSCOUT | LOSCOUT | WCI_RX |
| 1.8/3.3 | IOVCC2 | GPIO14 | | GPIO | | | | | UART0_TX | | | I2C0_CLK | | | PPI_Trigger0 | IISTX0_DOUT | | | | | | PWM0 | LCD_D0 | | | | BT_UART_TX | SPIMT0_SS3 | SPIMT1_MISO | I2CMT0_CLK | | | | |
| 1.8/3.3 | IOVCC2 | GPIO15 | | GPIO | | | | | UART0_RX | | | I2C0_DAT | | | PPI_Trigger1 | | IISRX0_DIN | | | | | PWM1 | LCD_D1 | | | | BT_UART_RX | SPIMT0_SS2 | SPIMT1_MOSI | I2CMT0_DAT | | | | |
| 1.8/3.3 | IOVCC2 | GPIO16 | | GPIO | | | SPI2_SS | | | UART1_TX | | | I2C1_DAT | | PPI_Trigger2 | IISG0_MCLK | IISG1_MCLK | | | | | PWM2 | LCD_D2 | | | | BT_REQ | | SPIMT1_SS1 | | I2CMT1_DAT | | | |
| 1.8/3.3 | IOVCC2 | GPIO17 | | GPIO | | | SPI2_CLK | | | UART1_RX | | | I2C1_CLK | | PPI_Trigger3 | IISG0_BCLK | IISG1_BCLK | | I2C3_CLK | | | PWM3 | LCD_D3 | | | | BT_ACCESS | | SPIMT1_SS0 | | I2CMT1_CLK | | | |
| 1.8/3.3 | IOVCC2 | GPIO30 | | GPIO | | | SPI2_SS | | | | | | | | PPI_Trigger0 | IISG0_LRCLK | IISG1_LRCLK | I2C2_DAT | I2C3_DAT | | DMIC01_CLK | | LCD_CS0 | | | | | | SPIMT1_CLK | | | | | |
| 1.8/3.3 | IOVCC2 | GPIO32 | | GPIO | | | SPI2_MISO | | | | | I2C0_CLK | | | PPI_Trigger1 | | | I2C2_CLK | | | | | LCD_RS_CD | | IISTX1_DOUT | | | SPIMT0_MISO | | I2CMT0_CLK | | HOSCOUT | LOSCOUT | |
| 1.8/3.3 | IOVCC2 | GPIO33 | | GPIO | | | SPI2_MOSI | | | | | I2C0_DAT | | | PPI_Trigger2 | | | | | | | | LCD_RD# | | | | | SPIMT0_MOSI | | I2CMT0_DAT | | | | IOVCC2_OUT |
| 1.8/3.3 | IOVCC2 | GPIO34 | | GPIO | | | SPI2_CLK | | | | | | I2C1_CLK | | PPI_Trigger3 | | | | | | DMIC01_CLK | | LCD_WR# | | | BT_UART_TX | SPIMT0_CLK | | | I2CMT1_CLK | | | | |
| 1.8/3.3 | IOVCC2 | GPIO35 | | GPIO | | | SPI2_MISO | | | | | | I2C1_DAT | | PPI_Trigger4 | | | | I2C3_CLK | | DMIC01_DAT | | LCD_TE | | | BT_UART_RX | SPIMT0_SS1 | | I2CMT0_CLK | I2CMT1_DAT | LRADC4 | | | |
| 3.3 | VCC | GPIO18 | | GPIO | | | SPI2_MISO | SPI3_MISO | | UART1_CTS | | I2C0_CLK | | SD0_DAT0 | PPI_Trigger4 | IISTX0_DOUT | IISRX0_DIN | | | | | PWM1 | LCD_D4 | SD1_DAT2 | IISTX1_DOUT | | PTA_GRANT | SPIMT0_MISO | SPIMT1_MISO | I2CMT0_CLK | I2CMT1_CLK | | | | |
| 3.3 | VCC | GPIO19 | | GPIO | | | SPI2_MOSI | SPI3_MOSI | | UART1_RTS | | I2C0_DAT | | SD0_DAT1 | PPI_Trigger5 | IISG0_MCLK | IISG1_MCLK | | | | | PWM2 | LCD_D5 | SD1_DAT3 | | | | SPIMT0_MOSI | SPIMT1_MOSI | I2CMT0_DAT | I2CMT1_DAT | | HOSCOUT | LOSCOUT | |
| 3.3 | VCC | GPIO20 | | GPIO | | | SPI2_SS | SPI3_SS | | UART1_TX | UART2_TX | | I2C1_CLK | SD0_DAT2 | PPI_Trigger6 | IISG0_BCLK | IISG1_BCLK | | | | | PWM3 | LCD_D6 | | | | | SPIMT0_SS0 | SPIMT1_SS0 | | I2CMT1_CLK | LRADC2 | | | |
| 3.3 | VCC | GPIO21 | | GPIO | | | SPI2_CLK | SPI3_CLK | | UART1_RX | UART2_RX | I2C0_CLK | I2C1_DAT | SD0_DAT3 | PPI_Trigger7 | IISG0_LRCLK | IISG1_LRCLK | | | | | PWM0 | LCD_D7 | | | | | SPIMT0_CLK | SPIMT1_CLK | | I2CMT1_DAT | LRADC3 | | | |
| 3.3 | VCC | GPIO24 | | GPIO | | | | SPI3_CLK | | UART1_TX | | I2C0_CLK | I2C1_DAT | SD0_CLK0 | PPI_Trigger10 | IISG0_MCLK | IISG1_MCLK | | I2C3_CLK | | | | LCD_D4 | | | | | SPIMT0_SS1 | SPIMT1_SS0 | I2CMT0_CLK | | | | | WCI_RX |
| 3.3 | VCC | GPIO25 | | GPIO | | | | SPI3_IO2 | | UART1_RX | | I2C0_DAT | | SD0_CMD | PPI_Trigger11 | IISG0_BCLK | IISG1_BCLK | I2C2_CLK | I2C3_DAT | | | | LCD_D5 | | | | | | SPIMT1_CLK | I2CMT0_DAT | | LRADC5 | HOSCOUT | LOSCOUT | WCI_TX |
| 3.3 | VCC | GPIO26 | | GPIO | | | | SPI3_SS | | | UART2_TX | | I2C1_CLK | SD0_DAT4 | PPI_Trigger0 | IISG0_LRCLK | IISG1_LRCLK | I2C2_DAT | | | | | LCD_D6 | SD1_CMD | | | | | SPIMT1_MISO | | I2CMT1_CLK | | | | VCC_OUT |
| 3.3 | VCC | GPIO27 | | GPIO | | | | SPI3_IO3 | | | UART2_RX | | I2C1_DAT | SD0_DAT5 | PPI_Trigger1 | IISTX0_DOUT | | | I2C3_CLK | | | | LCD_D7 | SD1_CLK | | | | | SPIMT1_MOSI | | I2CMT1_DAT | LRADC6 | | | VCC_OUT |
| 3.3 | VCC | GPIO28 | | GPIO | | | | SPI3_IO2 | UART0_TX | | | I2C0_CLK | | SD0_DAT6 | PPI_Trigger2 | IISTX0_DOUT | | I2C2_CLK | | | | | | SD1_DAT1 | | | BT_UART_TX | | SPIMT1_SS0 | I2CMT0_CLK | | | | | |
| 3.3 | VCC | GPIO29 | | GPIO | | | SPI2_SS | SPI3_IO3 | UART0_RX | | | I2C0_DAT | | SD0_DAT7 | PPI_Trigger3 | | IISRX0_DIN | I2C2_DAT | | | | | | SD1_DAT0 | IISTX1_DOUT | | BT_UART_RX | | SPIMT1_SS1 | I2CMT0_DAT | | | | | |
| 1.8 | IOVCC4=VC18 | GPIO49 | | GPIO | | | SPI2_SS | | | UART1_TX | | I2C0_CLK | | | PPI_Trigger0 | | | I2C2_CLK | | | | PWM0 | | SD1_CMD | | | BT_REQ | SPIMT0_SS0 | | I2CMT0_CLK | | | | | |
| 1.8 | IOVCC4=VC18 | GPIO50 | | GPIO | | | SPI2_CLK | | | UART1_RX | | I2C0_DAT | | | PPI_Trigger1 | | | I2C2_DAT | | | | PWM1 | | SD1_CLK | | | BT_ACCESS | SPIMT0_CLK | | I2CMT0_DAT | | | | | |
| 1.8 | IOVCC4=VC18 | GPIO51 | | GPIO | | | SPI2_MISO | | | UART1_CTS | | | I2C1_CLK | | PPI_Trigger2 | | | | I2C3_CLK | | | PWM2 | | SD1_DAT1 | | | PTA_GRANT | SPIMT0_MISO | | | I2CMT1_CLK | | | | |
| 1.8 | IOVCC4=VC18 | GPIO52 | | GPIO | | | SPI2_MOSI | | | UART1_RTS | | I2C0_CLK | I2C1_DAT | | PPI_Trigger3 | IISTX0_DOUT | | | I2C3_DAT | | | PWM3 | | SD1_DAT0 | IISTX1_DOUT | | | SPIMT0_MOSI | | | I2CMT1_DAT | | | | |
| 1.8 | IOVCC4=VC18 | GPIO53 | | GPIO | | | SPI2_SS | SPI3_SS | | | UART2_TX | I2C0_DAT | | SD0_DAT4 | PPI_Trigger4 | | IISG1_MCLK | I2C2_CLK | | | | PWM0 | | SD1_DAT2 | | | | | SPIMT1_SS0 | I2CMT0_CLK | | | HOSCOUT | LOSCOUT | |
| 1.8 | IOVCC4=VC18 | GPIO54 | | GPIO | | | SPI2_CLK | SPI3_CLK | | | UART2_RX | | I2C1_CLK | SD0_DAT5 | PPI_Trigger5 | | IISG1_BCLK | I2C2_DAT | | | | PWM1 | | SD1_DAT3 | | | | | SPIMT1_CLK | I2CMT0_DAT | | | | | |
| 1.8 | IOVCC4=VC18 | GPIO55 | | GPIO | | | SPI2_MISO | SPI3_MISO | | | UART2_CTS | | I2C1_DAT | SD0_DAT6 | PPI_Trigger6 | | IISG1_LRCLK | | I2C3_CLK | | | PWM2 | | SD1_CLK | | | | | SPIMT1_MISO | | I2CMT1_CLK | | | | |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1.8 | IOVCC4=VC18 | GPIO56 | | GPIO | | | SPI2_MOSI | SPI3_MOSI | | | UART2_RTS | | | SD0_DAT7 | PPI_Trigger7 | IISTX0_DOUT | IISRX0_DIN | I2C2_CLK | I2C3_DAT | | | PWM3 | | | | | | | SPIMT1_MOSI | | I2CMT1_DAT | | | | |
| 1.8 | IOVCC4=VC18 | GPIO57 | | GPIO | | | SPI2_SS | | | | | I2C0_CLK | | SD0_DAT2 | PPI_Trigger8 | | | I2C2_DAT | | | | PWM1 | | | IISTX1_DOUT | | | SPIMT0_SS2 | | I2CMT0_CLK | | | HOSCOUT | LOSCOUT | |
| 1.8 | IOVCC4=VC18 | GPIO58 | | GPIO | | | SPI2_CLK | | | | | I2C0_DAT | | SD0_DAT3 | PPI_Trigger9 | IISTX0_DOUT | | | I2C3_CLK | | | | | | | | | SPIMT0_SS3 | | I2CMT0_DAT | | | | | WCI_TX |
| 1.8 | IOVCC4=VC18 | GPIO59 | | GPIO | | | SPI2_MISO | | | | | | I2C1_CLK | SD0_CLK1 | PPI_Trigger10 | IISG0_MCLK | IISG1_MCLK | | I2C3_DAT | | | | | | | | | SPIMT0_SS1 | | | I2CMT1_CLK | | | | WCI_RX |
| 1.8 | IOVCC4=VC18 | GPIO60 | | GPIO | | | SPI2_MOSI | | | | | | I2C1_DAT | SD0_CLK0 | PPI_Trigger11 | IISG0_BCLK | IISG1_BCLK | | I2C3_CLK | | | | | | | | | | | | I2CMT1_DAT | | HOSCOUT | LOSCOUT | |
| 1.8 | IOVCC4=VC18 | GPIO61 | | GPIO | | | | SPI3_IO2 | | | | | | SD0_DAT0 | PPI_Trigger4 | IISG0_LRCLK | IISG1_LRCLK | I2C2_CLK | | | | | | | | | | | SPIMT1_SS3 | | | LRADC5 | HOSCOUT | LOSCOUT | |
| 1.8 | IOVCC4=VC18 | GPIO62 | | GPIO | | | | SPI3_IO3 | UART0_RX | UART1_RX | UART2_RX | | | SD0_DAT1 | PPI_Trigger5 | IISTX0_DOUT | IISRX0_DIN | I2C2_DAT | | | | | | | | | | | SPIMT1_SS1 | | | LRADC4 | | | |
| 1.8 | IOVCC4=VC18 | GPIO63 | | GPIO | | | SPI2_MISO | SPI3_MISO | UART0_TX | UART1_TX | UART2_TX | | | SD0_CMD | PPI_Trigger6 | IISG0_MCLK | IISG1_MCLK | | I2C3_DAT | | | | | | IISTX1_DOUT | | | | SPIMT1_SS2 | | | LRADC3 | HOSCOUT | LOSCOUT | IOVCC4_OUT |
| 2.5 | SVCC | WIO0 | | WIO | ONOFF | I2C1_CLK | LRADC1 | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 2.5 | SVCC | WIO1 | | WIO | | I2C1_DAT | LRADC1 | | | LOSCOUT | | | | | | | | | | | | | | | | | | | | | | | | | |
| 2.5 | SVCC | WIO2 | LOSCI | WIO | | I2C0_CLK | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 2.5 | SVCC | WIO3 | LOSCO | WIO | | I2C0_DAT | | | | LOSCOUT | | | | | | | | | | | | | | | | | | | | | | | | | |
| 1.7 | AVCC | AOUT_P | AOUT/AOUT_P | GPIO | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 1.7 | AVCC | AOUT_N | AOUT_N | GPIO | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 1.7 | AVCC | INPUT0N | INPUT0N | GPIO | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 1.7 | AVCC | INPUT0P | INPUT0P | GPIO | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 3.3 | VCC | GPIO75 | | GPIO | | | SPI2_MISO | SPI3_MISO | UART0_TX | | UART2_RX | | | | | IISG0_BCLK | IISG1_BCLK | I2C2_CLK | | USBDP | DMIC01_CLK | PWM2 | | SD1_DAT3 | | | BT_UART_TX | | SPIMT1_SS3 | I2CMT0_CLK | I2CMT1_CLK | | | | WCI_TX |
| 3.3 | VCC | GPIO76 | | GPIO | | | SPI2_MOSI | SPI3_MOSI | UART0_RX | | UART2_TX | | | | | IISG0_LRCLK | IISG1_LRCLK | I2C2_DAT | | USBDM | DMIC01_DAT | PWM3 | | SD1_CLK1 | | | BT_UART_RX | | SPIMT1_SS2 | I2CMT0_DAT | I2CMT1_DAT | | | | WCI_RX |
| 3.3 | VCC | GPIO22 | | GPIO | | | | SPI3_MISO | | | UART2_CTS | I2C0_DAT | | SD0_DAT6 | PPI_Trigger8 | | | I2C2_CLK | | | | PWM1 | | SD1_DAT0 | IISTX1_DOUT | | | SPIMT0_SS3 | SPIMT1_MISO | I2CMT0_CLK | | | HOSCOUT | LOSCOUT | |

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# 2 Bluetooth

## 2.1Features

- Support Bluetooth V5.3
- Support LE Audio(LC3 Codec)
- Supports basic rate and enhanced data rate
- Supports Secure Simple Pairing
- Supports Low Power Mode
- Bluetooth Dual Mode support: Simultaneous LE and BR / EDR
- Supports multiple Low Energy states
- Fast AGC control to improve receiving dynamic range
- Supports AFH to dynamically detect channel quality to improve transmission quality
- Supports GFSK, π/4 DQPSK and 8DPSK modulation
- Supports Power / Enhanced Power Control
- LE Data Packet Length Extension
- LE 2M PHY
- Support long range BLE
- Support AOA/AOD
- Channel Selection Algorithm #2
- Support PTA (Packet Traffic Arbiter) when co-existing with WI-FI
- Support WCI2 when co-existing with 4G

## 2.2Bluetooth Performance

- Max transmitting output power: BR 13dBm, BLE 14dBm
- Bluetooth receiving sensitivity: -96.5dBm@GFSK, -95dBm@π/4 DQPSK, -89.5dBm@8DPSK modulation
- Bluetooth receiving sensitivity (BLE mode): -99.5dBm@1Mbps, -96dBm@2Mbps

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# 3 Processor Core

## 3.1Features

- 192MHz(Max) ARM MSTAR processor Core with FPU
- Support extended DSP instruction
- 32 bits data, 16/32 bits mixed coding command
- Ultra-low power consumption with integrated sleep modes.
- Platform robustness with optional integrated memory protection.
- 1kB trace RAM.

## 3.2Nested Vectored Interrupt Controller

### 3.2.1 Features

This section describes the NVIC and the registers it uses. The NVIC supports:

- 1 to 64 interrupts.
- A programmable priority level of 0-255 for each interrupt. A higher level corresponds to a lower priority, so level 0 is the highest interrupt priority.
- Level and pulse detection of interrupt signals.
- Dynamic reprioritization of interrupts.
- Grouping of priority values into group priority and sub-priority fields.
- Interrupt tail-chaining.
- An external Non-maskable interrupt (NMI)
- Optional WIC, providing ultra-low power sleep mode support.

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# 4 DSP Core

- 192MHz(Max) CEVA TL420 DSP Core
- Audio Configuration Features set
  - High code compactness due to:
    -- Two instruction widths (16-bit or 32-bit)
    -- Powerful instruction set capabilities
  - 10 stage pipeline
  - All instructions can be conditional:
    -- Conditional execution
    -- Reduces cycle count and code size on control and overhead code
  - Computational units:
    -- One 32-bit x 32-bit Multiply-and-Accumulate (MAC) using 72-bit product
    -- One 32-bit x 16-bit MAC using 72-bit product
    -- One 32-bit x 32-bit MAC unit with automatic scaling
    -- One 32-bit x 16-bit MAC unit with automatic scaling
    -- One 36-bit arithmetic unit
    -- One 36-bit logical unit
    -- One 36-bit bit-manipulation unit, including a full barrel shifter and an exponent unit
    -- Four 36-bit accumulators
    -- Fully programmable product post-shifter for product scaling
  - 32-bit Scalar (SC) unit for integer operations
  - Two address generation units for two independent accesses to the data memories
    -- Up to two 32-bit load operations in parallel
    -- Single 32-bit store operation
  - Unaligned memory access for load and store operations
  - Multiple data addressing modes:
    -- Indirect addressing
    -- Modulo addressing
    -- Bit-reversed addressing
    -- Direct addressing
    -- Indexed addressing
    -- Stack addressing
  - Support for scaling data after multiplications or before storing to data memory
  - Three different saturation modes
  - Single-cycle exponent evaluation of 36-bit values
  - Four-cycle FFT double precision butterfly
  - Bits unpacking capabilities (for Huffman algorithms)
  - 2G byte program and data space
  - Word-addressable data space
  - Eight 32-bit memory addressing registers
  - Separate I/O space for peripherals
  - Fully registered memory interface
  - Direct memory access for upload and download to TCM
- Voice Configuration Feature Set
  - Two 16-bit x 16-bit MAC units
  - SIMD operations in word precisions
  - Advanced support for split operations in 16-bit precision
  - Two-cycle FFT single precision butterfly
  - Single-cycle division step support
- Legacy Compatibility Configuration Feature Set
  - Up to two 32-bit store operations in parallel
  - Viterbi instructions acceleration

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- Automatic context-switch operation on interrupts (task switching)
- Eight status and condition flags
- Legacy compatibility to previous TeakLite cores

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# 5 Memory Controller

## 5.1Features

- Internal 1168KB RAM for data and program
- Build in 32KB Execute-in-place(Xip) Cache to QSPI NOR flash and 32KB cache to HSPI/OSPI PSRAM
- Memory Protect Unit (MPU)
- DMA/SDMA can access all the RAM blocks
- CPU can access all the RAM and ROM block through CPU bus
- DSP's data bus and program bus can access DSPRAM blocks
- 2.5D GPU and 2D GPU can access part of the RAM for graphics processing

# 6 Secure Subsystem

## 6.1 OTFD

### 6.1.1 Features

- Support On-the-fly decryption of external memories (QSPI NorFlash/Octo-SPI PSRAM), part of which can be encrypted by AES-128 in counter mode

## 6.2 SecureEngine

### 6.2.1 Features

- Support hardware AES-128/192/256 Encrypt & Decrypt in ECB/CTR/CBC/CBC-CTS mode
- Support hardware TRNG used to collects random bits from the ring oscillator, 64 random bits per time.
- Support SHA-1/SHA-224/SHA-256
- Support hardware CRC-32/CRC-16
- Support 1024bits efuse
- Support 64bit CHIP unique ID
- Support RSA 2048bits signature

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# 7 DMA Controller

## 7.1Features

- DMA transmission is independent with the CPU and DSP
- Support for memory to memory, memory to peripheral, peripheral to memory transmission.
- 10-channel ordinary DMA, that supports for transmission in burst 8 mode or single mode. Only one of the ten DMA channels can transfer data at the same time.
- DMA0-9 transmission can be triggered on the occurrence of selected events
- Each channel can send two interrupts to the CPU on completion of certain operational events
- Transmission width includes 8-bit, 16-bit, and 32-bit, which is determined by DMA transmission type.

## 7.2Memory and Peripheral Access Description

### 7.2.1 Access memory

（1） Memory is accessed by DMA according to physical address.
（2） The following situations will lead to transfer error:
- Access rom
- Ram has been occupied by other modules
- The clock of ram is disabled
（3） Memory-to-memory transmission:
- The data in memory should be aligned by word (32-bit) when transferred.
（4） Memory-to-peripheral and peripheral-to-memory transmission:
- The data in memory should be aligned by word (32-bit) when the transfer width is 32-bit or 24-bit.
- The data in memory should be aligned by half-word (16-bit) when the transfer width is 16-bit.
- The data in memory should be aligned by byte (8-bit) when the transfer width is 8-bit.
（5） SRCTYPE/ DSTTYPE/SADDR/DADDR should be set when DMA access the peripherals.
（6） CPU cannot access the peripherals FIFO when the peripherals FIFO is accessed by DMA.

### 7.2.2 Access Peripheral Mode

| Peripheral | Single | Burst8 |
|---|---|---|
| SPI0/1/2/3 | Y | Y |
| UART0/1/2 | Y | Y |
| I2C0/1/2/3 | Y | Y |
| USB | N | Y |
| SD0 | N | Y |
| SD1 | N | Y |
| PWM | Y | Y |
| DAC | N | Y |
| ADC | N | Y |
| IIS | N | Y |

### 7.2.3 DMA channel priority

Each memory block can be accessed by only one master at the same time. The 10 channels of DMA share

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the same bus, so only one of these channels can use the bus to transfer data at the same moment.
While accessing one memory block, DMA submits an access request to the memory controller's arbiter. Meanwhile, CPU might send another request to access the same memory block. The arbiter grants the bus of this memory block according to the priority scheme. The priority scheme is known as round-robin algorithm. When DMA does not get the bus of this memory block, the memory controller will hold DMA. See memory controller specification for details.
Once DMA obtains the highest priority among the three masters, one of the 10 channels occupies the DMA bus according to the internal priority. The priority of DMA channel is DMA0 > DMA1 > DMA2 > DMA3 > DMA4 > DMA5 > DMA6 > DMA7>DMA8>DMA9.

### 7.2.4 DMA Access Channel

DMA supports memory-to-memory, memory-to-peripheral, and peripheral-to-memory transmission. Specific access paths are shown as follows:

| DST / SRC | RAM | SPI01 23TX FIFO | UART 012TX | USB FIFO | SD1 FIFO | SD0 FIFO | DAC FIFO | IIS TX FIFO | I2C0/1 /2/3 FIFO | PWM |
|---|---|---|---|---|---|---|---|---|---|---|
| RAM | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y |
| SPI0123 RX FIFO | Y | - | - | - | | - | | | | |
| UART012R X FIFO | Y | - | - | - | | - | | | | |
| USB FIFO | Y | - | - | - | | Y | | | | |
| SD1 FIFO | Y | | | | | | | | | |
| SD0 FIFO | Y | - | - | Y | | - | | | | |
| ADC FIFO | Y | - | - | | | - | | | | |
| IIS RX FIFO | Y | | | | | | | | | |
| I2C0/1/2/3 FIFO | Y | | | | | | | | | |

## 7.3DMA Register List

*Table 7-1 DMA controller base address*

| Name | Physical Base Address | KSEG1 Base Address |
|---|---|---|
| DMAController | 0x4001C000 | 0x4001C000 |

*Table 7-2 DMA controller register list*

| Offset | Register Name | Description |
|---|---|---|
| 0x00000000 | DMAIP | DMA interrupt pending register |
| 0x00000004 | DMAIE | DMA interrupt enable register |
| 0x00000100+0x100*x | DMAxCTL | DMAx control register (0<=x<=9) |
| 0x00000104+0x100*x | DMAxSTART | DMAx start register (0<=x<=9) |
| 0x00000108+0x100*x | DMAxSADDR0 | DMAx source address register0 (0<=x<=9) |
| 0x0000010c+0x100*x | DMAxSADDR1 | DMAx source address register1 (0<=x<=9) |
| 0x00000110+0x100*x | DMAxDADDR0 | DMAx destination address register0 (0<=x<=9) |
| 0x00000114+0x100*x | DMAxDADDR1 | DMAx destination address register1 (0<=x<=9) |

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| | | |
|---|---|---|
| 0x00000118+0x100*x | DMAxBC | DMAx byte counter register (0<=x<=9) |
| 0x0000011c+0x100*x | DMAxRC | DMAx remain counter register (0<=x<=9) |

## 7.4DMA Register Description

### 7.4.1 DMAIP

**DMAIP (DMA Interrupt Pending Register, offset = 0x00000000)**

| Bit (s) | Name | Description | Access | Reset |
|---|---|---|---|---|
| 31:26 | Reserved | Reserved | R | x |
| 25 | DMA9HFIP | DMA9 Half Transmission IRQ Pending<br>This bit can be written '1' to clear. | RW | 0x0 |
| 24 | DMA8HFIP | DMA8 Half Transmission IRQ Pending<br>This bit can be written '1' to clear. | RW | 0x0 |
| 23 | DMA7HFIP | DMA7 Half Transmission IRQ Pending<br>This bit can be written '1' to clear. | RW | 0x0 |
| 22 | DMA6HFIP | DMA6 Half Transmission IRQ Pending<br>This bit can be written '1' to clear. | RW | 0x0 |
| 21 | DMA5HFIP | DMA5 Half Transmission IRQ Pending<br>This bit can be written '1' to clear. | RW | 0x0 |
| 20 | DMA4HFIP | DMA4 Half Transmission IRQ Pending<br>This bit can be written '1' to clear. | RW | 0x0 |
| 19 | DMA3HFIP | DMA3 Half Transmission IRQ Pending<br>This bit can be written '1' to clear. | RW | 0x0 |
| 18 | DMA2HFIP | DMA2 Half Transmission IRQ Pending<br>This bit can be written '1' to clear. | RW | 0x0 |
| 17 | DMA1HFIP | DMA1 Half Transmission IRQ Pending<br>This bit can be written '1' to clear. | RW | 0x0 |
| 16 | DMA0HFIP | DMA0 Half Transmission IRQ Pending<br>This bit can be written '1' to clear. | RW | 0x0 |
| 15:10 | Reserved | Reserved | R | x |
| 9 | DMA9TCIP | DMA9 Transmission Complete IRQ Pending<br>This bit can be written '1' to clear. | RW | 0x0 |
| 8 | DMA8TCIP | DMA8 Transmission Complete IRQ Pending<br>This bit can be written '1' to clear. | RW | 0x0 |
| 7 | DMA7TCIP | DMA7 Transmission Complete IRQ Pending<br>This bit can be written '1' to clear. | RW | 0x0 |
| 6 | DMA6TCIP | DMA6 Transmission Complete IRQ Pending<br>This bit can be written '1' to clear. | RW | 0x0 |
| 5 | DMA5TCIP | DMA5 Transmission Complete IRQ Pending<br>This bit can be written '1' to clear. | RW | 0x0 |
| 4 | DMA4TCIP | DMA4 Transmission Complete IRQ Pending<br>This bit can be written '1' to clear. | RW | 0x0 |
| 3 | DMA3TCIP | DMA3 Transmission Complete IRQ Pending<br>This bit can be written '1' to clear. | RW | 0x0 |
| 2 | DMA2TCIP | DMA2 Transmission Complete IRQ Pending<br>This bit can be written '1' to clear. | RW | 0x0 |
| 1 | DMA1TCIP | DMA1 Transmission Complete IRQ Pending<br>This bit can be written '1' to clear. | RW | 0x0 |
| 0 | DMA0TCIP | DMA0 Transmission Complete IRQ Pending | RW | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | This bit can be written '1' to clear. | | |

## 7.4.2 DMAIE

**DMAIE (DMA Interrupt Enable Register, offset = 0x00000004)**

| Bit (s) | Name | Description | Access | Reset |
|---|---|---|---|---|
| 31:26 | Reserved | Reserved | R | x |
| 25 | DMA9HFIE | DMA9 Half Transmission Complete IRQ enable:<br>0: Disable Half Transmission Complete interrupt;<br>1: Enable Half Transmission Complete interrupt. | RW | 0x0 |
| 24 | DMA8HFIE | DMA8 Half Transmission Complete IRQ enable:<br>0: Disable Half Transmission Complete interrupt;<br>1: Enable Half Transmission Complete interrupt. | RW | 0x0 |
| 23 | DMA7HFIE | DMA7 Half Transmission Complete IRQ enable:<br>0: Disable Half Transmission Complete interrupt;<br>1: Enable Half Transmission Complete interrupt. | RW | 0x0 |
| 22 | DMA6HFIE | DMA6 Half Transmission Complete IRQ enable:<br>0: Disable Half Transmission Complete interrupt;<br>1: Enable Half Transmission Complete interrupt. | RW | 0x0 |
| 21 | DMA5HFIE | DMA5 Half Transmission Complete IRQ enable:<br>0: Disable Half Transmission Complete interrupt;<br>1: Enable Half Transmission Complete interrupt. | RW | 0x0 |
| 20 | DMA4HFIE | DMA4 Half Transmission Complete IRQ enable:<br>0: Disable Half Transmission Complete interrupt;<br>1: Enable Half Transmission Complete interrupt. | RW | 0x0 |
| 19 | DMA3HFIE | DMA3 Half Transmission Complete IRQ enable:<br>0: Disable Half Transmission Complete interrupt;<br>1: Enable Half Transmission Complete interrupt. | RW | 0x0 |
| 18 | DMA2HFIE | DMA2 Half Transmission Complete IRQ enable:<br>0: Disable Half Transmission Complete interrupt;<br>1: Enable Half Transmission Complete interrupt. | RW | 0x0 |
| 17 | DMA1HFIE | DMA1 Half Transmission Complete IRQ enable:<br>0: Disable Half Transmission Complete interrupt;<br>1: Enable Half Transmission Complete interrupt. | RW | 0x0 |
| 16 | DMA0HFIE | DMA0 Half Transmission Complete IRQ enable:<br>0: Disable Half Transmission Complete interrupt;<br>1: Enable Half Transmission Complete interrupt. | RW | 0x0 |
| 15:10 | Reserved | Reserved | R | x |
| 9 | DMA9TCIE | DMA9 Transmission Complete IRQ Enable:<br>0: disable DMA9 Transmission Complete interrupt<br>1: enable DMA9 Transmission Complete interrupt | RW | 0x0 |
| 8 | DMA8TCIE | DMA8 Transmission Complete IRQ Enable:<br>0: disable DMA8 Transmission Complete interrupt<br>1: enable DMA8 Transmission Complete interrupt | RW | 0x0 |
| 7 | DMA7TCIE | DMA7 Transmission Complete IRQ Enable:<br>0: disable DMA7 Transmission Complete interrupt<br>1: enable DMA7 Transmission Complete interrupt | RW | 0x0 |
| 6 | DMA6TCIE | DMA6 Transmission Complete IRQ Enable:<br>0: disable DMA6 Transmission Complete interrupt<br>1: enable DMA6 Transmission Complete interrupt | RW | 0x0 |
| 5 | DMA5TCIE | DMA5 Transmission Complete IRQ Enable:<br>0: disable DMA5 Transmission Complete interrupt<br>1: enable DMA5 Transmission Complete interrupt | RW | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 4 | DMA4TCIE | DMA4 Transmission Complete IRQ Enable:<br>0: disable DMA4 Transmission Complete interrupt<br>1: enable DMA4 Transmission Complete interrupt | RW | 0x0 |
| 3 | DMA3TCIE | DMA3 Transmission Complete IRQ Enable:<br>0: disable DMA3 Transmission Complete interrupt<br>1: enable DMA3 Transmission Complete interrupt | RW | 0x0 |
| 2 | DMA2TCIE | DMA2 Transmission Complete IRQ Enable:<br>0: disable DMA2 Transmission Complete interrupt<br>1: enable DMA2 Transmission Complete interrupt | RW | 0x0 |
| 1 | DMA1TCIE | DMA1 Transmission Complete IRQ Enable:<br>0: disable DMA1 Transmission Complete interrupt<br>1: enable DMA1 Transmission Complete interrupt | RW | 0x0 |
| 0 | DMA0TCIE | DMA0 Transmission Complete IRQ Enable:<br>0: disable DMA0 Transmission Complete interrupt<br>1: enable DMA0 Transmission Complete interrupt | RW | 0x0 |

### 7.4.3 DMAxCTL

**DMAxCTL (DMAx control Register, offset = 0x00000100+0x100*x, 0 <= x <= 9)**

| Bit (s) | Name | Description | Access | Reset |
|---|---|---|---|---|
| 31:22 | - | Reserved | - | 0 |
| 21:20 | TWS | Transmit data width select<br>0:32bit<br>1:16bit<br>2:8bit<br>3:reserved | RW | 0 |
| 19 | - | Reserved | - | 0 |
| 18 | reload | Reload the DMA controller registers and start DMA transmission after current DMA transmission is complete:<br>0: disable reload mode<br>1: enable reload mode | RW | 0 |
| 17 | TRM | 0: Burst8<br>1: Single | RW | 0 |
| 16 | AUDIOTYPE | The method of audio data stored of DMA-ADC, DMA-DAC, DMA-I2S transmission:<br>0: interleaved stored in the memory<br>1: separated stored in the memory | RW | 0x0 |
| 15 | DAM | Destination address mode<br>0: increment<br>1: constant | RW | 0 |
| 14:13 | - | Reserved | - | 0 |
| 12:8 | DSTSL | Destination select:<br>5'b00000: memory<br>5'b00001: UART0 TX FIFO<br>5'b00010: UART1 TX FIFO<br>5'b00011: UART2 TX FIFO<br>5'b00100: USB FIFO<br>5'b00101: SD0 FIFO<br>5'b00110: SD1 FIFO<br>5'b00111: SPI0 TX FIFO<br>5'b01000: SPI1 TX FIFO<br>5'b01001: SPI2 TX FIFO<br>5'b01010: SPI3 TX FIFO<br>5'b01011: DAC FIFO0 | RW | 0 |

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| | | | | |
|---|---|---|---|---|
| | | 5'b01110: IIS TX0 FIFO<br>5'b10011: I2C0 TX FIFO<br>5'b10100: I2C1 TX FIFO<br>5'b10101: PWM FIFO<br>5'b11000: I2C2 TX FIFO<br>5'b11001: I2C3 TX FIFO<br>Other: Reserved | | |
| 7 | SAM | Source address mode<br>0: increment<br>1: constant | RW | 0 |
| 6:5 | - | Reserved | - | 0 |
| 4:0 | SRCSL | Source Select:<br>5'b00000: memory<br>5'b00001: UART0 RX FIFO<br>5'b00010: UART1 RX FIFO<br>5'b00011: UART2 RX FIFO<br>5'b00100: USB FIFO<br>5'b00101: SD0 FIFO<br>5'b00110: SD1 FIFO<br>5'b00111: SPI0 RX FIFO<br>5'b01000: SPI1 RX FIFO<br>5'b01001: SPI2 RX FIFO<br>5'b01010: SPI3 RX FIFO<br>5'b01011: ADC FIFO<br>5'b01110: IIS RX FIFO<br>5'b10011: I2C0 RX FIFO<br>5'b10100: I2C1 RX FIFO<br>5'b11000: I2C2 RX FIFO<br>5'b11001: I2C3 RX FIFO<br>Other: Reserved | RW | 0 |

### 7.4.4 DMAxSTART

**DMAxSTART (offset = 0x00000104+0x100*x, 0 <= x <= 9)**

| Bit (s) | Name | Description | Access | Reset |
|---|---|---|---|---|
| 31:1 | Reserved | - | - | - |
| 0 | DMASTART | DMA start bit:<br>A low-to-high conversion of this bit will lead to load source address, destination address, DRQ type to the DMA controller. This bit will be automatically cleared by the DMAx controller if the DMAx transmission is complete or DMAx transmission error occurs.<br>This bit can be written '0' to abort DMAx transmission. | RW | 0x0 |

### 7.4.5 DMAxSADDR0

**DMAxSADDR0 (DMAx Source Address Register0, offset = 0x00000108+0x100*x, 0 <= x <= 9)**

| Bit (s) | Name | Description | Access | Reset |
|---|---|---|---|---|
| 31:0 | DMASADDR | The source address of DMA transmission.<br>The bit[0] is no effect if data width is 16-bit.<br>The bit[1..0] is no effect if data width is 32-bit. | RW | 0x0 |

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### 7.4.6 DMAxSADDR1

**DMAxSADDR1 (DMAx Source Address Register1, offset = 0x0000010c+0x100*x, 0 <= x <= 9)**

| Bit (s) | Name | Description | Access | Reset |
|---|---|---|---|---|
| 31:0 | DMASADDR | The source address of DMA transmission.<br>The bit[0] is no effect if data width is 16-bit.<br>The bit[1..0] is no effect if data width is 32-bit. | RW | 0x0 |

### 7.4.7 DMAxDADDR0

**DMAxDADDR0 (DMAx Destination Address Register0, offset = 0x00000110+0x100*x, 0 <= x <= 9)**

| Bit (s) | Name | Description | Access | Reset |
|---|---|---|---|---|
| 31:0 | DMADADDR | The destination address of DMA transmission.<br>The bit[0] is no effect if data width is 16-bit.<br>The bit[1..0] is no effect if data width is 32-bit. | RW | 0x0 |

### 7.4.8 DMAxDADDR1

**DMAxDADDR1 (DMAx Destination Address Register1, offset = 0x00000114+0x100*x, 0 <= x <= 9)**

| Bit (s) | Name | Description | Access | Reset |
|---|---|---|---|---|
| 31:0 | DMADADDR | The destination address of DMA transmission.<br>The bit[0] is no effect if data width is 16-bit.<br>The bit[1..0] is no effect if data width is 32-bit. | RW | 0x0 |

### 7.4.9 DMAxBC

**DMAxBC (DMAx BYTE Counter Register, offset = 0x00000118+0x100*x, 0 <= x <= 9)**

| Bit (s) | Name | Description | Access | Reset |
|---|---|---|---|---|
| 31:20 | Reserved | Reserved | R | x |
| 19:0 | DMABYTECOUNTER | The frame length of DMA transmission. | RW | 0x0 |

### 7.4.10 DMAxRC

**DMAxRC (DMAx Remain Counter Register, offset = 0x0000011c+0x100*x, 0 <= x <= 9)**

| Bit (s) | Name | Description | Access | Reset |
|---|---|---|---|---|
| 31:20 | Reserved | - | - | - |
| 19:0 | DMARemainCounterH | The Remain Counter of DMA transmission. | R | 0x0 |

# 8 SDMA Controller

## 8.1Features

- 5-channel SDMA
- SDMA support stride mode
- SDMA support Y address decrease. This is designed for LCD Y-flip
- SDMA0 only support memory to LCD
- SDMA1~3 support memory to memory, memory to JPEG, JPEG to memory
- SDMA4 support memory to memory

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- Each channel can send two interrupts to the CPU on completion of certain operational events
- Transmission width is 32-bit. Max Burst length is configurable (8,16,32)

## 8.2Using mode Description

### 8.2.1 Stride Mode

Stride mode is mainly used for window cutting application of image display.

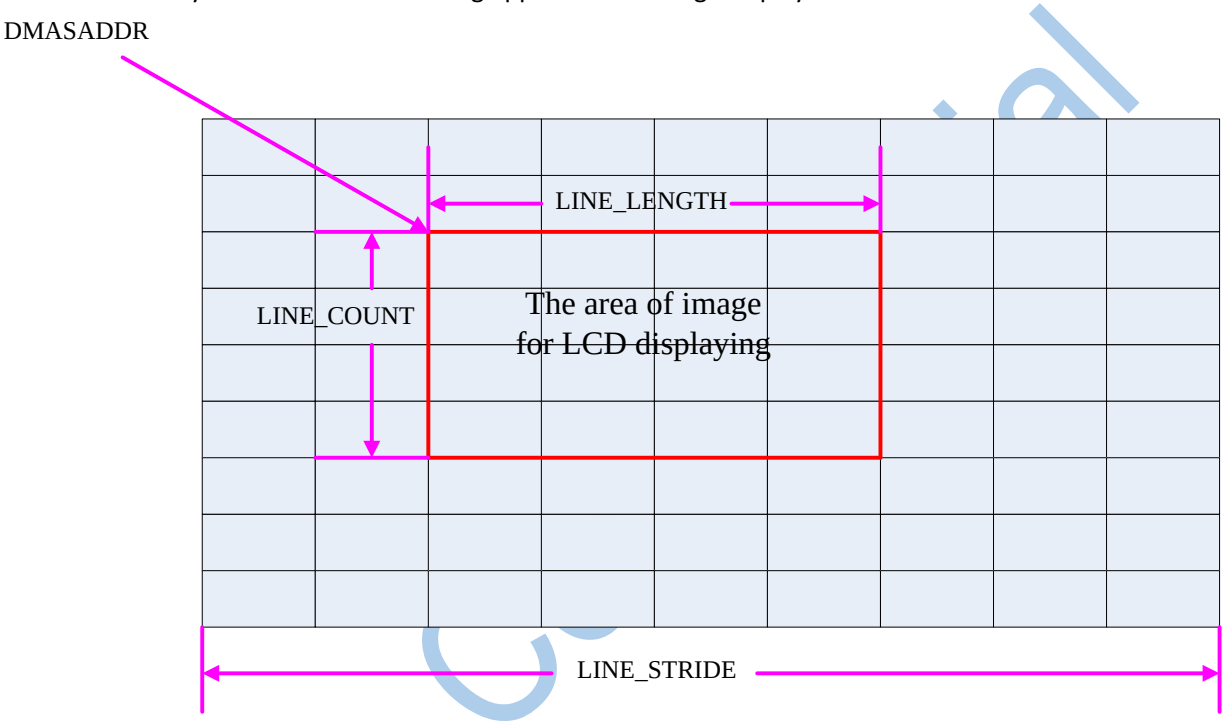



***Figure 8-1 Data storage diagram of stride mode***

When the image data is transmitted to LCD for displaying, As Figure 8-1 shown above, the red box is the part to be sent to LCD for displaying.
The DMASADDR register configures the start address of the image in the red box;
The LINE_ LENGTH register configures the length of the image for displaying;
The LINE_ COUNT register configures the height of the image for displaying;
The LINE_ STRIDE register configures the length of the original image.

When the image data is write to memory, the stride mode of the target address is the same as the above figure.

### 8.2.2 Couple Mode

The couple mode is used to improve the efficiency of JPEG decoding output to PSRAM. The basic principle is to set an intermediate buffer, and use two channels of SDMA to parallel the JPEG decoding and data writing to PSRAM through this intermediate buffer.
In couple mode, COUPLE_NUMBER is used to configure which two SDMA forms the couple mode. There are three optional groups, SDMA1 and SDMA4, SDMA2 and SDMA4, SDMA3 and SDMA4.
Couple mode requires a block of RAM as an intermediate buffer to speed up the write operation to PSRAM. The starting address and the size of this buffer are configured by register SDMA_COUPLE_BUF_ADDR and SDMA_COUPLE_BUF_SIZE.

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### 8.2.3 Color Fill Mode

Color Fill mode is mainly used to fill fixed colors in the image window quickly with SDMA.
Only SDMA1/2/3 can use Color Fill mode.

### 8.2.4 Priorities

The priority of SDMA controller is controlled by register SDMA_ PRIORITY, when the memory accessing commands of different master conflict. The SDMA controller select the higher priority command from SDMA1/SDMA2/SDMA3 as SDMA123 first, and then select the higher priority command from SDMA123, SDMA0 and SDMA4 to send to the memory device finally.

## 8.3 SDMA Register List

***Table 8-1 SDMA controller base address***

| Name | Physical Base Address | KSEG1 Base Address |
|---|---|---|
| SDMAController | 0x40078000 | 0x40078000 |

***Table 8-2 SDMA controller register list***

| Offset | Register Name | Description |
|---|---|---|
| 0x00000000 | SDMAIP | SDMA interrupt pending register |
| 0x00000004 | SDMAIE | SDMA interrupt enable register |
| 0x00000070 | SDMA_PRIORITY | SDMA Priority Register |
| 0x000000a0 | SDMA_COUPLE_CONFIG | SDMA Couple Configure Register |
| 0x000000a4 | SDMA_COUPLE_BUF_ADDR | SDMA Couple Buffer Address Register |
| 0x000000a8 | SDMA_COUPLE_BUF_SIZE | SDMA Couple Buffer Size Register |
| 0x000000ac | SDMA_COUPLE_START | SDMA Couple Start Register |
| 0x000000b0 | SDMA_COUPLE_WRITE_POINTER | SDMA Couple Write Pointer Register |
| 0x000000b4 | SDMA_COUPLE_READ_POINTER | SDMA Couple Read Pointer Register |
| 0x00000100 | SDMA0CTL | SDMA0 control register |
| 0x00000104 | SDMA0START | SDMA0 start register |
| 0x00000108 | SDMA0SADDR | SDMA0 source address0 register |
| 0x00000118 | SDMA0BC | SDMA0 byte counter register |
| 0x0000011c | SDMA0RC | SDMA0 remain counter register |
| 0x00000200 | SDMA1CTL | SDMA1 control register |
| 0x00000204 | SDMA1START | SDMA1 start register |
| 0x00000208 | SDMA1SADDR | SDMA1 source address0 register |
| 0x00000210 | SDMA1DADDR | SDMA1 destination address0 register |
| 0x00000218 | SDMA1BC | SDMA1 byte counter register |
| 0x0000021c | SDMA1RC | SDMA1 remain counter |

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| | | |
|---|---|---|
| | | register |
| 0x00000300 | SDMA2CTL | SDMA2 control register |
| 0x00000304 | SDMA2START | SDMA2 start register |
| 0x00000308 | SDMA2SADDR | SDMA2 source address0 register |
| 0x00000310 | SDMA2DADDR | SDMA2 destination address0 register |
| 0x00000318 | SDMA2BC | SDMA2 byte counter register |
| 0x0000031c | SDMA2RC | SDMA2 remain counter register |
| 0x00000400 | SDMA3CTL | SDMA3 control register |
| 0x00000404 | SDMA3START | SDMA3 start register |
| 0x00000408 | SDMA3SADDR | SDMA3 source address0 register |
| 0x00000410 | SDMA3DADDR | SDMA3 destination address0 register |
| 0x00000418 | SDMA3BC | SDMA3 byte counter register |
| 0x0000041c | SDMA3RC | SDMA3 remain counter register |
| 0x00000500 | SDMA4CTL | SDMA4 control register |
| 0x00000504 | SDMA4START | SDMA4 start register |
| 0x00000508 | SDMA4SADDR | SDMA4 source address0 register |
| 0x00000510 | SDMA4DADDR | SDMA4 destination address0 register |
| 0x00000518 | SDMA4BC | SDMA4 byte counter register |
| 0x0000051c | SDMA4RC | SDMA4 remain counter register |
| 0x00000c00 | LINE_LENGTH0 | Line Length Register |
| 0x00000c04 | LINE_COUNT0 | Line Count Register |
| 0x00000c08 | LINE_SRC_STRIDE0 | Line Stride Register |
| 0x00000c10 | LINE_REMAIN0 | Line Remain Register |
| 0x00000c14 | BYTE_REMAIN_IN_LINE0 | Byte Remain in Line Register |
| 0x00000c20 | LINE_LENGTH1 | Line Length Register |
| 0x00000c24 | LINE_COUNT1 | Line Count Register |
| 0x00000c28 | LINE_SRC_STRIDE1 | Line Stride Register |
| 0x00000c2c | LINE_DST_STRIDE1 | Line Stride Register |
| 0x00000c30 | LINE_REMAIN1 | Line Remain Register |
| 0x00000c34 | BYTE_REMAIN_IN_LINE1 | Byte Remain in Line Register |
| 0x00000c40 | LINE_LENGTH2 | Line Length Register |
| 0x00000c44 | LINE_COUNT2 | Line Count Register |
| 0x00000c48 | LINE_SRC_STRIDE2 | Line Stride Register |
| 0x00000c4c | LINE_DST_STRIDE2 | Line Stride Register |
| 0x00000c50 | LINE_REMAIN2 | Line Remain Register |
| 0x00000c54 | BYTE_REMAIN_IN_LINE2 | Byte Remain in Line Register |
| 0x00000c60 | LINE_LENGTH3 | Line Length Register |

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| | | |
|---|---|---|
| 0x00000c64 | LINE_COUNT3 | Line Count Register |
| 0x00000c68 | LINE_SRC_STRIDE3 | Line Stride Register |
| 0x00000c6c | LINE_DST_STRIDE3 | Line Stride Register |
| 0x00000c70 | LINE_REMAIN3 | Line Remain Register |
| 0x00000c74 | BYTE_REMAIN_IN_LINE3 | Byte Remain in Line Register |
| 0x00000c80 | LINE_LENGTH4 | Line Length Register |
| 0x00000c84 | LINE_COUNT4 | Line Count Register |
| 0x00000c88 | LINE_SRC_STRIDE4 | Line Stride Register |
| 0x00000c8c | LINE_DST_STRIDE4 | Line Stride Register |
| 0x00000c90 | LINE_REMAIN4 | Line Remain Register |
| 0x00000c94 | BYTE_REMAIN_IN_LINE4 | Byte Remain in Line Register |
| 0x00000d00 | COLOR_FILL_DATA0 | Color Fill Data0 Register |
| 0x00000d04 | COLOR_FILL_DATA1 | Color Fill Data1 Register |
| 0x00000d08 | COLOR_FILL_DATA2 | Color Fill Data2 Register |

## 8.4SDMA Register Description

### 8.4.1 SDMAIP

**SDMAIP (DMA Interrupt Pending Register, offset = 0x00000000)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:21 | - | Reserved | R | 0x0 |
| 20 | SDMA4HFIP | DMA4 Half Transmission IRQ Pending<br>This bit can be written '1' to clear. | R/W | 0x0 |
| 19 | SDMA3HFIP | DMA3 Half Transmission IRQ Pending<br>This bit can be written '1' to clear. | R/W | 0x0 |
| 18 | SDMA2HFIP | DMA2 Half Transmission IRQ Pending<br>This bit can be written '1' to clear. | R/W | 0x0 |
| 17 | SDMA1HFIP | DMA1 Half Transmission IRQ Pending<br>This bit can be written '1' to clear. | R/W | 0x0 |
| 16 | SDMA0HFIP | DMA0 Half Transmission IRQ Pending<br>This bit can be written '1' to clear. | R/W | 0x0 |
| 15:5 | - | Reserved | R | 0x0 |
| 4 | SDMA4TCIP | DMA4 Transmission Complete IRQ Pending<br>This bit can be written '1' to clear. | R/W | 0x0 |
| 3 | SDMA3TCIP | DMA3 Transmission Complete IRQ Pending<br>This bit can be written '1' to clear. | R/W | 0x0 |
| 2 | SDMA2TCIP | DMA2 Transmission Complete IRQ Pending<br>This bit can be written '1' to clear. | R/W | 0x0 |
| 1 | SDMA1TCIP | DMA1 Transmission Complete IRQ Pending<br>This bit can be written '1' to clear. | R/W | 0x0 |
| 0 | SDMA0TCIP | DMA0 Transmission Complete IRQ Pending<br>This bit can be written '1' to clear. | R/W | 0x0 |

### 8.4.2 SDMAIE

**SDMAIE (DMA Interrupt Enable Register, offset = 0x00000004)**

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| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:21 | - | Reserved | R | 0x0 |
| 20 | SDMA4HFIE | DMA4 Half Transmission Complete IRQ enable:<br>0: Disable Half Transmission Complete interrupt;<br>1: Enable Half Transmission Complete interrupt. | R/W | 0x0 |
| 19 | SDMA3HFIE | DMA3 Half Transmission Complete IRQ enable:<br>0: Disable Half Transmission Complete interrupt;<br>1: Enable Half Transmission Complete interrupt. | R/W | 0x0 |
| 18 | SDMA2HFIE | DMA2 Half Transmission Complete IRQ enable:<br>0: Disable Half Transmission Complete interrupt;<br>1: Enable Half Transmission Complete interrupt. | R/W | 0x0 |
| 17 | SDMA1HFIE | DMA1 Half Transmission Complete IRQ enable:<br>0: Disable Half Transmission Complete interrupt;<br>1: Enable Half Transmission Complete interrupt. | R/W | 0x0 |
| 16 | SDMA0HFIE | DMA0 Half Transmission Complete IRQ enable:<br>0: Disable Half Transmission Complete interrupt;<br>1: Enable Half Transmission Complete interrupt. | R/W | 0x0 |
| 15:5 | - | Reserved | R | 0x0 |
| 4 | SDMA4TCIE | DMA4 Transmission Complete IRQ Enable:<br>0: disable DMA4 Transmission Complete interrupt<br>1: enable DMA4 Transmission Complete interrupt | R/W | 0x0 |
| 3 | SDMA3TCIE | DMA3 Transmission Complete IRQ Enable:<br>0: disable DMA3 Transmission Complete interrupt<br>1: enable DMA3 Transmission Complete interrupt | R/W | 0x0 |
| 2 | SDMA2TCIE | DMA2 Transmission Complete IRQ Enable:<br>0: disable DMA2 Transmission Complete interrupt<br>1: enable DMA2 Transmission Complete interrupt | R/W | 0x0 |
| 1 | SDMA1TCIE | DMA1 Transmission Complete IRQ Enable:<br>0: disable DMA1 Transmission Complete interrupt<br>1: enable DMA1 Transmission Complete interrupt | R/W | 0x0 |
| 0 | SDMA0TCIE | DMA0 Transmission Complete IRQ Enable:<br>0: disable DMA0 Transmission Complete interrupt<br>1: enable DMA0 Transmission Complete interrupt | R/W | 0x0 |

### 8.4.3 SDMA_PRIORITY

**SDMA PRIORITY (DMA Priority Register, offset = 0x00000070)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:21 | - | Reserved | R | 0x0 |
| 20 | SDMA123_PRIORITY | 0: SDMA1>SDMA2>SDMA3<br>1: SDMA1/SDMA2/SDMA3 Round Robin | R/W | 0x0 |
| 19:18 | - | Reserved | R | 0x0 |
| 17:16 | SDMA1234_PRIORITY | 0：SDMA1~3 FIFO and SDMA4 FIFO Round Robin;<br>1：SDMA1~3 FIFO > SDMA4 FIFO;<br>2：SDMA4 FIFO > SDMA1~3 FIFO;<br>Others: Reserved | R/W | 0x0 |
| 15:7 | - | Reserved | R | 0x0 |
| 6:0 | FIFO_LEVEL | When the empty level of LCD FIFO is more than FIFO_LEVEL, the SDMA controller will set SDMA0 as highest priority. The | R/W | 0xc |

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| | | | | |
|---|---|---|---|---|
| | | purpose is to ensure the real-time performance of LCD display. | | |

### 8.4.4 SDMA_COUPLE_CONFIG

**SDMA_COUPLE_CONFIG (SDMA Couple Config Register, offset = 0x000000a0)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:20 | - | Reserved | R | 0x0 |
| 19:2 | BUF_THRESHOLD | The read and write threshold of the couple buffer. When SDMA1~3 writes more bytes to buffer than BUF_THRESHOLD, SDMA4 starts to read from the buffer. | R/W | 0x1f |
| 1:0 | COUPLE_NUMBER | 0: no couple mode<br>1: SDMA1 and SDMA4<br>2: SDMA2 and SDMA4<br>3: SDMA3 and SDMA4 | R/W | 0x0 |

### 8.4.5 SDMA_COUPLE_BUF_ADDR

**SDMA_COUPLE_BUF_ADDR (SDMA Couple Buffer Address Register, offset = 0x000000a4)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:30 | - | Reserved | R | 0x0 |
| 29:2 | ADDR | The address of SDMA couple buffer, which is the destination address for SDMA1~3 and the source address for SDMA4. | R/W | 0x0 |
| 1:0 | - | Reserved | R | 0x0 |

### 8.4.6 SDMA_COUPLE_BUF_SIZE

**SDMA_COUPLE_BUF_SIZE (SDMA Couple Buffer Size Register, offset = 0x000000a8)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:20 | - | Reserved | R | 0x0 |
| 19:2 | SIZE | The size of SDMA couple buffer, which is 4 bytes aligned. | R/W | 0x0 |
| 1:0 | - | Reserved | R | 0x0 |

### 8.4.7 SDMA_COUPLE_START

**SDMA_COUPLE_START (SDMA Couple Start Register, offset = 0x000000ac)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:1 | - | Reserved | R | 0x0 |
| 0 | START | If the bit be set 1, the SDMA couple transmission will be started. SDMA1/2/3 works first. When SDMA1/2/3 writes more data to the couple buffer than BUF_THRESHOLD, SDMA4 will start to work and read from the couple buffer.<br>After the SDMA couple transmission is completed, the bit clears to 0 automatically. | R/W | 0x0 |

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## 8.4.8 SDMA_COUPLE_WRITE_POINTER

**SDMA_COUPLE_WRITE_POINTER (SDMA Couple Write Pointer Register, offset = 0x000000b0)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:30 | - | Reserved | R | 0x0 |
| 29:0 | POINTER | Record the pointer of SDMA1/2/3 writing to the couple buffer. | R | 0x0 |

## 8.4.9 SDMA_COUPLE_READ_POINTER

**SDMA_COUPLE_READ_POINTER (SDMA Couple Read Pointer Register, offset = 0x000000b0)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:30 | - | Reserved | R | 0x0 |
| 29:0 | POINTER | Record the pointer of SDMA4 reading from the couple buffer. | R | 0x0 |

## 8.4.10 SDMA0CTL

**DMA0CTL (DMAx Control Register, offset = 0x00000100)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:25 | - | Reserved | R | 0x0 |
| 24 | STRME | Stride Mode Enable:<br>0: Disable<br>1: Enable | R/W | 0x0 |
| 23:19 | - | Reserved | R | 0x0 |
| 18 | RELOAD | Reload the DMA controller registers and start DMA transmission after current DMA transmission is complete:<br>0: disable reload mode<br>1: enable reload mode | R/W | 0x0 |
| 17:14 | - | Reserved | R | 0x0 |
| 13:8 | DSTSL | Destination select:<br>5'b00000: LCD | R | 0x0 |
| 7 | - | Reserved | R | 0x0 |
| 6 | YSAM | Y Source address mode<br>0: increment<br>1: decrement | R/W | 0x0 |
| 5:0 | SRCSL | Source Select:<br>5'b00000: memory | R | 0x0 |

## 8.4.11 SDMA0START

**DMA0START (DMA Start Register, offset = 0x00000104)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:1 | - | Reserved | R | 0x0 |
| 0 | DMASTART | DMA start bit:<br>A low-to-high conversion of this bit will lead to load source address, destination address, DRQ type to the DMA controller. | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | This bit will be automatically cleared by the DMAx controller if the DMAx transmission is complete or DMAx transmission error occurs.<br>This bit can be written '0' to abort DMAx transmission. | | |

## 8.4.12 SDMA0SADDR

**DMA0SADDR (DMA Source Address Register, offset = 0x00000108)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | DMASADDR | The source address of DMA transmission.<br>The bit[0] is no effect if data width is 16-bit.<br>The bit[1..0] is no effect if data width is 32-bit. | R/W | 0x0 |

## 8.4.13 SDMA0BC

**DMA0BC (DMA Byte Counter Address Register, offset = 0x00000118)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:20 | - | Reserved | R | 0x0 |
| 19:0 | DMABYTECOUNTER | The frame length of DMA transmission. | R/W | 0x0 |

## 8.4.14 SDMA0RC

**DMA0RC (DMA Remain Counter Address Register, offset = 0x0000011C)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:20 | - | Reserved | R | 0x0 |
| 19:0 | DMAREMAINCOUNTER | The Remain Counter of DMA transmission. | R | 0x0 |

## 8.4.15 SDMA1CTL

**DMA1CTL (DMAx Control Register, offset = 0x00000200)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:25 | - | Reserved | R | 0x0 |
| 24 | STRME | Stride Mode Enable:<br>0: Disable<br>1: Enable | R/W | 0x0 |
| 23:21 | - | Reserved | R | 0x0 |
| 20 | JPEG_OUT_FORMAT | 0: RGB888<br>1: RGB565 | R/W | 0x0 |
| 19:15 | - | Reserved | R | 0x0 |
| 14 | YDAM | Y Destination address mode<br>0: increment<br>1: decrement | R/W | 0x0 |
| 13:9 | - | Reserved | R | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 8 | DSTSL | Destination select:<br>0: memory<br>1: JPEG | R/W | 0x0 |
| 7 | - | Reserved | R | 0x0 |
| 6 | YSAM | Y Source address mode<br>0: increment<br>1: decrement | R/W | 0x0 |
| 5:2 | - | Reserved | R | 0x0 |
| 1:0 | SRCSL | Source Select:<br>0: memory<br>1: JPEG<br>2: color_fill<br>3: Reserved | R/W | 0x0 |

## 8.4.16 SDMA1START

**DMA1START (DMA Start Register, offset = 0x00000204)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:1 | - | Reserved | R | 0x0 |
| 0 | DMASTART | DMA start bit:<br>A low-to-high conversion of this bit will lead to load source address, destination address, DRQ type to the DMA controller. This bit will be automatically cleared by the DMAx controller if the DMAx transmission is complete or DMAx transmission error occurs.<br>This bit can be written '0' to abort DMAx transmission. | R/W | 0x0 |

## 8.4.17 SDMA1SADDR

**DMA1SADDR (DMA Source Address Register, offset = 0x00000208)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | DMASADDR | The source address of DMA transmission.<br>The bit[0] is no effect if data width is 16-bit.<br>The bit[1..0] is no effect if data width is 32-bit. | R/W | 0x0 |

## 8.4.18 SDMA1DADDR

**DMA1DADDR (DMA Destination Address Register, offset = 0x00000210)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | DMADADDR | The destination address of DMA transmission.<br>The bit[0] is no effect if data width is 16-bit.<br>The bit[1..0] is no effect if data width is 32-bit. | R/W | 0x0 |

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## 8.4.19 SDMA1BC

**DMA1BC (DMA Byte Counter Address Register, offset = 0x00000218)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:20 | - | Reserved | R | 0x0 |
| 19:0 | DMABYTECOUNTER | The frame length of DMA transmission. | R/W | 0x0 |

## 8.4.20 SDMA1RC

**DMA1RC (DMA Remain Counter Address Register, offset = 0x0000021C)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:20 | - | Reserved | R | 0x0 |
| 19:0 | DMAREMAINCOUNTER | The Remain Counter of DMA transmission. | R | 0x0 |

## 8.4.21 SDMA2CTL

**DMA2CTL (DMAx Control Register, offset = 0x00000300)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:25 | - | Reserved | R | 0x0 |
| 24 | STRME | Stride Mode Enable:<br>0: Disable<br>1: Enable | R/W | 0x0 |
| 23:21 | - | Reserved | R | 0x0 |
| 20 | JPEG_OUT_FORMAT | 0: RGB888<br>1: RGB565 | R/W | 0x0 |
| 19:15 | - | Reserved | R | 0x0 |
| 14 | YDAM | Y Destination address mode<br>0: increment<br>1: decrement | R/W | 0x0 |
| 13:9 | - | Reserved | R | 0x0 |
| 8 | DSTSL | Destination select:<br>0: memory<br>1: JPEG | R/W | 0x0 |
| 7 | - | Reserved | R | 0x0 |
| 6 | YSAM | Y Source address mode<br>0: increment<br>1: decrement | R/W | 0x0 |
| 5:2 | - | Reserved | R | 0x0 |
| 1:0 | SRCSL | Source Select:<br>0: memory<br>1: JPEG<br>2: color_fill<br>3: Reserved | R/W | 0x0 |

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## 8.4.22 SDMA2START

**DMA2START (DMA Start Register, offset = 0x00000304)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:1 | - | Reserved | R | 0x0 |
| 0 | DMASTART | DMA start bit:<br>A low-to-high conversion of this bit will lead to load source address, destination address, DRQ type to the DMA controller. This bit will be automatically cleared by the DMAx controller if the DMAx transmission is complete or DMAx transmission error occurs.<br>This bit can be written '0' to abort DMAx transmission. | R/W | 0x0 |

## 8.4.23 SDMA2SADDR

**DMA2SADDR (DMA Source Address Register, offset = 0x00000308)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | DMASADDR | The source address of DMA transmission.<br>The bit[0] is no effect if data width is 16-bit.<br>The bit[1..0] is no effect if data width is 32-bit. | R/W | 0x0 |

## 8.4.24 SDMA2DADDR

**DMA2DADDR (DMA Destination Address Register, offset = 0x00000310)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | DMADADDR | The destination address of DMA transmission.<br>The bit[0] is no effect if data width is 16-bit.<br>The bit[1..0] is no effect if data width is 32-bit. | R/W | 0x0 |

## 8.4.25 SDMA2BC

**DMA2BC (DMA Byte Counter Address Register, offset = 0x00000318)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:20 | - | Reserved | R | 0x0 |
| 19:0 | DMABYTECOUNTER | The frame length of DMA transmission. | R/W | 0x0 |

## 8.4.26 SDMA2RC

**DMA2RC (DMA Remain Counter Address Register, offset = 0x0000031C)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:20 | - | Reserved | R | 0x0 |
| 19:0 | DMAREMAINCOUNTER | The Remain Counter of DMA transmission. | R | 0x0 |

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## 8.4.27 SDMA3CTL

**DMA3CTL (DMAx Control Register, offset = 0x00000400)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:25 | - | Reserved | R | 0x0 |
| 24 | STRME | Stride Mode Enable:<br>0: Disable<br>1: Enable | R/W | 0x0 |
| 23:21 | - | Reserved | R | 0x0 |
| 20 | JPEG_OUT_FORMAT | 0: RGB888<br>1: RGB565 | R/W | 0x0 |
| 19:15 | - | Reserved | R | 0x0 |
| 14 | YDAM | Y Destination address mode<br>0: increment<br>1: decrement | R/W | 0x0 |
| 13:9 | - | Reserved | R | 0x0 |
| 8 | DSTSL | Destination select:<br>0: memory<br>1: JPEG | R/W | 0x0 |
| 7 | - | Reserved | R | 0x0 |
| 6 | YSAM | Y Source address mode<br>0: increment<br>1: decrement | R/W | 0x0 |
| 5:2 | - | Reserved | R | 0x0 |
| 1:0 | SRCSL | Source Select:<br>0: memory<br>1: JPEG<br>2: color_fill<br>3: Reserved | R/W | 0x0 |

## 8.4.28 SDMA3START

**DMA3START (DMA Start Register, offset = 0x00000404)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:1 | - | Reserved | R | 0x0 |
| 0 | DMASTART | DMA start bit:<br>A low-to-high conversion of this bit will lead to load source address, destination address, DRQ type to the DMA controller. This bit will be automatically cleared by the DMAx controller if the DMAx transmission is complete or DMAx transmission error occurs.<br>This bit can be written '0' to abort DMAx transmission. | R/W | 0x0 |

## 8.4.29 SDMA3SADDR

**DMA3SADDR (DMA Source Address Register, offset = 0x00000408)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 31:0 | DMASADDR | The source address of DMA transmission.<br>The bit[0] is no effect if data width is 16-bit.<br>The bit[1..0] is no effect if data width is 32-bit. | R/W | 0x0 |

## 8.4.30 SDMA3DADDR

**DMA3DADDR (DMA Destination Address Register, offset = 0x00000410)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | DMADADDR | The destination address of DMA transmission.<br>The bit[0] is no effect if data width is 16-bit.<br>The bit[1..0] is no effect if data width is 32-bit. | R/W | 0x0 |

## 8.4.31 SDMA3BC

**DMA3BC (DMA Byte Counter Address Register, offset = 0x00000418)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:20 | - | Reserved | R | 0x0 |
| 19:0 | DMABYTECOUNTER | The frame length of DMA transmission. | R/W | 0x0 |

## 8.4.32 SDMA3RC

**DMA3RC (DMA Remain Counter Address Register, offset = 0x0000041C)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:20 | - | Reserved | R | 0x0 |
| 19:0 | DMAREMAINCOUNTER | The Remain Counter of DMA transmission. | R | 0x0 |

## 8.4.33 SDMA4CTL

**DMA4CTL (DMAx Control Register, offset = 0x00000500)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:25 | - | Reserved | R | 0x0 |
| 24 | STRME | Stride Mode Enable:<br>0: Disable<br>1: Enable | R/W | 0x0 |
| 23:15 | - | Reserved | R | 0x0 |
| 14 | YDAM | Y Destination address mode<br>0: increment<br>1: decrement | R/W | 0x0 |
| 13:9 | - | Reserved | R | 0x0 |
| 8 | DSTSL | Destination select:<br>0: memory<br>1: Reserved | R | 0x0 |
| 7 | - | Reserved | R | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 6 | YSAM | Y Source address mode<br>0: increment<br>1: decrement | R/W | 0x0 |
| 5:1 | - | Reserved | R | 0x0 |
| 0 | SRCSL | Source Select:<br>0: memory | R | 0x0 |

## 8.4.34 SDMA4START

**DMA4START (DMA Start Register, offset = 0x00000504)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:1 | - | Reserved | R | 0x0 |
| 0 | DMASTART | DMA start bit:<br>A low-to-high conversion of this bit will lead to load source address, destination address, DRQ type to the DMA controller. This bit will be automatically cleared by the DMAx controller if the DMAx transmission is complete or DMAx transmission error occurs.<br>This bit can be written '0' to abort DMAx transmission. | R/W | 0x0 |

## 8.4.35 SDMA4SADDR

**DMA4SADDR (DMA Source Address Register, offset = 0x00000508)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | DMASADDR | The source address of DMA transmission.<br>The bit[0] is no effect if data width is 16-bit.<br>The bit[1..0] is no effect if data width is 32-bit. | R/W | 0x0 |

## 8.4.36 SDMA4DADDR

**DMA4DADDR (DMA Destination Address Register, offset = 0x00000510)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | DMADADDR | The destination address of DMA transmission.<br>The bit[0] is no effect if data width is 16-bit.<br>The bit[1..0] is no effect if data width is 32-bit. | R/W | 0x0 |

## 8.4.37 SDMA4BC

**DMA4BC (DMA Byte Counter Address Register, offset = 0x00000518)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:20 | - | Reserved | R | 0x0 |
| 19:0 | DMABYTECOUNTER | The frame length of DMA transmission. | R/W | 0x0 |

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## 8.4.38 SDMA4RC

**DMA4RC (DMA Remain Counter Address Register, offset = 0x0000051C)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:20 | - | Reserved | R | 0x0 |
| 19:0 | DMAREMAINCOUNTER | The Remain Counter of DMA transmission. | R | 0x0 |

## 8.4.39 LINE_LENGTH0

**LINE_LENGTH0(Line Length Address Register, offset = 0x00000C00)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | 0x0 |
| 15:0 | LENGTH | Line Length. | R/W | 0x0 |

## 8.4.40 LINE_COUNT0

**LINE_COUNT0(Line Counter Address Register, offset = 0x00000C04)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | 0x0 |
| 15:0 | COUNT | Line Count. | R/W | 0x0 |

## 8.4.41 LINE_SRC_STRIDE0

**LINE_STRIDE0(Line stride Address Register, offset = 0x00000C08)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | 0x0 |
| 15:0 | STRIDE | Line Stride. | R/W | 0x0 |

## 8.4.42 LINE_REMAIN0

**LINE_REMAIN0(Line remain Address Register, offset = 0x00000C10)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | 0x0 |
| 15:0 | COUNTER | Line remain counter in a transmission of stride mode | R | 0x0 |

## 8.4.43 BYTE_REMAIN_IN_LINE0

**BYTE_REMAIN_IN_LINE0(Line remain in line Address Register, offset = 0x00000C14)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 15:0 | COUNTER | Byte remain counter in the line that transmitting. | R | 0x0 |

## 8.4.44 LINE_LENGTH1

**LINE_LENGTH1(Line Length Address Register, offset = 0x00000C20)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | 0x0 |
| 15:0 | LENGTH | Line Length. | R/W | 0x0 |

## 8.4.45 LINE_COUNT1

**LINE_COUNT1(Line Counter Address Register, offset = 0x00000C24)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | 0x0 |
| 15:0 | COUNT | Line Count. | R/W | 0x0 |

## 8.4.46 LINE_SRC_STRIDE1

**LINE_STRIDE1(Line stride Address Register, offset = 0x00000C28)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | 0x0 |
| 15:0 | STRIDE | Line Stride. | R/W | 0x0 |

## 8.4.47 LINE_DST_STRIDE1

**LINE_STRIDE1(Line stride Address Register, offset = 0x00000C2C)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | 0x0 |
| 15:0 | STRIDE | Line Stride. | R/W | 0x0 |

## 8.4.48 LINE_REMAIN1

**LINE_REMAIN1(Line remain Address Register, offset = 0x00000C30)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | 0x0 |
| 15:0 | COUNTER | Line remain counter in a transmission of stride mode | R | 0x0 |

## 8.4.49 BYTE_REMAIN_IN_LINE1

**BYTE_REMAIN_IN_LINE1(Line remain in line Address Register, offset = 0x00000C34)**

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| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | 0x0 |
| 15:0 | COUNTER | Byte remain counter in the line that transmitting. | R | 0x0 |

## 8.4.50 LINE_LENGTH2

**LINE_LENGTH2(Line Length Address Register, offset = 0x00000C40)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | 0x0 |
| 15:0 | LENGTH | Line Length. | R/W | 0x0 |

## 8.4.51 LINE_COUNT2

**LINE_COUNT2(Line Counter Address Register, offset = 0x00000C44)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | 0x0 |
| 15:0 | COUNT | Line Count. | R/W | 0x0 |

## 8.4.52 LINE_SRC_STRIDE2

**LINE_STRIDE2(Line stride Address Register, offset = 0x00000C48)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | 0x0 |
| 15:0 | STRIDE | Line Stride. | R/W | 0x0 |

## 8.4.53 LINE_DST_STRIDE2

**LINE_STRIDE2(Line stride Address Register, offset = 0x00000C4C)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | 0x0 |
| 15:0 | STRIDE | Line Stride. | R/W | 0x0 |

## 8.4.54 LINE_REMAIN2

**LINE_REMAIN2(Line remain Address Register, offset = 0x00000C50)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | 0x0 |
| 15:0 | COUNTER | Line remain counter in a transmission of stride mode | R | 0x0 |

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## 8.4.55 BYTE_REMAIN_IN_LINE2

**BYTE_REMAIN_IN_LINE2(Line remain in line Address Register, offset = 0x00000C54)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | 0x0 |
| 15:0 | COUNTER | Byte remain counter in the line that transmitting. | R | 0x0 |

## 8.4.56 LINE_LENGTH3

**LINE_LENGTH3(Line Length Address Register, offset = 0x00000C60)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | 0x0 |
| 15:0 | LENGTH | Line Length. | R/W | 0x0 |

## 8.4.57 LINE_COUNT3

**LINE_COUNT3(Line Counter Address Register, offset = 0x00000C64)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | 0x0 |
| 15:0 | COUNT | Line Count. | R/W | 0x0 |

## 8.4.58 LINE_SRC_STRIDE3

**LINE_STRIDE3(Line stride Address Register, offset = 0x00000C68)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | 0x0 |
| 15:0 | STRIDE | Line Stride. | R/W | 0x0 |

## 8.4.59 LINE_DST_STRIDE3

**LINE_STRIDE3(Line stride Address Register, offset = 0x00000C6C)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | 0x0 |
| 15:0 | STRIDE | Line Stride. | R/W | 0x0 |

## 8.4.60 LINE_REMAIN3

**LINE_REMAIN3(Line remain Address Register, offset = 0x00000C70)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | 0x0 |
| 15:0 | COUNTER | Line remain counter in a transmission of stride mode | R | 0x0 |

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## 8.4.61 BYTE_REMAIN_IN_LINE3

**BYTE_REMAIN_IN_LINE3(Line remain in line Address Register, offset = 0x00000C74)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | 0x0 |
| 15:0 | COUNTER | Byte remain counter in the line that transmitting. | R | 0x0 |

## 8.4.62 LINE_LENGTH4

**LINE_LENGTH4(Line Length Address Register, offset = 0x00000C80)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | 0x0 |
| 15:0 | LENGTH | Line Length. | R/W | 0x0 |

## 8.4.63 LINE_COUNT4

**LINE_COUNT43(Line Counter Address Register, offset = 0x00000C84)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | 0x0 |
| 15:0 | COUNT | Line Count. | R/W | 0x0 |

## 8.4.64 LINE_SRC_STRIDE4

**LINE_STRIDE4(Line stride Address Register, offset = 0x00000C88)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | 0x0 |
| 15:0 | STRIDE | Line Stride. | R/W | 0x0 |

## 8.4.65 LINE_DST_STRIDE4

**LINE_STRIDE4(Line stride Address Register, offset = 0x00000C8C)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | 0x0 |
| 15:0 | STRIDE | Line Stride. | R/W | 0x0 |

## 8.4.66 LINE_REMAIN4

**LINE_REMAIN4(Line remain Address Register, offset = 0x00000C90)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | 0x0 |
| 15:0 | COUNTER | Line remain counter in a transmission of stride mode | R | 0x0 |

## 8.4.67 BYTE_REMAIN_IN_LINE4

**BYTE_REMAIN_IN_LINE4(Line remain in line Address Register, offset = 0x00000C94)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | 0x0 |
| 15:0 | COUNTER | Byte remain counter in the line that transmitting. | R | 0x0 |

## 8.4.68 COLOR_FILL_DATA0

**COLOR_FILL_DATA0(COLOR Fill Data0 Register, offset = 0x00000D00)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | DATA | Color fill data | R/W | 0x0 |

## 8.4.69 COLOR_FILL_DATA1

**COLOR_FILL_DATA1(COLOR Fill Data1 Register, offset = 0x00000D04)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | DATA | Color fill data | R/W | 0x0 |

## 8.4.70 COLOR_FILL_DATA2

**COLOR_FILL_DATA2(COLOR Fill Data2 Register, offset = 0x00000D08)**

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | DATA | Color fill data | R/W | 0x0 |

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# 9 PMU

## 9.1Features

The ATS3085Sx integrates a comprehensive power supply system, including the following features:

- Supports 4.2~4.45V Li-Ion battery and Integrate 400mA Linear battery charger with power path management.
- 2 buck converters output VC18, VD12 from BAT.
- Linear regulators output SVCC, VCC, VCC2, VDD.
- Integrated 6*Load switch (VPSRAM/2*VCC_OUT/IOVCC1_OUT/IOVCC2_OUT/IOVCC4_OUT), which can be used as power supply for peripheral.
- Internal bandgap reference.

## 9.2Module Description

### 9.2.1 Block Diagram

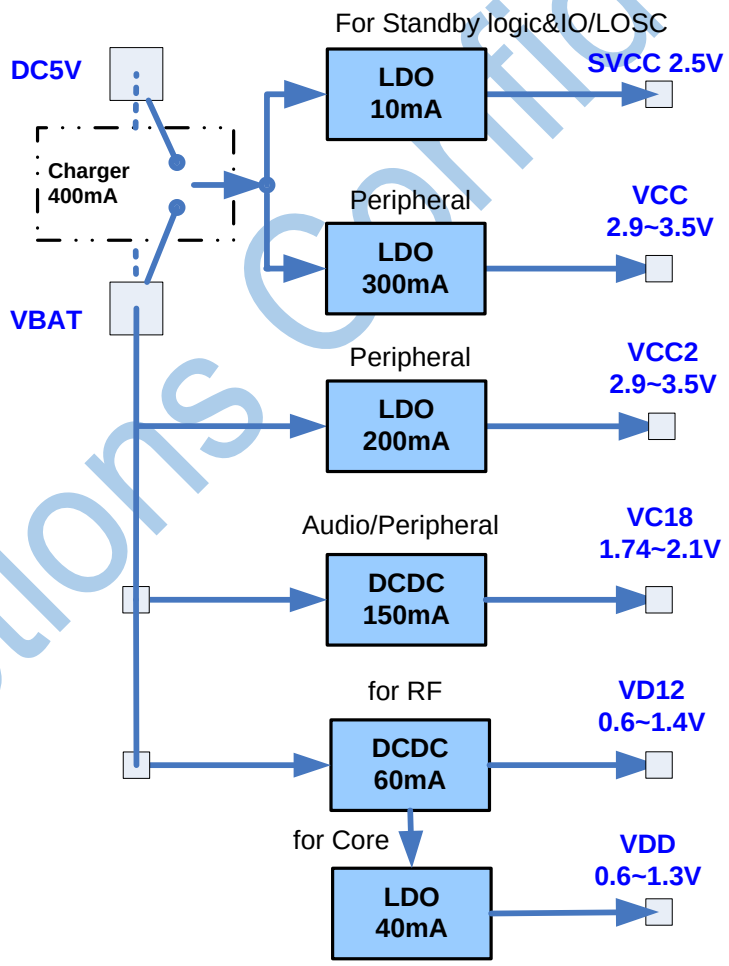



*Figure 9-1 Power Supply Block Diagram*

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### 9.2.2 State machine of PMU

## Description of PMU States:

- S1——active mode, Main CPU/BT is active
- S2——sensor transmit mode, all processors are in sleep, only sensor controller working
- S3——standby, all processors are in deep sleep, can be waked up by interrupt
- S4——shutdown, all regulators are power down except SVCC, can be waked up by DC5V/8HZ timer/WIO pins

*Table 9-1 The states of Regulators*

| PMU State | VCC/VDD/VD12 | VC18 | VCC2 | SVCC |
|---|---|---|---|---|
| S1/S2/S3 | Enable | Programmable | Programmable | Enable |
| S4 | Disable | | | Enable |

### 9.2.3 DC-DC Converter

ATS3085Sx integrates 2 buck DC-DC converters which are named VD12/VC18. VD12/VC18 can also be configured as LDO mode. The DC-DC converter efficiently scales battery voltage to the required supply voltage. The DC-DC converters include several advanced features:

- Input power from BAT
- Support 2.2uH power inductor (>300mA Isat).

*Table 9-2 The feature of DC-DC*

| DCDC | Output Range(V) | Inductor | Capacitor | Inductor Peak Current | Max Load | Output Voltage | Efficiency (Typical) | Ripple | Freq |
|---|---|---|---|---|---|---|---|---|---|
| VD12 | 1.74~2.1 | 2.2uH | ≥4.7uF | 200mA | 60mA | 0.6V | 50% (20uA) | - | - |
| | | | | | | 1.2V | 80% (10mA) | 20mV | ~2M |
| VC18 | 0.6~1.4 | 2.2uH | ≥4.7uF | 300mA | 150mA | 1.86V | 85% (10mA) | 20mV | ~2M |

### 9.2.4 Linear Regulators

ATS3085Sx integrates 6 linear regulators respectively generate SVCC, VCC, VCC2, VD12, VC18, VDD.
All regulators have soft start, SVCC/VCC/VCC2/VC18/VDD have low voltage protection and VCC/VCC2 has over current protection.

#### 9.2.4.1 Regulators Accurate and Maximum Output Current

The output voltages are precisely within ±2.5%, providing maximum output currents when voltages drop to 95%. Table9-3 shows data of maximum output current.

*Table 9-3 Regulators Maximum Output Current*

| Block Name | Output Voltage | Maximum Output Current |
|---|---|---|
| SVCC | 2.5V/2.0V(Shipping mode) | 10 mA |
| VCC | 2.9~3.5V | 300 mA |

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| | | |
|---|---|---|
| VCC2 | 2.9~3.5V | 200 mA |
| VC18 | 1.74~2.1 | 50 mA(LDO mode) |
| VD12 | 0.6~1.4V | 60 mA(LDO mode) |
| VDD | 0.6~1.3V | 40 mA |

### 9.2.5 Load switch

ATS3085Sx integrated 6*Load switch (VPSRAM/2*VCC_OUT/IOVCC1_OUT/IOVCC2_OUT/IOVCC4_OUT), which can be used as power supply for peripheral. VPSRAM is used for PSRAM power supply, while other power supplies can be sent out through specific GPIOs.

***Table 9-4 Load switch description***

| Block Name | Output Voltage | Power source |
|---|---|---|
| VCC_OUT | 2.9~3.5V | VCC |
| IOVCC1_OUT | 2.9~3.5V | VCC |
| IOVCC2_OUT | 2.9~3.5V | VCC/VC18 |
| IOVCC4_OUT | 1.74~2.1 | VC18 |
| VPSRAM | 1.74~2.1 | VC18 |

### 9.2.6 A/D Converters

There are 12 bits multi-channel SARADC for system monitor.

***Table 9-5 Input range of ADC channels***

| LRADC Channel | Input range |
|---|---|
| CHARGEI | 0~512mA |
| BAT | 0V~4.8V |
| DC5V | 0V~6.0V |
| SENSOR | -40~120 ℃ |
| SVCC | 0~3.6V |
| LRADC1 (SVCC domain) | 0~3.6V |
| VCCI | 0~512mA |
| LRADC2/3/4/5/6 (VCC/VC18 domain) | 0~3.6V/0~1.2V |

Note: The input voltage of LRADC can not exceed its power supply voltage.

## 9.3PMUVDD Register List

***Table 9-6 PMUVDD block base address***

| Name | Physical Base Address | KSEG1 Base Address |
|---|---|---|
| PMUVDD | 0x40004000 | 0x40004000 |

***Table 9-7 PMUVDD Block Configuration Registers List***

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| Offset | Register Name | Description | Power domain |
|---|---|---|---|
| 0x00 | VOUT_CTL0 | PMU control Register | VDD |
| 0x04 | VOUT_CTL1_S1 | PMU control under S1 | VDD |
| 0x08 | VOUT_CTL1_S2 | PMU control under S2 | VDD |
| 0X0C | VOUT_CTL1_S3 | PMU control under S3 | VDD |
| 0x10 | PMU_DET | PMU status Register | VDD |
| 0x14 | VOUT_CTL1_S1M | PMU control under S1M Register | VDD |
| 0X30 | PWRGATE_DIG | Digital module Power gating control | VDD |
| 0X34 | PWRGATE_DIG_ACK | Digital module Power up acknowledgement | VDD |
| 0x38 | RAM_SHUTDOWN | RAM Power gating control register | VDD |
| 0x40 | PMU_INTMASK | PMU interrupt set register | VDD |
| 0x44 | RAM_DEEPSLEEP | RAM Retention control register | VDD |
| 0x48 | RAM_LIGHTSLEEP | RAM low power control register | VDD |

## 9.4PMUVDD Register Description

### 9.4.1 VOUT_CTL0

Voltage set register (VDD), offset: 0x00

| Bit (s) | Name | Description | Access | Reset |
|---|---|---|---|---|
| 31 | VPSRAM_EN | VPSRAM output enable:<br>0：disable<br>1：enable | R/W | 0 |
| 30:22 | - | Reserved | R/W | X |
| 21:20 | VCC_SET_DC5V | VCC voltage level select when DC5V generate it.<br>0x0: 3.0V<br>0x1: 3.2V<br>0x2: 3.4V<br>0x3: 3.6V | R/W | 0x2 |
| 19 | - | Reserved | R/W | X |
| 18 | VC18_SW | The mode switch of VC18.<br>0：LDO<br>1：DCDC | R/W | 0 |
| 17 | VD12_SW | The mode switch of VD12.<br>0：LDO<br>1：DCDC | R/W | 0 |
| 16 | - | Reserved | R/W | 0 |
| 15 | VCC_OC_SET | VCC LDO Current limit:<br>0: 250mA<br>1: 450mA | R/W | 0 |
| 14 | VCC2_OC_SET | VCC2 LDO Current limit:<br>0: 250mA<br>1: 450mA | R/W | 0 |
| 13:12 | VCC_SET | VCC voltage level select when BAT generate it.<br>00 2.9V<br>01 3.1V<br>*10 3.3V<br>11 3.5V | R/W | 10 |
| 11:10 | VC18_SET | VC18 voltage level select | R/W | 01 |

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| | | | | |
|---|---|---|---|---|
| | | 00: 1.74V<br>*01: 1.86V<br>10: 1.98V<br>11: 2.1V | | |
| 9:8 | VCC2_SET | VCC2 voltage level select<br>00 2.9V<br>*01 3.1V<br>10 3.3V<br>11 3.5V | R/W | 01 |
| 7:4 | - | Reserved | R/W | 0xc |
| 3 | VCC2_EN | VCC2 LDO enable.<br>0：disable<br>1：enable | R/W | 0 |
| 2:0 | - | Reserved | R/W | 0x1 |

### 9.4.2 VOUT_CTL1_S1

Voltage set register (VDD), offset: 0x04

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:24 | - | Reserved | R/W | 0xc1 |
| 23:22 | VDD_BYPASS_S1BT | VDD LDO enable control in S1BT state<br>00: VDD LDO disable, bypass to VDD12<br>01：auto bypass when VD12 <=VDD<br>10：auto bypass when VD12 < =VDD+50mV<br>11：enable LDO OP | R/W | 11 |
| 21 | VC18_EN_S1 | VC18 enable：<br>0: disable<br>1: enable | | 1 |
| 20:12 | - | Reserved | R/W | 0x1f8 |
| 11:8 | VD12_SET_S1BT | VD12 voltage set by BT subsystem.<br>0000: 0.6V<br>0001: 0.65V<br>0010: 0.7V<br>0011: 0.75V<br>0100: 0.8V<br>0101: 0.85V<br>0110: 0.9V<br>0111: 0.95V<br>1000: 1.0V<br>1001: 1.05V<br>1010: 1.1V<br>1011: 1.15V<br>*1100: 1.2V<br>1101: 1.25V<br>1110: 1.3V<br>1111: 1.4V | R | X |
| 7:4 | VDD_SET_DSP | VDD voltage set by DSP subsystem.<br>0000: Reserved<br>0001: 0.60V<br>0010: 0.65V<br>0011: 0.70V<br>0100: 0.75V | R | X |

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| | | | | |
|---|---|---|---|---|
| | | 0101: 0.80V<br>0110: 0.85V<br>0111: 0.90V<br>1000: 0.95V<br>1001: 1.0V<br>1010: 1.05V<br>1011: 1.1V<br>1100: 1.15V<br>1101: 1.2V<br>1110: 1.25V<br>1111: 1.3V | | |
| 3:0 | VDD_SET_S1BT | VDD voltage set by BT subsystem.<br>0000: Reserved<br>0001: 0.60V<br>0010: 0.65V<br>0011: 0.70V<br>0100: 0.75V<br>0101: 0.80V<br>0110: 0.85V<br>0111: 0.9V<br>1000: 0.95V<br>1001: 1.0V<br>1010: 1.05V<br>1011: 1.1V<br>1100: 1.15V<br>1101: 1.2V<br>1110: 1.25V<br>1111: 1.3V | R | X |

### 9.4.3 VOUT_CTL1_S2

Voltage set register (VDD), offset: 0x08

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:24 | - | Reserved | R/W | 0x13 |
| 23:22 | VDD_BYPASS_S2 | VDD LDO enable control in S2 state：<br>00：bypass to VDD12, VDD LDO disable<br>01：auto bypass when VD12 =VDD<br>10：auto bypass when VD12 < =VDD+50mV<br>11：enable VDD LDO | R/W | 11 |
| 21 | VC18_EN_S2 | VC18 enable<br>0: disable<br>1: enable | R/W | 0x1 |
| 20:12 | - | Reserved | R/W | 0x0F8 |
| 11:8 | VD12_SET_S2 | VD12 voltage control of S2 state.<br>0000: 0.6V<br>0001: 0.65V<br>0010: 0.7V<br>0011: 0.75V<br>0100: 0.8V<br>0101: 0.85V<br>*0110: 0.9V<br>0111: 0.95V | R/W | 0110 |

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| | | | | |
|---|---|---|---|---|
| | | 1000: 1.0V<br>1001: 1.05V<br>1010: 1.1V<br>1011: 1.15V<br>1100: 1.2V<br>1101: 1.25V<br>1110: 1.3V<br>1111: 1.4V | | |
| 7:3 | - | Reserved | R/W | 01010 |
| 2:0 | VDD_SET_S2 | VDD voltage control of S2 state.<br>000 Reserved<br>001 0.60V<br>010 0.65V<br>011 0.70V<br>100 0.75V<br>*101 0.80V<br>110 0.85V<br>111 0.90V | R/W | 101 |

### 9.4.4 VOUT_CTL1_S3

Voltage set register (VDD), offset: 0x0C

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:24 | - | Reserved | R/W | 0x3 |
| 23:22 | VDD_BYPASS_S3 | VDD LDO enable control in S3 state：<br>00：bypass to VDD12, VDD LDO disable<br>01：auto bypass when VD12 =VDD<br>10：auto bypass when VD12 < =VDD+50mV<br>11：enable VDD LDO | R/W | 11 |
| 21 | VC18_EN_S3 | VC18 enable<br>0: disable<br>1: enable | R/W | 1 |
| 20:15 | - | Reserved | R/W | 0 |
| 14:12 | VD12_SET_S3_HOSCEN | VD12 voltage control of S3 state when HOSC is enable.<br>000: 0.6V<br>001: 0.65V<br>010: 0.7V<br>011: 0.75V<br>*100: 0.8V<br>101: 0.85V<br>110: 0.9V<br>111: 0.95V | R/W | 100 |
| 11 | - | Reserved | R | 0 |
| 10:8 | VD12_SET_S3 | VD12 voltage control of S3 state when HOSC is disable.<br>000: 0.6V<br>001: 0.65V<br>010: 0.7V<br>011: 0.75V<br>*100: 0.8V<br>101: 0.85V<br>110: 0.9V<br>111: 0.95V | R/W | 100 |

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| | | | | |
|---|---|---|---|---|
| 7 | - | Reserved | R | 0 |
| 6:4 | VDD_SET_S3_HOSCEN | VDD voltage control of S3 state when HOSC is enable.<br><br>000 Reserved<br>001 0.60V<br>010 0.65V<br>011 0.70V<br>100 0.75V<br>*101 0.80V<br>110 0.85V<br>111 0.90V | R/W | 101 |
| 3 | - | Reserved | R | 0 |
| 2:0 | VDD_SET_S3 | VDD voltage control of S3 state when HOSC is disable.<br>000 Reserved<br>001 0.60V<br>010 0.65V<br>*011 0.70V<br>100 0.75V<br>101 0.80V<br>110 0.85V<br>111 0.90V | R/W | 011 |

### 9.4.5 PMU_DET

Charger detect register (VDD), Offset: 0x10

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:13 | - | Reserved | R | 0 |
| 12 | COUNTER8HZ_PD | Counter8hz reset pending occurred<br>0: Pending not occurred<br>1: Pending occurred<br>Writing 1 to the bit will clear the bit | R/W | 0 |
| 11 | S1MAIN_REQ | PMU is in S1M state or not | R | X |
| 10 | S1BT_REQ | PMU is in S1BT state or not | R | X |
| 9 | - | reserved | R | 0 |
| 8 | S2_REQ | PMU is in S2 state or not | R | X |
| 7:1 | - | reserved | R | 0 |
| 0 | DC5VIN_DET | The indication of DC5V is plug in.<br>0: DC5V<BAT+0.033V<br>1: DC5V> BAT+0.080V | R | X |

### 9.4.6 VOUT_CTL1_S1M

Voltage set register (VDD) offset: 0x14

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:24 | - | Reserved | R | 0x21 |
| 23:22 | VDD_BYPASS_S1M | VDD LDO enable control in S1M state<br>00： bypass to VDD12, VDD LDO disable<br>01： auto bypass when VD12 =VDD<br>10： auto bypass when VD12 < =VDD+50mV<br>11： enable VDD LDO | R/W | 0x3 |

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| | | | | |
|---|---|---|---|---|
| 21:12 | - | Reserved | R/W | 0x3F9 |
| 11:8 | VD12_SET_S1M | 0000: 0.6V<br>0001: 0.65V<br>0010: 0.7V<br>0011: 0.75V<br>0100: 0.8V<br>0101: 0.85V<br>0110: 0.9V<br>0111: 0.95V<br>1000: 1.0V<br>1001: 1.05V<br>1010: 1.1V<br>1011: 1.15V<br>*1100: 1.2V<br>1101: 1.25V<br>1110: 1.3V<br>1111: 1.4V | R/W | 0xC |
| 7:4 | VDD_SET_S1ML | VDD voltage coarse control : (S1M light sleep)<br>0000: Reserved<br>0001: 0.60V<br>0010: 0.65V<br>0011: 0.70V<br>0100: 0.75V<br>0101: 0.80V<br>0110: 0.85V<br>0111: 0.90V<br>1000: 0.95V<br>*1001:1.0V<br>1010: 1.05V<br>1011: 1.1V<br>1100: 1.15V<br>1101: 1.2V<br>1110: 1.25V<br>1111: 1.3V | R/W | 0x9 |
| 3:0 | VDD_SET_S1M | VDD voltage coarse control ： (S1M)<br>0000: Reserved<br>0001: 0.60V<br>0010: 0.65V<br>0011: 0.70V<br>0100: 0.75V<br>0101: 0.80V<br>0110: 0.85V<br>0111: 0.90V<br>1000: 0.95V<br>1001: 1.0V<br>1010: 1.05V<br>*1011: 1.1V<br>1100: 1.15V<br>1101: 1.2V<br>1110: 1.25V<br>1111: 1.3V | R/W | 0xB |

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### 9.4.7 PWRGATE_DIG

Power gating Control Register (VDD), Offset: 0x30

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31 | MAINCPU_PG | Main CPU sub system power gating bit:<br>0：disable<br>1：enable<br>When disable gating, power will shut down | R/W | 1 |
| 30:29 | - | Reserved | - | 0 |
| 28 | BT_PG | BT sub system power gating bit:<br>0：disable<br>1：enable<br>When disable gating, power will shut down | R/W | 1 |
| 27 | - | Reserved | - | 0 |
| 26 | AUDSP_PG | AUDSP sub system power gating bit:<br>0：disable<br>1：enable<br>When disable gating, power will shut down | R/W | 1 |
| 25 | 2.5DGPU_PG | 2.5DGPU power gating bit:<br>0：disable<br>1：enable<br>When disable gating, power will shut down | R/W | 1 |
| 24:22 | - | reserved | R/W | X |
| 21 | DISPLAY_PG | Display system power gating bit:<br>0：disable<br>1：enable<br>When disable gating, power will shut down | R/W | 1 |
| 20:0 | - | Reserved | R | 0x 100000 |

### 9.4.8 PWRGATE_DIG_ACK

Power gating Control Register (VDD), Offset: 0x34

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31 | MAINCPU_ACK | Main CPU sub system power OK Acknowledgement:<br>0：NO ACK<br>1：ACK | R | X |
| 30:29 | - | Reserved | R | 0 |
| 28 | BT_ACK | BT sub system power OK Acknowledgement:<br>0：NO ACK<br>1：ACK | R | X |
| 27 | - | Reserved | R | 0 |
| 26 | AUDSP_ACK | AUDSP sub system power OK Acknowledgement:<br>0：NO ACK<br>1：ACK | R | X |
| 25 | 2.5DGPU_ACK | 2.5DGPU power OK Acknowledgement:<br>0：NO ACK<br>1：ACK | R | X |
| 24:22 | - | reserved | R | X |
| 21 | DISPLAY_ACK | Display system power OK Acknowledgement: | R | X |

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| | | | | |
|---|---|---|---|---|
| | | 0：NO ACK<br>1：ACK | | |
| 20:0 | - | RESERVED | R | 0 |

### 9.4.9 RAM_SHUTDOWN

RAM Power gating Control Register (VDD), Offset: 0x38

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:27 | - | Reserved | R | 0x0 |
| 26 | RAM_USB_SD | USB RAM<br>0：NOT<br>1：shut down | R/W | 0x0 |
| 25:19 | - | Reserved | R | 0x0 |
| 18 | JPEGRAM_SD | JPEG_OUTRAM<br>0：NOT<br>1：Shut Down | R/W | 0x0 |
| 17 | SHARERAM_SD | SHARERAM 32k<br>0：NOT<br>1：Shut Down | R/W | 0x0 |
| 16 | RAM16_SD | RAM16 48k<br>0：NOT<br>1：Shut Down | R/W | 0x0 |
| 15 | RAM15_SD | RAM15 32k<br>0：NOT<br>1：Shut Down | R/W | 0x0 |
| 14 | RAM14_SD | RAM14 64k<br>0：NOT<br>1：Shut Down | R/W | 0x0 |
| 13 | RAM13_SD | RAM13 16k<br>0：NOT<br>1：Shut Down | R/W | 0x0 |
| 12 | RAM12_SD | RAM12 64k<br>0：NOT<br>1：Shut Down | R/W | 0x0 |
| 11 | RAM11_SD | RAM11 64k<br>0：NOT<br>1：Shut Down | R/W | 0x0 |
| 10 | RAM10_SD | RAM10 64k<br>0：NOT<br>1：Shut Down | R/W | 0x0 |
| 9 | RAM9_SD | RAM9 64k<br>0：NOT<br>1：Shut Down | R/W | 0x0 |
| 8 | RAM8_SD | RAM8 64k<br>0：NOT<br>1：Shut Down | R/W | 0x0 |
| 7 | RAM7_SD | RAM7 64k<br>0：NOT<br>1：Shut Down | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 6 | RAM6_SD | RAM6 64k<br>0：NOT<br>1：Shut Down | R/W | 0x0 |
| 5 | RAM5_SD | RAM5 64k<br>0：NOT<br>1：Shut Down | R/W | 0x0 |
| 4 | RAM4_SD | RAM4 64k<br>0：NOT<br>1：Shut Down | R/W | 0x0 |
| 3 | RAM3_SD | RAM3 64k<br>0：NOT<br>1：Shut Down | R/W | 0x0 |
| 2 | RAM2_SD | RAM2 64k<br>0：NOT<br>1：Shut Down | R/W | 0x0 |
| 1 | RAM1_SD | RAM1 64k<br>0：NOT<br>1：Shut Down | R/W | 0x0 |
| 0 | RAM0_SD | RAM0 64k<br>0：NOT<br>1：Shut Down | R/W | 0x0 |

### 9.4.10 PMU_INTMASK

Power gating Control Register (VDD), Offset: 0x40

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:13 | - | RESERVED | R | 0 |
| 12 | REMOTE_INTEN | REMOTE interrupt enable<br>0: Do not send interrupt signal to CPU<br>1: Send interrupt signal to CPU | R/W | 0 |
| 11 | BATLV_INTEN | BAT low voltage interrupt enable<br>0: Do not send interrupt signal to CPU<br>1: Send interrupt signal to CPU | R/W | 0 |
| 10 | DC5VLV_INTEN | DC5V low voltage interrupt enable<br>0: Do not send interrupt signal to CPU<br>1: Send interrupt signal to CPU | R/W | 0 |
| 9:7 | - | reserved | R | 0 |
| 6 | WIO0LV_INTEN | WIO0 low voltage interrupt enable<br>0: Do not send interrupt signal to CPU<br>1: Send interrupt signal to CPU | R/W | 0 |
| 5 | - | reserved | R | 0 |
| 4 | ALARM8HZ_INTEN | ALARM8HZ interrupt enable<br>0: Do not send interrupt signal to CPU<br>1: Send interrupt signal to CPU | R/W | 0x0 |
| 3 | DC5VOUT_INTEN | DC5V plug out interrupt enable<br>0: Do not send interrupt signal to CPU<br>1: Send interrupt signal to CPU | R/W | 0 |
| 2 | DC5VIN_INTEN | DC5V plug in interrupt enable<br>0: Do not send interrupt signal to CPU<br>1: Send interrupt signal to CPU | R/W | 1 |

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| | | | | |
|---|---|---|---|---|
| 1 | ONOFF_S_INTEN | ONOFF short press interrupt enable<br>0: Do not send interrupt signal to CPU<br>1: Send interrupt signal to CPU | R/W | 0 |
| 0 | ONOFF_L_INTEN | ONOFF long press interrupt enable<br>0: Do not send interrupt signal to CPU<br>1: Send interrupt signal to CPU | R/W | 1 |

### 9.4.11 RAM_DEEPSLEEP

RAM Retention Control Register (VDD), Offset: 0x44

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:27 | - | Reserved | R | 0x0 |
| 26 | RAM_USB_DS | USB RAM<br>0：NOT<br>1：Deep sleep | R/W | 0x0 |
| 25:20 | - | Reserved | R | 0x0 |
| 18 | JPEGRAM_DS | JPEG_OUTRAM<br>0：NOT<br>1：Deep sleep | R/W | 0x0 |
| 17 | SHARERAM_DS | SHARERAM 32k<br>0：NOT<br>1：Deep sleep | R/W | 0x0 |
| 16 | RAM16_DS | RAM16 48k<br>0：NOT<br>1：Deep sleep | R/W | 0x0 |
| 15 | RAM15_DS | RAM15 32k<br>0：NOT<br>1：Deep sleep | R/W | 0x0 |
| 14 | RAM14_DS | RAM14 64k<br>0：NOT<br>1：Deep sleep | R/W | 0x0 |
| 13 | RAM13_DS | RAM13 16k<br>0：NOT<br>1：Deep sleep | R/W | 0x0 |
| 12 | RAM12_DS | RAM12 64k<br>0：NOT<br>1：Deep sleep | R/W | 0x0 |
| 11 | RAM11_DS | RAM11 64k<br>0：NOT<br>1：Deep sleep | R/W | 0x0 |
| 10 | RAM10_DS | RAM10 64k<br>0：NOT<br>1：Deep sleep | R/W | 0x0 |
| 9 | RAM9_DS | RAM9 64k<br>0：NOT<br>1：Deep sleep | R/W | 0x0 |
| 8 | RAM8_DS | RAM8 64k<br>0：NOT<br>1：Deep sleep | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 7 | RAM7_DS | RAM7 64k<br>0：NOT<br>1：Deep sleep | R/W | 0x0 |
| 6 | RAM6_DS | RAM6 64k<br>0：NOT<br>1：Deep sleep | R/W | 0x0 |
| 5 | RAM5_DS | RAM5 64k<br>0：NOT<br>1：Deep sleep | R/W | 0x0 |
| 4 | RAM4_DS | RAM4 64k<br>0：NOT<br>1：Deep sleep | R/W | 0x0 |
| 3 | RAM3_DS | RAM3 64k<br>0：NOT<br>1：Deep sleep | R/W | 0x0 |
| 2 | RAM2_DS | RAM2 64k<br>0：NOT<br>1：Deep sleep | R/W | 0x0 |
| 1 | RAM1_DS | RAM1 64k<br>0：NOT<br>1：Deep sleep | R/W | 0x0 |
| 0 | RAM0_DS | RAM0 64k<br>0：NOT<br>1：Deep sleep | R/W | 0x0 |

### 9.4.12 RAM_LIGHTSLEEP

Ram Lightsleep Control Register (VDD), Offset: 0x48

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:27 | - | Reserved | R | 0x0 |
| 26 | RAM_USB_LS | USB RAM<br>0：NOT<br>1：Light sleep | R/W | 0x0 |
| 25:19 | - | Reserved | R/W | 0x0 |
| 18 | JPEGRAM_LS | JPEG_OUTRAM<br>0：NOT<br>1：Light sleep | R/W | 0x0 |
| 17 | SHARERAM_LS | SHARERAM 32k<br>0：NOT<br>1：Light sleep | R/W | 0x0 |
| 16 | RAM16_LS | RAM16 48k<br>0：NOT<br>1：Light sleep | R/W | 0x0 |
| 15 | RAM15_LS | RAM15 32k<br>0：NOT<br>1：Light sleep | R/W | 0x0 |
| 14 | RAM14_LS | RAM14 64k<br>0：NOT<br>1：Light sleep | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 13 | RAM13_LS | RAM13 16k<br>0：NOT<br>1：Light sleep | R/W | 0x0 |
| 12 | RAM12_LS | RAM12 64k<br>0：NOT<br>1：Light sleep | R/W | 0x0 |
| 11 | RAM11_LS | RAM11 64k<br>0：NOT<br>1：Light sleep | R/W | 0x0 |
| 10 | RAM10_LS | RAM10 64k<br>0：NOT<br>1：Light sleep | R/W | 0x0 |
| 9 | RAM9_LS | RAM9 64k<br>0：NOT<br>1：Light sleep | R/W | 0x0 |
| 8 | RAM8_LS | RAM8 64k<br>0：NOT<br>1：Light sleep | R/W | 0x0 |
| 7 | RAM7_LS | RAM7 64k<br>0：NOT<br>1：Light sleep | R/W | 0x0 |
| 6 | RAM6_LS | RAM6 64k<br>0：NOT<br>1：Light sleep | R/W | 0x0 |
| 5 | RAM5_LS | RAM5 64k<br>0：NOT<br>1：Light sleep | R/W | 0x0 |
| 4 | RAM4_LS | RAM4 64k<br>0：NOT<br>1：Light sleep | R/W | 0x0 |
| 3 | RAM3_LS | RAM3 64k<br>0：NOT<br>1：Light sleep | R/W | 0x0 |
| 2 | RAM2_LS | RAM2 64k<br>0：NOT<br>1：Light sleep | R/W | 0x0 |
| 1 | RAM1_LS | RAM1 64k<br>0：NOT<br>1：Light sleep | R/W | 0x0 |
| 0 | RAM0_LS | RAM0 64k<br>0：NOT<br>1：Light sleep | R/W | 0x0 |

## 9.5PMUSVCC Register List

*Table 9-8 PMUSVCC block base address*

| Name | Physical Base Address | KSEG1 Base Address |
|---|---|---|
| PMUSVCC | 0x40004000 | 0x40004000 |

*Table 9-9 PMUSVCC Block Configuration Registers List*

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| Offset | Register Name | Description | Power domain |
|---|---|---|---|
| 0X100 | CHG_CTL_SVCC | CHARGING CONTROL | SVCC |
| 0X108 | SYSTEM_SET_SVCC | POWER SYSTEM & ONOFF SET | SVCC |
| 0X10C | POWER_CTL_SVCC | POWER OFF CONTROL | SVCC |
| 0X110 | WAKE_CTL_SVCC | WAKE UP & INTERRUPT SOURCE CONTROL | SVCC |
| 0X114 | WAKE_PD_SVCC | WAKE UP PENDING | SVCC |
| 0X118 | COUNTER8HZ_SVCC | 8-HZ COUNTER | SVCC |
| 0X11C | ALARM8HZ_SVCC | COUNTER ALARM | SVCC |

## 9.6PMUSVCC Register Description

### 9.6.1 CHG_CTL_SVCC

Charger control register (SVCC), Offset: 0x100

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:11 | - | reserved | R/W | X |
| 10 | CHG_EN | CHARGER ENABLE BIT<br>0：DISABLE<br>1：ENABLE | R/W | 1 |
| 9:5 | CV_OFFSET | The voltage setting when charger in CV mode.<br>00000:4.1375V<br>.....<br>11111:4.525V<br>Step=12.5mV | R/W | 01001 |
| 4 | - | reserved | R/W | 0 |
| 3:0 | CHG_CURRENT | The current setting when charger in CC mode.<br>*0000: 20mA<br>0001: 30mA<br>0010: 40mA<br>0011: 50mA<br>0100: 60mA<br>0101: 70mA<br>0110: 80mA<br>0111: 90mA<br>1000: 100mA<br>1001: 150mA<br>1010: 200mA<br>1011: 250mA<br>1100: 300mA<br>1101: 350mA<br>1110: 400mA<br>1111: 450mA | R/W | 0x0 |

### 9.6.2 SYSTEM_SET_SVCC

System set Register (SVCC), Offset: 0x108

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:28 | - | reserved | R | 0 |

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| | | | | |
|---|---|---|---|---|
| 27 | DC5VRST_EN | DC5V reset enable<br>0:disable<br>1:enable | R/W | 0 |
| 26:24 | - | reserved | R/W | 010 |
| 23:21 | ONOFF_PRESS_TIME | ONOFF press time setting<br>000: 50ms < t < 0.125s, regarded as short press;<br>t >=0.125s, regarded as long press;<br><br>001: 50ms < t < 0.25s, regarded as short press;<br>t >=0.25s, regarded as long press;<br><br>010: 50ms < t < 0.5s, regarded as short press;<br>t >=0.5s, regarded as long press;<br><br>011: 50ms < t < 1s, regarded as short press;<br>t >=1s, regarded as long press;<br><br>100: 50ms < t < 1.5s, regarded as short press;<br>t >=1.5s, regarded as long press;<br><br>101: 50ms < t < 2s, regarded as short press;<br>t >=2s, regarded as long press;<br><br>110: 50ms < t < 3s, regarded as short press;<br>t >=3s, regarded as long press;<br><br>111: 50ms < t < 4s, regarded as short press;<br>t >=4s, regarded as long press; | R/W | 001 |
| 20:19 | ONOFF_RST_EN | ONOFF long press function selection.<br>00：no action<br>01：reset the chip<br>10：restart the chip | R/W | 01 |
| 18 | ONOFF_RST_TIME | ONOFF long press reset/restart time configuration.<br>0:8s<br>1:12s | R/W | 0 |
| 17 | ONOFF_PRES | ONOFF press state indication.<br>0: ONOFF is not press down<br>1: ONOFF is press down | R | 0 |
| 16 | PULLR_SEL | ONOFF press down voltage select.<br>0：Regarded as ONOFF pin press down When ONOFF detect high<br>1：Regarded as ONOFF pin press down When ONOFF detect low | R/W | 0 |
| 15:12 | - | Reserved | R/W | 0 |
| 11 | LB_EN | Entry shutdown mode when low battery.<br>0：disable<br>1：enable | R/W | 1 |
| 10:9 | LB_VOL | LB (Low battery) voltage setting<br>00: 2.8V<br>01: 2.9V<br>*10: 3.0V<br>11: 3.3V | R/W | 10 |
| 8 | OC_EN | VCC over current protection enable bit. | R/W | 1 |

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| | | | | |
|---|---|---|---|---|
| | | 0：disable<br>1：enable | | |
| 7:0 | - | Reserved | R/W | X |

### 9.6.3 POWER_CTL_SVCC

Power Control Register (SVCC), Offset: 0x10C

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:1 | - | Reserved | R | 0 |
| 0 | ENTER_S4 | 0: enter S4<br>1: default | R/W | 1 |

### 9.6.4 WAKE_CTL_SVCC

Wake up source enable Register (SVCC), Offset: 0x110

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:30 | - | Reserved | R/W | 0 |
| 29:27 | BATLV_VOL | The threshold of BAT low voltage wakeup.<br>000：3.4V<br>001：3.5V<br>010：3.6V<br>011：4.1V<br>100：4.2V<br>101：4.25V<br>110：4.3V<br>111：4.35V | R/W | 000 |
| 26:24 | DC5VLV_VOL | The threshold of DC5V low voltage wakeup.<br>000：0.2V<br>001：0.5V<br>010：1.0V<br>011：1.5V<br>100：2.0V<br>101：3V<br>110：4V<br>111：4.5V | R/W | 000 |
| 23:22 | - | Reserved | R/W | 00 |
| 21:20 | WIO0LV_VOL | WIO0 low voltage selection.<br>00：<0.45V;<br>01：<1V;<br>10：<1.25V;<br>11：<1.5V; | R/W | 01 |
| 19:13 | - | Reserved | R/W | X |
| 12 | REMOTE_WKEN | Remote wakeup enable<br>0：disable<br>1：enable | R/W | 0 |
| 11 | BATLV_WKEN | BAT low voltage wakeup enable<br>0：disable<br>1：enable | R/W | 0 |

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| | | | | |
|---|---|---|---|---|
| 10 | DC5VLV_WKEN | DC5V low voltage wakeup enable<br>0：disable<br>1：enable | R/W | 0 |
| 9 | - | Reserved | R/W | 0 |
| 8 | WIO0LV_DETEN | WIO0 low voltage detect enable<br>0：disable<br>1：enable | R/W | 1 |
| 7 | - | Reserved | R/W | 0 |
| 6 | WIO0LV_WKEN | WIO0 low voltage wakeup enable<br>0：disable<br>1：enable | R/W | 0 |
| 5 | WIO_WKEN | WIO0 wakeup enable<br>0：disable<br>1：enable | R/W | 0 |
| 4 | ALARM8HZ_WKEN | ALARM8hz wakeup enable<br>0：disable<br>1：enable | R/W | 0 |
| 3 | DC5VOUT_WKEN | DC5V plug out wakeup enable<br>0：disable<br>1：enable | R/W | 0 |
| 2 | DC5VIN_WKEN | DC5V plug in wakeup enable<br>0：disable<br>1：enable | R/W | 1 |
| 1 | SHORT_WKEN | ONOFF short press wakeup enable<br>0: disable<br>1: enable | R/W | 0 |
| 0 | LONG_WKEN | ONOFF long press wakeup enable<br>0: disable<br>1: enable | R/W | 1 |

### 9.6.5 WAKE_PD_SVCC

WAKE up source pending Register (SVCC), Offset: 0x114

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:23 | - | Reserved | R/W | 0 |
| 22 | ONOFF_8S_RESET | ONOFF_8S reset pending occurred<br>0: Pending not occurred<br>1: Pending occurred<br>Writing 1 to the bit will clear the bit | R/W | 0 |
| 21 | WD_RST_PD | WD reset pending occurred<br>0: Pending not occurred<br>1: Pending occurred<br>Writing 1 to the bit will clear the bit | R/W | 0 |
| 20 | DC5VIN_RST_PD | DC5VIN reset pending：<br>0: No DC5VIN reset occurred<br>1: DC5VIN reset occurred | R/W | 0 |
| 19 | - | Reserved | R/W | 0 |
| 18 | LB_PD | Low battery protection pending.<br>0: No Low battery occurred<br>1: No Low battery occurred | R/W | 0 |

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| | | | | |
|---|---|---|---|---|
| 17 | OC_PD | Over current pending<br>0: No Over current occurred<br>1: Over current occurred | R/W | 0 |
| 16 | LVPRO_PD | Low voltage protect pending.<br>0: Low voltage protect is not occurred<br>1: Low voltage protect is occurred | R/W | 0 |
| 15:13 | - | Reserved | R | 0 |
| 12 | REMOTE_PD | Remote pending.<br>0: Remote pending is not occurred<br>1: Remote pending is occurred | R/W | 0 |
| 11 | BATLV_PD | BAT low voltage pending.<br>0: BAT low voltage is not occurred<br>1: BAT low voltage is occurred | R/W | 0 |
| 10 | DC5VLV_PD | DC5V low voltage pending.<br>0: DC5V low voltage is not occurred<br>1: DC5V low voltage is occurred | R/W | 0 |
| 9:7 | - | Reserved | R | 0 |
| 6 | WIO0LV_PD | WIO0 low voltage pending.<br>0: WIO0 low voltage is not occurred<br>1: WIO0 low voltage is occurred | R/W | 0 |
| 5 | WIO_PD | WIO wakeup pending.<br>0: WIO wakeup is not occurred<br>1: WIO wakeup is occurred | R/W | 0 |
| 4 | ALARM8HZ_PD | ALARM8HZ pending.<br>0: ALARM8HZ is not occurred<br>1: ALARM8HZ is occurred | R/W | 0 |
| 3 | DC5VOUT_PD | DC5V plug out pending.<br>0: DC5V plug out is not occurred<br>1: DC5V plug out is occurred | R/W | 0 |
| 2 | DC5VIN_PD | DC5V plug in pending.<br>0: DC5V plug in is not occurred<br>1: DC5V plug in is occurred | R/W | 0 |
| 1 | ONOFF_S_PD | ONOFF short press pending.<br>0: ONOFF short press is not occurred<br>1: ONOFF short press is occurred | R/W | 0 |
| 0 | ONOFF_L_PD | ONOFF long press pending.<br>0: ONOFF long press is not occurred<br>1: ONOFF long press is occurred | R/W | 0 |

### 9.6.6 COUNTER8HZ_SVCC

SVCC 8hz counter Register (SVCC), offset: 0x118

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:28 | - | Reserved | R | 0x0 |
| 27 | CLK_SEL | Counter8hz clock source select.<br>0：RC32k<br>1：LOSC | RW | 0x0 |
| 26 | COUNTER_EN | Counter enable<br>0: disable<br>1: enable | RW | 0x0 |
| 25:0 | COUNTER_VAL | 8hz counter value. | R | 0x0 |

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### 9.6.7 ALARM8HZ_SVCC

SVCC 8hz Alarm Register (SVCC), offset: 0x11c

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:27 | - | Reserved | R | 0x0 |
| 26 | ALARM8HZ_PD | ALARM8HZ pending | R | 0 |
| 25:0 | ALARM_VAL | 8hz Alarm value setting | RW | 0x0 |

## 9.7PMUADC Register List

*Table 9-10 PMUADC block base address*

| Name | Physical Base Address | KSEG1 Base Address |
|---|---|---|
| PMUADC | 0x40004000 | 0x40004000 |

*Table 9-11 PMUADC Block Configuration Registers List*

| Offset | Register Name | Description | Power domain |
|---|---|---|---|
| 0x200 | PMUADC_CTL | SARADC control | VDD |
| 0X204 | PMUADC_INTMASK | SARADC sample interrupt set | VDD |
| 0X208 | PMUADC_PD | SARADC sample pending | VDD |
| 0X20C | PMUADCDIG_CTL | SARADC digital control | VDD |
| 0x210 | CHARGI_DATA | Measure Charging current | VDD |
| 0x214 | BATADC_DATA | Measure battery voltage | VDD |
| 0x218 | DC5VADC_DATA | Measure DC5V voltage | VDD |
| 0x21C | SENSADC_DATA | IC Temperature sensor | VDD |
| 0x220 | SVCCADC_DATA | Measure SVCC voltage | VDD |
| 0x224 | LRADC1_DATA | Measure IO voltage | VDD |
| 0x228 | VCCI_DATA | Measure BAT to VCC current voltage | VDD |
| 0x22C | LRADC2_DATA | Measure IO voltage | VDD |
| 0x230 | LRADC3_DATA | Measure IO voltage | VDD |
| 0x234 | LRADC4_DATA | Measure IO voltage | VDD |
| 0x238 | LRADC5_DATA | Measure IO voltage | VDD |
| 0x23C | LRADC6_DATA | Measure IO voltage | VDD |

## 9.8PMUADC Register Description

### 9.8.1 PMUADC_CTL

PMUADC Control Register (VDD), Offset = 0x200

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:24 | - | Reserved | - | X |
| 23 | PMUADC_EN | ADC enable：<br>0：disable<br>1：enable | R/W | 0 |
| 22:18 | - | Reserved | - | 0x0a |

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| | | | | |
|---|---|---|---|---|
| 17 | LRADC6_SCAL | The voltage range of ADC6 input.<br>0: 0~VCC, enable the voltage dividing resistor;<br>1: 0~1.2V, bypass the voltage dividing resistor; | R/W | 0 |
| 16 | LRADC5_SCAL | The voltage range of ADC5 input.<br>0: 0~VCC, enable the voltage dividing resistor;<br>1: 0~1.2V, bypass the voltage dividing resistor; | R/W | 0 |
| 15 | LRADC4_SCAL | The voltage range of ADC4 input.<br>0: 0~VCC, enable the voltage dividing resistor;<br>1: 0~1.2V, bypass the voltage dividing resistor; | R/W | 0 |
| 14 | LRADC3_SCAL | The voltage range of ADC3 input.<br>0: 0~VCC, enable the voltage dividing resistor;<br>1: 0~1.2V, bypass the voltage dividing resistor; | R/W | 0 |
| 13 | LRADC2_SCAL | The voltage range of ADC2 input.<br>0: 0~VCC, enable the voltage dividing resistor;<br>1: 0~1.2V, bypass the voltage dividing resistor; | R/W | 0 |
| 12 | - | Reserved | - | 00 |
| 11 | LRADC6_CHEN | LRADC6 A/D channel enable.<br>0: Disable<br>1: Enable | R/W | 0 |
| 10 | LRADC5_CHEN | LRADC5 A/D channel enable.<br>0: Disable<br>1: Enable | R/W | 0 |
| 9 | LRADC4_CHEN | LRADC4 A/D channel enable.<br>0: Disable<br>1: Enable | R/W | 0 |
| 8 | LRADC3_CHEN | LRADC3 A/D channel enable.<br>0: Disable<br>1: Enable | R/W | 0 |
| 7 | LRADC2_CHEN | LRADC2 A/D channel enable.<br>0: Disable<br>1: Enable | R/W | 0 |
| 6 | VCCI_CHEN | BAT-VCC current A/D channel enable.<br>0: Disable<br>1: Enable | R/W | 0 |
| 5 | LRADC1_CHEN | LRADC1 A/D channel enable.<br>0: Disable<br>1: Enable | R/W | 1 |
| 4 | SVCC_CHEN | SVCC A/D channel enable<br>0: Disable<br>1: Enable | R/W | 0 |
| 3 | SENSOR_CHEN | Sensor A/D channel enable<br>0: Disable, sensor circuit and output disable<br>1: Enable, sensor circuit and output enable | R/W | 0 |
| 2 | DC5V_CHEN | DC5V A/D channel enable<br>0: Disable<br>1: Enable | R/W | 0 |
| 1 | BATV_CHEN | Battery A/D channel enable<br>0: Disable<br>1: Enable | R/W | 1 |
| 0 | CHARGI_CHEN | CHARGI A/D channel enable<br>0: Disable<br>1: Enable | R/W | 0 |

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## 9.8.2 PMUADC_INTMASK

PMUADC Control Register (VDD), Offset = 0x204

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:12 | - | Reserved | R | 0x0 |
| 11 | LRADC6_INTEN | LRADC6 A/D interrupt enable.<br>0: Disable<br>1: Enable | R/W | 0x0 |
| 10 | LRADC5_INTEN | LRADC5 A/D interrupt enable.<br>0: Disable<br>1: Enable | R/W | 0x0 |
| 9 | LRADC4_INTEN | LRADC4 A/D interrupt enable.<br>0: Disable<br>1: Enable | R/W | 0x0 |
| 8 | LRADC3_INTEN | LRADC3 A/D interrupt enable.<br>0: Disable<br>1: Enable | R/W | 0x0 |
| 7 | LRADC2_INTEN | LRADC2 A/D interrupt enable.<br>0: Disable<br>1: Enable | R/W | 0x0 |
| 6 | VCCI_INTEN | VCCI A/D interrupt enable.<br>0: Disable<br>1: Enable | R/W | 0x0 |
| 5 | LRADC1_INTEN | LRADC1 A/D interrupt enable.<br>0: Disable<br>1: Enable | R/W | 0x0 |
| 4 | SVCC_INTEN | SVCC A/D interrupt enable<br>0: Disable<br>1: Enable | R/W | 0x0 |
| 3 | SENSOR_INTEN | Sensor A/D interrupt enable<br>0: Disable<br>1: Enable | R/W | 0x0 |
| 2 | DC5V_INTEN | DC5V A/D interrupt enable<br>0: Disable<br>1: Enable | R/W | 0x0 |
| 1 | BATV_INTEN | Battery A/D interrupt enable<br>0: Disable<br>1: Enable | R/W | 0x0 |
| 0 | CHARGI_INTEN | CHARGI A/D interrupt enable<br>0: Disable<br>1: Enable | R/W | 0x0 |

## 9.8.3 PMUADC_PD

PMUADC Control Register (VDD), Offset = 0x208

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:21 | - | Reserved | R | 0 |
| 11 | LRADC6_PD | LRADC6 A/D data ready pending<br>Writing 1 to the bit will clear the bit | R/W | 0 |
| 10 | LRADC5_PD | LRADC5 A/D data ready pending | R/W | 0 |

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| | | | | |
|---|---|---|---|---|
| | | Writing 1 to the bit will clear the bit | | |
| 9 | LRADC4_PD | LRADC4 A/D data ready pending<br>Writing 1 to the bit will clear the bit | R/W | 0 |
| 8 | LRADC3_PD | LRADC3 A/D data ready pending<br>Writing 1 to the bit will clear the bit | R/W | 0 |
| 7 | LRADC2_PD | LRADC2 A/D data ready pending<br>Writing 1 to the bit will clear the bit | R/W | 0 |
| 6 | VCCI_PD | VCCI A/D data ready pending<br>Writing 1 to the bit will clear the bit | R/W | 0 |
| 5 | LRADC1_PD | LRADC1 A/D data ready pending<br>Writing 1 to the bit will clear the bit | R/W | 0 |
| 4 | SVCC_PD | SVCC A/D data ready pending<br>Writing 1 to the bit will clear the bit | R/W | 0 |
| 3 | SENSOR_PD | Sensor A/D data ready pending<br>Writing 1 to the bit will clear the bit | R/W | 0 |
| 2 | DC5V_PD | DC5V A/D data ready pending<br>Writing 1 to the bit will clear the bit | R/W | 0 |
| 1 | BATV_PD | Battery A/D data ready pending<br>Writing 1 to the bit will clear the bit | R/W | 0 |
| 0 | CHARGI_PD | CHARGI A/D data ready pending<br>Writing 1 to the bit will clear the bit | R/W | 0 |

### 9.8.4 PMUADCDIG_CTL

PMUADC Control Register (VDD), Offset = 0x20C

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:30 | LRADC6_AVG | LRADC6 over sample setting.<br>00：do not oversampling<br>01：average after 4 oversampling<br>10：average after 8 oversampling<br>11：average after 16 oversampling | R/W | 0x0 |
| 29:28 | LRADC5_AVG | LRADC5 over sample setting.<br>00：do not oversampling<br>01：average after 4 oversampling<br>10：average after 8 oversampling<br>11：average after 16 oversampling | R/W | 0x0 |
| 27:26 | LRADC4_AVG | LRADC4 over sample setting.<br>00：do not oversampling<br>01：average after 4 oversampling<br>10：average after 8 oversampling<br>11：average after 16 oversampling | R/W | 0x0 |
| 25:24 | LRADC3_AVG | LRADC3 over sample setting.<br>00：do not oversampling<br>01：average after 4 oversampling<br>10：average after 8 oversampling<br>11：average after 16 oversampling | R/W | 0x0 |
| 23:22 | LRADC2_AVG | LRADC2 over sample setting.<br>00：do not oversampling<br>01：average after 4 oversampling<br>10：average after 8 oversampling | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 11：average after 16 oversampling | | |
| 21:20 | VCCI_AVG | VCCI over sample setting.<br>00：do not oversampling<br>01：average after 4 oversampling<br>10：average after 8 oversampling<br>11：average after 16 oversampling | R/W | 0x0 |
| 19:18 | LRADC1_AVG | LRADC1 over sample setting.<br>00：do not oversampling<br>01：average after 4 oversampling<br>10：average after 8 oversampling<br>11：average after 16 oversampling | R/W | 0x0 |
| 17:16 | SVCCADC_AVG | SVCCADC over sample setting.<br>00：do not oversampling<br>01：average after 4 oversampling<br>10：average after 8 oversampling<br>11：average after 16 oversampling | R/W | 0x0 |
| 15:14 | SENSADC_AVG | SENSADC over sample setting.<br>00：do not oversampling<br>01：average after 4 oversampling<br>10：average after 8 oversampling<br>11：average after 16 oversampling | R/W | 0x0 |
| 13:12 | DC5VADC_AVG | DC5VADC over sample setting.<br>00：do not oversampling<br>01：average after 4 oversampling<br>10：average after 8 oversampling<br>11：average after 16 oversampling | R/W | 0x0 |
| 11:10 | BAT_AVG | BATADC over sample setting.<br>00：do not oversampling<br>01：average after 4 oversampling<br>10：average after 8 oversampling<br>11：average after 16 oversampling | R/W | 0x0 |
| 9:8 | CHARGI_AVG | CHARGI over sample setting.<br>00：do not oversampling<br>01：average after 4 oversampling<br>10：average after 8 oversampling<br>11：average after 16 oversampling | R/W | 0x0 |
| 7:6 | - | Reserved | R/W | 0x0 |
| 5 | LRADC2_R_SEL | LRADC2 VCC pull up resistance enable. The precision of this resistance is 100k±1%(TYP).<br>0: disable<br>1: enable | R/W | 0x0 |
| 4:0 | - | Reserved | R/W | X |

## 9.8.5 CHARGI_DATA

Charger current ADC DATA Register (VDD), Offset = 0x210

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:12 | - | RESERVED | R | 0 |

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| | | | | |
|---|---|---|---|---|
| 12:0 | CHARGI | Charger current ADC, used to detect Battery charging current. | R | xx |

### 9.8.6 BATADC_DATA

BAT Voltage ADC DATA Register (VDD), Offset = 0x214

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:14 | - | RESERVED | R | 0 |
| 13:0 | BATV | BAT Voltage ADC, used to detect Battery voltage.<br>The ADC sampling values are aligned with high bits, and the extra 2 low bits are used when oversampling.<br>Input voltage range is:<br>Li-ion: 0-4.8V | R | xx |

### 9.8.7 DC5VADC_DATA

DC5V ADC DATA Register (VDD), Offset = 0x218

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:12 | - | RESERVED | R | 0 |
| 11:0 | DC5V | DC5V Voltage ADC, used to detect DC5V voltage.<br>Input voltage range is: 0-6V using half range | R | xx |

### 9.8.8 SENSADC_DATA

Sensor ADC DATA Register (VDD), Offset = 0x21C

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:12 | - | RESERVED | R | 0 |
| 11:0 | SENSOR | TEMPSENSOR ADC, used to detect TEMPSENSOR voltage. | R | xx |

### 9.8.9 SVCCADC_DATA

SVCCADC DATA Register (VDD), Offset = 0X220

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:12 | - | RESERVED | R | 0 |
| 11:0 | SVCC | SVCC voltage output<br>SVCCADC input voltage range is from 0 to 3.6V | R | xx |

### 9.8.10 LRADC1_DATA

LRADC1 DATA Register (VDD), Offset = 0x224

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:14 | - | RESERVED | R | 0 |
| 13:0 | LRADC1 | LRADC1 data output.<br>The ADC sampling values are aligned with high bits, and | R | xx |

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| | | | | |
|---|---|---|---|---|
| | | the extra 2 low bits are used when oversampling. | | |

### 9.8.11 VCCI_DATA

VCCI DATA Register (VDD), Offset = 0x228

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:14 | - | RESERVED | R | 0 |
| 12:0 | VCCI | VCCI data output.<br>The ADC sampling values are aligned with high bits, and the extra 2 low bits are used when oversampling. | R | xx |

### 9.8.12 LRADC2_DATA

LRADC2 DATA Register (VDD), Offset = 0x22C

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:14 | - | RESERVED | R | 0 |
| 13:0 | LRADC2 | LRADC2 data output.<br>The ADC sampling values are aligned with high bits, and the extra 2 low bits are used when oversampling. | R | xx |

### 9.8.13 LRADC3_DATA

LRADC3 DATA Register (VDD), Offset = 0x230

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:14 | - | RESERVED | R | 0 |
| 13:0 | LRADC3 | LRADC3 data output.<br>The ADC sampling values are aligned with high bits, and the extra 2 low bits are used when oversampling. | R | xx |

### 9.8.14 LRADC4_data

LRADC4 DATA Register (VDD), Offset = 0x234

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:14 | - | RESERVED | R | 0 |
| 13:0 | LRADC4 | LRADC4 data output.<br>The ADC sampling values are aligned with high bits, and the extra 2 low bits are used when oversampling. | R | xx |

### 9.8.15 LRADC5_DATA

LRADC5 DATA Register (VDD), Offset = 0x238

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:14 | - | RESERVED | R | 0 |
| 13:0 | LRADC5 | LRADC5 data output.<br>The ADC sampling values are aligned with high bits, and the extra 2 low bits are used when oversampling. | R | xx |

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### 9.8.16 LRADC6_DATA

LRADC6 DATA Register (VDD), Offset = 0x23C

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:14 | - | RESERVED | R | 0 |
| 13:0 | LRADC6 | LRADC6 data output.<br>The ADC sampling values are aligned with high bits, and the extra 2 low bits are used when oversampling. | R | xx |

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# 10 System Control

## 10.1 RMU

### 10.1.1 Features

The RMU Controller of ATS3085Sx has following features:

- The RMU (Reset Management Unit) can reset all the peripherals.
- The MCU can enter power-saving mode by setting the registers of RMU.

### 10.1.2 RMU Register List

*Table 10-1 RMU base address*

| Name | Physical Base Address | KSEG1 Base Address |
|---|---|---|
| RMU | 0x40000000 | 0x40000000 |

*Table 10-2 RMU register list*

| Offset | Register Name | Description |
|---|---|---|
| 0x00000000 | MRCR0 | Module Reset Control Register0 |
| 0x00000004 | MRCR1 | Module Reset Control Register1 |
| 0x00000008 | RST_CTL | watch dog Reset Control Register |

### 10.1.3 RMU Register Description

#### 10.1.3.1 MRCR0

MRCR0 (Module Reset Control Register0, offset = 0x00000000)

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:30 | - | - | R | 0x0 |
| 29 | I2C2RESET | I2C2 controller reset bit:<br>0: reset<br>1: normal | R/W | 0x0 |
| 28 | I2C1RESET | I2C1 controller reset bit:<br>0: reset<br>1: normal | R/W | 0x0 |
| 27 | I2C0RESET | I2C0 controller reset bit:<br>0: reset<br>1: normal | R/W | 0x0 |
| 26 | UART2RESET | UART2 controller reset bit:<br>0: reset<br>1: normal | R/W | 0x0 |
| 25 | UART1RESET | UART1 controller reset bit:<br>0: reset<br>1: normal | R/W | 0x0 |
| 24 | UART0RESET | UART0 controller reset bit:<br>0: reset<br>1: normal | R/W | 0x0 |
| 23 | I2C3RESET | I2C3 controller reset bit: | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0: reset<br>1: normal | | |
| 22 | - | Reserved | R | 0x0 |
| 21 | SDMA_RESET | Special DMA controller RESET:<br>0: reset<br>1: normal | R/W | 0x0 |
| 20 | - | Reserved | R | 0x0 |
| 19 | LRADCRESET | LRADC Controller reset bit:<br>0: reset<br>1: normal | R/W | 0x1 |
| 18 | - | Reserved | R | 0x0 |
| 17 | PWMRESET | PWM reset bit:<br>0: reset<br>1: normal | R/W | 0x0 |
| 16 | - | Reserved | R | 0x0 |
| 15 | 2.5GPU_RESET | 2.5DGPU RESET:<br>0: reset<br>1: normal | R/W | 0x0 |
| 14 | LCDRESET | LCD controller reset bit:<br>0: reset<br>1: normal | R/W | 0x0 |
| 13 | JPEG_RESET | JPEG controller RESET:<br>0: reset<br>1: normal | R/W | 0x0 |
| 12 | 2DGPURESET | 2DGPU Controller reset bit:<br>0: reset<br>1: normal | R/W | 0x0 |
| 11 | USBRESET2 | USB Controller reset2 bit:<br>0: reset<br>1: normal | R/W | 0x0 |
| 10 | USBRESET | USB Controller reset bit:<br>0: reset<br>1: normal | R/W | 0x0 |
| 9 | SPI1DCACHERESET | SPI1DCACHE Controller reset bit:<br>0: reset<br>1: normal | R/W | 0x0 |
| 8 | SPI0CACHERESET | SPICACHE Controller reset bit:<br>0: reset<br>1: normal | R/W | 0x0 |
| 7 | SPI3RESET | SPI3 controller reset bit:<br>0: reset<br>1: normal | R/W | 0x0 |
| 6 | SPI2RESET | SPI2 controller reset bit:<br>0: reset<br>1: normal | R/W | 0x0 |
| 5 | SPI1RESET | SPI1 controller reset bit:<br>0: reset<br>1: normal | R/W | 0x0 |
| 4 | SPI0RESET | SPI0 controller reset bit:<br>0: reset<br>1: normal | R/W | 0x0 |
| 3 | - | Reserved | R | 0x0 |
| 2 | SD1RESET | SD1 card controller reset bit: | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0: reset<br>1: normal | | |
| 1 | SD0RESET | SD0 card controller reset bit:<br>0: reset<br>1: normal | R/W | 0x0 |
| 0 | DMARESET | DMA reset bit:<br>0: reset<br>1: normal | R/W | 0x0 |

### 10.1.3.2 MRCR1

MRCR1 (Module Reset Control Register1, offset = 0x00000004)

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:25 | - | Reserved | R | 0x0 |
| 24 | BTHUBRESET | BluetoothHub reset bit:<br>0: reset<br>1: normal | R/W | 0x0 |
| 23:20 | - | Reserved | R | 0x0 |
| 19 | I2CMT1RESET | I2CMT1 controller reset bit:<br>0: reset<br>1: normal | R/W | |
| 18 | I2CMT0RESET | I2CMT0 controller reset bit:<br>0: reset<br>1: normal | R/W | |
| 17 | SPIMT1RESET | SPIMT1 controller reset bit:<br>0: reset<br>1: normal | R/W | |
| 16 | SPIMT0RESET | SPIMT0 controller reset bit:<br>0: reset<br>1: normal | R/W | 0x0 |
| 15 | TIMER5_RESET | TIMER5 controller RESET:<br>0: reset<br>1: normal | R/W | 0x0 |
| 14 | TIMER4_RESET | TIMER4 controller RESET:<br>0: reset<br>1: normal | R/W | 0x0 |
| 13 | TIMER3_RESET | TIMER3 controller RESET:<br>0: reset<br>1: normal | R/W | 0x0 |
| 12 | TIMER2_RESET | TIMER2 controller RESET:<br>0: reset<br>1: normal | R/W | 0x0 |
| 11 | TIMER1_RESET | TIMER1 controller RESET:<br>0: reset<br>1: normal | R/W | 0x0 |
| 10 | TIMER0_RESET | TIMER0 controller RESET:<br>0: reset<br>1: normal | R/W | 0x0 |
| 9:7 | - | Reserved | R/W | 0x0 |
| 6 | I2STX1RESET | I2STX1 Controller RESET:<br>0: reset | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 1: normal | | |
| 5 | I2SRXRESET | I2SRX reset bit:<br>0: reset<br>1: normal | R/W | 0x0 |
| 4 | I2STX0RESET | I2STX0 reset bit:<br>0: reset<br>1: normal | R/W | 0x0 |
| 3 | ADCRESET | ADC reset bit<br>0: reset<br>1: normal | R/W | 0x0 |
| 2 | DACRESET | DAC reset bit:<br>0: reset<br>1: normal | R/W | 0x0 |
| 1 | - | Reserved | R | 0x1 |
| 0 | AUDIODSP_ALLRESET | All DSP reset<br>0: reset all dsp<br>1: normal | R/W | 0x0 |

#### 10.1.3.3 RST_CTL

WDRST_CTL (watch dog reset Control Register, offset = 0x00000008)

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:4 | | - | R | 0 |
| 3 | SYSRESETREQ | 0: DISABLE<br>1: ENABLE | RW | 1 |
| 2 | - | - | RW | 0 |
| 1 | WD_RST_BTCPU | 0: DISABLE<br>1: ENABLE | RW | 0 |
| 0 | - | - | RW | 0 |

## 10.2 UID

### 10.2.1 Features

- Support 64bit CHIP unique ID.

### 10.2.2 UID Register List

***Table 10-3*** **UID Registers Address**

| Name | Physical Base Address | KSEG1 Base Address |
|---|---|---|
| UID | 0x400000A0 | 0x400000A0 |

***Table 10-4 UID Registers*** **List**

| Offset | Register Name | Description |
|---|---|---|
| 0x0000 | UID0 | Unique Device ID Register0 |
| 0x0004 | UID1 | Unique Device ID Register1 |

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### 10.2.3 UID Register Description

#### 10.2.3.1 UID0

Unique Device ID Register0
Offset = 0x0000

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | UID0 | Unique Device ID Information | R | 0x0 |

#### 10.2.3.2 UID1

Unique Device ID Register1
Offset = 0x0004

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | UID1 | Unique Device ID Information | R | 0x0 |

## 10.3 CMU Analog

### 10.3.1 Features

- Support only one oscillator inputs: 32M.
- Supply 1 HOSC circuit.
- Supply 1 LOSC as an option.
- Supply 3 RC circuits: RC4M, RC64M, and RC192M.
- Supply 3 PLLs and special clocks of all modules. The 3 PLLs are CORE PLL, AUDIOPLL and SPLL. Special clocks may be from the LOSC, RC32K, RC4M or RC64M and RC192M.
- COREPLL supports spread spectrum.
- The source of all PLL are HOSC.

### 10.3.2 CMU Analog Register List

*Table 10-5 CMU Analog Controller Registers Address*

| Name | Physical Base Address | KSEG1 Base Address |
|---|---|---|
| CMU_ANALOG_REGISTER | 0x40000100 | 0x40000100 |

*Table 10-6 CMU Analog Controller Registers*

| Offset | Register Name | Description |
|---|---|---|
| 0x00 | HOSC_CTL | HOSC Control Register |
| 0x04 | HOSCLDO_CTL | HOSC LDO Control Register |
| 0x08 | RC64M_CTL | RC64M Control Register |
| 0x10 | LOSC_CTL | LOSC Control Register |
| 0x14 | RC4M_CTL | RC4M Control Register |
| 0x18 | RC192M_CTL | RC128M Control Register |
| 0x1C | AVDDLDO_CTL | AVDD LDO Control Register |
| 0x20 | COREPLL_CTL | COREPLL Control Register |
| 0x24 | SPLL_CTL | SPLL Control Reigster |

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| | | |
|---|---|---|
| 0x28 | AUDIOPLL_CTL | AUDIOPLL Control Register |
| 0x64 | RC32K_CTL | RC32K Control Register |
| 0x68 | RC32K_CAL | RC32K Calibration Register |
| 0x6C | RC32K_COUNT | RC32K Calibration Count Register |
| 0x74 | HGLFCLK_CTL | HOSC Generate Low Frequency Clock Control Register |

## 10.3.3 CMU Analog Register Description

### 10.3.3.1 HOSC_CTL

HOSC control register.
Offset = 0x00(VDD domain)

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:29 | - | Reserved | R | 0x0 |
| 28 | HOSC_READY | HOSC READY Signal | R | 0x0 |
| 27:25 | - | Reserved | R/W | 0x2 |
| 24:17 | HOSC_CAP | HOSCO PAD base cap select, range from 0pF to 20pF:<br>Trim cap = 0.1pF * HOSCI_CAP | R/W | 0x64 |
| 16:15 | - | Reserved | R/W | 0x0 |
| 14:13 | HOSC_AMPLMT | HOSC amplitude Adjust<br>00：450mV<br>01：500mV<br>10：550mV<br>11：600mV | R/W | 0x3 |
| 12 | HOSC_AMPLMT_EN | HOSC amplitude Adjust Enable:<br>0: disabled<br>1: enable | R/W | 0x0 |
| 11:9 | HGMC | High Frequency crystal Oscillator GMMIN select bits | R/W | 0x7 |
| 8:0 | - | Reserved | R/W | 0x75 |

### 10.3.3.2 HOSCLDO_CTL

HOSC LDO Control Register
Offset = 0x04(VDD domain)

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:6 | -- | reserved | R | X |
| 5 | OSCVDD0_PD_BIASSET | OSCVDD LDO0 ipd set:<br>0: disable<br>1: enable, save 1/2 bias current | R/W | 0x0 |
| 4 | OSCVDD0_PD | OSCVDD0 pd set:<br>0: disable<br>1: enable | R/W | 0x1 |
| 3:2 | OSCVDD0_VSET | OSCVDD0 output voltage set:<br>00: 1.0V<br>01: 1.1V****<br>10: 1.2V<br>11: 1.3V | R/W | 0x1 |
| 1 | -- | reserved | R/W | 0x0 |
| 0 | OSCVDD0_EN | OSCVDD0 LDO0 enable: | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0: disable<br>1: enable | | |

Note: OSCVDDLDO should be enabled or disabled by hardware according to OSCVDDx_EN and HOSCEN in different PMU State.

### 10.3.3.3 RC64M_CTL

RC64M Clock Control Register
Offset = 0x08(VDD domain)

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:25 | FRQCAL | RC64M Calibration Frequency<br>The frequency calibration result, read only. | R | 0x40 |
| 24 | CALDONE | RC64M FREQ Calibration done<br>0: uncalibrated or calibrating<br>1: calibration done<br>Cleared when CALEN/RC64M_EN is cleared, or when the FREQ of RC64M exceeds INRANGE | R | 0x0 |
| 23:14 | FRQASET | RC64M Hardware Auto Calibration frequency setting<br>When CALMODE is 1, hardware may adjust the calibration reference freq of RC64M through this field, one step is 500KHz,<br>FOSC=500KHz*N | R/W | 0x80 |
| 13:11 | - | Reserved. | R/W | 010 |
| 10:4 | FRQMSET | RC64M Manual Calibration frequency setting<br>When CALMODE is 0, software may adjust the freq of RC64M through this field. | R/W | 0x40 |
| 3 | CALMODE | RC64M Calibration Mode<br>0: manual calibration by FRQMSET<br>1: auto calibration by FRQASET | R/W | 0x0 |
| 2 | CALEN | RC64M Calibration Enable:<br>0: disable<br>1: enable | R/W | 0x0 |
| 1:0 | - | Reserved, be read as zero. | R/W | 0x0 |

### 10.3.3.4 LOSC_CTL

LOSC Control Register.
Offset = 0x10(SVCC domain)

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:21 | - | Reserved, be read as zero. | R | 0x0 |
| 20:16 | LOSCI_CAP | LOSCI PAD Capacitance select:<br>00000: 0pf<br>00001: 0.8pf<br>00010: 1.6pf<br>00011: 2.4pf<br>...<br>10000: 12.8pf**<br>...<br>11111: 24.8pf | R/W | 0x19 |
| 15 | LOSC_RDY | LOSC Status: | R | 0 |

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| | | | | |
|---|---|---|---|---|
| | | 0: LOSC not Ready<br>1: LOSC ready | | |
| 14:13 | - | Reserved, be read as zero. | R/W | 0 |
| 12:8 | LOSCO_CAP | LOSCO PAD Capacitance select:<br>00000: 0pf<br>00001: 0.8pf<br>00010: 1.6pf<br>00011: 2.4pf<br>…<br>10000: 12.8pf**<br>…<br>11111: 24.8pf | R/W | 0x19 |
| 7 | - | Reserved | R/W | 0x0 |
| 6:4 | LGMC | Low Frequency crystal Oscillator GMMIN select bits | R/W | 0x7 |
| 3:1 | - | Reserved | R/W | 0x0 |
| 0 | LOSC_EN | LOSC enable:<br>0: disable<br>1: enable | R/W | 0x0 |

### 10.3.3.5 RC4M_CTL

RC4M Clock Control Register
Offset = 0x14(VDD domain)

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:7 | - | Reserved | R/W | 0x0 |
| 6:1 | RC4M_FRQSET | OSC4M frequency set bits: | R/W | 0x2C |
| 0 | - | Reserved, be read as zero. | R/W | 0x0 |

Note:

### 10.3.3.6 RC192M_CTL

RC192M Clock Control Register
Offset = 0x18(VDD domain)

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:25 | FRQCAL | RC192M Calibration Frequency<br>The frequency calibration result, read only. | R | 0x40 |
| 24 | CALDONE | RC192M FREQ Calibration done<br>0: uncalibrated or calibrating<br>1: calibration done<br>Cleared when CALEN/RC192M _EN is cleared, or when the FREQ of RC192M exceeds INRANGE | R | 0x0 |
| 23:14 | FRQASET | RC192M Hardware Auto Calibration frequency setting<br>000: 128M<br>001: 160M<br>010: 192M *****<br>011: 224M<br>100: 256M<br>Others: reserved | R/W | 0x2 |
| 13:11 | - | Reserved | R/W | 010 |

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| | | | | |
|---|---|---|---|---|
| 10:4 | FRQMSET | RC192M Manual Calibration frequency setting<br>When CALMODE is 0, software may adjust the freq of RC192M through this field. | R/W | 0x48 |
| 3 | CALMODE | RC192M Calibration Mode<br>0: manual calibration by FRQMSET<br>1: auto calibration by FRQASET | R/W | 0x0 |
| 2 | CALEN | RC192M Calibration Enable:<br>0: disable<br>1: enable | R/W | 0x0 |
| 1 | - | Reserved, be read as zero. | R/W | 0x0 |
| 0 | RC192MEN | RC192M Enable bit:<br>0: Disable<br>1: Enable | R/W | 0x0 |

Note:

### 10.3.3.7 AVDDLDO_CTL

AVDD LDO Control Register
Offset = 0x1C(VDD domain)

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:6 | -- | reserved | R | 0x0 |
| 5 | AVDD_PD_BIASSET | AVDD LDO ipd set:<br>0: disable<br>1: enable, save 1/2 bias current | R/W | 0x0 |
| 4 | AVDD_PD_EN | AVDD pd set:<br>0: disable<br>1: enable | R/W | 0x1 |
| 3:2 | AVDD_VSET | AVDD output voltage set:<br>00: 1.0V<br>01: 1.1V****<br>10: 1.2V<br>11: 1.3V | R/W | 0x1 |
| 1 | -- | reserved | R/W | 0x0 |
| 0 | AVDD_EN | AVDD LDO enable:<br>0: disable<br>1: enable | R/W | 0x0 |

### 10.3.3.8 COREPLL_CTL

COREPLL Control Register
Offset = 0x20(VDD domain)

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:9 | - | Reserved, be read as zero. | R | 0x0 |
| 8 | COREPLLPMD | COREPLL phase match detect:<br>0: CORE PLL phase not match<br>1: CORE PLL phase match | R | X |
| 7 | COREPLLEN | COREPLL Enable：<br>0: Disable<br>1: Enable | R/W | 0x0 |
| 6 | - | Reserved | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 5:0 | COREPLLSCORE | COREPLL Frequency Select: Formula: 8M* SCORE<br>Range:40 ~ 320M<br>Value must be bigger than 4<br>0-4: reserved<br>5: 5*8M=40M<br>……<br>40: 40*8M=320M<br>Others reserved. | R/W | 0x05 |

### 10.3.3.9 SPLL_CTL

SPLL Control Register
Offset = 0x24(VDD domain)

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:9 | - | Reserved | R | X |
| 8 | SPLL_LOCK | SPLL phase match detect:<br>0: SPLL phase not match<br>1: SPLL phase match<br>This bit is only used for debug. Software must not detect PLL state according to this bit. | R | X |
| 7:1 | - | Reserved | R/W | 0x0 |
| 0 | SPLL_EN | SPLL Enable:<br>0: disable<br>1: enable | R/W | 0x0 |

### 10.3.3.10 AUDIOPLL_CTL

AUDIOPLL Control Register
Offset = 0x28(VDD domain)

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:9 | - | Reserved | R/W | 0x0 |
| 8 | AUDPLL_LOCK | AUDIO PLL phase match detect:<br>0: AUDIO PLL phase not match<br>1: AUDIO PLL phase match<br>This bit is only used for debug. Software must not detect PLL state according to this bit.<br>This bit is updated by master CPU.<br>This bit could only be read by master DSP. | R | X |
| 7 | AUDPLLEN | Audio PLL Enable<br>0：Disable<br>1：Enable | R/W | 0x0 |
| 6 | AUDPLL_MODE | AUDIO PLL mode select:<br>0: integer-N<br>1: fractional-N | R/W | 0x0 |
| 5:0 | AUDPLL_APS | AUDIO PLL Clock Select<br>Refer to the following table to see the configuration. | R/W | 0x0 |

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### 10.3.3.11 RC32K_CTL

RC32K Clock Control Register
Offset = 0x64(SVCC domain)

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:7 | - | Reserved, be read as zero. | R | 0x0 |
| 6:1 | RC32K_FRQSET | RC32K frequency set bits | R/W | 0x2C |
| 0 | - | Reserved, be read as zero. | R | 0x0 |

### 10.3.3.12 RC32K_CAL

RC32K Clock Calibration Register
Offset = 0x68(VDD domain)

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:24 | - | Reserved, be read as zero. | R | 0x0 |
| 23:8 | CALCYCLE | RC32K Calibration Cycles<br>The calibration internal = CALCYCLE*Frequency of RC32K | R/W | 0x0 |
| 7:5 | - | Reserved, be read as zero. | R | 0x0 |
| 4 | CALDONE | RC32K calibration Done<br>0: not calibration<br>1: calibration done | R | 0x0 |
| 3:2 | - | Reserved, be read as zero. | R | 0x0 |
| 1 | CALSRC | RC32K calibration source:<br>0: RC32K<br>1: LOSC | R/W | 0x0 |
| 0 | CALEN | RC32K Calibration Start Enable:<br>0: disable<br>1: enable | R/W | 0x0 |

### 10.3.3.13 RC32K_COUNT

RC32K Clock Calibration Count Register
Offset = 0x6C(VDD domain)

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | CALCOUNT | RC32K Calibration Count | R | 0x0 |

### 10.3.3.14 HGLFCLK_CTL

HOSC Generate Low Frequency Clock Control Register
Offset = 0x74(VDD domain)

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:2 | - | Reserved, be read as zero. | R | 0x0 |
| 1 | CK32K768SRC | CK32K768 Source:<br>0: LOSC<br>1: HGLFCLK | R/W | 0x0 |
| 0 | HGLFCLKEN | HOSC Generate Low Frequency Clock Enable: | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0: disable<br>1: enable | | |

## 10.4 CMU Digital

### 10.4.1 Features

The CMU Controller of ATS3085Sx has following features:

- The CMU (Clock Management Unit) can select RC4M, HOSC, CORE_PLL, RC64M and RC192M as the clock of each peripheral.
- The clock sources of the memory blocks can be selected by CMU.
- 4 CMU states: S1, S1BT, S2 and S3.

### 10.4.2 CMU Digital Register List

*Table 10-7 CMU Digital Controller Registers Address*

| Name | Physical Base Address | KSEG1 Base Address |
|---|---|---|
| CMU_DIGITAL_REGISTER | 0x40001000 | 0x40001000 |

*Table 10-8 CMU Digital Controller Registers*

| Offset | Register Name | Description |
|---|---|---|
| 0x0000 | CMU_SYSCLK | SYSCLK Control Register |
| 0x0004 | CMU_DEVCLKEN0 | DEVCLKEN Control Register0 |
| 0x0008 | CMU_DEVCLKEN1 | DEVCLKEN Control Register1 |
| 0x0010 | CMU_SD0CLK | SD0 CLK Control Register |
| 0x0014 | CMU_SD1CLK | SD1 CLK Control Register |
| 0x001C | CMU_JPEGCLK | JPEG Clock Control Register |
| 0x0020 | CMU_SPI0CLK | SPI0 CLK Control Register |
| 0x0024 | CMU_SPI1CLK | SPI1 CLK Control Register |
| 0x0028 | CMU_SPI2CLK | SPI2 CLK Control Register |
| 0x002C | CMU_SPI3CLK | SPI3 CLK Control Register |
| 0x0030 | CMU_2DGPUCLK | 2DGPUCLK Control Register |
| 0x0034 | CMU_LCDCLK | LCDCLK Control Register |
| 0x0038 | CMU_25DGPUCLK | 2.5DGPU Clock Control Register |
| 0x0040 | CMU_TIMER0CLK | TIMER0 Clock Control Register |
| 0x0044 | CMU_TIMER1CLK | TIMER1 Clock Control Register |
| 0x0048 | CMU_TIMER2CLK | TIMER2 Clock Control Register |
| 0x004c | CMU_TIMER3CLK | TIMER3 Clock Control Register |
| 0x0050 | CMU_TIMER4CLK | TIMER4 Clock Control Register |
| 0x0054 | CMU_TIMER5CLK | TIMER5 Clock Control Register |
| 0x0068 | CMU_PWM0CLK | PWMCLK0 Control Register |
| 0x006C | CMU_PWM1CLK | PWMCLK1 Control Register |
| 0x0070 | CMU_PWM2CLK | PWMCLK2 Control Register |
| 0x0074 | CMU_PWM3CLK | PWMCLK3 Control Register |
| 0x0078 | CMU_LRADCCLK | LRADC CLK Control Register |
| 0x0080 | CMU_I2C0CLK | I2C0CLK Control Register |
| 0x0084 | CMU_I2C1CLK | I2C1CLK Control Register |
| 0x0088 | CMU_I2C2CLK | I2C2CLK Control Register |
| 0x008C | CMU_I2C3CLK | I2C3CLK Control Register |

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| | | |
|---|---|---|
| 0x0090 | CMU_AUDDSPCLK | Audio DSPCLK Control Register |
| 0x0098 | CMU_DACCLK | DACCLK Control Register |
| 0x009c | CMU_ADCCLK | ADCCLK Control Register |
| 0x00A0 | CMU_I2STXCLK | I2STXCLK Control Register |
| 0x00A4 | CMU_I2SRXCLK | I2SRXCLK Control Register |
| 0x00B0 | CMU_MEMCLKEN0 | MEMCLKEN Control Register0 |
| 0x00B4 | CMU_MEMCLKEN1 | MEMCLKEN Control Register1 |
| 0x00C0 | CMU_MEMCLKSRC0 | MEMCLKSRC Control Register0 |
| 0x00C4 | CMU_MEMCLKSRC1 | MEMCLKSRC Control Register1 |
| 0x00D0 | CMU_S1CLKCTL | S1 Clock Control Register |
| 0x00D4 | CMU_S1BTCLKCTL | S1BT Clock Control Register |
| 0x00DC | CMU_S2CLKCTL | S2 Clock Control Register |
| 0x00E0 | CMU_S3CLKCTL | S3 Clock Control Register |
| 0x00E4 | CMU_WDCLK | CMU WatchDog Clock Control Register |
| 0x00E8 | CMU_GPIOCLKCTL | CMU GPIO interrupt Clock Control register |
| 0x00EC | CMU_PMUWKUPCLKCTL | PMU Wakeup Clock Control Register |
| 0x0100 | CMU_SPIMT0CLK | CMU SPIMT0 Clock Control Register |
| 0x0104 | CMU_SPIMT1CLK | CMU SPIMT1 Clock Control Register |
| 0x0110 | CMU_I2CMT0CLK | CMU I2CMT0 Clock Control Register |
| 0x0114 | CMU_I2CMT1CLK | CMU I2CMT1 Clock Control Register |

## 10.4.3 CMU Digital Register Description

### 10.4.3.1 CMU_SYSCLK

CMU_SYSCLK Control register
Offset = 0x00

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:25 | - | Reserved | R | 0x0 |
| 24 | CPUTIMERCLKSRC | CPU Timer Clock Source:<br>0: HOSC/32<br>1: RC4M/4 | R/W | 0x0 |
| 23:10 | - | Reserved | R | 0x0 |
| 9:8 | AHBCLKDIV | HCLK divisor<br>00: /2<br>01: /4<br>10/11: /1 | R/W | 0x0 |
| 7:4 | CPUCLKDIV | CPU_CLK divisor:<br>0: /1<br>1: /2<br>2: /3<br>3: /4<br>4: /5<br>…<br>13: /14<br>14: /1.5<br>15: /2.5 | R/W | 0x0 |
| 3 | - | Reserved | R | 0x0 |
| 2:0 | CORECLKSRC | CORE_CLK Source:<br>000: RC4M<br>001: HOSC | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 010: COREPLL<br>011: RC64M<br>100: reserved<br>101: RC192M<br>110: RC32K | | |

### 10.4.3.2 CMU_DEVCLKEN0

CMU_DEVCLKEN Control register0
Offset = 0x04

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31 | - | Reserved | R/W | 0x0 |
| 30 | EXINTCLKEN | External Interrupt clock enable bit:<br>0: disable<br>1: enable<br>The source clock of external interrupt is RC3M | R/W | 0x1 |
| 29 | I2C2CLKEN | I2C2 controller clock enable bit:<br>0: disable<br>1: enable | R/W | 0x0 |
| 28 | I2C1CLKEN | I2C1 controller clock enable bit:<br>0: disable<br>1: enable | R/W | 0x0 |
| 27 | I2C0CLKEN | I2C0 controller clock enable bit:<br>0: disable<br>1: enable | R/W | 0x0 |
| 26 | UART2CLKEN | UART2 controller clock enable bit:<br>0: disable<br>1: enable | R/W | 0x0 |
| 25 | UART1CLKEN | UART1 controller clock enable bit:<br>0: disable<br>1: enable | R/W | 0x0 |
| 24 | UART0CLKEN | UART0 controller clock enable bit:<br>0: disable<br>1: enable | R/W | 0x0 |
| 23 | I2C3CLKEN | I2C3 controller clock enable bit:<br>0: disable<br>1: enable | R/W | 0x0 |
| 22 | - | Reserved | R/W | 0x0 |
| 21 | SDMACLKEN | SPECIAL DMA Clock Enable bit :<br>0: disable<br>1: enable | R/W | 0x0 |
| 20 | CPUTIMERCLKEN | CPU Timer Clock Enable bit:<br>0: disable<br>1: enable | R/W | 0x0 |
| 19 | LRADCCLKEN | LRADC Controller clock enable bit:<br>0: disable<br>1: enable | R/W | 0x1 |
| 18 | - | Reserved | R/W | 0x0 |
| 17 | PWM0CLKEN | PWM0 clock enable bit:<br>0: disable<br>1: enable<br>This bit controls all the clock gating of PWM0 | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 16 | - | Reserved | R/W | 0x0 |
| 15 | 25DGPUCLKEN | 25DGPU clock enable bit:<br>0: disable<br>1: enable | R/W | 0x0 |
| 14 | LCDCLKEN | LCD controller clock enable bit:<br>0: disable<br>1: enable | R/W | 0x0 |
| 13 | JPEGCLKEN | JPEG Controller clock enable bit:<br>0: disable<br>1: enable | R/W | 0x0 |
| 12 | 2DGPUCLKEN | 2DGPU Controller clock enable bit:<br>0: disable<br>1: enable | R/W | 0x0 |
| 11 | USBHCLKEN | USB Controller HCLK enable bit:<br>0: disable<br>1: enable<br>This bit controls the HCLK and RC32K to the USB controller. | R/W | 0x0 |
| 10 | USBCLKEN | USB Controller clock enable bit:<br>0: disable<br>1: enable | R/W | 0x0 |
| 9 | SPI1DCACHECLKEN | SPI1DCACHE Controller clock enable bit:<br>0: disable<br>1: enable | R/W | 0x0 |
| 8 | SPI0CACHECLKEN | SPICACHE Controller clock enable bit:<br>0: disable<br>1: enable | R/W | 0x0 |
| 7 | SPI3CLKEN | SPI3 controller clock enable bit:<br>0: disable<br>1: enable | R/W | 0x0 |
| 6 | SPI2CLKEN | SPI2 controller clock enable bit:<br>0: disable<br>1: enable | R/W | 0x0 |
| 5 | SPI1CLKEN | SPI1 controller clock enable bit:<br>0: disable<br>1: enable | R/W | 0x0 |
| 4 | SPI0CLKEN | SPI0 controller clock enable bit:<br>0: disable<br>1: enable | R/W | 0x0 |
| 3 | - | Reserved | R/W | 0x0 |
| 2 | SD1CLKEN | SD1 card controller clock enable bit:<br>0: disable<br>1: enable | R/W | 0x0 |
| 1 | SD0CLKEN | SD0 card controller clock enable bit:<br>0: disable<br>1: enable | R/W | 0x0 |
| 0 | DMACLKEN | DMA clock enable bit:<br>0: disable<br>1: enable | R/W | 0x0 |

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### 10.4.3.3 CMU_DEVCLKEN1

CMU_DEVCLKEN Control register1
Offset = 0x08

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31 | 32KCLKOUTEN | 32K Clock Output Enable<br>0: disable<br>1: enable | R/W | 0x0 |
| 30 | 32MCLKOUTEN | 32M Clock Output Enable<br>0: disable<br>1: enable | R/W | 0x0 |
| 29:23 | - | Reserved | R/W | 0x0 |
| 22 | PWM3CLKEN | PWM3 clock enable bit:<br>0: disable<br>1: enable | R/W | 0x0 |
| 21 | PWM2CLKEN | PWM2 clock enable bit:<br>0: disable<br>1: enable | R/W | 0x0 |
| 20 | PWM1CLKEN | PWM1 clock enable bit:<br>0: disable<br>1: enable | R/W | 0x0 |
| 19 | I2CMT1CLKEN | I2CMT1 controller clock enable bit:<br>0: disable<br>1: enable | R/W | 0x0 |
| 18 | I2CMT0CLKEN | I2CMT0 controller clock enable bit:<br>0: disable<br>1: enable | R/W | 0x0 |
| 17 | SPIMT1CLKEN | SPIMT1 controller clock enable bit:<br>0: disable<br>1: enable | R/W | 0x0 |
| 16 | SPIMT0CLKEN | SPIMT0 controller clock enable bit:<br>0: disable<br>1: enable | R/W | 0x0 |
| 15 | TIMER5CLKEN | Timer5 controller clock:<br>0: disable<br>1: enable<br>This bit controls the clock gating of TIMER5_CLK | R/W | 0x0 |
| 14 | TIMER4CLKEN | Timer4 controller clock:<br>0: disable<br>1: enable<br>This bit controls the clock gating of TIMER4_CLK | R/W | 0x0 |
| 13 | TIMER3CLKEN | Timer3 controller clock:<br>0: disable<br>1: enable<br>This bit controls the clock gating of TIMER3_CLK | R/W | 0x0 |
| 12 | TIMER2CLKEN | Timer2 controller clock:<br>0: disable<br>1: enable<br>This bit controls the clock gating of TIMER2_CLK | R/W | 0x0 |
| 11 | TIMER1CLKEN | Timer1 controller clock:<br>0: disable<br>1: enable<br>This bit controls the clock gating of TIMER1_CLK | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 10 | TIMER0CLKEN | Timer0 controller clock:<br>0: disable<br>1: enable<br>This bit controls the clock gating of TIMER0_CLK | R/W | 0x0 |
| 9 | AUDDSPDMACLKEN | Audio DSP DMA Clock Enable bit:<br>0: disable<br>1: enable | R/W | 0x0 |
| 8 | DACANACLKEN | DAC Analog Clock Enable bit:<br>0: disable<br>1: enable | R/W | 0x0 |
| 7 | I2SHCLKEN | I2STX/RX AHB clock enable bit:<br>0: disable<br>1: enable | R/W | 0x0 |
| 6 | I2SSRDCLKEN | I2SSRD Mclock enable bit:<br>0: disable<br>1: enable | R/W | 0x0 |
| 5 | I2SG1CLKEN | I2SG1 clock enable bit:<br>0: disable<br>1: enable | R/W | 0x0 |
| 4 | I2SG0CLKEN | I2SG0 clock enable bit:<br>0: disable<br>1: enable | R/W | 0x0 |
| 3 | ADCCLKEN | ADC controller clock enable bit:<br>0: disable<br>1: enable | R/W | 0x0 |
| 2 | DACCLKEN | DAC controller clock enable bit:<br>0: disable<br>1: enable | R/W | 0x0 |
| 1 | AUDDSPTIMERCLKEN | Audio DSP Timer clock enable bit:<br>0: disable<br>1: enable | R/W | 0x0 |
| 0 | AUDDSPCLKEN | Audio DSP clock enable bit:<br>0: disable<br>1: enable | R/W | 0x0 |

### 10.4.3.4 CMU_SD0CLK

SD0 Clock Control Register
Offset = 0x0010

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:10 | - | Reserved | R | 0x0 |
| 9:8 | SD0CLKSRC | SD0 Card Controller Clock Source Select:<br>00: HOSC<br>01: CORE_PLL<br>10: RC192M<br>11: Reserved | R/W | 0x0 |
| 7 | - | Reserved | R | 0x0 |
| 6 | SD0CLKPOSTDIV | SD0 Card Controller Clock Post-Divisor | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0: /1<br>1: /128 | | |
| 5:4 | - | Reserved | R | 0x0 |
| 3:0 | SD0CLKDIV | SD0 Card Controller Clock Divisor<br>0000: /1<br>0001: /2<br>0010: /3<br>0011: /4<br>0100: /5<br>0101: /6<br>0110: /7<br>0111: /8<br>1000: /9<br>1001: /10<br>1010: /11<br>1011: /12<br>1100: /13<br>1101: /14<br>1110: /15<br>1111: /16 | R/W | 0x0 |

### 10.4.3.5 CMU_SD1CLK

SD1 Clock Control Register
Offset = 0x0014

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:10 | - | Reserved | R | 0x0 |
| 9:8 | SD1CLKSRC | SD1 Card Controller Clock Source Select:<br>00: HOSC<br>01: CORE_PLL<br>10: RC192M<br>11: Reserved | R/W | 0x0 |
| 7 | - | Reserved | R | 0x0 |
| 6 | SD1CLKPOSTDIV | SD1 Card Controller Clock Post-Divisor<br>0: /1<br>1: /128 | R/W | 0x0 |
| 5:4 | - | Reserved | R | 0x0 |
| 3:0 | SD1CLKDIV | SD1 Card Controller Clock Divisor<br>0000: /1<br>0001: /2<br>0010: /3<br>0011: /4<br>0100: /5<br>0101: /6<br>0110: /7<br>0111: /8<br>1000: /9<br>1001: /10<br>1010: /11<br>1011: /12 | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 1100: /13<br>1101: /14<br>1110: /15<br>1111: /16 | | |

### 10.4.3.6 CMU_JPEGCLK

JPEG Clock Control Register
Offset = 0x001C

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:10 | - | Reserved | R | 0x0 |
| 9:8 | JPEGCLKSRC | JPEG Controller Clock Source:<br>00: Reserved<br>01: CORE_PLL<br>10: RC192M<br>11: RC64M | R/W | 0x1 |
| 7:4 | - | Reserved | R | 0x0 |
| 3:0 | JPEGCLKDIV | JPEG Clock Divisor:<br>0: /1<br>1: /2<br>2: /3<br>3: /4<br>4: /5<br>…<br>13: /14<br>14: /1.5<br>15: /2.5 | R/W | 0x0 |

### 10.4.3.7 CMU_SPI0CLK

SPI0 Clock Control Register
Offset = 0x0020

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:13 | - | Reserved | R | 0x0 |
| 12 | SPI0MODE | SPI0 Controller Mode Select:<br>0: Normal Mode<br>1: DDR Mode | R/W | 0x0 |
| 11:10 | - | Reserved | R | 0x0 |
| 9:8 | SPI0CLKSRC | SPI0 Controller Clock Source:<br>00: HOSC<br>01: CORE_PLL<br>10: RC192M<br>11: RC64M | R/W | 0x0 |
| 7:4 | - | Reserved | R | 0x0 |
| 3:0 | SPI0CLKDIV | SPI0 Clock Divisor: | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0: /1<br>1: /2<br>2: /3<br>3: /4<br>4: /5<br>…<br>13: /14<br>14: /1.5<br>15: /2.5 | | |

### 10.4.3.8 CMU_SPI1CLK

SPI1 Clock Control Register
Offset = 0x0024

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:10 | - | Reserved | R | 0x0 |
| 9:8 | SPI1CLKSRC | SPI1 Controller Clock Source:<br>00: HOSC<br>01: CORE_PLL<br>10: RC192M<br>11: RC64M | R/W | 0x0 |
| 7:4 | - | Reserved | R | 0x0 |
| 3:0 | SPI1CLKDIV | SPI1 Clock Divisor:<br>0: /1<br>1: /2<br>2: /3<br>3: /4<br>4: /5<br>…<br>13: /14<br>14: /1.5<br>15: /2.5 | R/W | 0x0 |

### 10.4.3.9 CMU_SPI2CLK

SPI2 Clock Control Register
Offset = 0x0028

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:10 | - | Reserved | R | 0x0 |
| 9:8 | SPI2CLKSRC | SPI2 Controller Clock Source:<br>00: HOSC<br>01: CORE_PLL<br>10: RC192M<br>11: RC64M | R/W | 0x0 |
| 7:4 | - | Reserved | R | 0x0 |
| 3:0 | SPI2CLKDIV | SPI2 Clock Divisor: | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0: /1<br>1: /2<br>2: /3<br>3: /4<br>4: /5<br>…<br>13: /14<br>14: /1.5<br>15: /2.5 | | |

### 10.4.3.10 CMU_SPI3CLK

SPI3 Clock Control Register
Offset = 0x002C

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:10 | - | Reserved | R | 0x0 |
| 9:8 | SPI3CLKSRC | SPI3 Controller Clock Source:<br>00: HOSC<br>01: CORE_PLL<br>10: RC192M<br>11: RC64M | R/W | 0x0 |
| 7:4 | - | Reserved | R | 0x0 |
| 3:0 | SPI3CLKDIV | SPI3 Clock Divisor:<br>0: /1<br>1: /2<br>2: /3<br>3: /4<br>4: /5<br>…<br>13: /14<br>14: /1.5<br>15: /2.5 | R/W | 0x0 |

### 10.4.3.11 CMU_2DGPUCLK

2DGPU Clock Control Register
Offset = 0x0030

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:10 | - | Reserved | R | 0x0 |
| 9:8 | 2DGPUCLKSRC | 2DGPU Controller Clock Source:<br>00: HOSC<br>01: CORE_PLL<br>10: RC192M<br>11: RC64M | R/W | 0x0 |
| 7:4 | - | Reserved | R | 0x0 |
| 3:0 | 2DGPUCLKDIV | 2DGPU Controller Clock Divisor:<br>0: /1<br>1: /2<br>2: /3<br>3: /4 | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 4: /5<br>…<br>13: /14<br>14: /1.5<br>15: /2.5 | | |

### 10.4.3.12 CMU_LCDCLK

LCD Clock Control Register
Offset = 0x0034

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:10 | - | Reserved | R | 0x0 |
| 9:8 | LCDCLKSRC | LCD Controller Clock Source:<br>00: HOSC<br>01: CORE_PLL<br>10: RC192M<br>11: RC64M | R/W | 0x0 |
| 7:5 | - | Reserved | R | 0x0 |
| 4 | LCDCLKPREDIV | LCD Controller Clock PRE-divisor:<br>0: /1<br>1: /6 | R/W | 0x0 |
| 3:0 | LCDCLKDIV | LCD Controller Clock Divisor:<br>0: /1<br>1: /2<br>2: /3<br>3: /4<br>4: /5<br>…<br>11: /12<br>Others: Reserved | R/W | 0x0 |

### 10.4.3.13 CMU_25DGPUCLK

2.5DGPU Clock Control Register
Offset = 0x0038 default value: 0x00001100

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:13 | - | Reserved | R | 0x0 |
| 12 | 25DGPURAMCLKGATING | 25DGPU RAM Clock Gating Disable:<br>0: enable<br>1: disable | R/W | 0x1 |
| 11:10 | - | Reserved | R | 0x0 |
| 9:8 | 25DGPUCLKSRC | 25DGPU Controller Clock Source:<br>00: Reserved<br>01: CORE_PLL<br>10: RC192M<br>11: RC64M | R/W | 0x1 |
| 7:4 | - | Reserved | R | 0x0 |
| 3:0 | 25DGPUCLKDIV | 25DGPU Clock Divisor:<br>0: /1 | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 1: /2<br>2: /3<br>3: /4<br>4: /5<br>…<br>13: /14<br>14: /1.5<br>15: /2.5 | | |

### 10.4.3.14 CMU_TIMER0CLK

TIMER0 Clock Control Register
Offset = 0x0040

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:3 | - | Reserved | R | 0x0 |
| 2:0 | TIMER0CLKSRC | Timer0 clock Source:<br>000: HOSC<br>001: HOSC/32<br>010: CK32K768<br>011: Reserved<br>100: RC32K | R/W | 0x0 |

### 10.4.3.15 CMU_TIMER1CLK

TIMER1 Clock Control Register
Offset = 0x0044

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:3 | - | Reserved | R | 0x0 |
| 2:0 | TIMER1CLKSRC | Timer1 clock Source:<br>000: HOSC<br>001: HOSC/32<br>010: CK32K768<br>011: Reserved<br>100: RC32K | R/W | 0x0 |

### 10.4.3.16 CMU_TIMER2CLK

TIMER2 Clock Control Register
Offset = 0x0048

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:3 | - | Reserved | R | 0x0 |
| 2:0 | TIMER2CLKSRC | Timer2 clock Source:<br>000: HOSC<br>001: HOSC/32<br>010: CK32K768<br>011: HOSC/32000<br>100: RC32K | R/W | 0x0 |

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### 10.4.3.17 CMU_TIMER3CLK

TIMER3 Clock Control Register
Offset = 0x004C

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:3 | - | Reserved | R | 0x0 |
| 2:0 | TIMER3CLKSRC | Timer3 clock Source:<br>000: HOSC<br>001: HOSC/32<br>010: CK32K768<br>011: HOSC/32000<br>100: RC32K | R/W | 0x0 |

### 10.4.3.18 CMU_TIMER4CLK

TIMER4 Clock Control Register
Offset = 0x0050

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:3 | - | Reserved | R | 0x0 |
| 2:0 | TIMER4CLKSRC | Timer4 clock Source:<br>000: HOSC<br>001: HOSC/32<br>010: RC4M<br>011: HOSC/32000<br>100: RC32K | R/W | 0x0 |

### 10.4.3.19 CMU_TIMER5CLK

TIMER5 Clock Control Register
Offset = 0x0054

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:3 | - | Reserved | R | 0x0 |
| 2:0 | TIMER5CLKSRC | Timer5 clock Source:<br>000: HOSC<br>001: HOSC/32<br>010: CK32K768<br>011: HOSC/32000<br>100: RC32K | R/W | 0x0 |

### 10.4.3.20 CMU_PWM0CLK

CMU_PWM0CLK Control register
Offset = 0x0068

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:11 | - | Reserved | R | 0x0 |
| 10:9 | PWM0CLKSRC | PWM0 Controller Clock Source<br>00: RC32K<br>01: HOSC | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 10/11: RC64M | | |
| 8:0 | PWM0CLKDIV | PWM0 Controller Clock Divisor:<br>0: /1<br>1: /2<br>...<br>255: /256<br>256: /512<br>257: /1024<br>258: /2048<br>259: /4096<br>260: /8192<br>261~511: reserved | R/W | 0x0 |

### 10.4.3.21 CMU_PWM1CLK

CMU_PWM1CLK Control register
Offset = 0x006C

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:11 | - | Reserved | R | 0x0 |
| 10:9 | PWM1CLKSRC | PWM1 Controller Clock Source<br>00: RC32K<br>01: HOSC<br>10/11: RC64M | R/W | 0x0 |
| 8:0 | PWM1CLKDIV | PWM1 Controller Clock Divisor:<br>0: /1<br>1: /2<br>...<br>255: /256<br>256: /512<br>257: /1024<br>258: /2048<br>259: /4096<br>260: /8192<br>261~511: reserved | R/W | 0x0 |

### 10.4.3.22 CMU_PWM2CLK

CMU_PWM2CLK Control register
Offset = 0x0070

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:11 | - | Reserved | R | 0x0 |
| 10:9 | PWM2CLKSRC | PWM2 Controller Clock Source<br>00: RC32K<br>01: HOSC<br>10/11: RC64M | R/W | 0x0 |
| 8:0 | PWM2CLKDIV | PWM2 Controller Clock Divisor:<br>0: /1<br>1: /2<br>...<br>255: /256<br>256: /512<br>257: /1024 | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 258: /2048<br>259: /4096<br>260: /8192<br>261~511: reserved | | |

### 10.4.3.23 CMU_PWM3CLK

CMU_PWM3CLK Control register
Offset = 0x0074

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:11 | - | Reserved | R | 0x0 |
| 10:9 | PWM3CLKSRC | PWM3 Controller Clock Source<br>00: RC32K<br>01: HOSC<br>10/11: RC64M | R/W | 0x0 |
| 8:0 | PWM3CLKDIV | PWM3 Controller Clock Divisor:<br>0: /1<br>1: /2<br>…<br>255: /256<br>256: /512<br>257: /1024<br>258: /2048<br>259: /4096<br>260: /8192<br>261~511: reserved | R/W | 0x0 |

### 10.4.3.24 CMU_LRADCCLK

CMU_LRADCCLK Control register
Offset = 0x0078

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:7 | - | Reserved | R | 0x0 |
| 6:4 | LRADCCLKSRC | LRADC Clock Source<br>00: RC32K<br>01: CK32K768<br>10: RC4M /16<br>11: HOSC/128 | R/W | 0x0 |
| 3:2 | - | Reserved | R | 0x0 |
| 1:0 | LRADCCLKDIV | LRADC Clock Divisor<br>00: /1<br>01: /2<br>10: /4<br>11: /8 | R/W | 0x0 |

### 10.4.3.25 CMU_I2C0CLK

CMU I2C0CLK Control register
Offset = 0x0080

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| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:5 | - | Reserved | R | 0x0 |
| 4 | I2C0CLKSRC | I2C0 Clock Source<br>0: HOSC<br>1: RC192M/6 | R/W | 0x0 |
| 1:0 | I2C0CLKDIV | I2C0 Controller Clock Divisor:<br>00: /1<br>01: /2<br>10: /4<br>11: /8 | R/W | 0x0 |

### 10.4.3.26 CMU_I2C1CLK

CMU I2C1CLK Control register
Offset = 0x0084

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:5 | - | Reserved | R | 0x0 |
| 4 | I2C1CLKSRC | I2C1 Clock Source<br>0: HOSC<br>1: RC192M/6 | R/W | 0x0 |
| 1:0 | I2C1CLKDIV | I2C1 Controller Clock Divisor:<br>00: /1<br>01: /2<br>10: /4<br>11: /8 | R/W | 0x0 |

### 10.4.3.27 CMU_I2C2CLK

CMU I2C2CLK Control register
Offset = 0x0088

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:5 | - | Reserved | R | 0x0 |
| 4 | I2C2CLKSRC | I2C2 Clock Source<br>0: HOSC<br>1: RC192M/6 | R/W | 0x0 |
| 1:0 | I2C2CLKDIV | I2C2 Controller Clock Divisor:<br>00: /1<br>01: /2<br>10: /4<br>11: /8 | R/W | 0x0 |

### 10.4.3.28 CMU_I2C3CLK

CMU I2C3CLK Control register
Offset = 0x008C

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:5 | - | Reserved | R | 0x0 |
| 4 | I2C3CLKSRC | I2C3 Clock Source | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0: HOSC<br>1: RC192M/6 | | |
| 1:0 | I2C3CLKDIV | I2C3 Controller Clock Divisor:<br>00: /1<br>01: /2<br>10: /4<br>11: /8 | R/W | 0x0 |

### 10.4.3.29 CMU_AUDDSPCLK

CMU_AUDDSPCLK Control register
Offset = 0x90

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:10 | - | Reserved | R | 0x0 |
| 9:8 | DSPAPBCLKDIV | DSP_APB_CLK divisor:<br>00: /1<br>01: /2<br>10: /4<br>11: /8 | R/W | 0x0 |
| 7:4 | DSPCLKDIV | DSP Clock Divisor:<br>0: /1<br>1: /2<br>2: /3<br>3: /4<br>4: /5<br>…<br>13: /14<br>14: /1.5<br>15: /2.5 | R/W | 0x0 |
| 3 | - | Reserved | R | 0x0 |
| 2:0 | DSPCLKSRC | DSP CLK Source:<br>000: Reserved<br>001: HOSC<br>010: COREPLL<br>011: RC64M<br>100: Reserved<br>101: RC192M<br>110: Reserved | R/W | 0x1 |

### 10.4.3.30 CMU_DACCLK

DAC Clock Control Register
Offset = 0x0098

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:25 | - | Reserved | R | 0x0 |
| 24 | DACFIFOCLKEN | DAC FIFO Access Clock Enable:<br>0: disable<br>1: enable | R/W | 0x0 |
| 23:22 | - | Reserved | R | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 21 | DACCICFIRCLKEN | DAC CIC FIR Clock Enable:<br>0: disable<br>1: enable | R/W | 0x0 |
| 20 | DACSDMCLKEN | DAC SDM Clock Enable:<br>0: disable<br>1: enable | R/W | 0x0 |
| 19:18 | - | Reserved | R | 0x0 |
| 17:16 | DACFIR2XCLKDIV | DAC 2X FIR Clock Divisor<br>00: /1<br>01: /2<br>10: /3<br>11: reserved | R/W | 0x0 |
| 15 | - | Reserved | R | 0x0 |
| 14 | DACFIRCLKDIV | DAC FIR Clock Divisor<br>0: /1<br>1: /2 | R/W | 0x0 |
| 13 | DACHUMDIV | DAC Hnum Divisor<br>0: /1<br>1: /2 | R/W | 0x0 |
| 12 | DACCICCLKDIV | DAC CICCLK and SDMLCK Clock Divisor<br>0: /1<br>1: /2 | R/W | 0x0 |
| 11:9 | - | Reserved | R | 0x0 |
| 8 | DACCLKSRC | DAC Clock Source Selection:<br>0: Audiopll/4<br>1: HOSC | R/W | 0x0 |
| 7:2 | - | Reserved | R | 0x0 |
| 1:0 | DACCLKDIV | DAC Clock Divisor.<br>00: /1<br>01: /2<br>10: /3<br>11: /4 | R/W | 0x0 |

### 10.4.3.31 CMU_ADCCLK

ADC Clock Control Register
Offset = 0x009C

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31 | - | Reserved | R/W | 0x0 |
| 30 | ADCSWITCHRC4M | ADC Clock Switch to RC4M:<br>0: ADC Clock from Audiopll<br>1: ADC Clock from RC4M | R/W | 0x0 |
| 29:25 | - | Reserved | R | 0x0 |
| 24 | ADCFIFOCLKEN | ADC FIFO Access Clock Enable:<br>0: disable<br>1: enable | R/W | 0x0 |
| 23 | ADCFIREN | ADC FIR Filter CLK enable：<br>0： disable<br>1： enable | RW | 0x0 |
| 22 | ADCCICEN | ADC CIC Filter enable：<br>0： disable | RW | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 1： enable | | |
| 21 | ADCANAEN | ADC Analog CLK enable：<br>0： disable<br>1： enable | RW | 0x0 |
| 20 | ADCDMICEN | DMIC CLK enable：<br>0： disable<br>1： enable | RW | 0x0 |
| 19 | ADCFIRCLKRVS | ADC FIR Clock Reverse control<br>0： Normal<br>1： Reverse | RW | 0x0 |
| 18 | ADCCICCLKRVS | ADC CIC Clock Reverse control<br>0： Normal<br>1： Reverse | RW | 0x0 |
| 17 | ADCANACLKRVS | ADC Analog Clock Reverse control<br>0： Normal<br>1： Reverse | RW | 0x0 |
| 16 | ADCDMICCLKRVS | ADC DMIC Clock Reverse control<br>0： Normal<br>1： Reverse | RW | 0x0 |
| 15 | ADCFIRCLKDIV | ADC FIR Clock Divisor:<br>0: /1<br>1: /3 | RW | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | ADCOVFSCLKDIV | ADC OVFS Clock Divisor:<br>00: /1<br>01: /2<br>10: /4<br>11: reserved | RW | 0x0 |
| 11:9 | - | Reserved | R | 0x0 |
| 8 | ADCCLKSRC | ADC Clock Source<br>0: AudioPLL0<br>1: AudioPLL1 | R/W | 0x0 |
| 7:5 | - | Reserved | R | 0x0 |
| 4 | ADCCLKPREDIV | ADC Clock Pre-Divisor<br>0: /2<br>1: /4 | R/W | 0x0 |
| 3 | - | Reserved | R | 0x0 |
| 2:0 | ADCCLKDIV | ADC Clock Divisor<br>000: /1<br>001: /2<br>010: /3<br>011: /4<br>100: /6<br>101: /8<br>110: /12<br>111: --- | R/W | 0x0 |

### 10.4.3.32 CMU_I2STXCLK

I2STX Clock Control register
Offset = 0x00A0

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| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31 | I2SG0LRCLKOEN | I2SG0 LRCLK Output to PAD enable<br>0: disable<br>1: enable<br>This bit enables the LRCLK sending to the I2SG0 clock output PAD. | R/W | 0x0 |
| 30 | I2SG0BCLKOEN | I2SG0 BCLK Output to PAD enable<br>0: disable<br>1: enable<br>This bit enables the BCLK sending to the I2SG0 clock output PAD. | R/W | 0x0 |
| 29 | I2SG0MCLKOEN | I2SG0 MCLK Output to PAD enable<br>0: disable<br>1: enable<br>This bit enables the MCLK sending to the I2SG0 clock output PAD. | R/W | 0x0 |
| 28 | I2STX0CLKSRC | I2STX0 Clock Source:<br>0: I2SG0<br>1: I2SG1 | R/W | 0x0 |
| 27:25 | - | Reserved | R | 0x0 |
| 24:23 | I2SG0LRCLKPROC | I2SG0 LRCLK Process:<br>00: 50% DUTY for I2S<br>01: 1 BCLK for I2S<br>10: 50% DUTY for PCM<br>11: 1 BCLK for PCM<br>This bit is valid when I2SG0BLRCLKSRC is set to 0. | R/W | 0x0 |
| 22:20 | I2SG0LRCLKDIV | I2SG0 LRCLK Divisor：<br>000: /32<br>001: /64<br>010: /128<br>011: /256<br>100: /16<br>Others: reserved<br>This bit is valid when I2SG0BLRCLKSRC is set to 0. | R/W | 0x0 |
| 19:18 | I2SG0BCLKDIV | I2SG0 BCLK Divisor：<br>00: /4<br>01: /6<br>10: /8<br>11: /12<br>This bit is valid when I2SG0BLRCLKSRC is set to 0. | R/W | 0x0 |
| 17 | - | Reserved | R | |
| 16 | I2SG0BLRCLKSRC | I2SG0 BCLK/LRCLK Source：<br>0: internal from I2SG0_MCLK<br>1: external from PAD | R/W | 0x0 |
| 15 | - | Reserved | R | 0x0 |
| 14 | I2SG0MCLKEXTREV | I2SG0_MCLK_EXT Reverse：<br>0：Normal<br>1：Reversed | R/W | 0x1 |
| 13:12 | I2SG0MCLKSRC | I2SG0_MCLK source：<br>00：I2SG0CLK<br>01：DAC_256fs_CLK<br>10：DAC_128fs_CLK<br>11：I2SG0_MCLK_EXT | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 11 | I2STX0FIFOCLKEN | I2STX0 FIFO Clock Enable<br>0: disable<br>1: enable | R/W | 0x0 |
| 10:4 | - | Reserved | R | 0x0 |
| 3:0 | I2SG0CLKDIV | I2SG0 Clock Divisor.<br>0000: /1<br>0001: /2<br>0010: /3<br>0011: /4<br>0100: /6<br>0101: /8<br>0110: /12<br>0111: /16<br>1000: /24<br>1001: /32<br>1010: /48<br>1011: /96<br>Others Reserved. | R/W | 0x0 |

### 10.4.3.33 CMU_I2SRXCLK

I2STX Clock Control register
Offset = 0x00A4

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31 | I2SG1LRCLKOEN | I2SG1 LRCLK Output to PAD enable<br>0: disable<br>1: enable<br>This bit enables the LRCLK sending to the I2SG1 clock output PAD. | R/W | 0x0 |
| 30 | I2SG1BCLKOEN | I2SG1 BCLK Output to PAD enable<br>0: disable<br>1: enable<br>This bit enables the BCLK sending to the I2SG1 clock output PAD. | R/W | 0x0 |
| 29 | I2SG1MCLKOEN | I2SG1 MCLK Output to PAD enable<br>0: disable<br>1: enable<br>This bit enables the MCLK sending to the I2SG1 clock output PAD. | R/W | 0x0 |
| 28 | I2SRX0CLKSRC | I2SRX Clock Source:<br>0: I2SG0<br>1: I2SG1 | R/W | 0x0 |
| 27:25 | - | Reserved | R | 0x0 |
| 24:23 | I2SG1LRCLKPROC | I2SG1 LRCLK Process:<br>00: 50% DUTY for I2S<br>01: 1 BCLK for I2S<br>10: 50% DUTY for PCM<br>11: 1 BCLK for PCM<br>This bit is valid when I2SG1BLRCLKSRC is set to 0. | R/W | 0x0 |
| 22:20 | I2SG1LRCLKDIV | I2SG1 LRCLK Divisor：<br>000: /32<br>001: /64 | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 010: /128<br>011: /256<br>100: /16<br>Others: reserved<br>This bit is valid when I2SG1BLRCLKSRC is set to 0. | | |
| 19:18 | I2SG1BCLKDIV | I2SG1 BCLK Divisor：<br>00: /4<br>01: /6<br>10: /8<br>11: /12<br>This bit is valid when I2SG1BLRCLKSRC is set to 0. | R/W | 0x0 |
| 17 | - | Reserved | R | 0x0 |
| 16 | I2SG1BLRCLKSRC | I2SG1 BCLK/LRCLK Source：<br>0: internal from I2SG1_MCLK<br>1: external from PAD | R/W | 0x0 |
| 15 | - | Reserved | R | 0x0 |
| 14 | I2SG1MCLKEXTREV | I2SG1_MCLK_EXT Reverse：<br>0：Normal<br>1：Reversed | R/W | 0x1 |
| 13:12 | I2SG1MCLKSRC | I2SG1_MCLK source：<br>00：I2SG1CLK<br>01：reserved<br>10：reserved<br>11：I2SG1_MCLK_EXT | R/W | 0x0 |
| 11 | I2SRX0FIFOCLKEN | I2SRX0 FIFO Clock Enable<br>0: disable<br>1: enable | R/W | 0x0 |
| 10:4 | - | Reserved | R | 0x0 |
| 3:0 | I2SG1CLKDIV | I2SG1 Clock Divisor.<br>0000: /1<br>0001: /2<br>0010: /3<br>0011: /4<br>0100: /6<br>0101: /8<br>0110: /12<br>0111: /16<br>1000: /24<br>1001: /32<br>1010: /48<br>1011: /96<br>Others Reserved. | R/W | 0x0 |

## 10.4.3.34 CMU_MEMCLKEN0

CMU_MEMCLKEN Control register0
Offset = 0x00B0

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31 | SPI1CACHERAMCLKEN | SPI1CACHE RAM clock enable bit:<br>0: disable<br>1: enable | R/W | 0x1 |

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| | | | | |
|---|---|---|---|---|
| 30 | SPI0CACHERAMCLKEN | SPI0CACHE RAM clock enable bit:<br>0: disable<br>1: enable | R/W | 0x1 |
| 29:26 | - | Reserved | R/W | 0x0 |
| 25 | SRAMCLKEN | Share RAM clock enable bit :<br>0: disable<br>1: enable | R/W | 0x1 |
| 24:23 | - | Reserved | R/W | 0x0 |
| 22 | RAM16CLKEN | RAM16 clock enable bit :<br>0: disable<br>1: enable | R/W | 0x1 |
| 21 | - | Reserved | R/W | 0x0 |
| 20 | RAM15CLKEN | RAM15 clock enable bit :<br>0: disable<br>1: enable | R/W | 0x1 |
| 19 | - | Reserved | R/W | 0x0 |
| 18 | RAM14CLKEN | RAM14 clock enable bit :<br>0: disable<br>1: enable | R/W | 0x1 |
| 17 | - | Reserved | R/W | 0x0 |
| 16 | RAM13CLKEN | RAM13 clock enable bit :<br>0: disable<br>1: enable | R/W | 0x1 |
| 15 | - | Reserved | R/W | 0x0 |
| 14 | RAM12CLKEN | RAM12 clock enable bit :<br>0: disable<br>1: enable | R/W | 0x1 |
| 13 | - | Reserved | R/W | 0x0 |
| 12 | RAM11CLKEN | RAM11 clock enable bit :<br>0: disable<br>1: enable | R/W | 0x1 |
| 11 | RAM10CLKEN | RAM10 clock enable bit :<br>0: disable<br>1: enable | R/W | 0x1 |
| 10 | RAM9CLKEN | RAM9 clock enable bit :<br>0: disable<br>1: enable | R/W | 0x1 |
| 9 | RAM8CLKEN | RAM8 clock enable bit :<br>0: disable<br>1: enable | R/W | 0x1 |
| 8 | RAM7CLKEN | RAM7 clock enable bit :<br>0: disable<br>1: enable | R/W | 0x1 |
| 7 | RAM6CLKEN | RAM6 clock enable bit :<br>0: disable<br>1: enable | R/W | 0x1 |
| 6 | RAM5CLKEN | RAM5 clock enable bit :<br>0: disable<br>1: enable | R/W | 0x1 |
| 5 | RAM4CLKEN | RAM4 clock enable bit :<br>0: disable<br>1: enable | R/W | 0x1 |
| 4 | RAM3CLKEN | RAM3 clock enable bit : | R/W | 0x1 |

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| | | | | |
|---|---|---|---|---|
| | | 0: disable<br>1: enable | | |
| 3 | RAM2CLKEN | RAM2 clock enable bit :<br>0: disable<br>1: enable | R/W | 0x1 |
| 2 | RAM1CLKEN | RAM1 clock enable bit :<br>0: disable<br>1: enable | R/W | 0x1 |
| 1 | RAM0CLKEN | RAM0 clock enable bit :<br>0: disable<br>1: enable | R/W | 0x1 |
| 0 | ROMCLKEN | ROM clock enable bit :<br>0: disable<br>1: enable | R/W | 0x1 |

### 10.4.3.35 CMU_MEMCLKEN1

CMU_MEMCLKEN Control register1
Offset = 0x00B4

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:29 | - | Reserved | R/W | 0x0 |
| 28 | JPEGOUTRAM3CLKEN | JPEG OUTRAM3 Clock Enable:<br>0: disable<br>1: enable | R/W | 0x0 |
| 27 | JPEGOUTRAM2CLKEN | JPEG OUTRAM2 Clock Enable:<br>0: disable<br>1: enable | R/W | 0x0 |
| 26 | JPEGOUTRAM1CLKEN | JPEG OUTRAM1 Clock Enable:<br>0: disable<br>1: enable | R/W | 0x0 |
| 25 | JPEGOUTRAM0CLKEN | JPEG OUTRAM0 Clock Enable:<br>0: disable<br>1: enable | R/W | 0x0 |
| 24 | JPEGRAMCLKEN | JPEG RAM Clock Enable:<br>0: disable<br>1: enable | R/W | 0x0 |
| 23:9 | - | Reserved | R/W | X |
| 8 | USBRAMCLKEN | URAM0,1 Clock Enable:<br>0: disable<br>1: enable | R/W | 0x1 |
| 7:1 | - | Reserved | R/W | 0x0 |
| 0 | AUDDSPRAMCLKEN | AUDDSPDTCRAM, PTCRAM Clock Enable:<br>0: disable<br>1: enable | R/W | 0x1 |

### 10.4.3.36 CMU_MEMCLKSRC0

CMU_MEMCLK Source Control register0
Offset = 0x00C0

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:30 | - | Reserved | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 29:28 | SRAMCLKDIV | Share RAM Clock Divisor:<br>00: /1<br>01: /2<br>10: /4<br>11: /8 | R/W | 0x0 |
| 27:25 | SRAMCLKSRC | Share RAM Clock Source:<br>000: RC4M<br>001: HOSC<br>010: COREPLL<br>011: RC192M<br>100: RC64M | R/W | 0x0 |
| 24 | - | Reserved | R/W | 0x0 |
| 23:22 | RAM16CLKSRC | RAM16 Clock Source:<br>00: CPUCLK<br>01: AUDIODSPCLK<br>Others Reserved. | R/W | 0x0 |
| 21:20 | RAM15CLKSRC | RAM15 Clock Source:<br>00: CPUCLK<br>01: AUDIODSPCLK<br>Others Reserved. | R/W | 0x0 |
| 19:18 | RAM14CLKSRC | RAM14 Clock Source:<br>00: CPUCLK<br>01: AUDIODSPCLK<br>Others Reserved. | R/W | 0x0 |
| 17:16 | RAM13CLKSRC | RAM13 Clock Source:<br>00: CPUCLK<br>01: AUDIODSPCLK<br>Others Reserved. | R/W | 0x0 |
| 15:14 | RAM12CLKSRC | RAM12 Clock Source:<br>00: CPUCLK<br>01: AUDIODSPCLK<br>Others Reserved. | R/W | 0x0 |
| 13:12 | RAM11CLKSRC | RAM11 Clock Source:<br>00: CPUCLK<br>01: AUDIODSPCLK<br>Others Reserved. | R/W | 0x0 |
| 11 | RAM10CLKSRC | RAM10 Clock Source:<br>0: CPUCLK<br>1: AUDIODSPCLK | R/W | 0x0 |
| 10 | RAM9CLKSRC | RAM9 Clock Source:<br>0: CPUCLK<br>1: AUDIODSPCLK | R/W | 0x0 |
| 9 | RAM8CLKSRC | RAM8 Clock Source:<br>0: CPUCLK<br>1: AUDIODSPCLK | R/W | 0x0 |
| 8 | RAM7CLKSRC | RAM7 Clock Source:<br>0: CPUCLK<br>1: AUDIODSPCLK | R/W | 0x0 |
| 7 | RAM6CLKSRC | RAM6 Clock Source:<br>0: CPUCLK<br>1: AUDIODSPCLK | R/W | 0x0 |
| 6 | RAM5CLKSRC | RAM5 Clock Source:<br>0: CPUCLK | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 1: AUDIODSPCLK | | |
| 5 | RAM4CLKSRC | RAM4 Clock Source:<br>0: CPUCLK<br>1: AUDIODSPCLK | R/W | 0x0 |
| 4 | RAM3CLKSRC | RAM3 Clock Source:<br>0: CPUCLK<br>1: AUDIODSPCLK | R/W | 0x0 |
| 3 | RAM2CLKSRC | RAM2 Clock Source:<br>0: CPUCLK<br>1: AUDIODSPCLK | R/W | 0x0 |
| 2 | RAM1CLKSRC | RAM1 Clock Source:<br>0: CPUCLK<br>1: AUDIODSPCLK | R/W | 0x0 |
| 1 | RAM0CLKSRC | RAM0 Clock Source:<br>0: CPUCLK<br>1: AUDIODSPCLK | R/W | 0x0 |
| 0 | - | Reserved | R/W | 0x0 |

### 10.4.3.37 CMU_MEMCLKSRC1

CMU_MEMCLK Source Control register1
Offset = 0x00C4

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:29 | - | Reserved | R/W | 0x0 |
| 28 | JPEG_OUTRAM3CLKSRC | JPEG OUTRAM3 Clock Source:<br>0: CPUCLK<br>1: JPEGCLK | R/W | 0x0 |
| 27 | JPEG_OUTRAM2CLKSRC | JPEG OUTRAM2 Clock Source:<br>0: CPUCLK<br>1: JPEGCLK | R/W | 0x0 |
| 26 | JPEG_OUTRAM1CLKSRC | JPEG OUTRAM1 Clock Source:<br>0: CPUCLK<br>1: JPEGCLK | R/W | 0x0 |
| 25 | JPEG_OUTRAM0CLKSRC | JPEG OUTRAM0 Clock Source:<br>0: CPUCLK<br>1: JPEGCLK | R/W | 0x0 |
| 24 | JPEGRAMCLKSRC | JPEG internal RAM Clock Source:<br>0: CPUCLK<br>1: JPEGCLK | R/W | 0x0 |
| 23:14 | - | Reserved | R/W | 0x0 |
| 13:12 | USBRAMPRECLKDIV | USBRAM0, RAM1 Pre-Clock Divisor:<br>00: /1<br>01: /2<br>10: /4<br>11: /8 | R/W | 0x0 |
| 11:9 | USBRAMPRECLKSRC | USBRAM0, RAM1 Pre-Clock Source:<br>000: RC4M<br>001: HOSC<br>010: COREPLL<br>011: RC192M<br>100: RC64M | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | Others: Reserved | | |
| 8 | USBRAMCLKSRC | USBRAM0, RAM1 Clock Source:<br>0: USBRAMPRECLK<br>1: USBctl_URAM_CLK | R/W | 0x0 |
| 7:6 | - | Reserved | R/W | 0x0 |
| 5:4 | AUDDSPTCRAMPRECLKDIV | AUDDSPDTCRAM, PTCRAM Pre-Clock Divisor:<br>00: /1<br>01: /2<br>10: /4<br>11: /8 | R/W | 0x0 |
| 3:1 | AUDDSPTCRAMPRECLKSRC | AUDDSPDTCRAM, PTCRAM Pre-Clock Source:<br>000: RC4M<br>001: HOSC<br>010: COREPLL<br>011: RC192M<br>100: RC64M | R/W | 0x0 |
| 0 | AUDDSPTCRAMCLKSRC | AUDDSPDTCRAM, PTCRAM Clock Source:<br>0: AUDDSPTCRAMPRECLK<br>1: AUDDSPCLK | R/W | 0x0 |

### 10.4.3.38 CMU_S1CLKCTL

CMU S1CLK Control register
Offset = 0x00D0

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:3 | - | Reserved | R | 0x0 |
| 2 | RC64M_S1EN | RC64M S1 Entering Control:<br>0: disable<br>1: enable | R/W | 0x0 |
| 1 | HOSC_S1EN | HOSC S1 Entering Control:<br>0: disable<br>1: enable | R/W | 0x1 |
| 0 | RC4M_S1EN | RC4M S1 Entering Control:<br>0: disable<br>1: enable | R/W | 0x1 |

### 10.4.3.39 CMU_S1BTCLKCTL

CMU S1BTCLK Control register
Offset = 0x00D4

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:3 | - | Reserved | R | 0x0 |
| 2 | RC64M_S1BTEN | RC64M S1BT Entering Control:<br>0: disable<br>1: enable | R/W | 0x0 |
| 1 | HOSC_S1BTEN | HOSC S1BT Entering Control:<br>0: disable<br>1: enable<br>If this bit is set, HOSC will be enabled all the S1BT time. | R/W | 0x1 |
| 0 | RC4M_S1BTEN | RC4M S1BT Entering Control:<br>0: disable | R/W | 0x1 |

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| | | | | |
|---|---|---|---|---|
| | | 1: enable | | |

### 10.4.3.40 CMU_S2CLKCTL

CMU S2SCLK Control register
Offset = 0x00DC

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:3 | - | Reserved | R | 0x0 |
| 2 | RC64M_S2EN | RC64M S2 Entering Control:<br>0: disable<br>1: enable | R/W | 0x0 |
| 1 | HOSC_S2EN | HOSC S2 Entering Control:<br>0: disable<br>1: enable<br>If this bit is set, HOSC will be enabled all the S2S time. | R/W | 0x0 |
| 0 | RC4M_S2EN | RC4M S2 Entering Control:<br>0: disable<br>1: enable | R/W | 0x0 |

### 10.4.3.41 CMU_S3CLKCTL

CMU S3CLK Control register
Offset = 0x00E0

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:4 | - | Reserved | R | 0x0 |
| 3 | S3RAM4CLKON | S3 ShareRAMCLK, SPIMTxCLK and I2CMTxCLK on:<br>0: disable<br>1: enable | R/W | 0x0 |
| 2 | - | Reserved | R | 0x0 |
| 1 | HOSC_S3EN | HOSC S3 Entering Control:<br>0: disable<br>1: enable<br>If this bit is set, HOSC will be enabled all the S3 time. | R/W | 0x0 |
| 0 | RC4M_S3EN | RC4M S3 Entering Control:<br>0: disable<br>1: enable | R/W | 0x0 |

### 10.4.3.42 CMU_WDCLK

CMU Watchdog CLK Control register
Offset = 0x00E4

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:2 | - | Reserved | R | 0x0 |
| 1 | CPUDS_WDCLKEN | CPU Deepsleep WatchDog Clock Enable<br>0: disable<br>1: enable | R/W | 0x0 |
| 0 | WDCLKSRC | WatchDog Clock Source<br>0: RC32K | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 1: CK32K768 | | |

### 10.4.3.43 CMU_GPIOCLKCTL

CMU GPIO interrupt Clock Control register
Offset = 0x00E8

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:1 | - | Reserved | R | 0x0 |
| 0 | GPIOINTCLKSRC | GPIO interrupt Clock Source:<br>0: RC32K<br>1: RC4M | R/W | 0x1 |

### 10.4.3.44 CMU_PMUWKUPCLKCTL

CMU PMU STATE And WAKEUP Clock Control register
Offset = 0x00EC

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:1 | - | Reserved | R | 0x0 |
| 0 | PMUWKUPCLKSRC | PMU Wakeup Clock Source:<br>0: RC32K<br>1: RC4M/4 | R/W | 0x1 |

### 10.4.3.45 CMU_SPIMT0CLK

SPIMT0 Clock Control Register
Offset = 0x0100

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:10 | - | Reserved | R | 0x0 |
| 9:8 | SPIMT0CLKSRC | SPIMT0 Controller Clock Source:<br>00: RC4M<br>01: RC64M/2<br>10: HOSC<br>Others: Reserved | R/W | 0x0 |
| 7:4 | - | Reserved | R | 0x0 |
| 3:0 | SPIMT0CLKDIV | SPIMT0 Clock Divisor:<br>0: /1<br>1: /2<br>2: /3<br>3: /4<br>4: /5<br>…<br>13: /14<br>14: /1.5<br>15: /2.5 | R/W | 0x0 |

### 10.4.3.46 CMU_SPIMT1CLK

SPIMT1 Clock Control Register
Offset = 0x0104

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| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:10 | - | Reserved | R | 0x0 |
| 9:8 | SPIMT1CLKSRC | SPIMT1 Controller Clock Source:<br>00: RC4M<br>01: RC64M/2<br>10: HOSC<br>Others: Reserved | R/W | 0x0 |
| 7:4 | - | Reserved | R | 0x0 |
| 3:0 | SPIMT1CLKDIV | SPIMT1Clock Divisor:<br>0: /1<br>1: /2<br>2: /3<br>3: /4<br>4: /5<br>…<br>13: /14<br>14: /1.5<br>15: /2.5 | R/W | 0x0 |

### 10.4.3.47 CMU_I2CMT0CLK

I2CMT0 Clock Control Register
Offset = 0x0110

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:10 | - | Reserved | R | 0x0 |
| 9:8 | I2CMT0CLKSRC | I2CMT0 Controller Clock Source:<br>00: RC4M<br>01: RC64M/2<br>10: HOSC<br>Others: Reserved | R/W | 0x0 |
| 7:2 | - | Reserved | R | 0x0 |
| 1:0 | I2CMT0CLKDIV | I2CMT0 Clock Divisor:<br>0: /1<br>1: /2<br>2: /4<br>3: /8 | R/W | 0x0 |

### 10.4.3.48 CMU_I2CMT1CLK

I2CMT1 Clock Control Register
Offset = 0x0114

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:10 | - | Reserved | R | 0x0 |
| 9:8 | I2CMT1CLKSRC | I2CMT1 Controller Clock Source:<br>00: RC4M<br>01: RC64M/2<br>10: HOSC<br>Others: Reserved | R/W | 0x0 |
| 7:2 | - | Reserved | R | 0x0 |
| 1:0 | I2CMT1CLKDIV | I2CMT1Clock Divisor:<br>0: /1<br>1: /2 | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 2: /4<br>3: /8 | | |

## 10.5 RTC

This part have 3 individual modules: Calendar, Watch Dog (WD) and Timer0/1/2/3/4/5.

### 10.5.1 Features

- An individual power supply: VDD
- Internal oscillator, LOSC optional
- Calendar with an alarm IRQ which can wake up the PMU
- Interval Timer can wake up CPU periodically
- Six Timers with IRQS, all of them are low power timers
- A watch dog which can be configured optional as IRQ or Reset

### 10.5.2 RTC Register List

*Table 10-9 RTC block base address*

| Name | Physical Base Address | KSEG1 Base Address |
|---|---|---|
| RTC | 0x4000C000 | 0x4000C000 |

*Table 10-10 RTC Controller Registers*

| Offset | Register Name | Description |
|---|---|---|
| 0x00 | RTC_CTL | RTC Control Register |
| 0x08 | RTC_MSALM | RTC MilliSecond Alarm Register |
| 0x0C | RTC_HMSALM | RTC Day Hour Minute and Second Alarm Register |
| 0x10 | RTC_YMDALM | RTC Year Month Date Alarm Register |
| 0x14 | RTC_MS | RTC MilliSecond Register |
| 0x18 | RTC_HMS | RTC Day Hour Minute and Second Register |
| 0x1C | RTC_YMD | RTC Year Month Date Register |
| 0x20 | WD_CTL | Watch Dog Control register |
| 0x24 | RTC_IT_VAL | RTC Interval Timer Value Register |

### 10.5.3 RTC Register Description

#### 10.5.3.1 RTC_CTL

Calendar Control Register
Offset=0x0000

| Bits | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:30 | - | Reserved | R | 0x0 |
| 29 | IT_IRQ_PD | Interval Timer IRQ Pending<br>0:normal<br>1: Pending<br>Write 1 auto clean | R/W | 0x0 |
| 28 | IT_IRQ_EN | Interval Timer IRQ Enable<br>0:disable<br>1:enable | R/W | 0x0 |
| 27:25 | - | Reserved | R | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 24 | IT_EN | Interval Timer Enable<br>0:disable<br>1:enable | R/W | 0x0 |
| 23:18 | - | Reserved | R | 0x0 |
| 17 | CAL_CLK_SEL | CAL clock Select<br>0:4HZ Select<br>1:100HZ Select | R/W | 0x0 |
| 16 | RTC32768HZ_SEL | RTC32768HZ Clock select<br>0 : LO32768Hz<br>1 : EXT_32768HZ | R/W | 0x0 |
| 15:8 | - | Reserved | R | 0x0 |
| 7 | LEAP | RTC Leap Year bit<br>1: leap year<br>0: not leap year | R | 0x1 |
| 6:5 | - | Reserved | R | 0x0 |
| 4 | CAL_EN | Calendar Enable<br>1: Enable<br>0: Disable | R/W | 0x0 |
| 3:2 | 4HZ_CLK_SEL | 01: 32768_division<br>10: HOSC_division<br>00: HCL_division<br>Others: reserved | R/W | 0x0 |
| 1 | ALIE | Alarm IRQ Enable<br>1: Enable<br>0: Disable | R/W | 0x0 |
| 0 | ALIP | Alarm IRQ Pending bit, writing 1 to this bit will clear it | R/W | 0x0 |

NOTE: The CAL_EN bit must be disabled when The RTC_HMS/RTC_YMD/RTC_MS register being written Or error will occur.

### 10.5.3.2 RTC_MSALM

Offset=0x0008

| Bits | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:7 | - | Reserved | R | 0x0 |
| 6:0 | MS | MilliSecond setting<br>00H – 1FH | R/W | 0x0 |

Note: MilliSecond =MS*10, ignore when RTC_4HZ Select.

### 10.5.3.3 RTC_HMSALM

Offset=0x000C

| Bits | Name | Description | Access | Reset |
|---|---|---|---|---|
| 31:21 | - | Reserved | R | 0 |
| 20:16 | HOURALM | Alarm hour setting<br>00H – 17H | R/W | 0 |
| 15:14 | - | Reserved | R | 0 |
| 13:8 | MINALM | Alarm minute setting<br>00H – 3BH | R/W | 0 |
| 7:6 | - | Reserved | R | 0 |

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| | | | | |
|---|---|---|---|---|
| 5:0 | SECALM | Alarm second setting<br>00H – 3BH | R/W | 0 |

### 10.5.3.4 RTC_YMDALM

Offset=0x0010

| Bits | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:23 | - | Reserved | R | 0x0 |
| 22:16 | YEARALM | Alarm year setting<br>00H – 63H | R/W | 0x0 |
| 15:12 | - | Reserved | R | 0x0 |
| 11:8 | MONALM | Alarm month setting<br>01H – 0CH | R/W | 0x1 |
| 7:5 | - | Reserved | R | 0x0 |
| 4:0 | DATEALM | Alarm day setting<br>01H – 1FH | R/W | 0x1 |

Note: When this register is set to zero, the RTC alarm alarms daily. Otherwise when it is set to nonzero, the RTC alarm alarms on the specific date and time.

### 10.5.3.5 RTC_MS

Offset=0x0014

| Bits | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:7 | - | Reserved | R | 0x0 |
| 6:0 | MS | MilliSecond setting<br>00H –63H | R/W | 0x0 |

Note: MilliSecond=MS*10, ignore when RTC_4HZ Select.

### 10.5.3.6 RTC_HMS

Offset=0x0018

| Bits | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:21 | - | Reserved | R | 0x0 |
| 20:16 | HOUR | Time hour setting<br>00H – 17H | R/W | 0x0 |
| 15:14 | - | Reserved | R | 0x0 |
| 13:8 | MIN | Time minute setting<br>00H – 3BH | R/W | 0x0 |
| 7:6 | - | Reserved | R | 0x0 |
| 5:0 | SEC | Time second setting<br>00H – 3BH | R/W | 0x0 |

### 10.5.3.7 RTC_YMD

Offset=0x001C

| Bits | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:23 | - | Reserved | R | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 22:16 | YEAR | Time year setting<br>00H – 63H | R/W | 0x0 |
| 15:12 | - | Reserved | R | 0x0 |
| 11:8 | MON | Time month setting<br>01H – 0CH | R/W | 0x1 |
| 7:5 | - | Reserved | R | 0x0 |
| 4:0 | DATE | Time day setting<br>01H – 1FH | R/W | 0x1 |

### 10.5.3.8 WD_CTL

Offset=0x0020（VDD）

| Bits | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31..7 | - | reserved | R | 0x0 |
| 6 | IRQP | Watch dog IRQ pending bit, writing 1 to this bit will clear it | RW | 0x0 |
| 5 | SIGS | Watchdog Signal (IRQ or Reset-) Select.0: Reset,1: irq-.<br>0:Send Reset signal when watchdog overflow.<br>1:Send IRQ signal when watchdog overflow. | RW | 0x0 |
| 4 | WDEN | Watch Dog timer enable, when WD timer is enabled and the WD timer overflows, an internal reset (WDRST-) is generated to force the system into reset status and then reboot.<br>1: Enable<br>0: Disable | RW | 0x0 |
| 3..1 | CLKSEL | Watch Dog timer Clock Select,<br>WDCKS Clock Selected Watch Dog Length<br>The watch dog's overflow value is 180.<br>000 1khz 176 ms<br>001 512hz 352 ms<br>010 256hz 703ms<br>011 128hz 1.4 s<br>100 64hz 2.8s<br>101 32hz 5.6 s<br>110 16hz 11.2s<br>111 10ms | RW | 0x0 |
| 0 | CLR | Clear bit, write 1 to clear WD timer, cleared automatically | RW | 0x0 |

### 10.5.3.9 RTC_IT_VAL

RTC Interval Timer Value Register (VDD)
Offset=0x24

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | 0x0 |
| 15:0 | VAL | Read or write Interval Timer/Counter value register<br>Note: If set IT_VAL=n, IRQ will happen after n+1 RTC_CLK. | R/W | 0x0 |

## 10.5.4 TIMER Register List

***Table 10-11 TIMER block base address***

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| Name | Physical Base Address | KSEG1 Base Address |
|---|---|---|
| TIMER_REGISTER | 0x4000C100 | 0x4000C100 |

*Table 10-12 TIMER Controller Registers*

| Offset | Register Name | Description |
|---|---|---|
| 0x00 | T0_CTL | Timer0 Control register |
| 0x04 | T0_VAL | Timer0 Value |
| 0x08 | T0_CNT | Timer0 current counter register |
| 0x20 | T1_CTL | Timer1 Control register |
| 0x24 | T1_VAL | Timer1 Value |
| 0x28 | T1_CNT | Timer1 current counter register |
| 0x40 | T2_CTL | Timer2 Control register |
| 0x44 | T2_VAL | Timer2 Value |
| 0x48 | T2_CNT | Timer2 current counter register |
| 0x60 | T3_CTL | Timer3 Control register |
| 0x64 | T3_VAL | Timer3 Value |
| 0x68 | T3_CNT | Timer3 current counter register |
| 0x80 | T4_CTL | Timer4 Control register |
| 0x84 | T4_VAL | Timer4 Value |
| 0x88 | T4_CNT | Timer4 current counter register |
| 0xA0 | T5_CTL | Timer5 Control register |
| 0xA4 | T5_VAL | Timer5 Value register |
| 0xA8 | T5_CNT | Timer5 current counter register |

## 10.5.5 TIMER Register Description

### 10.5.5.1 T0_CTL

Timer0 control register
Offset=0x0000(VDD)

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:12 | - | reserved | R | 0x0 |
| 11 | DIR | Timer Counting direction set:<br>0: down<br>1: up | R/W | 0x0 |
| 10:9 | MODE_SEL | Timer mode select:<br>00: normal timer<br>11: continue mode<br>Others : reserved | R/W | 0x0 |
| 8:6 | - | Reserved | R | 0x0 |
| 5 | EN | Timer 0 Enable<br>0: Disable<br>1: Enable | R/W | 0x0 |
| 4 | - | Reserved | R | 0x0 |
| 3 | PAUSE | PAUSE Counter<br>0: normal<br>1: pause | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 2 | RELO | Timer 0 Reload.<br>0: Not reload<br>1: Reload | R/W | 0x0 |
| 1 | ZIEN | Timer0 IRQ Enable<br>When this bit is enabled, Timer0_Zero_IRQ sent out the irq signal until the pending bit was cleared. | R/W | 0x0 |
| 0 | ZIPD | Timer0 IRQ Pending,<br>Writing 1 to clear this bit. | R/W | 0x0 |

### 10.5.5.2 T0_VAL

Timer0 value register
Offset=0x0004(VDD)

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | VAL | Read or write Timer/Counter value register<br>Note: If set Tx_VAL=n, irq would cause after n+1 Tx_CLK. | R/W | 0x0 |

### 10.5.5.3 T0_CNT

Timer0 current counter register
Offset=0x0008(VDD)

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | CNT | Read or write current Timer0 value | R | 0x0 |

### 10.5.5.4 T1_CTL

Timer1 control register
Offset=0x0020(VDD)

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:12 | - | reserved | R | 0x0 |
| 11 | DIR | Timer Counting direction set:<br>0: down<br>1: up | R/W | 0x0 |
| 10:9 | MODE_SEL | Timer mode select:<br>00: normal timer<br>11: continue mode<br>Others : reserved | R/W | 0x0 |
| 8:6 | - | Reserved | R | 0x0 |
| 5 | EN | Timer 1 Enable<br>0: Disable<br>1: Enable | R/W | 0x0 |
| 4 | - | Reserved | R | 0x0 |
| 3 | PAUSE | PAUSE Counter<br>0: normal<br>1: pause | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 2 | RELO | Timer 1 Reload.<br>0: Not reload<br>1: Reload | R/W | 0x0 |
| 1 | ZIEN | Timer1 IRQ Enable<br>When this bit is enabled, Timer1_Zero_IRQ sent out the irq signal until the pending bit was cleared. | R/W | 0x0 |
| 0 | ZIPD | Timer1 IRQ Pending,<br>Writing 1 to clear this bit. | R/W | 0x0 |

#### 10.5.5.5 T1_VAL

Timer1 value register
Offset=0x0024(VDD)

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | VAL | Read or write current Timer1 value<br>Note: If set Tx_VAL=n, irq would cause after n+1 Tx_CLK. | R/W | 0x0 |

#### 10.5.5.6 T1_CNT

Timer1 current counter register
Offset=0x0028(VDD)

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | CNT | Read or write current Timer1 value | R | 0x0 |

#### 10.5.5.7 T2_CTL

Timer2 control register
Offset=0x0040(VDD)

| Bits | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:12 | - | Reserved | R | 0x0 |
| 11 | DIR | Timer Counting direction set:<br>0: down<br>1: up | R/W | 0x0 |
| 10:9 | MODE_SEL | Timer mode select:<br>00: normal timer<br>11: continue mode<br>Others : reserved | R/W | 0x0 |
| 8:6 | - | reserved | R | 0x0 |
| 5 | EN | Timer2 Enable<br>0: Disable<br>1: Enable | R/W | 0x0 |
| 4 | - | reserved | R | 0x0 |
| 3 | PAUSE | PAUSE Counter<br>0: normal<br>1: pause | R/W | 0x0 |
| 2 | RELO | Timer2 Reload enable<br>0: Not reload<br>1: Reload | R/W | 0x0 |
| 1 | ZIEN | Timer2 IRQ Enable<br>When this bit is enabled, Timer2_IRQ sent out the irq signal until the pending bit was cleared. | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | **If DIR='0',T2_CNT compare with ZERO**<br>**If DIR='1',T2_CNT compare with T2_VAL.**<br>**For more detail reference to timer2/3 block diagram** | | |
| 0 | ZIPD | Timer2 IRQ Pending,<br>Writing 1 to clear this bit.<br>**If occurred a timer overflow or zero, this pending would be set to '1'** | R/W | 0x0 |

### 10.5.5.8 T2_VAL

Timer2 value register
Offset=0x0044(VDD)

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | VAL | Set timer counter value.<br>If timer setting in count up mode, time would count up from zero to the value set in Tx_VAL. when the current timer counter equal to Tx_VAL, it would cause an irq. If timer reload mode was set, Tx_VAL would be reload auto and timer current value was reset to zero.<br>If timer setting in countdown mode, timer would countdown form Tx_VAL to zero. When the current timer counter equal to zero. it would cause an irq. If timer reload mode was set, zero would be reload auto and timer current value was reset to Tx_VAL.<br>When in counter mode,Tx_VAL[7:0] was used.<br>Note: If set Tx_VAL=n, irq would cause after n+1 Tx_CLK. | R/W | 0x0 |

### 10.5.5.9 T2_CNT

Timer2 current counter register
Offset=0x0048(VDD)

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | CNT | Timer current value registers. | R | 0x0 |

### 10.5.5.10 T3_CTL

Timer3 control register
Offset=0x0060(VDD)

| Bits | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:12 | - | Reserved | R | 0x0 |
| 11 | DIR | Timer Counting direction set:<br>0: down<br>1: up | R/W | 0x0 |
| 10:9 | MODE_SEL | Timer mode select:<br>00: normal timer<br>11: continue mode<br>Others : reserved | R/W | 0x0 |
| 8:6 | - | reserved | R | 0x0 |
| 5 | EN | Timer3 Enable | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0: Disable<br>1: Enable | | |
| 4 | - | reserved | R | 0x0 |
| 3 | PAUSE | PAUSE Counter<br>0: normal<br>1: pause | R/W | 0x0 |
| 2 | RELO | Timer3 Reload enable<br>0: Not reload<br>1: Reload | R/W | 0x0 |
| 1 | ZIEN | Timer3 IRQ Enable<br>When this bit is enabled, Timer3_IRQ sent out the irq signal until the pending bit was cleared.<br>If DIR='0', T3_CNT compare with ZERO<br>If DIR='1', T3_CNT compare with T3_VAL. | R/W | 0x0 |
| 0 | ZIPD | Timer3 IRQ Pending,<br>Writing 1 to clear this bit.<br>If occurred a timer overflow or zero, this pending would be set to '1' | R/W | 0x0 |

### 10.5.5.11 T3_VAL

Timer3 value register
Offset=0x0064(VDD)

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | VAL | Set timer counter value.<br>If timer setting in count up mode, time would count up from zero to the value set in Tx_VAL. when the current timer counter equal to Tx_VAL, it would cause an irq. If timer reload mode was set, Tx_VAL would be reload auto and timer current value was reset to zero.<br>If timer setting in countdown mode, timer would countdown form Tx_VAL to zero. When the current timer counter equal to zero. it would cause an irq. If timer reload mode was set, zero would be reload auto and timer current value was reset to Tx_VAL.<br>When in counter mode,Tx_VAL[7:0] was used.<br>Note: If set Tx_VAL=n, irq would cause after n+1 Tx_CLK. | R/W | 0x0 |

### 10.5.5.12 T3_CNT

Timer3 current counter register
Offset=0x0068(VDD)

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | CNT | Timer current value registers. | R | 0x0 |

### 10.5.5.13 T4_CTL

Timer4 control register
Offset=0x0080(VDD)

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 31:12 | - | reserved | R | 0x0 |
| 11 | DIR | Timer Counting direction set:<br>0: down<br>1: up | R/W | 0x0 |
| 10:9 | MODE_SEL | Timer mode select:<br>00: normal timer<br>11: continue mode<br>Others : reserved | R/W | 0x0 |
| 8:6 | - | Reserved | R | 0x0 |
| 5 | EN | Timer 0 Enable<br>0: Disable<br>1: Enable | R/W | 0x0 |
| 4 | - | Reserved | R | 0x0 |
| 3 | PAUSE | PAUSE Counter<br>0: normal<br>1: pause | R/W | 0x0 |
| 2 | RELO | Timer 4 Reload.<br>0: Not reload<br>1: Reload | R/W | 0x0 |
| 1 | ZIEN | Timer4 IRQ Enable<br>When this bit is enabled, Timer4_Zero_IRQ sent out the irq signal until the pending bit was cleared. | R/W | 0x0 |
| 0 | ZIPD | Timer4 IRQ Pending,<br>Writing 1 to clear this bit. | R/W | 0x0 |

### 10.5.5.14 T4_VAL

Timer4 value register
Offset=0x0084(VDD)

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | VAL | Set timer counter value.<br>If timer setting in count up mode, time would count up from zero to the value set in Tx_VAL. when the current timer counter equal to Tx_VAL, it would cause an irq. If timer reload mode was set, Tx_VAL would be reload auto and timer current value was reset to zero.<br>If timer setting in countdown mode, timer would countdown form Tx_VAL to zero. When the current timer counter equal to zero. it would cause an irq. If timer reload mode was set, zero would be reload auto and timer current value was reset to Tx_VAL.<br>When in counter mode,Tx_VAL[7:0] was used.<br>Note: If set Tx_VAL=n, irq would cause after n+1 Tx_CLK. | R/W | 0x0 |

### 10.5.5.15 T4_CNT

Timer4 current counter register
Offset=0x0088(VDD)

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| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | CNT | Timer current value registers. | R | 0x0 |

### 10.5.5.16 T5_CTL

Timer5 control register
Offset=0xA0

| Bits | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:12 | - | Reserved | R | 0x0 |
| 11 | DIR | Timer Counting direction set:<br>0: down<br>1: up | R/W | 0x0 |
| 10:9 | MODE_SEL | Timer mode select:<br>00: normal timer<br>11: continue mode<br>Others : reserved | R/W | 0x0 |
| 8:6 | - | Reserved | R | 0x0 |
| 5 | EN | Timer Enable<br>0:Disable<br>1:Enable | R/W | 0x0 |
| 4 | - | reserved | R | 0x0 |
| 3 | PAUSE | PAUSE Counter<br>0: normal<br>1: pause | R/W | 0x0 |
| 2 | RELO | Timer Reload enable<br>0:Not reload<br>1:Reload | R/W | 0x0 |
| 1 | ZIEN | Timer IRQ Enable<br>When this bit is enabled, Timer5_Zero_IRQ sent out the irq signal until the pending bit was cleared.<br>**If DIR='0', T5_CNT compare with ZERO**<br>**If DIR='1', T5_CNT compare with T5_VAL.**<br>**For more detail reference to block diagram** | R/W | 0x0 |
| 0 | ZIPD | Timer IRQ Pending,<br>Writing 1 to clear this bit. | R/W | 0x0 |

### 10.5.5.17 T5_VAL

Timer5 value register
Offset=0xA4

| Bits | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | VAL | Set timer counter value.<br>If timer setting in count up mode, time would count up from zero to the value set in T5_VAL. When the current timer counter equal to T5_VAL, it would cause an irq. If timer reload mode was set, T5_VAL would be reload auto and timer current value was reset to zero.<br>If timer setting in countdown mode, timer would countdown form T5_VAL to zero. When the current timer counter equal to zero. It would cause an irq. If timer reload mode was set, zero | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | would be reload auto and timer current value was reset to T5_VAL.<br>When in counter mode,T5_VAL[7:0] was used.<br>Note: If set T5_VAL=n, irq would cause after n+1 T5_CLK. | | |

#### 10.5.5.18 T5_CNT

Timer5 current counter register
Offset=0xA8

| Bits | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | CNT | current Timer5 value | R | 0x0 |

## 10.6 PPI & Wake up Controller

### 10.6.1 Features

The Programmable peripheral interconnect (PPI)

- 34 triggers to interact autonomously with each tasks independent of the CPU
- 16 tasks can be used

Wake up Controller

- Eight interrupts as wake up source
- Four power status：S3, S2, S1
- Eight interrupts can be configured as wake up source for each power status
- Wake up Controller can change the power status by the interrupt

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# 11 Storage

- ATS3085Sx integrates two SD/MMC controller: SD0 and SD1
- SIP 64Mbit/32Mbit OPI PSRAM for gorgeous UI performance through the SPI1 interface.

## 11.1 SD/MMC

**SD/MMC Card Controller Features**

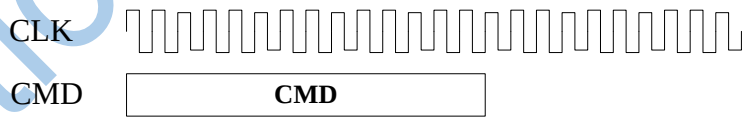
- Fully compliant with MMC Specification 4.3
- Support SDcard up to 2.0
- Support SDIO up to 2.0
- Integrated Clock Delay Chain Technique to Regulate Card Interface Timing: Latching Delay Chain for input Signal, Output Delay Chain for output signal.
- Integrated Watchdog timeout Counter to report Exception happening.
- Integrated Pull up resistance (value 50Komh) for Data and CMD Line.
- Integrated CRC calculate and check circuit.
- Send continuous clock to support SDIO card.
- Support 3.1/1.8V CLK PAD voltage.
- Support 3.1/1.8V CMD PAD voltage.
- Support 3.1/1.8V DAT PAD voltage.
- Band Width: 50MByte/S
- Maximal SD interface Clock: 50MHz

### 11.1.1 Function Description

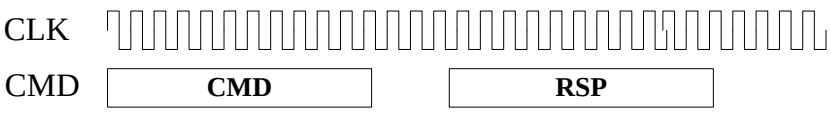
#### 11.1.1.1 Transfer Mode Description

We summarized all SD card operation and ranged them into 10 transfer modes, which the design of SD card controller is based on. Use the sequence set of propriety transfer modes, it can manage all SD bus operations. These transfer modes are described as following:

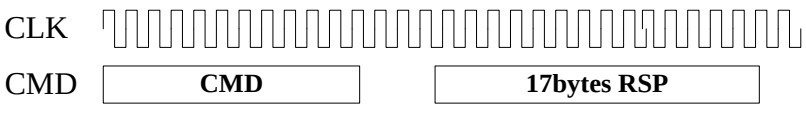
TM_0: Transfer only CMD




TM_1: Send CMD and receive 6 byte RSP:



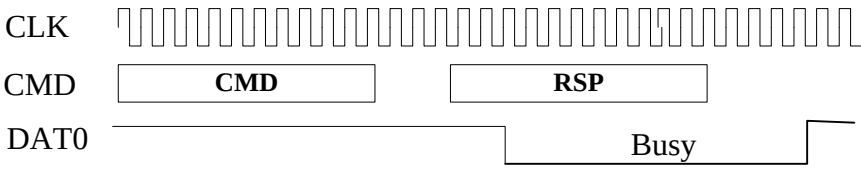

TM_2: Send CMD and receive 17 byte RSP:



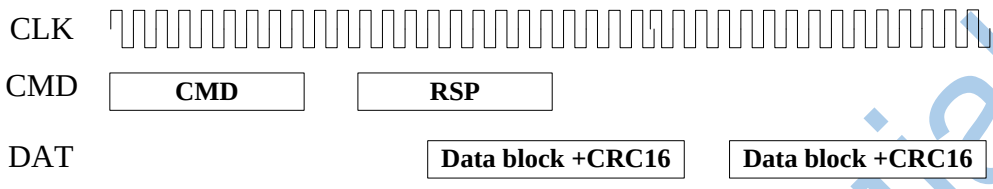

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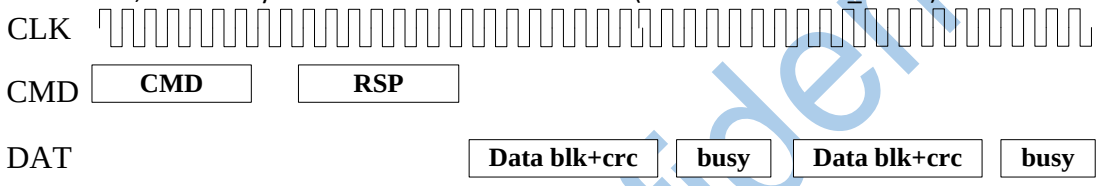
TM_3: Send CMD, receive 6byte RSP and wait busy signal on data0:



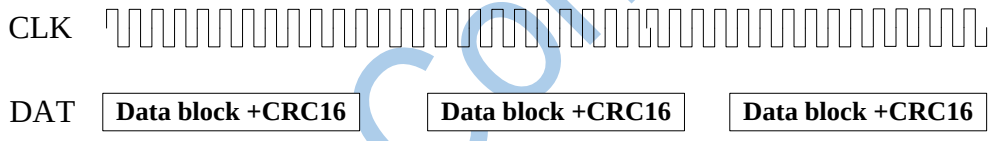

TM_4: Send CMD, receive 6byte RSP and receive data in (in sight of SD controller) mode:



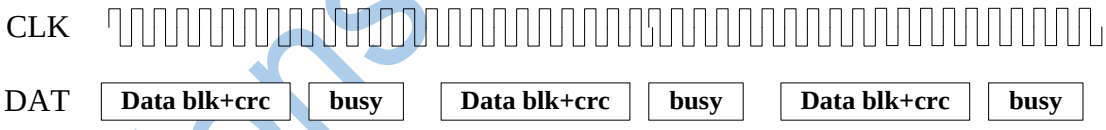

TM_5: Send CMD, receive6byte RSP and send data out mode (include check RCR_status).



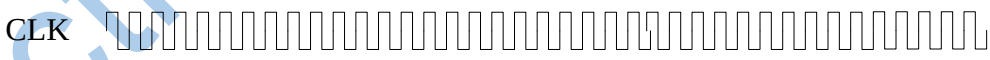

TM_6: Transfer data in without command and response:




TM_7: Transfer data out without command and response:

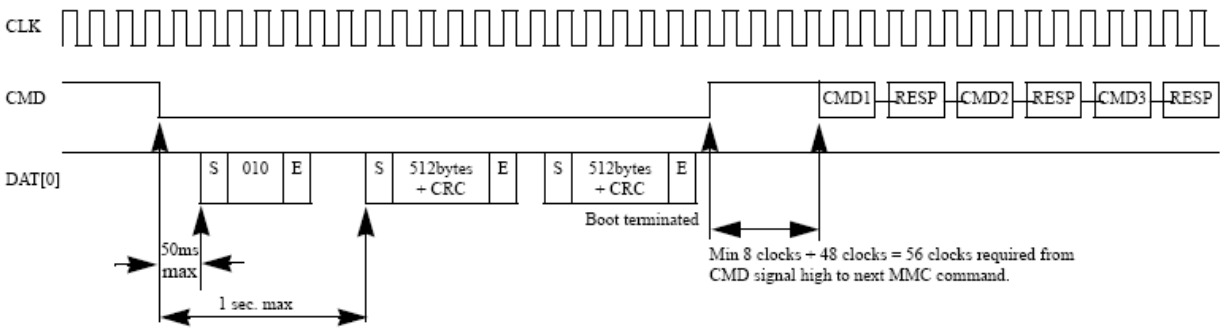



TM_8: Transfer only Clocks (with a cycle number determined by TCN in SD_CTL bit 11~8 ):




Note: the clock number will be a multiplicative of 16

TM_9&TM_10: BOOT mode transfer data in (TM_Mode9). (Hardware check data crc16, do not check boot mode acknowledge.):




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BOOT mode Transfer data in mode (TM_mode10). (Hardware check data crc16, check boot mode acknowledge).TM_mode10, the same as mode9 except the controller need to check boot acknowledge.

We manage the SD card based on it's state machine, so before every operation the driver developer must know the current state of the card, then select a certain CMD to send to change the state of card or read/write it. Every CMD has an ARG, a RSP type, and may has data in or out operation.
Frist, select a correct transfer mode, and configure the ARG register. When there are data need to be transmitted, configure the data transfer master (DMA or CPU) as needed.
Then, Start the transfer. When the data to transfer is ready, start data transfer master, and then wait both of them to be finished.
Finally, check every state that related to the transfer: CRC7_ERROR, CMD_NO_RSP, CRC16_ERROR, TIMEOUT.

#### 11.1.1.2 Hardware watchdog timeout counter

Since there are lots of waiting situation of transfer mode, like wait until a card quit the busy state, wait until a card to send back data for a read CMD. To simplification the Software timeout configuration, we build a watchdog timeout counter in SDC module. When the transfer is no data, or just has one sector to transfer, this timeout counter behave normally. When transfer multiple sectors (TM_4 or TM_5) the value of the counter is reset at every beginning of each sector, it called the "watchdog" timeout counter.
If Transfer Watchdog timeout function enabled (SD_CTL bit31 is set), then Hardware Watchdog timer starts counting after transfer start. The value of the counter is keep counting until the transfer is finished, auto cleared at the end of every transfer. When the counter's value equals to the value set in TOUTCNT (SD_CTL, bits 30~24), then a timeout error will be set in SD_STATE bit15, and the current transfer will stop at the same time.
The time out counter running clock is described as following:

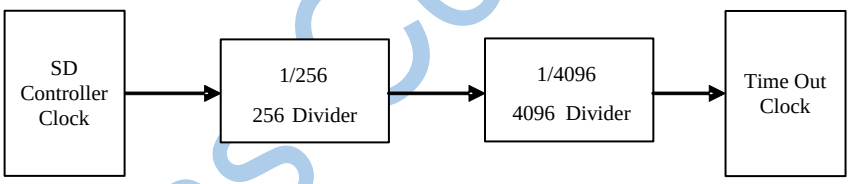



***Figure 11-1 timeout clock generating diagram.***

Hardware Timeout was only used in data in/data out mode/busy checked mode/boot mode, such as mode3, 4, 5, 6, 7, 9, 10.
If Hardware Timeout Counter's value equals '0', the timeout time is 128*(Timeout Clock Period), is the Max Timeout time.
If Hardware Timeout Counter equal '1', the timeout time is 1*(Timeout Clock Period),The Minimum Timeout time.
As Example, If SD Card Clock frequency is 10MHz, SD Card Clock period is 100ns, Hardware Timeout clock frequency is 10/256/4096MHz, Timeout clock period is 100*256*4096ns =104.8576ms. If Hardware Timeout Counter's value equals 0, the timeout time is 128*104.8576ms; If Hardware Timeout Counter's value equals 1, the timeout time is 1*104.8576ms.
If the SD clock is another frequency, the timeout period can be analogy in a simple way like table below:

***Table11-1 The relationship of Timeout & Clock***

| Card clk | Min timeout time (1) | Max timeout time (128) |
|---|---|---|
| 10Mhz | 104.8576ms | 13.4218s |
| 20Mhz | 104.8576ms/2 | 13.4218s /2 |
| 30Mhz | 104.8576ms/3 | 13.4218s /3 |

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| | | |
|---|---|---|
| 40Mhz | 104.8576ms/4 | 13.4218s /4 |
| 50Mhz | 104.8576ms/5 | 13.4218s /5 |

## 11.1.2 Operation Manual

### 11.1.2.1 Typical software operation sequence

1, SDC send CMD and received RSP operation sequence

1) Configure SD_CMD with CMD_index.
2) Configure SD_ARG with Appropriate argument.
3) Select one of the 10 Transfer mode (Configure SD_CTL Bits0～3)
4) Set SD_CTL bit7, start transfer.
5) Wait until SD_CTL bit7 is clear.
6) Check NO_RSP/CRC7/CRC16 in SD_STATE register.

2, SDC send CMD received RSP and transfer data operation sequence

1) Configure SD_CMD with CMD_index.
2) Configure SD_ARG with appropriate argument (card data LBA).
3) Configure SD_BLK_SZ/SD_BLK_NUM with appropriate transfer block size and block number.
4) Select one of the 4 data trans Transfer modes (Configure SD_CTL BIT0～3), one of TM4,5,6,7.
5) Configure DMA Source/destination/transfer mode/transfer number eg.
6) Start DMA transfer
7) Start SDC transfer (set SD_CTL bit7)
8) Wait until DMA transfer finished.
9) Wait until SDC transfer finished. (SD_CTL bit7 is clear)
10) Check NO_RSP/CRC7/CRC16/TIMEOUT status of SDC_STATE register.

3, eMMC BOOT mode operation sequence

1) Configure SD_BLK_SZ/SD_BLK_NUM with appropriate transfer block size and block number.
2) Select one of the 2 BOOT modes (Configure SD_CTL BIT0～3), as one of TM9、10.
3) Configure DMA Source/destination/transfer mode/transfer number eg.
4) Pull CMD line LOW by set CMD_LOW_EN.
5) Start SDC transfer. (Set SD_CTL bit7)
6) Wait DMA transfer finish
7) Wait SDC transfer finished. (SD_CTL bit7 is clear)
8) Check BOOT_ACK/CRC16/TIME_OUT status of SD_STATE

## 11.1.3 SDC Register List

***Table 11-2 SD Controller Registers Address***

| Name | Physical Base Address | KSEG1 Base Address |
|---|---|---|
| SD0_CONTROLLER_REGISTERS | 0x40054000 | 0x40054000 |
| SD1_CONTROLLER_REGISTERS | 0x40058000 | 0x40058000 |

***Table 11-3 SD Controller Registers***

| Offset | Register Name | Description |
|---|---|---|
| 0x00 | SDx_EN | SD MMC card enable register |
| 0x04 | SDx_CTL | SD MMC card control register |
| 0x08 | SDx_STATE | SD MMC card STATU register |
| 0x0C | SDx_CMD | SD MMC card send command register |

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| | | |
|---|---|---|
| 0x10 | SDx_ARG | SD MMC argument register |
| 0x14 | SDx_RSPBUF0 | SD MMC RSP Buffer0 (Bit31~0)register |
| 0x18 | SDx_RSPBUF1 | SD MMC RSP Buffer1 (Bit63~32)register |
| 0x1C | SDx_RSPBUF2 | SD MMC RSP Buffer2 (Bit95~64)register |
| 0x20 | SDx_RSPBUF3 | SD MMC RSP Buffer3 (Bit127~96)register |
| 0x24 | SDx_RSPBUF4 | SD MMC RSP Buffer4 (Bit135~128)register |
| 0x28 | SDx_DAT | SD MMC DATA register |
| 0x2C | SDx_BLK_SIZE | SD Block size |
| 0x30 | SDx_BLK_NUM | SD Block number |

## 11.1.4 SD0 Controller Register Description

### 11.1.4.1 SD0_EN

SD/MMC controller enable register
Offset = 0x0000

| Bit | Name | Description | RW | Reset |
|---|---|---|---|---|
| 31:10 | - | Reserved | R | x |
| 9 | CSS | Clock signal status select in idle mode<br>0:drive to low<br>1:drive to high | RW | 0 |
| 8 | CLKS | SD MMC clock output pad select<br>0: CLK to pad SD_CLK0<br>1: CLK to pad SD_CLK1 | RW | 0 |
| 7 | EN | SD module Enable<br>1: Enable<br>0: Disable | RW | 0 |
| 6 | AHB_DMA_SEL | SDC CHANEL(AHB OR DMA BUS)SELECTION<br>1: DMA BUS<br>0: AHB BUS | RW | 0 |
| 5 | FIFO_WIDTH | FIFO width selection<br>1: 8bit<br>0: 32bit | RW | 0 |
| 4 | - | Reserved | R | 0 |
| 3 | SDIOEN | SDIO function enable<br>1:enable<br>0:disable | RW | 0 |
| 2 | - | Reserved | R | 0 |
| 1:0 | DATAWID | SD Interface Data bus Width selection<br>00b :1 bit<br>01b: 4 bit<br>10b: 8 bit<br>11b: Reserved | RW | 00 |

### 11.1.4.2 SD0_CTL

SD/MMC controller control register
Offset = 0x0004

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| Bit | Name | Description | RW | Reset |
|---|---|---|---|---|
| 31 | TOUTEN | Enable /disable hardware r/w time out function.<br>0: Disable<br>1: Enable<br>The timeout period is configured by Data Timeout Counter | RW | 0 |
| 30:24 | TOUTCNT | Hardware Time out counter value: This counter determine the timeout time of SD card data output. (used in TM_mode3~7 and boot mode ), for details in section 9.1.1.2 | RW | 0 |
| 23:20 | RDELAY | DATA input (Read) delay time selection:<br>When host latch data delay the inside latching data clock to compensate signal transmission delay. | RW | 0x1 |
| 19:16 | WDELAY | DATA output(write) delay Time selection:<br>When sending data, controller will delay the output signals relative to clock ,this will guarantee a suitable hold time for the signal | RW | 0xF |
| 15:14 | - | Reserved | R | 0 |
| 13 | CMDLEN | CMD LOW Enable<br>Enable CMD line drive low level by Software<br>1: Drive CMD line to low.<br>0: Don't drives CMD line to low.<br>NOTE: Can Used in boot mode. | RW | 0 |
| 12 | SCC | Sending continuous clock.<br>1: Enable<br>0: Disable. Write 0 to this bit, stop continuance sending clock<br>(should clear it after transfer start bit is set) | RW | 0 |
| 11:8 | TCN | Transfer clock number. The clock number is 16 times of this field value and a '0' means 256 cycles.<br>(used with TM_MODE8) | RW | 0 |
| 7 | TS | Transfer Start:<br>1:<br>When set 1 by software, SDC transfer starts according to The Transfer Mode, access mode, and other control situations .It will automatically cleared after transfer complete or any Err occurred.<br><br>0 :<br>When cleared by software, The controller needs a few CLOCKs to stop transfer and reset the state machine fully so the software must check the result after writing "0" to it.<br><br>Note:<br>Software can not start a new transfer until it comes to "0" actually. | RW | 0 |
| 6 | LBE | Data in mode the last block send 8 more clocks enable bit<br>0: Disable<br>1: Enable<br>When it's set, at the end of the read last block, the SDC will send 8 more clocks when normally transfer is finished. | RW | 0 |

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| | | | | |
|---|---|---|---|---|
| | | It's for the card to complete the operation. | | |
| 5 | C7EN | Command CRC Check<br>1: Disable CRC7 checked<br>0: Enable CRC7 checked<br>When it's set, it's indicates the SDC don't check CRC7, and don't generate the CRC7_ERROR. | RW | 0 |
| 4 | - | Reserved | R | 0 |
| 3:0 | TM | Transfer Mode[3:0] Specifies the transfer mode when transfer start bit is set<br>0000: Transfer command without response<br>0001: Transfer command with 6 bytes response (not including Data transfer)<br>0010: Transfer command with 17 bytes response (not including Data transfer)<br>0011: Transfer command with 6 bytes response, and with busy (not including Data transfer)<br>0100: Transfer data in mode with command and response (include crc16 checked)<br>0101: Transfer data out mode with command and response(include CRC and busy checked)<br>0110: Transfer data in mode without command and response<br>0111: Transfer data out mode without command and response (include CRC and busy checked)<br>1000: Transfer only clock (without any command, response, and data)<br>1001: BOOT mode ,Transfer data in mode with Command (Hardware check data CRC16, do not check boot mode acknowledge.)<br>1010: BOOT mode ,Transfer data in mode with Command (Hardware check data CRC16, check boot mode acknowledge).<br>1011~1111: Reserved<br><br>NOTE:<br>After each transfer the value of these bits will be reset to "0xf" by hardware. | RW | 1111 |

### 11.1.4.3 SD0_STATE

SD/MMC controller STATE register
Offset = 0x0008

| Bits | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | 0 |
| 15 | TOUTE | Time out error<br>If set means a timeout error has happened. Next transfer started will clear it. | R | 0 |
| 14 | BAEP | Boot mode acknowledge error | R | 0 |

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| | | | | |
|---|---|---|---|---|
| | | When in TM_MODE9, (boot mode with hardware check acknowledge), if set means a acknowledge error or timeout has happened. Next transfer started will clear it. | | |
| 13 | - | Reserved | R | 0 |
| 12 | MEMRDY | Memory ready<br>1: memory ready for read or write<br>0: memory is not ready for read and write | R | 0 |
| 11 | CMDS | CMD Status: This bit reflects the level of the CMD Signal of SD/MMC card | R | 1 |
| 10 | DAT1S | DAT1 Status: This bit reflects the level of the DAT1 Signal of SD/MMC card | R | 1 |
| 9 | SDIOP | SDIO IRQ pending bit<br>1: One SDIO interrupt has happened<br>0: No SDIO interrupt happened.<br>Write1 to this bit will clear it. | RW | 0 |
| 8 | SDIOEN | SDIO IRQ enable<br>1: SDIO IRQ enable (the interrupt trigger form DAT1 PIN)<br>0: SDIO interrupt disable | RW | 0 |
| 7 | DAT0S | DAT0 Status:<br>This bit reflects the level of the DAT0 Signal of SD/MMC card | R | 1 |
| 6 | TEIE | Transfer end IRQ enable:<br>When set, enable interrupt request. | RW | 0 |
| 5 | TEI | Transfer end IRQ pending.<br>Write1 to this bit will clear it. | R | 0 |
| 4 | CLNR | Command Line No response<br>(only for command with response)<br>This bit is auto cleared when Transfer Start is set. | R | 0 |
| 3 | CLC | Command Line transfer Complete:<br>This bit is auto cleared when Transfer Start is set, and is set when command line transfer is complete. | R | 0 |
| 2 | WC16ER | CRC Write data Error:<br>When set, this indicated a CRC write error detected over the data line. This bit is auto cleared when Transfer Start is set. | R | 0 |
| 1 | RC16ER | CRC Read data Error:<br>When set, this indicated a CRC16 error detected over the received data. This bit is auto cleared when Transfer Start is set. | R | 0 |
| 0 | CRC7ER | CRC command response Error:<br>When set, this indicated CRC7 error detected over the response. This bit is auto cleared when Transfer Start is set. | R | 0 |

### 11.1.4.4 SD0_CMD

SD/MMC controller send command register
Offset=0x000c

| Bits | Name | Description | RW | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 31:8 | - | Reserved | | 0 |
| 7:0 | CMD | CMD-index to be transmitted<br>Command register. Fixed the bit7 is '0'<br>The bit6 is '1' | RW | 01xxxxxxb |

**Note**: The value in this register will be kept until next be written

### 11.1.4.5 SD0_ARG

SD/MMC controller CMD argument register
Offset=0x0010

| Bits | Name | Description | RW | Reset |
|---|---|---|---|---|
| 31:0 | ARG | CMD's arguments:<br>Be written before SD_CMD | RW | 0 |

**Note:** The value in this register will be kept until next be written

### 11.1.4.6 SD0_RSPBUF0

SD/MMC controller RSP Buffer0 (Bit31~0) register
Offset=0x0014

| Bits | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | RSP0 | CMD's response Bit31~0 including the CRC and end bit | R | 0 |

### 11.1.4.7 SD0_RSPBUF1

SD/MMC controller RSP Buffer1 (Bit63~32) register
Offset=0x0018

| Bits | Name | Description | RW | Reset |
|---|---|---|---|---|
| 31:0 | RSP1 | Bit63~32 | R | 0 |

### 11.1.4.8 SD0_RSPBUF2

SD/MMC controller RSP Buffer2 (Bit95~64) register
Offset=0x001c

| Bits | Name | Description | RW | Reset |
|---|---|---|---|---|
| 31:0 | RSP2 | Bit95~64 | R | 0 |

### 11.1.4.9 SD0_RSPBUF3

SD/MMC controller RSP Buffer3 (Bit127~96) register
Offset=0x0020

| Bits | Name | Description | RW | Reset |
|---|---|---|---|---|
| 31:0 | RSP3 | Bit127~96 | R | 0 |

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### 11.1.4.10 SD0_RSPBUF4

SD/MMC controller RSP Buffer4 (Bit135~128) register
Offset=0x0024

| Bits | Name | Description | RW | Reset |
|---|---|---|---|---|
| 31:8 | - | Reserved | R | 0 |
| 7:0 | RSP4 | Bit135~128 | R | 0 |

### 11.1.4.11 SD0_DAT

SD/MMC controller DATA register
Offset=0x0028

| Bits | Name | Description | RW | Reset |
|---|---|---|---|---|
| 31:0 | DATA | Data register | R | xx |

**Note1**: This register map to RAM access address. Can't be accessed except data transfer
**Note2**: When SD_EN[5] set to 0, SDC RAM width fix to 32bit.
For example:
If CPU or DMA need to sequential access 6(Not a multiple of 4) byte data: B1, B2, B3, B4, B5, B6.
Transfer data out: please write SD_DAT twice: B4B3B2B1 and XXB6B5. ('X' means don't care)
Transfer data in : CPU or DMA will read out B4B3B2B1 and XXB6B5 in twice from RAM.
Both master and slave should ignore these invalid bytes XX in the last 32-bit data.
**Note3**: When SD_EN[5] set to 1, only low 8bit of SD_DAT is valid. When SD in DMA mode, the developer should configure the width of DMA to be 8bit too.

### 11.1.4.12 SD0_BLK_SIZE

SD/MMC controller Block size Register
Offset=0x002c

| Bits | Name | Description | RW | Reset |
|---|---|---|---|---|
| 31:12 | - | Reserved | R | 0 |
| 11:0 | BC | Block Count[9:0].<br>This field determines a block size, which is how many bytes founded a block.<br>In SDR50/SDR100 mode, only 512bytes/block can be set. | RW | 0 |

### 11.1.4.13 SD0_BLK_NUM

SD/MMC controller Block number Register
Offset=0x0030

| Bits | Name | Description | RW | Reset |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | 0 |
| 15:0 | BN | This field determines block number in one read or write operations. Default value is 1. '0' means no block will be transferred. | RW | 01B |

**Note:** The host hardware do not support multiple blocks transfer when the Byte counter is not the multiple of 4 bytes.

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## 11.1.5 SD1 Controller Register Description

### 11.1.5.1 SD1_EN

SD/MMC controller enable register
Offset = 0x0000

| Bits | Name | Description | RW | Reset |
|---|---|---|---|---|
| 31:10 | - | Reserved | R | x |
| 9 | CSS | Clock signal status select in idle mode<br>0:drive to low<br>1:drive to high | RW | 0 |
| 8 | - | Reserved | R | x |
| 7 | EN | SD module Enable<br>1: Enable<br>0: Disable | RW | 0 |
| 6 | AHB_DMA_SEL | SDC CHANEL（AHB OR DMA BUS) SELECTION<br>1:DMA BUS<br>0:AHB BUS | RW | 0 |
| 5:4 | - | Reserved | R | x |
| 3 | SDIOEN | SDIO function enable<br>1:enable<br>0:disable | RW | 0 |
| 2 | - | Reserved | R | 0 |
| 1:0 | DATAWID | SD Interface Data bus Width selection<br>00b :1 bit<br>01b: 4 bit<br>Others: Reserved | RW | 00 |

### 11.1.5.2 SD1_CTL

SD/MMC controller control register
Offset = 0x0004

| Bit | Name | Description | RW | Reset |
|---|---|---|---|---|
| 31:24 | - | Reserved | R | x |
| 23:20 | RDELAY | DATA input (Read) delay time selection:<br>when host latch data delay the inside latching data clock to compensate signal transmission delay. | RW | 0x8 |
| 19:16 | WDELAY | DATA output(write) delay Time selection:<br>When sending data, controller will delay the output signals relative to clock , this will guarantee a suitable hold time for the signal. | RW | 0xF |
| 15:13 | - | Reserved | R | x |

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| | | | | |
|---|---|---|---|---|
| 12 | SCC | Sending continuous clock.<br>1: Enable<br>0: Disable. Write 0 to this bit, stop continuance sending clock<br>（should clear it after transfer start bit is set） | RW | 0 |
| 11:8 | TCN | Transfer clock number. The clock number is 16 times of this field value and a '0' means 256 cycles.<br>(used with TM_MODE8) | RW | 0 |
| 7 | TS | Transfer Start:<br>1:<br>When set 1 by software, SDC transfer starts according to The Transfer Mode, access mode, and other control situations .It will automatically cleared after transfer complete or any Err occurred.<br><br>0 :<br>When cleared by software, The controller needs a few CLOCKs to stop transfer and reset the state machine fully so the software must check the result after writing "0" to it.<br><br>Note:<br>Software can not start a new transfer until it comes to "0" actually.. | RW | 0 |
| 6 | LBE | Data in mode the last block send 8 more clocks enable bit<br>0: Disable<br>1: Enable<br>When set , at the end of the read last block, the SDC will send 8 more clocks when normally transfer is finished. It's for the card to complete the operation. | RW | 0 |
| 5:4 | - | Reserved | R | x |
| 3:0 | TM | Transfer Mode[3:0] Specifies the transfer mode when transfer start bit is set<br>0000: Transfer command without response<br>0001: Transfer command with 6 bytes response (not including Data transfer)<br>0010: Transfer command with 17 bytes response (not including Data transfer)<br>0011: Transfer command with 6 bytes response, and with busy (not including Data transfer)<br>0100: Transfer data in mode with command and response (include crc16 checked)<br>0101: Transfer data out mode with command and response(include CRC and busy checked)<br>0110: Transfer data in mode without command and response<br>0111: Transfer data out mode without command and response (include CRC and busy checked)<br>1000: Transfer only clock (without any command, response, and data) | RW | 1111 |

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| | | | | |
|---|---|---|---|---|
| | | NOTE:<br>After each transfer the value of these bits will be reset to “0xf” by hardware. | | |

### 11.1.5.3 SD1_STATE

SD/MMC controller STATE register
Offset = 0x0008

| Bits | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:13 | - | Reserved | R | x |
| 12 | MEMRDY | Memory ready<br>1: memory ready for read or write<br>0: memory is not ready for read and write | R | 0 |
| 11 | CMDS | CMD Status: This bit reflects the level of the CMD Signal of SD/MMC card | R | 1 |
| 10 | DAT1S | DAT1 Status: This bit reflects the level of the DAT1 Signal of SD/MMC card | R | 1 |
| 9 | SDIOP | SDIO IRQ pending bit<br>1: One SDIO interrupt has happened<br>0: No SDIO interrupt happened.<br>Write1 to this bit will clear it. | RW | 0 |
| 8 | SDIOEN | SDIO IRQ enable<br>1: SDIO IRQ enable (the interrupt trigger form DAT1 PIN)<br>0: SDIO interrupt disable | RW | 0 |
| 7 | DAT0S | DAT0 Status:<br>This bit reflects the level of the DAT0 Signal of SD/MMC card | R | 1 |
| 6 | TEIE | Transfer end IRQ enable:<br>When set, enable interrupt request. | RW | 0 |
| 5 | TEI | Transfer end IRQ pending.<br>Write1 to this bit will clear it. | R | 0 |
| 4 | CLNR | Command Line No response<br>(only for command with response)<br>This bit is auto cleared when Transfer Start is set. | R | 0 |
| 3 | CLC | Command Line transfer Complete:<br>This bit is auto cleared when Transfer Start is set, and is set when command line transfer is complete. | R | 0 |
| 2 | WC16ER | CRC Write data Error:<br>When set, this indicated a CRC write error detected over the data line. This bit is auto cleared when Transfer Start is set. | R | 0 |
| 1 | RC16ER | CRC Read data Error:<br>When set, this indicated a CRC16 error detected over the received data. This bit is auto cleared when Transfer Start is set. | R | 0 |
| 0 | CRC7ER | CRC command response Error:<br>When set, this indicated CRC7 error detected over the response. This bit is auto cleared when Transfer Start is set. | R | 0 |

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### 11.1.5.4 SD1_CMD

SD/MMC controller send command register
Offset=0x000c

| Bits | Name | Description | RW | Reset |
|---|---|---|---|---|
| 31:8 | - | Reserved | | x |
| 7:0 | CMD | CMD-index to be transmitted<br>Command register. Fixed the bit7 is '0'<br>The bit6 is '1' | RW | 01000000b |

**Note**: The value in this register will be kept until next be written

### 11.1.5.5 SD1_ARG

SD/MMC controller CMD argument register
Offset=0x0010

| Bits | Name | Description | RW | Reset |
|---|---|---|---|---|
| 31:0 | ARG | CMD's arguments:<br>Be written before SD_CMD | RW | 0 |

**Note:** The value in this register will be kept until next be written

### 11.1.5.6 SD1_RSPBUF0

SD/MMC controller RSP Buffer0 (Bit31~0) register
Offset=0x0014

| Bits | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | RSP0 | CMD's response Bit31~0 including the CRC and end bit | R | 0 |

### 11.1.5.7 SD1_RSPBUF1

SD/MMC controller RSP Buffer1 (Bit63~32) register
Offset=0x0018

| Bits | Name | Description | RW | Reset |
|---|---|---|---|---|
| 31:0 | RSP1 | Bit63~32 | R | 0 |

### 11.1.5.8 SD1_RSPBUF2

SD/MMC controller RSP Buffer2 (Bit95~64) register
Offset=0x001c

| Bits | Name | Description | RW | Reset |
|---|---|---|---|---|
| 31:0 | RSP2 | Bit95~64 | R | 0 |

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### 11.1.5.9 SD1_RSPBUF3

SD/MMC controller RSP Buffer3 (Bit127~96) register
Offset=0x0020

| Bits | Name | Description | RW | Reset |
|---|---|---|---|---|
| 31:0 | RSP3 | Bit127~96 | R | 0 |

### 11.1.5.10 SD1_RSPBUF4

SD/MMC controller RSP Buffer4 (Bit135~128) register
Offset=0x0024

| Bits | Name | Description | RW | Reset |
|---|---|---|---|---|
| 31:8 | - | Reserved | R | 0 |
| 7:0 | RSP4 | Bit135~128 | R | 0 |

### 11.1.5.11 SD1_DAT

SD/MMC controller DATA register
Offset=0x0028

| Bits | Name | Description | RW | Reset |
|---|---|---|---|---|
| 31:8 | - | Reserved | R | xx |
| 7:0 | DATA | Data register | R | xx |

**Note1**: This register map to RAM access address. Can't be accessed except data transfer
**Note2**: SDC1 RAM width fix to 8bit.

### 11.1.5.12 SD1_BLK_SIZE

SD/MMC controller Block size Register
Offset=0x002c

| Bits | Name | Description | RW | Reset |
|---|---|---|---|---|
| 31:10 | - | Reserved | R | x |
| 9:0 | BC | Block Count[9:0].<br>This field determines a block size, which is how many bytes founded a block. | RW | 0 |

### 11.1.5.13 SD1_BLK_NUM

SD/MMC controller Block number Register
Offset=0x0030

| Bits | Name | Description | RW | Reset |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | x |
| 15:0 | BN | This field determines block number in one read or write operations. Default value is 1. "0" means no block will be transferred. | RW | 01B |

**Note:** The host hardware do not support multiple blocks transfer when the Byte counter is not the multiple of 4 bytes.

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# 12 Transfer and Communication

## 12.1 USB

### 12.1.1 Features

- Complies with the USB2.0 full speed Specification
- Supports point-to-point communication with one full-speed device in Host mode(no HUB support).
- Supports full-speed in peripheral mode.
- Supports 6 IN endpoints and 6 OUT endpoint except endpoint0.
- Supports bulk Isochronous and Interrupt transfer.
- Partially configurable endpoint type and single, double triple or quad buffering.
- Integrated synchronous RAM as endpoint FIFOs.
- Supports suspend, resume and power managements function.
- Support remote wakeup for device mode.
- Support DMA transfer for endpoint in1 and out2.

### 12.1.2 USB Register List

***Table 12-1 USB Controller Registers Address***

| Name | Physical Base Address | KSEG1 Base Address |
|---|---|---|
| USB_CONTROLLER_REGISTERS | 0x40050000 | 0x40050000 |

***Table 12-2 USB Controller Registers***

| Offset | Register Name | Description |
|---|---|---|
| 0x421 | DPDMCTRL | DPDM control register |
| 0x422 | LINESTATUS | Line status register |

### 12.1.3 USB Register Description

#### 12.1.3.1 DPDMCTRL

DP DM control register
Offset = 0x421

| Bit (s) | Name | Description | Access | Reset |
|---|---|---|---|---|
| 7 | - | Reserved | R | 0B |
| 6 | PLUGIN | This bit Indicated the usb connection status when Linedeten is enable.<br>1: connect<br>0: disconnect | R | x |
| 5 | - | Reserved | R | 0B |
| 4 | LSDETEN | Line status detect enable<br>1：enable 0：disable | R/W | 1B |
| 3 | DMPUEN | 500Kohm DM pull up resistor enable.<br>1：enable 0：disable | R/W | 0B |
| 2 | DPPUEN | 500Kohm DP pull up resistor enable.<br>1：enable 0：disable | R/W | 0B |
| 1 | DMPDDIS | DM pull down disable.<br>1：disable 0：enable | R/W | 0B |

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| | | | | |
|---|---|---|---|---|
| 0 | DPPDDIS | DP pull down disable.<br>1：disable 0：enable | R/W | 0B |

### 12.1.3.2 LINESTATUS

Line status register
Offset = 0x422

| Bit (s) | Name | Description | Access | Reset |
|---|---|---|---|---|
| 7 | CEASESUSPEND | When in suspend state and USB clock has been suspended (CLKGATE.GOSUSPEND is set to 1), Once the controller is ready to resume (for Host) or remotely wakeup (for Peripheral) actively, software should firstly set this bit to resume USB clock(UTMISUSPENDM is driven to high). Then the other relative registers can be configured to generate Resume Signal on USB bus.<br>This bit can be set by hardware or by software when UTMISUSPENDM needs driving to high.<br>This bit will be cleared by hardware, when UTMISUSPENDM is driven to low.<br>When read, this bit reflects the UTMISUSPENDM signal state. | R/W | 0 |
| 6:5 | - | Reserved | R | 0H |
| 4:3 | LINESTATE | USB linestate[1:0]<br>Linestate0:DP<br>Linestase1:DM | R | 0B |
| 2:1 | - | Reserved | R | 0H |
| 0 | OTGRESET | USB OTG reset. If AOTG is in reset state, this bit will be set, otherwise it will be 0. | R/W | 0B |

## 12.2 I2C

### 12.2.1 Features

ATS3085Sx contains four normal I2C interfaces:

- Both master and slave functions supported
- Support for standard mode (100 kHz), Fast mode (400 kHz), and Fast mode+ (1 MHz)
- Both master and slave supports DMA mode
- Only 7-bit address mode support
- 8 bit x 8 TX FIFO and 8bit x 8 RX FIFO
- Supports general call
- Pull-up resistors are required on both of the I2C signal lines as the I2C drivers are open drain. Typically external 2.2k-Ohm resisters are used to pull the signals up to VCC if not select internal Pull-Up resistor.

### 12.2.2 Function Description

An Inter-IC bus, used to communicate across circuit-board distances. At the low end of the spectrum of communication options for "inside the box" communication is I2C.
I2C provides support for communication with various slow, on-board peripheral devices that are accessed intermittently, while being extremely modest in its hardware resource needs. It is a simple,

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low-bandwidth, short-distance protocol. Most available I2C devices operate at speeds up to 400Kbit/s, with some venturing up into the low megahertz range. I2C is easy to use to link multiple devices together since it has a built-in addressing scheme.

**Note:**

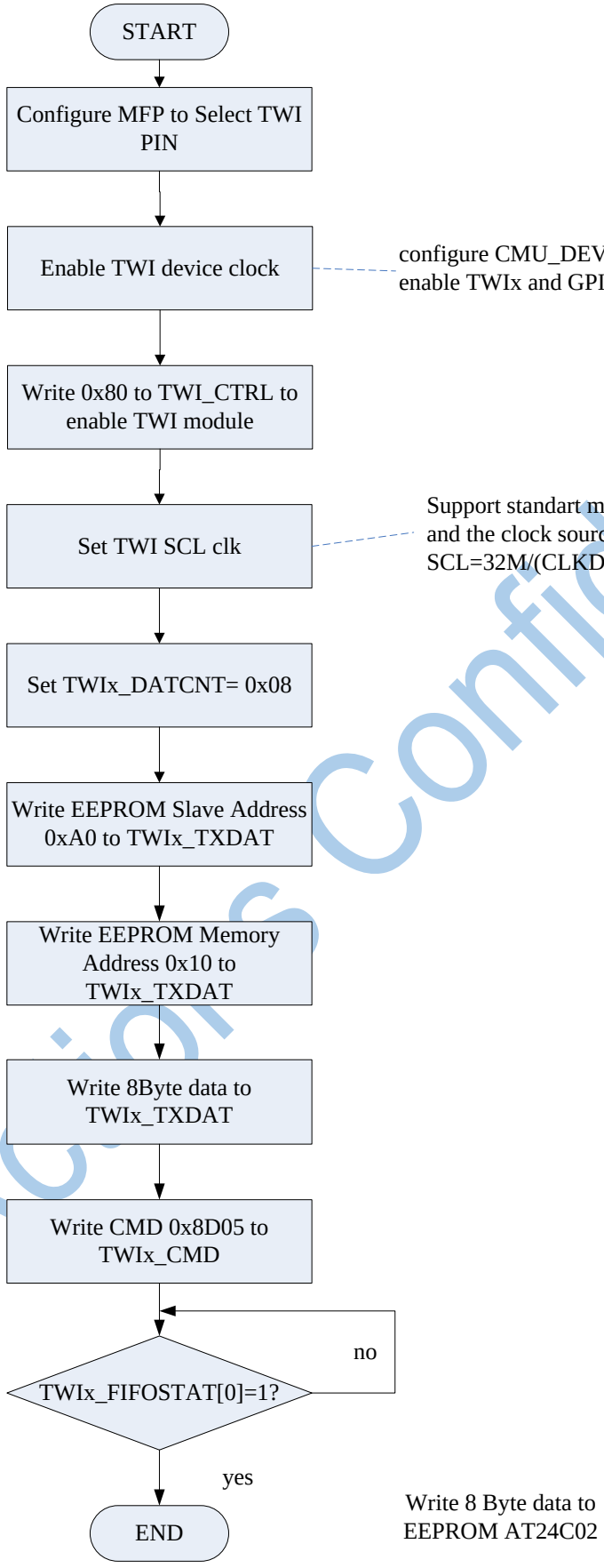
1. The I2C module is Slave mode when in IDLE status.
2. When write start command, the I2C module change from slave to master mode; and after a stop command change to slave mode.
3. Generate the IRQ while the bus statuses change.
   - transfer complete
   - detect normal stop bit ( no bus error )

   If the address received don't match the content of I2C address, don't generate the IRQ and reset to IDLE when detect the start bit followed the slave address in slave mode.
4. Release the bus by software after receive data or address.
5. In multi-master application, when miss the control of the bus, ignore the bus command before a stop command detected. After a stop command, the device start an arbitration procedure. The arbitration will end with one of the following:
   - The I2C module enters master mode ( it won the arbitration)
   - The I2C module enters slave mode (it lost the arbitration or the other master addresses the I2C module)
   - The I2C module enter master-in-waiting mode, when the I2C module is waiting for the other master (that won control) to complete its transaction so that it can once again try to grain the arbitration of the bus.

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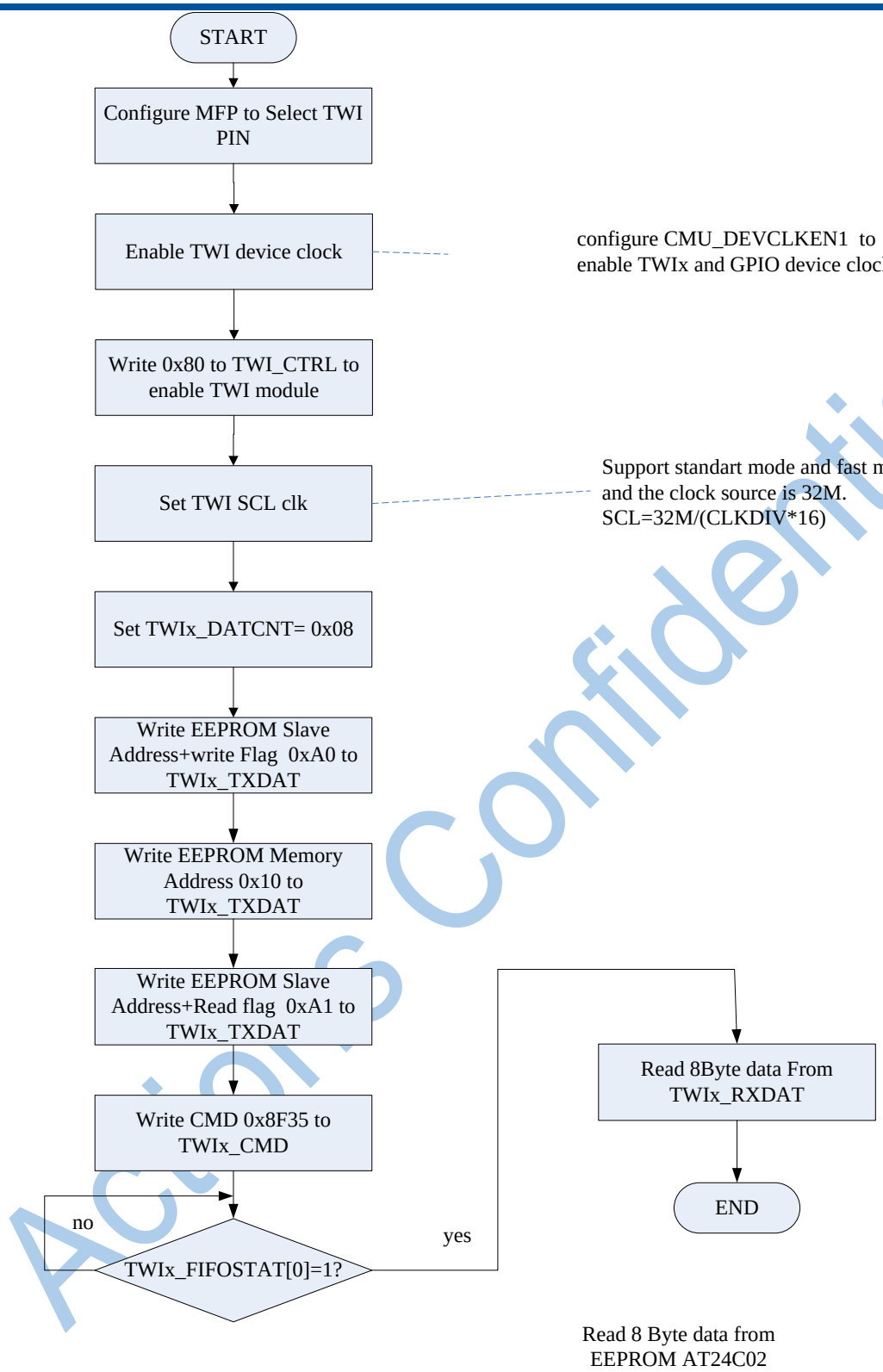
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## 12.2.3 Operation Manual




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## 12.2.4 I2C Register List

*Table 12-3 I2C Register Block Base Address*

| Name | Physical Base Address | KSEG1 Base Address |
|---|---|---|
| I2C0 | 0x40044000 | 0x40044000 |

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| | | |
|---|---|---|
| I2C1 | 0x40048000 | 0x40048000 |
| I2C2 | 0x4004C000 | 0x4004C000 |
| I2C3 | 0x40074000 | 0x40074000 |

*Table 12-4 I2C Registers Offset Address*

| Offset | Register Name | Description |
|---|---|---|
| 0x0000 | I2C_CTL | I2C Control Register |
| 0x0004 | I2C_CLKDIV | I2C Clock Divider Register |
| 0x0008 | I2C_STAT | I2C Status Register |
| 0x000C | I2C_ADDR | I2C Address Register |
| 0x0010 | I2C_TXDAT | I2C TX Data Register |
| 0x0014 | I2C_RXDAT | I2C RX Data Register |
| 0x0018 | I2C_CMD | I2C Command Register |
| 0x001C | I2C_FIFOCTL | I2C FIFO Control Register |
| 0x0020 | I2C_FIFOSTAT | I2C FIFO Status Register |
| 0x0024 | I2C_DATCNT | I2C Data Transmit Counter Register |
| 0x0028 | I2C_RCNT | I2C Data Transmit Remain Counter Register |

## 12.2.5 I2C Register Description

### 12.2.5.1 I2Cx_CTL

I2C Control Register
Offset=0x00

| Bits | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:21 | - | Reserved | R | 0x0 |
| 20 | TCP_FIFO_IRQ_EN | Transfer Complete in fifo mode IRQ Enable<br>0:disable<br>1:enable | RW | 0x0 |
| 19 | STD_IRQ_EN | Stop Detect IRQ Enable<br>0:disable<br>1:enable | RW | 0x0 |
| 18 | - | Reserved | R | 0x0 |
| 17 | NACK_IRQ_EN | NACK Detect IRQ Enable<br>0:disable<br>1:enable | RW | 0x0 |
| 16 | ADM_IRQ_EN | Address match as a slave IRQ Enable<br>0:disable<br>1:enable | RW | 0x0 |
| 15:11 | - | Reserved | R | 0x0 |
| 10 | IRQC | I2C IRQ Control<br>0: set when RX FIFO received 1 bytes data in IRQ mode<br>1: set when RX FIFO received 4 bytes data in IRQ mode | RW | 0x0 |
| 9 | DRQTE | TX DRQ enable<br>0：disable<br>1：enable | RW | 0x0 |
| 8 | DRQRE | RX DRQ enable<br>0：disable<br>1：enable | RW | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 7 | BUSSEL | TX/RX FIFO Bus select<br>0: AHB.<br>1: DMA | RW | 0x0 |
| 6 | IRQE | IRQ Enable.<br>0: Disable<br>1: Enable | RW | 0x0 |
| 5 | EN | Enable. When enable, reset the status machine to IDLE<br>0: Disable<br>1: Enable | RW | 0x0 |
| 4 | - | Reserved | RW | 0x0 |
| 3:2 | GBCC | Generating Bus Control Condition (only for master mode).<br>00: No effect<br>01: Generating START condition<br>10: Generating STOP condition<br>11: Generating Repeated START condition<br>Write the slave address to the I2C_DAT register, select start or restart, and then the start or restart command follow by the slave address will occur on the bus. | RW | 0x0 |
| 1 | RB | Release Bus.<br>Write 1 to this bit will release the bus. | RW | 0x0 |
| 0 | GRAS | Generate ACK or NACK Signal.<br>When receive data<br>0: generate the ACK signal at 9th clock of SCL<br>1: generate the NACK signal at 9th clock of SCL | RW | 0x0 |

### 12.2.5.2 I2Cx_CLKDIV

I2C Clock Divide Control Register
Offset=0x04

| Bits | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:8 | - | Reserved | R | 0x0 |
| 7:0 | CLKDIV | Clock Divider Factor (only for master mode).<br>I2C clock (SCL) can select standard (100kbps) mode and fast (400kbps) mode. Calculating SCL is as following:<br>SCL=I2Cx_CLK/(CLKDIV*16) | RW | 0x0 |

### 12.2.5.3 I2Cx_STAT

I2C Status Register
Offset=0x08

| Bits | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:11 | - | Reserved | R | 0x0 |
| 10 | SRGC | Slave receive general call<br>0: not receive a general call<br>1: receive a general call | R | 0x0 |
| 9 | SAMB | Slave address match bit<br>0: slave address not match<br>1: slave address match | R | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 8 | LBST | Last Byte Status Bit.<br>0: Indicate the last byte received or transmitted is address<br>1: Indicate the last byte received or transmitted is data | R | 0x0 |
| 7 | TCB | Transfer complete bit<br>0: not finish transfer<br>1: In normal mode:<br>A byte transfer finish, include transfer the ACK or NACK bit<br>Writing 1 to this bit will clear it. | RW | 0x0 |
| 6 | BBB | Bus busy bit<br>0: Not busy<br>1: Busy<br>This bit will set to 1 while the start command detected, and set to 0 after the stop command | R | 0x0 |
| 5 | STAD | Start detect bit, include restart.<br>The bit is clear when the I2C module is disable or when the STOP condition is detected. Writing 1 to the bit will clear it.<br>0: Start bit is not detected<br>1: Start bit is detected | RW | 0x0 |
| 4 | STPD | Stop detect bit<br>The bit is clear when the I2C module is disable or when the START condition is detected. Writing 1 to the bit will clear it.<br>0: Stop bit is not detected<br>1: Stop bit is detected | RW | 0x0 |
| 3 | - | Reserved | RW | 0x0 |
| 2 | IRQP | IRQ Pending Bit.<br>1: IRQ<br>0: No IRQ<br><br>Set condition:<br>1. transfer complete in fifo mode<br>2. detect normal stop bit ( no bus error )<br>3. transfer complete in cmd mode<br>4. Receive NACK when not overlook the NACK<br>5. Address match as a slave<br><br>Clear condition:<br>Writing 1 to this bit will clear it. | RW | 0x0 |
| 1 | BEB | Bus error bit<br>0: No error occur<br>1: Bus error occur<br>Write “1” to clear this bit<br>The below conditions occur generate error bit:<br>Detect stop bit right after detect start/restart bit.<br>Detect stop, start bit when sending or receiving data. | RW | 0x0 |
| 0 | RACK | Receive ACK or NACK when transmit data or address<br>0: NACK | R | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 1: ACK<br>The bit will be updated when the 9th of next byte clock arrived | | |

### 12.2.5.4 I2Cx_ADDR

I2C Address Register
Offset=0x0c

| Bits | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:8 | - | Reserved | R | 0x0 |
| 7:1 | SDAD | Own Slave Device Address.<br>Only use in slave mode. I2C_Addr contains the own address of the module when the device is use in slave mode.<br>Content of the register is irrelevant when the I2C module is functioning as a master. | RW | 0x0 |
| 0 | - | Reserved. | R | 0x0 |

### 12.2.5.5 I2Cx_TXDAT

I2C Data Register
Offset=0x10

| Bits | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:8 | - | Reserved | R | 0x0 |
| 7:0 | DA | The registers of Data or address to be transfer, or received to.<br>I2CDAT contains the byte to be transmitted on the I2C-bus or a byte that has been received from the I2C-bus. In master mode, along with the data byte to be transmitted, it also includes the slave address. The seven MSB's are the slave I2C device address while the LSB is the Read/Write bit.<br>8 level FIFO, 8 x 8bit | W | 0x0 |

### 12.2.5.6 I2Cx_RXDAT

I2C receive Data Register
Offset=0x14

| Bits | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:8 | - | Reserved | R | 0x0 |
| 7:0 | DA | The Receive data Register<br>8 level FIFO, 8 x 8bit | R | 0x0 |

### 12.2.5.7 I2Cx_CMD

I2C Command Register
Offset=0x18

| Bits | Name | Description | R/W | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | 0x0 |
| 15 | SECL | Start to execute the command list<br>0: not execute<br>1: execute command | RW | 0x0 |
| 14:13 | - | Reserved | R | 0x0 |
| 12 | WRS | Write or Read select<br>0: write<br>1: read<br>This bit only used in Slave mode. | RW | 0x0 |
| 11 | MSS | Master or slave mode select<br>0: slave mode<br>1: Master mode | RW | 0x0 |
| 10 | SE | Stop enable<br>0: disable<br>1: enable | RW | 0x0 |
| 9 | NS | NACK select<br>0: not select<br>1: select<br>generate the NACK signal at 9th clock of SCL of the last byte when read data | RW | 0x0 |
| 8 | DE | Data enable<br>0: disable<br>1: enable<br>The counts of data transmitted depend on the I2Cx_CNT register. | RW | 0x0 |
| 7:5 | SAS | Second address select<br>000: no address<br>001: 1 byte address<br>010: 2 byte address<br>011: 3 byte address<br>100: 4 byte address<br>101: 5 byte address<br>110: 6 byte address<br>111: 7 byte address | RW | 0x0 |
| 4 | RBE | Restart bit enable<br>0: not send restart bit<br>1: send restart bit | RW | 0x0 |
| 3:1 | AS | Address select<br>000: no address<br>001: 1 byte address<br>010: 2 byte address<br>011: 3 byte address<br>100: 4 byte address<br>101: 5 byte address<br>110: 6 byte address<br>111: 7 byte address<br>The address include slave address and slave internal memory address. | RW | 0x0 |
| 0 | SBE | Start bit enable<br>0: not send start bit<br>1: send start bit | RW | 0x0 |

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### 12.2.5.8 I2Cx_FIFOCTL

I2C FIFO control Register
Offset=0x1c

| Bits | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:3 | - | Reserved | R | 0x0 |
| 2 | TFR | TX FIFO reset bit<br>Write 1 to reset TX FIFO, auto clear to 0 when Tx FIFO reset complete. | RW | 0x0 |
| 1 | RFR | RX FIFO reset bit<br>Write 1 to reset RX FIFO, auto clear to 0 when Rx FIFO reset complete. | RW | 0x0 |
| 0 | NIB | NACK Ignore Bit<br>0: not ignore, when receive NACK when write, generate Error, do not continue the command list execute, generate IRQ<br>1: ignore NACK, when receive NACK, don't generate error, and will continue the command list execute | RW | 0x0 |

### 12.2.5.9 I2Cx_FIFOSTAT

I2C FIFO status Register
Offset=0x20

| Bits | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | 0x0 |
| 15:12 | TFD | Tx FIFO level display<br>This field indicate the current Tx FIFO level | R | 0x0 |
| 11:8 | RFD | Rx FIFO level display<br>This field indicate the current Rx FIFO level | R | 0x0 |
| 7 | - | Reserved | R | 0x0 |
| 6 | WRS | Write or read status bit when acts as slave, used only in FIFO mode<br>0: master write to slave<br>1: master read from slave | R | 0x0 |
| 5 | TFF | TX FIFO full bit<br>0: not full<br>1: full | R | 0x0 |
| 4 | TFE | TX FIFO empty bit<br>1: empty<br>0: not empty | R | 0x1 |
| 3 | RFF | RX FIFO full bit<br>0: not full<br>1: full | R | 0x0 |
| 2 | RFE | RX FIFO empty bit<br>1: empty<br>0: not empty | R | 0x1 |
| 1 | RNB | Receive NACK Error bit<br>0: not receive NACK<br>1: receive NACK when write data<br>Write 1 to clear this bit | RW | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 0 | CECB | Command Execute Complete bit<br>0: not complete<br>1: complete | R | 0x1 |

#### 12.2.5.10 I2Cx_DATCNT

I2C data Counter Register
Offset=0x24

| Bits | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | 0x0 |
| 15:0 | TC | Data Transmit counter | RW | 0x0 |

#### 12.2.5.11 I2Cx_RCNT

I2C remain Counter Register
Offset=0x28

| Bits | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | 0x0 |
| 15:0 | TC | Remain counter | R | 0x0 |

## 12.3 I2CMT

### 12.3.1 Features

ATS3085Sx contains two I2CMT interfaces:

- Normal mode and Auto mode.
- I2C master compatible at Normal mode.
- I2C master read/write one/multi-byte operation executed automatically at auto mode.
- Standard mode 100 kbps, up to fast mode 400 kbps.
- Support for clock stretching.
- Dedicated AHB-like DMA Bus for data access.
- Internal Pull-Up Resistor (50K) optional.

### 12.3.2 Function Description

I2CMT is a two-wire half-duplex master which can communicate with multiple slave devices connected to the same bus. The I2CMT master can be started by one of the four tasks. The I2CMT master implements dedicated AHB-like Bus for reading and writing to and from the RAM.

I2CMT master define four tasks, which triggered by events from other peripherals, such as Timer, interrupt. And each task can start a transmission (write or read one/multiple bytes).

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# 12.4 UART

## 12.4.1 Features

ATS3085Sx contains three UART interfaces: UART0/1/2. UART0/1/2has the following features:

- 5-8 Data Bits and LSB first in Transmit and Received
- 1-2 Stop Bits
- Even, Odd, or No Parity
- 16 Byte Transmit and Receive FIFOs
- Capable of speeds up to 4Mbps to enable connections with Bluetooth and other peripherals
- Support IRQ and DMA mode to transmit data
- Support RTS/CTS Automatic Hardware Flow Control to reduce interrupts to host system
- UART RX Support DMA single mode
- Add UART RX DMA counter for valid data in RAM.
- RX Baud Rate tolerance≤±2%
- Support Rx timeout IRQ pending

## 12.4.2 UART Register List

*Table 12-5 UART Registers Block Base Address*

| Name | Physical Base Address | KSEG1 Base Address |
|---|---|---|
| UART0 | 0x40038000 | 0x40038000 |
| UART1 | 0x4003C000 | 0x4003C000 |
| UART2 | 0x40040000 | 0x40040000 |

*Table 12-6 UART Registers Offset Address*

| Offset | Register Name | Description |
|---|---|---|
| 0x0000 | UARTx_CTL | UARTx Control Register |
| 0x0004 | UARTx_RXDAT | UARTx Receive FIFO Data Register |
| 0x0008 | UARTx_TXDAT | UARTx Transmit FIFO Data Register |
| 0x000C | UARTx_STA | UARTx Status Register |
| 0x0010 | UARTx_BR | UARTx BAUDRATE divider Register |
| 0x0014 | UARTx_CTL2 | UART Control2 Register |

## 12.4.3 UARTX Register Description

## 12.4.3.1 UARTx_CTL

**UART Control Register**

Offset=0x0000

| Bits | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31 | RXENABLE | UART RX disable<br>1:normal<br>0:disable | R/W | 0x0 |
| 30 | TXENABLE | UART TX disable<br>1:normal<br>0:disable | RW | 0x0 |
| 29 | TX_FIFO_EN | UART TX FIFO enable:<br>0: Disable | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 1: Enable | | |
| 28 | RX_FIFO_EN | UART RX FIFO enable:<br>0: Disable<br>1: Enable | R/W | 0x0 |
| 27:26 | TX_FIFO_SEL | UART TX FIFO Input Select.<br>0: From CPU<br>1: From DMA<br>2:Reserved<br>3:From DSP<br>Others: Reserved | RW | 0x0 |
| 25 | - | Reserved | RW | 0x0 |
| 24 | RX_FIFO_SEL | UART RX FIFO Output Select.<br>0: To CPU<br>1: To DMA | RW | 0x0 |
| 23:22 | AFL | Level for trigger RTS to high in autoflow when the autoflow bit is set<br>00:14byte<br>01:12byte<br>10:8byte<br>11:4byte | RW | 0x0 |
| 21:20 | - | Reserved | RW | 0x0 |
| 19 | TXIE | UART TX IRQ Enable.<br>0: Disable<br>1: Enable | RW | 0x0 |
| 18 | RXIE | UART RX IRQ Enable.<br>0: Disable<br>1: Enable | RW | 0x0 |
| 17 | TXDE | UART TX DRQ Enable.<br>0: Disable<br>1: Enable | RW | 0x0 |
| 16 | RXDE | UART RX DRQ Enable.<br>0: Disable<br>1: Enable | RW | 0x0 |
| 15 | EN | UART Enable.<br>When this bit is clear, the UART clock source is inhibited. This can be used to place the module in a low power standby state.<br>0:disable<br>1: enable | RW | 0x0 |
| 14 | - | Reserved | RW | 0x0 |
| 13 | RTSE | RTS Enable.<br>When this bit is set, request to send data.<br>Note: This bit has no effect if Autoflow enable bit is set.<br>0:no request<br>1: request to send data | RW | 0x0 |
| 12 | AFE | Autoflow mode Enable<br>Setting this bit enables automatic hardware flow control. Enabling this mode overrides software control of the signals.<br>0: Autoflow mode disable (normal mode)<br>1: Autoflow mode enable | RW | 0x0 |
| 11:10 | RDIC | UART RX DRQ/IRQ Control<br>00: set when RX FIFO received at least one byte data in | RW | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | IRQ/DRQ mode.<br>01: set when RX FIFO received 4 bytes data in IRQ mode<br>10: set when RX FIFO received 8 bytes data in IRQ/DRQ mode<br>11: set when RX FIFO received 12 bytes data in IRQ/DRQ mode<br>In DMA burst mode (normal DMA), DO not set 00,01 because at least 8 bytes necessary.<br>In DMA single mode (special DMA), DO set 00 for 1 Bytes transfer for each DRQ. | | |
| 9:8 | TDIC | UART TX DRQ/IRQ Control<br>00: set when TX FIFO is at least 1 byte empty in IRQ/DRQ mode.<br>01: set when TX FIFO is 4 bytes empty in IRQ mode.<br>10: set when TX FIFO is 8 bytes empty in IRQ/DRQ mode.<br>11: set when TX FIFO is 12 bytes empty in IRQ/DRQ mode.<br>In DMA burst mode (normal DMA), DO not set 00,01 because at least 8 bytes necessary.<br>In DMA single mode (special DMA), DO set 00 for 1 Bytes transfer for each DRQ. | RW | 0x0 |
| 7 | CTSE | CTS Enable.<br>If this bit is 1,the transmitter check CTS before send the next byte,<br>Note: no effect if autoflow enable bit is set<br>0:don't check CTS before send<br>1:check CTS before send | RW | 0x0 |
| 6:4 | PRS | Parity Select.<br>Bit 6: PEN, Parity enable<br>Bit 5: STKP, Stick parity<br>Bit 4: EPS, Even parity<br>PEN STKP EPS Selected Parity<br>0 x x None<br>1 0 0 Odd<br>1 0 1 logic 1<br>1 1 0 Even<br>1 1 1 logic 0 | RW | 0x0 |
| 3 | - | Reserved | R | 0 |
| 2 | STPS | STOP Select.<br>If this bit is 0, 1 stop bit is generated in transmission. If this bit is 1, 2 stop bits are generated.<br>0:1 stop bit<br>1:2 stop bit | RW | 0x0 |
| 1:0 | DWLS | Data Width Length Select.<br>00: 5 bits<br>01: 6 bits<br>10: 7 bits<br>11: 8 bits | RW | 0x0 |

### 12.4.3.2 UARTx_RXDAT

**UART Receive FIFO Data Register**

Offset=0x0004

| Bits | Name | Description | R/W | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 31:8 | Reserved | | R | 0x0 |
| 7:0 | RXDAT | Received Data.<br>The depth of FIFO is 8bit×16levels. | R | x |

### 12.4.3.3 UARTx_TXDAT

**UART Transmit FIFO Data Register**

Offset=0x0008

| Bits | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:8 | Reserved | | R | 0 |
| 7:0 | TXDAT | Transmitted Data.<br>The depth of FIFO is 8bit×16 levels | W | 0 |

### 12.4.3.4 UARTx_STA

**UART Status Register**

Offset=0x000c

| Bits | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:25 | Reserved | | R | 0 |
| 24 | RXTO_PD | Rx Timeout IRQ Pending bit<br>0:NO IRQ<br>1:IRQ<br>Write 1 auto clean | R/W | 0x0 |
| 23 | PAER | Parity Status.<br>0: Parity OK<br>1: Parity error.<br>Writing 1 to the bit will clear the bit.<br>When parity error. | R/W | 0x0 |
| 22 | STER | Stop Status.<br>0: Stop OK<br>1: Stop error.<br>Writing 1 to the bit will clear the bit.<br>When stop bit detect error. | R/W | 0x0 |
| 21 | UTBB | UART TX busy bit<br>0:not busy, TX FIFO is empty and all data be shift out<br>1:busy | R | 0 |
| 20:16 | TXFL | TX FIFO Level.<br>The field indicates the current TX FIFO empty level. | R | 10000 |
| 15:11 | RXFL | RX FIFO Level.<br>The field indicates the current RX FIFO level of valid data. | R | 0 |
| 10 | TFES | TX FIFO empty Status<br>0: not empty<br>1: empty | R | 1 |
| 9 | RFFS | RX FIFO full Status<br>0: not full<br>1: full | R | 0 |
| 8 | RTSS | RTS Status.<br>The bit reflects the status of the external RTS- pin. | R | 0 |
| 7 | CTSS | CTS Status.<br>The bit reflects the status of the external CTS- pin. | R | x |

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| | | | | |
|---|---|---|---|---|
| 6 | TFFU | TX FIFO Full.<br>1: Full<br>0: No Full | R | 0 |
| 5 | RFEM | RX FIFO Empty.<br>1: Empty<br>0: No Empty | R | 1 |
| 4 | RXST | Receive Status.<br>0: receive OK<br>1: receive error.<br>Writing 1 to the bit will clear the bit.<br>When stop bit detect error, or parity error, or clock error | RW | 0 |
| 3 | TFER | TX FIFO Error.<br>0: No Error<br>1: Error<br>Writing 1 to the bit will clear the bit and reset the TX FIFO. | RW | 0 |
| 2 | RXER | RX FIFO Error.<br>0: No Error<br>1: Error<br>Writing 1 to the bit will clear the bit and reset the RX FIFO. | RW | 0 |
| 1 | TIP | TX IRQ Pending Bit.<br>0: No IRQ<br>1: IRQ<br>Writing 1 to the bit to clear the bit. | RW | 1 |
| 0 | RIP | RX IRQ Pending Bit.<br>0: No IRQ<br>1: IRQ<br>Writing 1 to the bit to clear it. | RW | 0 |

### 12.4.3.5 UARTx_BR

**UART BAUDRATE divider register**

Offset=0x0010

| Bits | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | TXBRDIV | UART TX BAUDRATE divider<br>BaudRate = Colck_source/BaudRate divider<br>Clock_source=HOSC | R/W | 0x028 |
| 15:0 | RXBRDIV | UART RX BAUDRATE divider<br>BaudRate = Colck_source/BaudRate divider<br>Clock_source=HOSC | R/W | 0x028 |

### 12.4.3.6 UARTx_CTL2

**UART Control register2**

Offset=0x0014

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31 | RXTO_IRQ_EN | Rx Timeout IRQ Pending Enable<br>0：disable<br>1：enable<br>The IRQ will happen when the RXTO time pass and the UARTx Rx in IDLE. | R/W | 0x0 |
| 30:8 | - | Reserved | R | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 7:0 | RXTO | UARTx Rx IDLE Timeout(in byte time) | R/W | 0x10 |

## 12.5 SPI0

### 12.5.1 Features

SPI0 is a combination module which includes conventional SPI and SPI_Cache interface.

- Only support master mode
- Support mode0, mode1, mode2 and mode3
- Support AHB and Cache access SPI and DMA interface
- Only Support 8bit transfer mode
- Support 32 bit seed 8bit randomize
- Support 3wire
- Support 64 level delay chain, 0.25ns/step
- Support 1x/2x/dual/4x/quad/ DTR 4x access SPI NOR
- Support 3-byte/4-byte address modes access SPINOR
- Support 16bit CRC only when Cache access fifo, not support CRC when AHB access fifo
- Support QPI 4x/quad/DTR 4x for Cache access SPI NOR
- Support DTR(Double Transfer Rate)
- Support cache abort
- Support 106MHz spi_clk as highest speed
- Support 3.1V/1.8V IO voltage

## 12.6 SPI1

### 12.6.1 Features

- Support SPI normal mode: mode 0/1/2/3
- Support SPI/QPI PSRAM and Double-Data-Rate Octal SPI PSRAM(X8/X16)
- Support 1x/2x/dual/4x/quad access SPI NOR
- Support IRQ and DMA mode to transmit data
- Support 64 level delay chain, 0.15ns/step
- Support slave mode
- Support 140MHz spi_clk as highest speed

## 12.7 SPI2

### 12.7.1 Features

- Support SPI normal mode: mode 0/1/2/3
- Support normal 4 wire mode/3 wire mode
- Support IRQ and DMA mode to transmit data
- Support 32 level delay chain, 0.5ns/step
- Support slave mode
- Support 96MHz spi_clk as highest speed

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## 12.7.2 SPI2 Register List

*Table 12-7 SPI2 Registers Block Base Address*

| Name | Physical Base Address | KSEG1 Base Address |
|---|---|---|
| SPI2 | 0x40030000 | 0x40030000 |

*Table 12-8 SPI2 Registers Offset Address*

| Offset | Register Name | Description |
|---|---|---|
| 0x0000 | SPI2_CTL | SPI2 Control Register |
| 0x0004 | SPI2_STA | SPI2 Status Register |
| 0x0008 | SPI2_TXDAT | SPI2 Transmit FIFO Data Register |
| 0x000C | SPI2_RXDAT | SPI2 Receive FIFO Data Register |
| 0x0010 | SPI2_BC | SPI2 Byte Counter Register |

## 12.7.3 SPI2 Register Description

### 12.7.3.1 SPI2_CTL

**SPI2 Control Register**
Offset=0x0000

| Bits | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31 | CLKSEL | FIFO write or read clock select<br>0: use CPU clock<br>1: use DMA clock | R/W | 0x0 |
| 30 | FWS | FIFO width select<br>0: 8bit<br>1: 32bit | R/W | 0x0 |
| 29:28 | SPI_MODE_SELECT | SPI Mode Select<br>00 : Mode0<br>01 : Mode1<br>10 : Mode2<br>11 : Mode3 | R/W | 0x3 |
| 27 | - | Reserved | R/W | 0x0 |
| 26 | RX_WRITE_SEL | SPI Rx Write Select,<br>Select suitable cycle To sample the right rx data<br>0: delay 2 spi_clk cycle (used when SPI_DELAY <= 4'b1000)<br>1: delay 3 spi_clk cycle (used when SPI_DELAY <= 4'b1111) | R/W | 0x0 |
| 25 | DMS | DMA transmit mode select<br>0: burst8 mode<br>1: single mode | R/W | 0x0 |
| 24 | TXCEB | TX Convert Endian bit, only used in 32Bit mode:<br>0: not convert Endian<br>0x76543210 ->0x76543210<br>1: convert Endian<br>32bit mode:<br>0x76543210 ->0x10325476<br>When in 8 bit mode, this bit have no effect | R/W | 0x0 |
| 23 | RXCEB | RX Convert Endian bit, only used in 32Bit mode:<br>0: not convert Endian | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0x76543210 ->0x76543210<br>1: convert Endian<br>32bit mode:<br>0x76543210 ->0x10325476<br>When in 8 bit mode, this bit have no effect | | |
| 22 | MSS | Master or Slave mode select<br>0: Master mode<br>1: Slave mode | R/W | 0x0 |
| 21 | MSB | SPI LSB/MSB First Select<br>0: SPI transmit and receive MSB first<br>1: SPI transmit and receive LSB first | R/W | 0x0 |
| 20 | RILS | RX IRQ Level select<br>0: RX FIFO not empty, generate IRQ<br>1: RX FIFO at least 8 level data, generate IRQ<br>Note: this bit have no effect when SPI_RIRQ_EN is disable, SPI2_CTL[8]. | R/W | 0x0 |
| 19:15 | SPI_DELAY | **SPI2 Master read clock delay time (valid when SPI_WR select write/read and read mode)**<br>00000: no delay<br>00001: delay 0.5 ns<br>00010: delay 1 ns<br>00011: delay 1.5 ns<br>00100: delay 2 ns<br>00101: delay 2.5 ns<br>00110: delay 3 ns<br>00111: delay 3.5 ns<br>01000: delay 4 ns<br>01001: delay 4.5 ns<br>01010: delay 5 ns<br>01011: delay 5.5 ns<br>01100: delay 6 ns<br>01101: delay 6.5 ns<br>01110: delay 7 ns<br>01111: delay 7.5 ns<br>10000: delay 8 ns<br>10001: delay 8.5 ns<br>10010: delay 9 ns<br>10011: delay 9.5 ns<br>10100: delay 10 ns<br>10101: delay 10.5 ns<br>10110: delay 11 ns<br>10111: delay 11.5 ns<br>11000: delay 12 ns<br>11001: delay 12.5 ns<br>11010: delay 13 ns<br>11011: delay 13.5 ns<br>11100: delay 14 ns<br>11101: delay 14.5 ns<br>11110: delay 15 ns<br>11111: delay 15.5 ns | R/W | 0x0 |
| 14:11 | - | Reserved | R/W | 0x0 |
| 10 | SPI_3WIRE | SPI2 Master 3-wire mode enable<br>0:disable<br>1:enable<br>use 3-wire mode only in master read only or write only mode. | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 9 | SPI_TIRQ_EN | SPI TX IRQ Enable, trigger SPI TX IRQ when SPI TX FIFO at least 8 level empty<br>0: disable<br>1: enable | R/W | 0x0 |
| 8 | SPI_RIRQ_EN | SPI RX IRQ Enable, this trigger of SPI RX IRQ controlled by SPI2_CTL[20].<br>0: disable<br>1: enable | R/W | 0x0 |
| 7 | SPI_TDRQ_EN | SPI TX DRQ Enable, trigger DRQ when SPI TX FIFO at least 8 level empty; When DMA remain counter < 8, trigger DRQ until all data transfer completely;<br>0: disable<br>1: enable | R/W | 0x0 |
| 6 | SPI_RDRQ_EN | SPI RX DRQ Enable, trigger DRQ when SPI RX FIFO at least 8 level full; When DMA remain counter < 8, trigger DRQ until all data received completely;<br>0: disable<br>1: enable | R/W | 0x0 |
| 5 | SPI_TX_FIFO_EN | SPI Tx FIFO Enable<br>0: Disable<br>1: Enable | R/W | 0x0 |
| 4 | SPI_RX_FIFO_EN | SPI Rx FIFO Enable<br>0: Disable<br>1: Enable | R/W | 0x0 |
| 3 | SPI_SS | SPI NSS pin control output<br>0: output low<br>1: output high | R/W | 0x1 |
| 2 | - | Reserved | R/W | 0x0 |
| 1:0 | SPI_WR | SPI Read/Write Mode (AHB Interface Only)<br>00: disable<br>01: Read only<br>10: Write only<br>11: Read and Write | R/W | 0x0 |

### 12.7.3.2 SPI2_STA

SPI2 Status Register
Offset=0x0004

| Bits | Name | Description | Access | Reset |
|---|---|---|---|---|
| 31:14 | - | Reserved | R | 0x0 |
| 13 | SPI_RXHF | SPI RX FIFO Half Full<br>0: not half full<br>1: half full | R | 0x0 |
| 12 | SPI_TXHF | SPI TX FIFO Half Full<br>0: not half full<br>1: half full | R | 0x0 |
| 11 | TFWO | TX FIFO error pending if Write FIFO overflow occur<br>0: no error<br>1: error occur<br>Writing '1' to clear this bit | R/W | 0x0 |
| 10 | - | Reserved | R | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 9 | RFWO | RX FIFO error pending if Write FIFO overflow occur<br>0: no error<br>1: error occur<br>Writing '1' to clear this bit | R/W | 0x0 |
| 8 | TFRO | RX FIFO error pending if Read FIFO overflow occur<br>0: no error<br>1: error occur<br>Writing '1' to clear this bit | R/W | 0x0 |
| 7 | SPI_RXFU | SPI2 RX FIFO Full<br>0: not full<br>1: full | R | 0x0 |
| 6 | SPI_RXEM | SPI2 RX FIFO Empty<br>0: not empty<br>1: empty | R | 0x1 |
| 5 | SPI_TXFU | SPI2 TX FIFO Full<br>0: not full<br>1: full | R | 0x0 |
| 4 | SPI_TXEM | SPI2 TX FIFO Empty<br>0: not empty<br>1: empty | R | 0x1 |
| 3 | SPI_TIRQ_PD | SPI2 TX IRQ Pending, Writing '1' to this bit will clear it.<br>0: No TX IRQ Pending<br>1: TX IRQ Pending. | R/W | 0x0 |
| 2 | SPI_RIRQ_PD | SPI2 RX IRQ Pending, Writing '1' to this bit will clear it.<br>0: No RX IRQ Pending<br>1: RX IRQ Pending. | R/W | 0x0 |
| 1 | - | Reserved | R | 0x0 |
| 0 | SPI_BUSY | SPI2 master busy status bit.<br>0: SPI idle status<br>1: SPI busy status (Clock is transmiting or Rx Remain Counter doesn't reduced to zero) | R | 0x0 |

### 12.7.3.3 SPI2_TXDAT

SPI2 Transmit FIFO Data Register
Offset=0x0008

| Bits | Name | Description | Access | Reset |
|---|---|---|---|---|
| 31:0 | SPI2_TXDAT | SPI2 TX FIFO, 32bitx16 levels<br>When SPI2_CTL[30] select 8bit width, bit[7:0] is valid.<br>8bitx64levels<br>When SPI2_CTL[30] select 32bit width, bit[31:0] is valid.<br>Be read as zero. | W | 0x0 |

### 12.7.3.4 SPI2_RXDAT

SPI2 Receive FIFO Data Register
Offset=0x000C

| Bits | Name | Description | Access | Reset |
|---|---|---|---|---|
| 31:0 | SPI2_RXDAT | SPI RX FIFO,<br>When SPI2_CTL[30] select 8bit width, bit[7:0] is valid.<br>8bitx64levels<br>When SPI2_CTL[30] select 32bit width, bit[31:0] is valid. | R | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 32bitx16 levels | | |

#### 12.7.3.5 SPI2_BC

SPI2 Bytes Count Register, this register is used for setting SPI2 bytes counter bits in the SPI read mode only.

Offset=0x0010

| Bits | Name | Description | Access | Reset |
|---|---|---|---|---|
| 31:16 | REMAIN_CNT | Indicate how many bytes need to be received, only use in master mode | R | 0x0 |
| 15:0 | SPI2_BC | Bytes Counter [15: 0] | R/W | 0x0 |

## 12.8 SPI3

### 12.8.1 Features

- Support SPI normal mode: mode 0/1/2/3
- support normal 4 wire mode
- Support 1x/2x/dual/4x/quad access SPI NAND
- Support IRQ and DMA mode to transmit data
- Support 32 level delay chain, 0.5ns/step
- Support slave mode
- Support 96MHz spi_clk as highest speed
- Support auto check RB

### 12.8.2 SPI3 Register List

*Table 12-9 SPI3 Registers Block Base Address*

| Name | Physical Base Address | KSEG1 Base Address |
|---|---|---|
| SPI3 | 0x40034000 | 0x40034000 |

*Table 12-10 SPI3 Registers Offset Address*

| Offset | Register Name | Description |
|---|---|---|
| 0x0000 | SPI3_CTL | SPI3 Control Register |
| 0x0004 | SPI3_STA | SPI3 Status Register |
| 0x0008 | SPI3_TXDAT | SPI3 Transmit FIFO Data Register |
| 0x000C | SPI3_RXDAT | SPI3 Receive FIFO Data Register |
| 0x0010 | SPI3_BC | SPI3 Byte Counter Register |
| 0x0018 | SPI3_CHECKRB | SPI3 CHECKRB configure |
| 0x001c | SPI3_CHECKRB_COUNTER | SPI3 auto check RB counter register |

### 12.8.3 SPI3 Register Description

#### 12.8.3.1 SPI3_CTL

**SPI3 Control Register**

Offset=0x0000

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| Bits | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31 | CLKSEL | FIFO write or read clock select<br>0: use CPU clock<br>1: use DMA clock | R/W | 0x0 |
| 30 | FWS | FIFO width select<br>0: 8bit<br>1: 32bit | R/W | 0x0 |
| 29:28 | SPI_MODE_SELECT | SPI Mode Select<br>00 : Mode0<br>01 : Mode1<br>10 : Mode2<br>11 : Mode3 | R/W | 0x3 |
| 27 | - | Reserved | R/W | 0x0 |
| 26 | RX_WRITE_SEL | SPI Rx Write Select,<br>Select suitable cycle To sample the right rx data<br>0: delay 2 spi_clk cycle (used when SPI_DELAY <= 4'b1000)<br>1: delay 3 spi_clk cycle (used when SPI_DELAY <= 4'b1111) | R/W | 0x0 |
| 25 | DMS | DMA transmit mode select<br>0: burst8 mode<br>1: single mode | R/W | 0x0 |
| 24 | TXCEB | TX Convert Endian bit, only used in 32Bit mode:<br>0: not convert Endian<br>0x76543210 ->0x76543210<br>1: convert Endian<br>32bit mode:<br>0x76543210 ->0x10325476<br>When in 8 bit mode, this bit have no effect | R/W | 0x0 |
| 23 | RXCEB | RX Convert Endian bit, only used in 32Bit mode:<br>0: not convert Endian<br>0x76543210 ->0x76543210<br>1: convert Endian<br>32bit mode:<br>0x76543210 ->0x10325476<br>When in 8 bit mode, this bit have no effect | R/W | 0x0 |
| 22 | MSS | Master or Slave mode select<br>0: Master mode<br>1: Slave mode | R/W | 0x0 |
| 21 | MSB | SPI LSB/MSB First Select<br>0: SPI transmit and receive MSB first<br>1: SPI transmit and receive LSB first | R/W | 0x0 |
| 20 | RILS | RX IRQ Level select<br>0: RX FIFO not empty, generate IRQ<br>1: RX FIFO at least 8 level data, generate IRQ<br>Note: this bit have no effect when SPI_RIRQ_EN is disable, SPI3_CTL[8]. | R/W | 0x0 |
| 19:15 | SPI_DELAY | **SPI3 Master read clock delay time (valid when SPI_WR select write/read and read mode)**<br>00000: no delay<br>00001: delay 0.5 ns<br>00010: delay 1 ns<br>00011: delay 1.5 ns<br>00100: delay 2 ns | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 00101: delay 2.5 ns<br>00110: delay 3 ns<br>00111: delay 3.5 ns<br>01000: delay 4 ns<br>01001: delay 4.5 ns<br>01010: delay 5 ns<br>01011: delay 5.5 ns<br>01100: delay 6 ns<br>01101: delay 6.5 ns<br>01110: delay 7 ns<br>01111: delay 7.5 ns<br>10000: delay 8 ns<br>10001: delay 8.5 ns<br>10010: delay 9 ns<br>10011: delay 9.5 ns<br>10100: delay 10 ns<br>10101: delay 10.5 ns<br>10110: delay 11 ns<br>10111: delay 11.5 ns<br>11000: delay 12 ns<br>11001: delay 12.5 ns<br>11010: delay 13 ns<br>11011: delay 13.5 ns<br>11100: delay 14 ns<br>11101: delay 14.5 ns<br>11110: delay 15 ns<br>11111: delay 15.5 ns | | |
| 14:13 | - | Reserved | R/W | 0x0 |
| 12 | SPI_3WIRE | SPI3 Master 3-wire mode enable<br>0:disable<br>1:enable<br>use 3-wire mode only in master read only or write only mode. | R/W | 0x0 |
| 11:10 | SPI_IO_MODE | SPI data I/O mode select<br>00: 1x I/O mode select<br>01: 1x I/O mode select<br>10: 2x I/O mode select<br>11: 4x I/O mode select | R/W | 0x0 |
| 9 | SPI_TIRQ_EN | SPI TX IRQ Enable, trigger SPI TX IRQ when SPI TX FIFO at least 8 level empty<br>0: disable<br>1: enable | R/W | 0x0 |
| 8 | SPI_RIRQ_EN | SPI RX IRQ Enable, this trigger of SPI RX IRQ controlled by SPI3_CTL[20].<br>0: disable<br>1: enable | R/W | 0x0 |
| 7 | SPI_TDRQ_EN | SPI TX DRQ Enable, trigger DRQ when SPI TX FIFO at least 8 level empty; When DMA remain counter < 8, trigger DRQ until all data transfer completely;<br>0: disable<br>1: enable | R/W | 0x0 |
| 6 | SPI_RDRQ_EN | SPI RX DRQ Enable, trigger DRQ when SPI RX FIFO at least 8 level full.; When DMA remain counter < 8, trigger DRQ until all data received completely;<br>0: disable | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 1: enable | | |
| 5 | SPI_TX_FIFO_EN | SPI Tx FIFO Enable<br>0: Disable<br>1: Enable | R/W | 0x0 |
| 4 | SPI_RX_FIFO_EN | SPI Rx FIFO Enable<br>0: Disable<br>1: Enable | R/W | 0x0 |
| 3 | SPI_SS | SPI NSS pin control output<br>0: output low<br>1: output high | R/W | 0x1 |
| 2 | - | Reserved | R/W | 0x0 |
| 1:0 | SPI_WR | SPI Read/Write Mode (AHB Interface Only)<br>00: disable<br>01: Read only<br>10: Write only<br>11: Read and Write | R/W | 0x0 |

### 12.8.3.2 SPI3_STA

SPI3 Status Register
Offset=0x0004

| Bits | Name | Description | Access | Reset |
|---|---|---|---|---|
| 31:14 | - | Reserved | R | 0x0 |
| 13 | SPI_RXHF | SPI RX FIFO Half Full<br>0: not half full<br>1: half full | R | 0x0 |
| 12 | SPI_TXHF | SPI TX FIFO Half Full<br>0: not half full<br>1: half full | R | 0x0 |
| 11 | TFWO | TX FIFO error pending if Write FIFO overflow occur<br>0: no error<br>1: error occur<br>Writing '1' to clear this bit | R/W | 0x0 |
| 10 | - | Reserved | R | 0x0 |
| 9 | RFWO | RX FIFO error pending if Write FIFO overflow occur<br>0: no error<br>1: error occur<br>Writing '1' to clear this bit | R/W | 0x0 |
| 8 | TFRO | RX FIFO error pending if Read FIFO overflow occur<br>0: no error<br>1: error occur<br>Writing '1' to clear this bit | R/W | 0x0 |
| 7 | SPI_RXFU | SPI3 RX FIFO Full<br>0: not full<br>1: full | R | 0x0 |
| 6 | SPI_RXEM | SPI3 RX FIFO Empty<br>0: not empty<br>1: empty | R | 0x1 |
| 5 | SPI_TXFU | SPI3 TX FIFO Full<br>0: not full<br>1: full | R | 0x0 |
| 4 | SPI_TXEM | SPI3 TX FIFO Empty | R | 0x1 |

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| | | | | |
|---|---|---|---|---|
| | | 0: not empty<br>1: empty | | |
| 3 | SPI_TIRQ_PD | SPI3 TX IRQ Pending, Writing '1' to this bit will clear it.<br>0: No TX IRQ Pending<br>1: TX IRQ Pending. | R/W | 0x0 |
| 2 | SPI_RIRQ_PD | SPI3 RX IRQ Pending, Writing '1' to this bit will clear it.<br>0: No RX IRQ Pending<br>1: RX IRQ Pending. | R/W | 0x0 |
| 1 | - | Reserved | R | x |
| 0 | SPI_BUSY | SPI3 master busy status bit.<br>0: SPI idle status<br>1: SPI busy status (Clock is transmiting or Rx Remain Counter doesn't reduced to zero) | R | 0x0 |

#### 12.8.3.3 SPI3_TXDAT

SPI3 Transmit FIFO Data Register
Offset=0x0008

| Bits | Name | Description | Access | Reset |
|---|---|---|---|---|
| 31:0 | SPI3_TXDAT | SPI3 TX FIFO, 32bitx16 levels<br>When SPI3_CTL[30] select 8bit width, bit[7:0] is valid.<br>8bitx64levels<br>When SPI3_CTL[30] select 32bit width, bit[31:0] is valid.<br>Be read as zero. | W | 0x0 |

#### 12.8.3.4 SPI3_RXDAT

SPI3 Receive FIFO Data Register
Offset=0x000C

| Bits | Name | Description | Access | Reset |
|---|---|---|---|---|
| 31:0 | SPI3_RXDAT | SPI RX FIFO,<br>When SPI3_CTL[30] select 8bit width, bit[7:0] is valid.<br>8bitx64levels<br>When SPI3_CTL[30] select 32bit width, bit[31:0] is valid.<br>32bitx16 levels | R | 0x0 |

#### 12.8.3.5 SPI3_BC

SPI3 Bytes Count Register, this register is used for setting SPI3 bytes counter bits in the SPI read mode only.
Offset=0x0010

| Bits | Name | Description | Access | Reset |
|---|---|---|---|---|
| 31:16 | REMAIN_CNT | Indicate how many bytes need to be received, only use in master mode | R | 0x0 |
| 15:0 | SPI3_BC | Bytes Counter [15:0] | R/W | 0x0 |

#### 12.8.3.6 SPI3_CHECKRB

**SPI3 Check RB Register**
Offset=0x0018

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 31:9 | - | Reserved | R | 0x0 |
| 8 | RB | SNAND RB status bit<br>0: not busy<br>1: busy | R | 0x0 |
| 7 | CHECKRB_PD | SNAND auto check RB finish pending bit. After check RB finish, this bit will keep 1. Write 1 to clear this bit.<br>0: no check RB finish pending<br>1: check RB finish pending | R/W | 0x0 |
| 6:2 | - | Reserved | R | 0x0 |
| 1 | CHECKRB_IRQ_EN | SNAND check RB IRQ enable bit<br>SPI will set check RB finish pending when SNAND auto check RB finish.<br>0: Disable<br>1: Enable | R/W | 0x0 |
| 0 | CHECKRB | SNAND auto check RB start bit<br>0: disable auto check RB<br>1: start auto check RB | R/W | 0x0 |

#### 12.8.3.7 SPI3_CHECKRB_COUNTER

**SPI3 auto check RB counter Register**
Offset=0x001C

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:12 | - | Reserved | R | 0x0 |
| 11:0 | RBCOUNTER | Auto check RB counter, count by SPICLK | R/W | 0xFFF |

## 12.9 SPIMT

### 12.9.1 Features

ATS3085Sx contains two SPIMT interfaces:

- Normal mode and Auto mode.
- SPI master compatible at Normal mode.
- SPI master read/write one/multi-byte operation executed automatically at auto mode.
- 4 chip select signal.
- Clock may be 4MHz or 32MHz.
- Dedicated AHB-like DMA Bus for data access.

## 12.10 PWM

### 12.10.1 Features

- Independent of 4 PWM
- The frequency of PWM comes from the division of 32KHz/HOSC/CK64M
- Support normal mode , breath mode and programmable mode
  - Normal mode: can output 65536 kinds of duty
  - Breath mode: supports breathing lights with various flicker frequencies

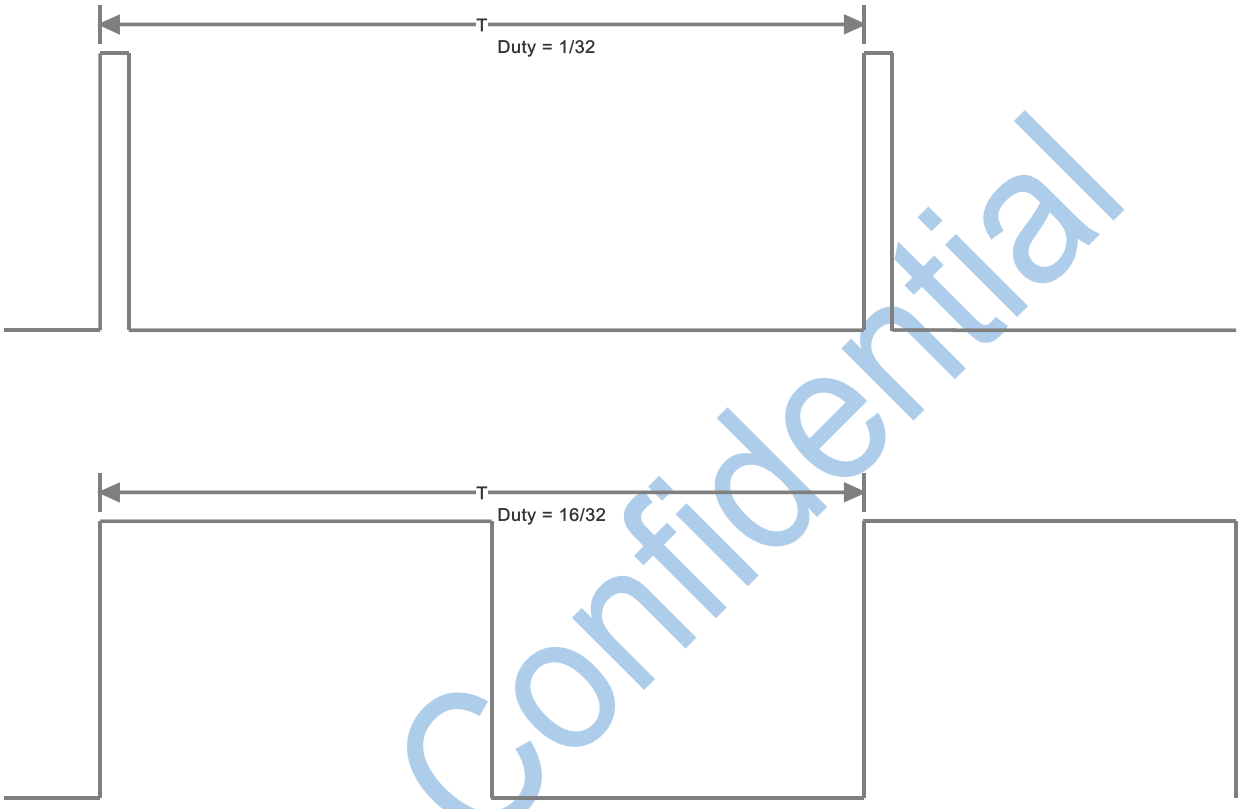
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- Programmable mode: The duty of output signal can be changed according the ram data.

## 12.10.2 Module Description

### 12.10.2.1 Normal Mode Timing

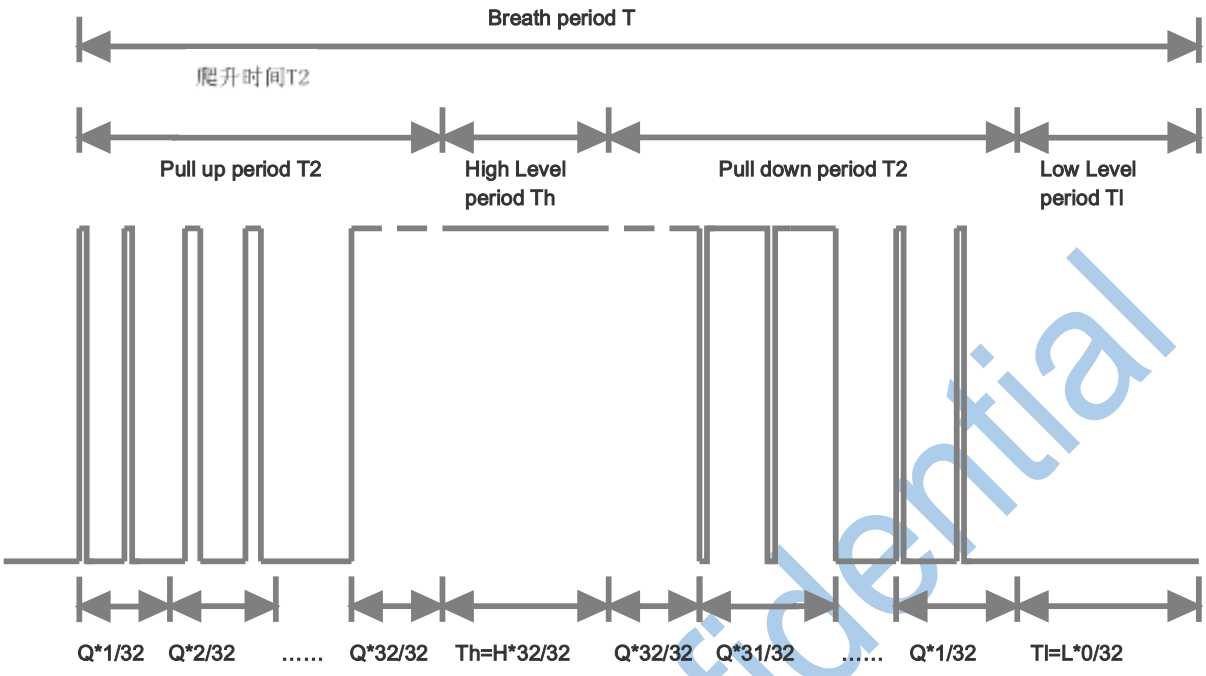



PWM can be used as backlights enable signal of LCD. The brightness of the backlights is decided by the duty cycle of PWM.

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### 12.10.2.2 Breath Mode Timing




The breath mode of PWM can be used for driver of breathing light. For example, if PWM CLK from CMU is f=1/t and we need the time of pull up or pull down to be T2=Q*32*32t=0.5s, and the time of high level Th=H*32T=0.5s and the time of low level Tl=L*32t=2s, when we set Q=2, the f should be 4096 and H is 64, L is 256.

### 12.10.2.3 Programmable Mode Timing

**Example used DMA transmission:**

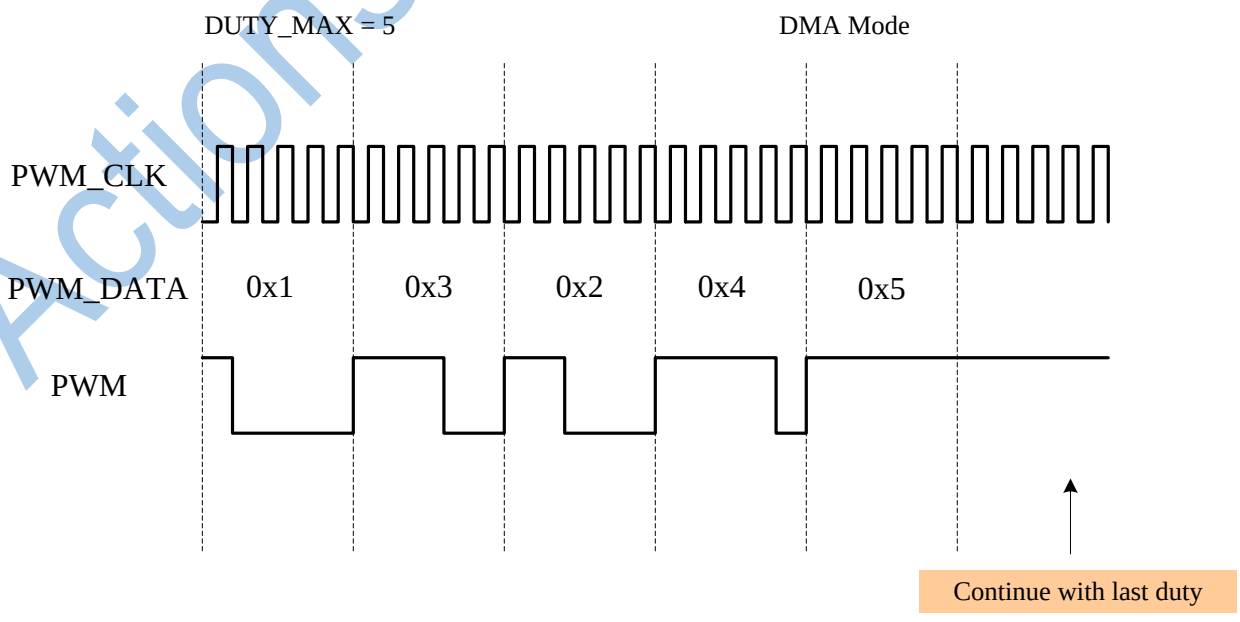



***Figure12-1 Programmable mode example used DMA transmission***

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## 12.10.3 Operation Manual

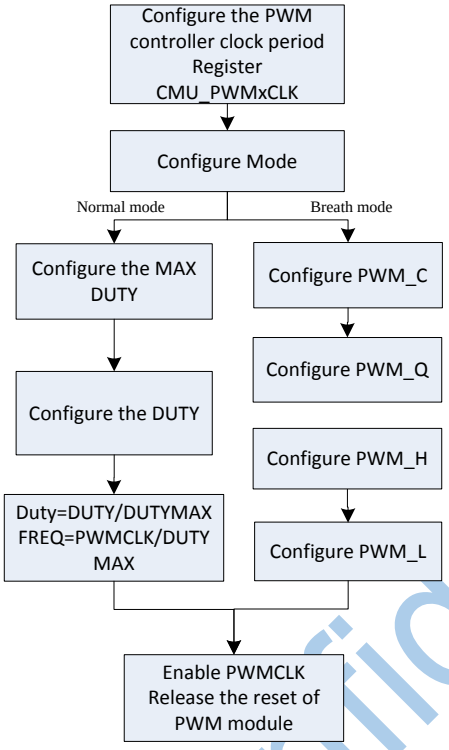



*Figure 12-2 PWM Configuration*

## 12.10.4 PWM Register List

*Table 12-11 PWM Registers Block Base Address*

| Name | Physical Base Address | KSEG1 Base Address |
|---|---|---|
| PWM | 0x40060000 | 0x40060000 |

*Table 12-12 PWM Registers Offset Address*

| Offset | Register Name | Description |
|---|---|---|
| 0x000+0x100*x | PWMx_CTL | PWM Output Control register, 0<=x<=3 |
| 0x004+0x100*x | PWMx_C | PWM breath mode C register, 0<=x<=3 |
| 0x008+0x100*x | PWMx_Q | PWM breath mode Q register, 0<=x<=3 |
| 0x00c+0x100*x | PWMx_H | PWM breath mode H register, 0<=x<=3 |
| 0x010+0x100*x | PWMx_L | PWM breath mode L register, 0<=x<=3 |
| 0x014+0x100*x | PWMx_DUTYMAX | PWM Duty Max register, 0<=x<=3 |
| 0x018+0x100*x | PWMx_DUTY | PWM Duty register, 0<=x<=3 |
| 0xf00 | PWM_DMACTL | PWM Enable Control register |
| 0xf04 | PWM_FIFODAT | PWM FIFO Data register |
| 0xf08 | PWM_FIFOSTA | PWM FIFO Status register |

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## 12.10.5 PWM Register Description

### 12.10.5.1 PWMx_CTL

PWMx Output Control Register, 0<=x<=3
Offset = 0x000 + 0x100*x

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:5 | Reserved | Reserved | R | 0x0 |
| 4 | CH_START | PWMx Channel start<br>0: stop after this PWM wave period<br>1: PWM start | RW | 0x0 |
| 3 | CH_EN | PWMx Channel Enable<br>0: disable<br>1: enable | RW | 0x0 |
| 2 | POL_SEL | Polarity select:<br>0:PWMx low voltage level active<br>1:PWMx high voltage level active<br>Active in fixed Mode and Breath Mode | RW | 0x0 |
| 1:0 | MODE_SEL | PWMx Mode Select:<br>0: Fixed Mode<br>1: Breath Mode<br>2: Programmable Mode<br>3: Reserved | RW | 0x0 |

### 12.10.5.2 PWMx_C

PWMx Breath mode C configuration Register, 0<=x<=3
Offset = 0x004 + 0x100*x

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:8 | Reserved | Reserved | R | 0x0 |
| 7:0 | C | Counter Top of PWM Duty<br>For Example: In Breath Mode if C = 32,<br>The PWM Duty is 1/32, 2/32, .. , 32/32 | RW | 0x20 |

### 12.10.5.3 PWMx_Q

PWMx Breath mode Q configuration Register, 0<=x<=3
Offset = 0x008 + 0x100*x

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:24 | Reserved | Reserved | R | 0x0 |
| 23:16 | QD | Time of Every Duty =C/C …1/C:<br>Falling time T2=QD*C*C*t<br>t is the period of CMU_PWM<br>C*t is the period of PWM Wave<br>Only Active in Breath Mode | RW | 0x0 |
| 15:8 | Reserved | Reserved | R | 0x0 |
| 7:0 | QU | Time of Every Duty =1/C …C/C:<br>Rising time T1=QU*C*C*t<br>t is the period of CMU_PWM<br>C*t is the period of PWM Wave | RW | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | Only Active in Breath Mode | | |

### 12.10.5.4 PWMx_H

PWMx Breath mode H/L configuration Register, 0<=x<=3
Offset = 0x00c + 0x100*x

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | Reserved | Reserved | R | 0x0 |
| 15:0 | H | Time of Duty =C/C :<br>High Level Time = H*C*t<br>t is the period of CMU_PWM\<br>C*t is the period of PWM Wave<br>Only Active in Breath Mode | RW | 0x0 |

### 12.10.5.5 PWMx_L

PWMx Breath mode H/L configuration Register, 0<=x<=3
Offset = 0x010 + 0x100*x

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | Reserved | Reserved | R | 0x0 |
| 15:0 | L | Time of Duty =0/C:<br>Low Level Time = L*C*t<br>t is the period of CMU_PWM<br>C*t is the period of PWM Wave<br>Only Active in Breath Mode | RW | 0x0 |

### 12.10.5.6 PWMx_DUTYMAX

PWMx Duty Max Register, 0<=x<=3
Offset = 0x014 + 0x100*x

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | Reserved | Reserved | R | 0x0 |
| 15:0 | DUTYMAX | Configurable maximum duty.<br>PWM period = PWM_CLK period * DUTY_MAX<br>Note: DUTYMAX >= 0x1 | RW | 0x1 |

### 12.10.5.7 PWMx_DUTY

PWMx Duty Register, 0<=x<=3
Offset = 0x018 + 0x100*x

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | Reserved | Reserved | R | 0x0 |
| 15:0 | DUTY | PWM DUDY = DUTY / DUTY_MAX<br>Only Active in Normal Mode | RW | 0x0 |

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### 12.10.5.8 PWM_DMACTL

PWM Module Enable Register
Offset = 0xf00

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:1 | Reserved | Reserved | R | 0x0 |
| 7:4 | PWM_FIFO_CLK_SEL | 0000:PWM0<br>0001:PWM1<br>0010:PWM2<br>0011:PWM3<br>Other: Reserved | RW | 0x0 |
| 3:1 | - | - | RW | 0x0 |
| 0 | START | PWM DMA START<br>0: disable<br>1: enable | RW | 0x0 |

### 12.10.5.9 PWM_FIFODAT

PWM FIFO Data Register
Offset = 0xf04

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | Reserved | Reserved | R | 0x0 |
| 15:0 | DATA | DUTY DATA<br>PWM DUDY = DUTY / DUTY_MAX<br>Only Active in programmable Mode | R | 0x0 |

Note: DUTY must be configured ≤ (DUTY_MUX), failing to do so may result in unpredictable behavior.

### 12.10.5.10 PWM_FIFOSTA

PWM FIFO Status Register
Offset = 0xf08

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:5 | Reserved | Reserved | R | 0x0 |
| 4:3 | LEVEL | PWM FIFO level<br>This field indicates pwm fifo empty level | R | 0x2 |
| 2 | EMPTY | PWM FIFO empty status:<br>0: not empty<br>1: empty | R | 0x1 |
| 1 | FULL | PWM FIFO full status:<br>0: not full<br>1: full | R | 0x0 |
| 0 | ERROR | PWM FIFO error status:<br>0: no error<br>1: error<br>Writing 1 to the bit is clear | RW | 0x0 |

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# 13 Audio Interface

## 13.1 DAC

### 13.1.1 Features

- Build in single channel 24-bit input sigma-delta DAC, SNR<=113dB, THD+N>=-98dB, DR>=111dB(A-weighting)
- DAC supports sample rate 8k/12k/11.025k/16k/22.05k/24k/32k/44.1k/48k/88.2k/96kHz
- Support digital volume of 256 steps with zero cross detection
- Build in single channel 20mW PA (Power Amplifier). PA output supports traditional mode, and differential mode. The noise floor of Audio PA on A-WT is 2.7uVrms (Typical) when in differential mode.
- An anti-pop circuit for suppressing noise of PA when enable and disable

### 13.1.2 Module Description

#### 13.1.2.1 Data Block Diagram

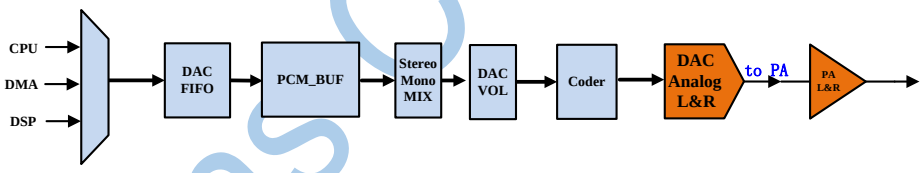



*Figure 13-1 DAC Main Block*

### 13.1.3 Signal List

*Table 13-1 DAC Signals*

| SIGNAL Name | I/O Type | Short Description |
|---|---|---|
| AOUT/AOUTP | AO | output of PA / Positive output of PA |
| AOUTN | AO | Negative output of PA |

### 13.1.4 Register List

*Table 13-2 DAC Controller Registers Address*

| Name | Physical Base Address | KSEG1 Base Address |
|---|---|---|

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| | | |
|---|---|---|
| DAC_Control_Register | 0x4005C000 | 0x4005C000 |

*Table 13-3 DAC Controller Registers*

| Offset | Register Name | Description |
|---|---|---|
| 0x00 | DAC_DIGCTL | DAC Digital Control Register |
| 0x08 | DAC_STAT | DAC State Register |
| 0x10 | PCM_BUF_CTL | PCM Buffer Control Register |
| 0x14 | PCM_BUF_STAT | PCM Buffer State Register |
| 0x18 | PCM_BUF_THRES_HE | Threshold to generate half empty signal |
| 0x1C | PCM_BUF_THRES_HF | Threshold to generate half full signal |
| 0x28 | VOL_LCH | Volume Left Channel Control Register |
| 0x2C | VOL_RCH | Volume Right Channel Control Register |

## 13.1.5 Register Description

### 13.1.5.1 DAC_DIGCTL

DAC Digital Control Register
Offset = 0x00

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:1 | - | R/W forbidden | - | - |
| 0 | DDEN | DAC Digital Enable.<br>0x0: Disable<br>0x1: Enable | RW | 0x0 |

### 13.1.5.2 DAC_STAT

DAC State Register
Offset = 0x08

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:8 | - | R/W forbidden | - | - |
| 7 | DAF0EIP | DAC FIFO0 half-Empty IRQ Pending Bit.<br>0x0: No IRQ<br>0x1: IRQ<br>Writing 1 to the bit is clear it. | RW | 0x0 |
| 6 | DAF0F | DAC FIFO0 Full Flag.<br>0x0: Not Full<br>0x1: Full | R | 0x0 |
| 5 | - | Reserved | R | 0x0 |
| 4:0 | DAF0S | DAC FIFO0 Status<br>These 5 bits shows how many sample data can be filled to the FIFO.<br>For example, when read as 11, means CPU/DMA can write 11 samples to FIFO.<br>Display one by one | R | x |

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### 13.1.5.3 PCM_BUF_CTL

PCM BUF Control Register
Offset = 0x10

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:8 | - | R/W forbidden | - | - |
| 7 | PCMBEPIE | PCM_BUF **Empty** IRQ Enable.<br>0x0: Disable<br>0x1: Enable | RW | 0x0 |
| 6 | PCMBFUIE | PCM_BUF **Full** IRQ Enable.<br>0x0: Disable<br>0x1: Enable | RW | 0x0 |
| 5 | PCMBHEIE | PCM_BUF **Half Empty** IRQ Enable.<br>0x0: Disable<br>0x1: Enable | RW | 0x0 |
| 4 | PCMBHFIE | PCM_BUF **Half Full** IRQ Enable.<br>0x0: Disable<br>0x1: Enable | RW | 0x0 |
| 3:0 | - | R/W forbidden | - | - |

### 13.1.5.4 PCM_BUF_STAT

PCM BUF State Register
Offset = 0x14

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:20 | - | R/W forbidden | - | - |
| 19 | PCMBEIP | PCM BUF **Empty** IRQ Pending Bit.<br>0x0: No IRQ<br>0x1: IRQ<br>Writing 1 to the bit is clear it. | RW | 0x0 |
| 18 | PCMBFIP | PCM BUF **Full** IRQ Pending Bit.<br>0x0: No IRQ<br>0x1: IRQ<br>Writing 1 to the bit is clear it. | RW | 0x0 |
| 17 | PCMBHEIP | PCM BUF **Half Empty** IRQ Pending Bit.<br>0x0: No IRQ<br>0x1: IRQ<br>Writing 1 to the bit is clear it. | RW | 0x0 |
| 16 | PCMBHFIP | PCM BUF **Half Full** IRQ Pending Bit.<br>0x0: No IRQ<br>0x1: IRQ<br>Writing 1 to the bit is clear it. | RW | 0x0 |
| 15:11 | - | R/W forbidden | - | - |
| 10:0 | PCMBS | PCM_BUF Status<br>These 11 bits shows how many samples FIFO are not filled.<br>For example, when read as 512, means CPU/DMA can write 512 samples to FIFO.<br>Display one sample | R | 0x400 |

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### 13.1.5.5 PCM_BUF_THRES_HE

PCM BUF Threshold of Half Empty Register
Offset = 0x18

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:12 | - | R/W forbidden | - | - |
| 11:0 | THRESHOLD | The threshold to generate **half empty** signal:<br>When the number of samples equal to the half empty threshold, pcm_buf generates half-empty IRQ. (**The Threshold value should be set even number, and not zero**). | RW | 0x100 |

### 13.1.5.6 PCM_BUF_THRES_HF

PCM BUF Threshold of Half Full Register
Offset = 0x1C

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:12 | - | R/W forbidden | - | - |
| 11:0 | THRESHOLD | The threshold to generate **half full** signal:<br>When the number of samples equal to the half full threshold, pcm_buf generates half-full IRQ. (**The Threshold value should be set even number, and not zero**). | RW | 0x300 |

### 13.1.5.7 VOL_LCH

Volume Left Channel Control Register
Offset = 0x28

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:12 | - | R/W forbidden | - | - |
| 11 | DONE_STA | DAC volume left channel soft stepping gain reach target set status：<br>0 : marching<br>1 : done | R | 0x1 |
| 10 | SOFT_STEP_EN | DAC volume left channel soft stepping gain control enable<br>0 : disable<br>1 : enable | RW | 0x0 |
| 9:8 | - | R/W forbidden | - | - |
| 7:0 | VOLL | Volume Left Channel Control ( step:0.375dB):<br>0xff: +24.375 dB<br>0xfe: +24dB<br>0xfd: +23.625dB<br>…<br>0xbf: +0.375dB<br>0xbe: 0dB<br>…<br>0x00: mute | RW | 0xbf |

Note: Only VOL_RCH is valid when DAC is working. VOL_LCH and VOL_RCH are valid both when IIS TX is working.

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### 13.1.5.8 VOL_RCH

Volume Right Channel Control Register
Offset = 0x2C

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:12 | - | R/W forbidden | - | - |
| 11 | DONE_STA | DAC volume right channel soft stepping gain reach target set status：<br>0 : marching<br>1 : done | R | 0x1 |
| 10 | SOFT_STEP_EN | DAC volume right channel soft stepping gain control enable<br>0 : disable<br>1 : enable | RW | 0x0 |
| 9:8 | - | R/W forbidden | - | - |
| 7:0 | VOLR | Volume right Channel Control ( step:0.375dB):<br>0xff: +24.375 dB<br>0xfe: +24dB<br>0xfd: +23.625dB<br>…<br>0xbf: +0.375dB<br>0xbe: 0dB<br>…<br>0x00: mute | RW | 0xbf |

Note: Only VOL_RCH is valid when DAC is working. VOL_LCH and VOL_RCH are valid both when IIS TX is working.

## 13.2 ADC

### 13.2.1 Features

- Build in stereo 24 bit input sigma-delta ADCs, SNR<=98dB, THD+N<=-88dB(A-weighting)
- ADC supports sample rate 8k/12k/11.025k/16k/22.05k/24k/32k/44.1k/48k/88.2k/96kHz
- A digital high-pass filter can be used to remove dc offsets when ADC use
- Every ADC has independent Automatic Amplitude Limitation
- Supports single-ended input analog microphones and full differential input microphone
- Supports 2 Digital microphones

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## 13.2.2 Module Description

### 13.2.2.1 Block Diagram

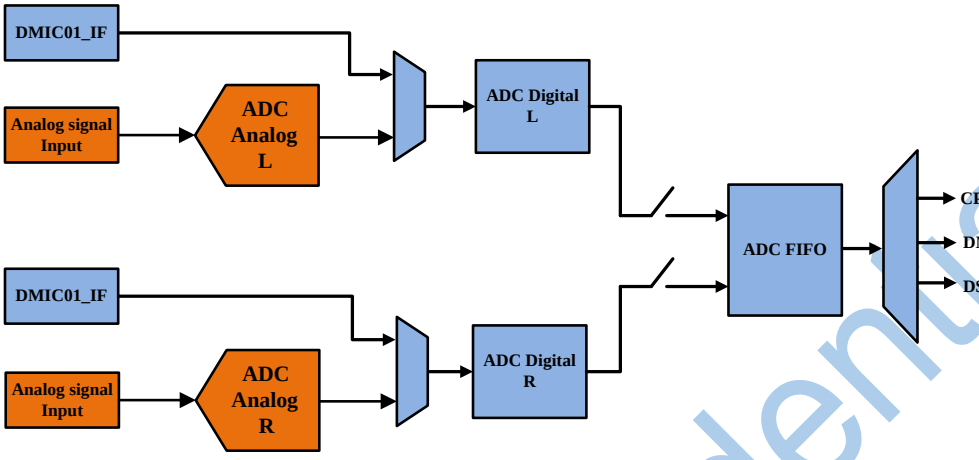



*Figure 13-2 ADC Main Block*

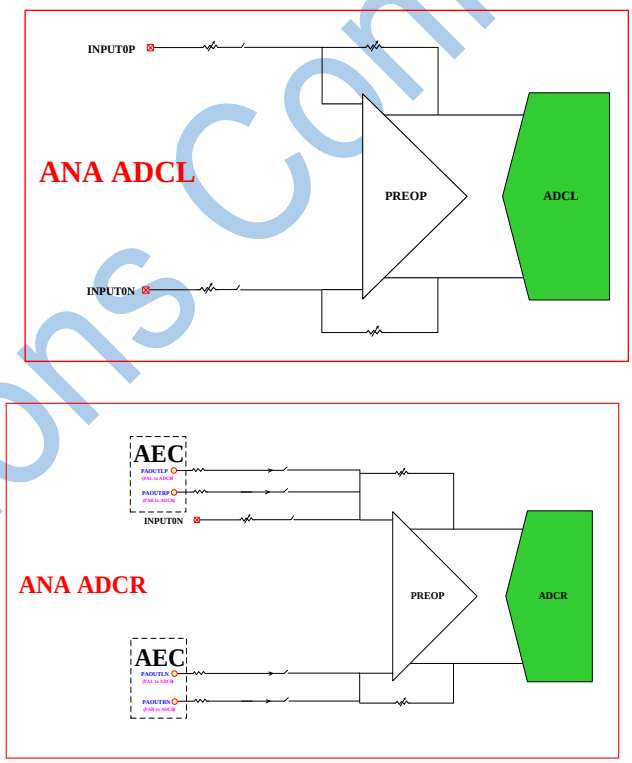



*Figure 13-3 ADC analog block*

### 13.2.2.2 ADC frequency response curves

ATS3085Sx provides 3 frequency response curves for use.

*Table 13-4 The parameters of 3 frequency response curves*

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| Mode Select | Pass Band Ripple (db) | Stopband Attenuation (db) | -1db Cutoff Frequency(Hz) |
|---|---|---|---|
| mode A | -0.04~0.06 | -71 | -0.463 |
| mode B | -0.09~0.00 | -82 | -0.449 |
| mode C | -0.04~0.08 | -71 | -0.235 |

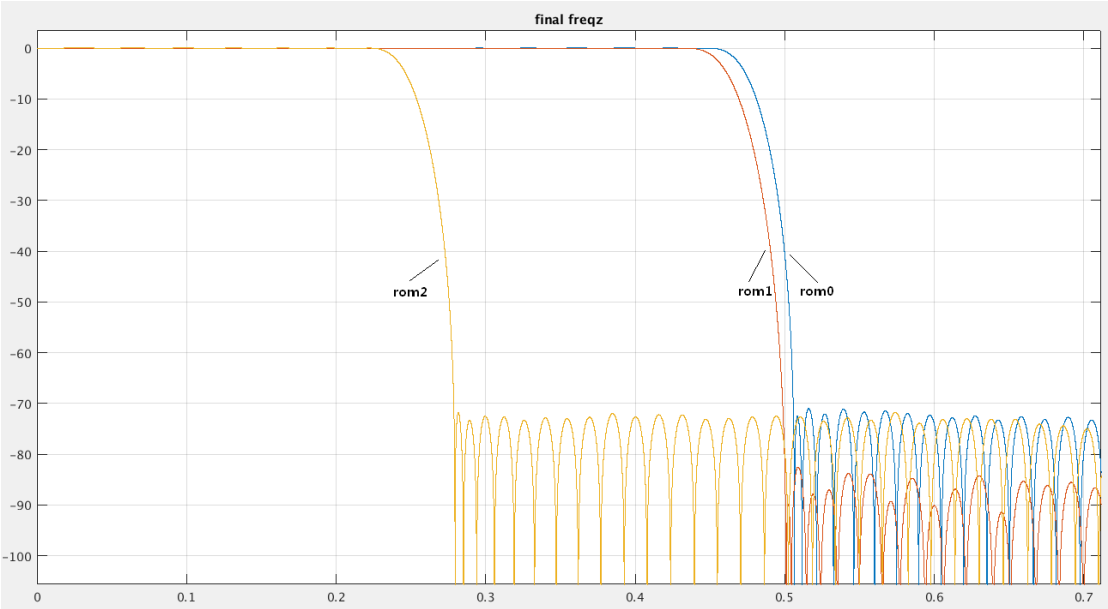



*Figure 13-4 The frequency response curves*

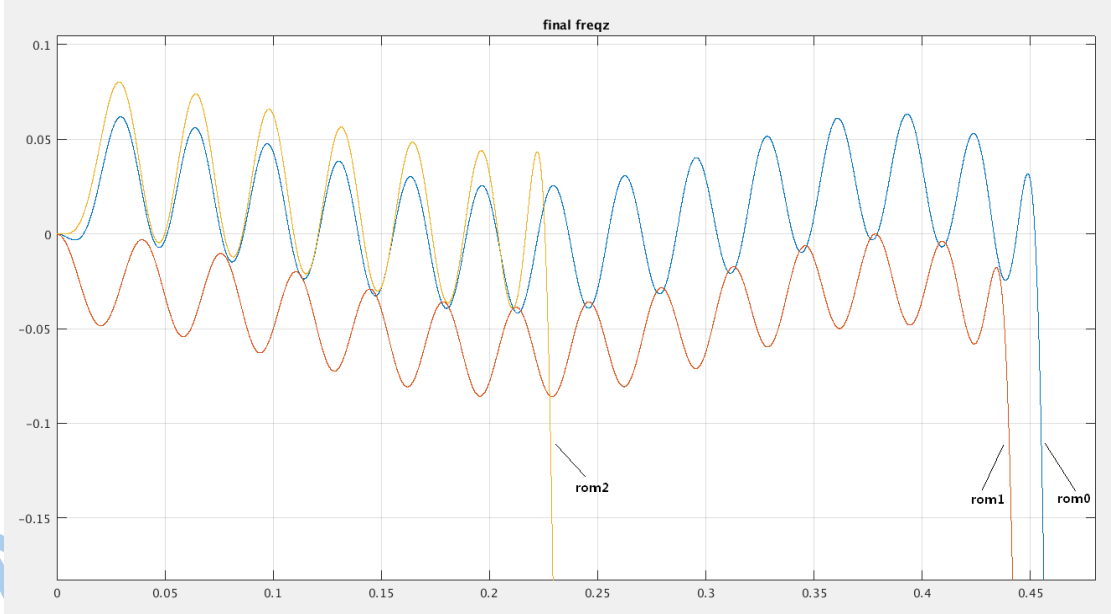



*Figure 13-5 In-band ripple*

### 13.2.2.3 Signal List

*Table 13-5 ADC Signals*

| SIGNAL Name | I/O Type | Short Description |
|---|---|---|
| INPUT0P | AI | Analog single-end input to ADC0, or can form differential input（ADC0） with INPUT0N |

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| | | |
|---|---|---|
| INPUT0N | AI | Analog single-end input to ADC1, or can form differential input（ADC0） with INPUT0P |
| DMIC_CLK | DO | DMIC clock output |
| DMIC_DATA | DI | DMIC data input |

#### 13.2.2.4 DMIC

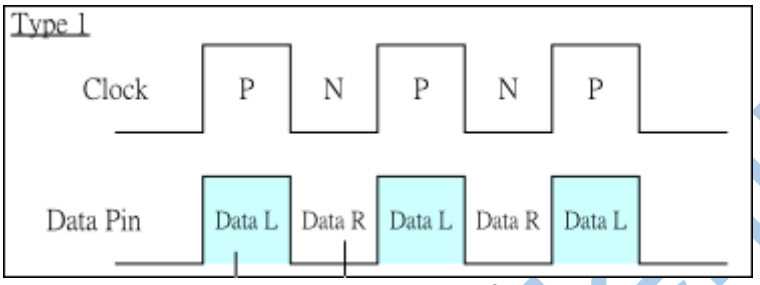



*Figure 13-6 DMIC Interface*

ATS3085Sx has 4 DMIC which can form 2 stereo voice inputs.

### 13.2.3 Register List

*Table 13-6 ADC Controller Registers Address*

| Name | Physical Base Address | KSEG1 Base Address |
|---|---|---|
| ADC_Control_Register | 0x4005C100 | 0x4005C100 |

*Table 13-7 ADC Controller Registers*

| Offset | Register Name | Description |
|---|---|---|
| 0x00 | ADC_DIGCTL | ADC Digital Control Register |
| 0x04 | CH0_DIGCTL | CH0 Digital Control Register |
| 0x08 | CH1_DIGCTL | CH1 Digital Control Register |
| 0x10 | ADC_STAT | ADC State Register |
| 0x20 | ADC0_CTL | ADC0 analog control Register |
| 0x24 | ADC1_CTL | ADC1 analog control Register |
| 0x2C | ADC_VMIC_CTL | ADC VMIC control Register |

### 13.2.4 Register Description

#### 13.2.4.1 ADC_DIGCTL

ADC Digital Control Register
Offset = 0x00

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:14 | - | R/W forbidden | - | - |
| 13 | ADC1_DIG_EN | ADC1 digital enable:<br>0:disable<br>1:enable | RW | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 12 | ADC0_DIG_EN | ADC0 digital enable:<br>0:disable<br>1:enable | RW | 0x0 |
| 11 | - | R/W forbidden | - | - |
| 10:8 | DMIC_PRE_GAIN | Pre-Gain control for DMIC<br>000 : 1x<br>001 : 2x<br>010 : 4x<br>011 : 8x<br>100 : 16x<br>101 : 32x<br>110 : 63x<br>111 : 63x<br>for all DMIC interface<br>If CHx data select DMIC and CHx_DIG_EN was enable channel x DMIC interface was enable. | RW | 0x0 |
| 7 | - | Reserved | R/W | 0x0 |
| 6 | DMIC01_CHS | DMIC01 channel select：<br>0：L/R<br>1：R/L<br>When set to 0, DMIC will latch L channel data in DMIC_CLK rising edge. When set to 1,DMIC will latch R channel data in DMIC_CLK rising edge. | RW | 0x0 |
| 5:4 | ADC_MD_SEL | mode select<br>00 : FIR mode A<br>01 : FIR mode B<br>10 : FIR mode C<br>11 : FIR mode D | RW | 0x0 |
| 3:0 | - | R/W forbidden | - | - |

### 13.2.4.2 CH0_DIGCTL

CH0 Digital Control Register
Offset = 0x04

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:4 | - | R/W forbidden | - | - |
| 3:0 | ADCGC | ADC0 Digital Gain Control<br>0x0: 0dB<br>0x1: 3.5dB<br>0x2: 7dB<br>0x3: 10.5dB<br>0x4: 14dB<br>0x5: 17.5dB<br>0x6: 21dB<br>0x7: 24.5dB<br>0x8: 28dB<br>0x9: 31.5dB<br>0xa: 35dB<br>0xb: 38.5dB<br>0xc: 42dB<br>0xd: 45.5dB | RW | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0xe: 49dB<br>0xf: 52.5dB | | |

### 13.2.4.3 CH1_DIGCTL

CH1 Digital Control Register
Offset = 0x08

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:4 | - | R/W forbidden | - | - |
| 3:0 | ADCGC | ADC1 Digital Gain Control<br>0x0: 0dB<br>0x1: 3.5dB<br>0x2: 7dB<br>0x3: 10.5dB<br>0x4: 14dB<br>0x5: 17.5dB<br>0x6: 21dB<br>0x7: 24.5dB<br>0x8: 28dB<br>0x9: 31.5dB<br>0xa: 35dB<br>0xb: 38.5dB<br>0xc: 42dB<br>0xd: 45.5dB<br>0xe: 49dB<br>0xf: 52.5dB | RW | 0x0 |

### 13.2.4.4 ADC_STAT

ADC State Register
Offset = 0x10

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:10 | - | R/W forbidden | - | - |
| 9 | FIFO0_ER | FIFO0 error<br>0 : no error<br>1 : error<br>When fifo0 was full, system bus write ADCFIFO0 would cause error<br>Writing 1 to the bit will clear it | RW | 0 |
| 8 | ADF0EF | ADC FIFO0 Empty Flag.<br>0x0: Not Empty<br>0x1: Empty | R | 0x1 |
| 7 | ADF0IP | ADC FIFO half-Full IRQ Pending Bit.<br>0x0: No IRQ<br>0x1: IRQ<br>Writing 1 to the bit is clear it. | RW | 0x0 |
| 6 | - | Reserved | R | 0x0 |
| 5:0 | ADF0S | ADC FIFO Status.<br>These 6 bits shows how many samples fifo0 filled.<br>For example, when read as 9, means CPU can read 9 samples from fifo0. | R | 0x0 |

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### 13.2.4.5 ADC0_CTL

ADC0 Control Register (all input can be used as mic/aux)
Offset = 0x20

<table>
<tr><th>Bit(s)</th><th>Name</th><th>Description</th><th>R/W</th><th>Reset</th></tr>
<tr><td>31:19</td><td>-</td><td>R/W forbidden except bit[20]</td><td>-</td><td>-</td></tr>
<tr><td>18:16</td><td>FDBUF0_IRS</td><td>FDBUF0 input resistor select<br>0xx: not connect, disable<br>100:40k<br>101:20k<br>110:10k<br>For se mode, this res should be connected. If only one channel input ,then set the res the same as AUX input res, the gain is calculated as PREAM_PG.</td><td>RW</td><td>0x0</td></tr>
<tr><td>15:12</td><td>PREAM0_PG</td><td>PREAMP0 OP feedback res select to form different gain with different input res:<br>For use, first select input mode and input res, then select preamp feedback res, then gain is fixed.<br>For diff mode (se mode -6db)
<table>
<tr><td></td><td>40k</td><td>20k</td><td>10k</td></tr>
<tr><td>0000</td><td>-12db</td><td>-6db</td><td>0db</td></tr>
<tr><td>0001</td><td>-9db</td><td>-3db</td><td>3db</td></tr>
<tr><td>0010</td><td>-6db</td><td>0db</td><td>6db</td></tr>
<tr><td>0011</td><td>-3db</td><td>3db</td><td>9db</td></tr>
<tr><td>0100</td><td>0db</td><td>6db</td><td>12db</td></tr>
<tr><td>0101</td><td>3db</td><td>9db</td><td>15db</td></tr>
<tr><td>0110</td><td>6db</td><td>12db</td><td>18db</td></tr>
<tr><td>0111</td><td>7.5db</td><td>13.5db</td><td>19.5db</td></tr>
<tr><td>1000</td><td>9db</td><td>15db</td><td>21db</td></tr>
<tr><td>1001</td><td>10.5db</td><td>16.5db</td><td>22.5db</td></tr>
<tr><td>1010</td><td>12db</td><td>18db</td><td>24db</td></tr>
<tr><td>1011</td><td>13.5db</td><td>19.5db</td><td>25.5db</td></tr>
<tr><td>1100</td><td>15db</td><td>21db</td><td>27db</td></tr>
<tr><td>1101</td><td>16.5db</td><td>22.5db</td><td>28.5db</td></tr>
<tr><td>1110</td><td>18db</td><td>24db</td><td>30db</td></tr>
<tr><td>1111</td><td>19.5db</td><td>25.5db</td><td>31.5db</td></tr>
</table></td><td>RW</td><td>0x0</td></tr>
<tr><td>11</td><td>-</td><td>R/W forbidden</td><td>-</td><td>-</td></tr>
<tr><td>10</td><td>FDBUF0_EN</td><td>FD BUF OP 0 enable for SE mode<br>0:disable<br>1:enable</td><td>RW</td><td>0x0</td></tr>
<tr><td>9</td><td>PREOP0_EN</td><td>PREOP 0 enable<br>0:disable<br>1:enable</td><td>RW</td><td>0x0</td></tr>
<tr><td>8</td><td>ADC0_EN</td><td>ADC0 channel SDM enable<br>0:disable<br>1:enable<br>When enable this bit, must enable bit[20]</td><td>RW</td><td>0x0</td></tr>
<tr><td>7:6</td><td>INPUT0N_EN</td><td>INPUT0N pad to ADC0/1 channel input enable<br>00~01: disable<br>11: enable channel</td><td>RW</td><td>0x0</td></tr>
<tr><td>5:4</td><td>INPUT0P_EN</td><td>INPUT0P pad to ADC0 channel input enable<br>00~01: disable<br>11: enable channel</td><td>RW</td><td>0x0</td></tr>
</table>

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| | | | | |
|---|---|---|---|---|
| 3 | - | R/W forbidden | - | - |
| 2 | INPUT0_IN_MODE | INPUT0 input mode select<br>1: single end input mode: INPUT0P to ADCL channel, INPUT0N to ADCR channel;<br>0: differential input mode: INPUT0P and INPUT0N form the differential input to ADCL channel<br>INPUT0P is the positive end ; INPUT0N is the negative end. | RW | 0x0 |
| 1:0 | INPUT0_IRS | INPUT0 input resistor select<br>00: 40k<br>01:20k<br>10:10k | RW | 0x0 |

### 13.2.4.6 ADC1_CTL

ADC1 Control Register (all input can be used as mic/aux)
Offset = 0x24

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:29 | - | R/W forbidden | - | - |
| 28 | MIX2AD1FDSE | Mixout to ADC1 input mode select<br>0:fd mode<br>1:se mode | RW | 0x0 |
| 27 | PAR2AD1_EN | PA OUTR mix to ADC1 channel<br>0:disable<br>1:enable<br>Input res is 40k. | RW | 0x0 |
| 26 | PAL2AD1_EN | PA OUTL mix to ADC1 channel<br>0:disable<br>1:enable<br>Input res is 40k. | RW | 0x0 |
| 25 | PAR2AD1_PD_EN | PA OUTR mix to ADC1 channel pull down<br>0:disable<br>1:enable | RW | 0x0 |
| 24 | PAL2AD1_PD_EN | PA OUTL mix to ADC1 channel pull down<br>0:disable<br>1:enable | RW | 0x0 |
| 23:19 | - | R/W forbidden except bit[20] | - | - |
| 18:16 | FDBUF1_IRS | FDBUF1 input resistor select<br>0xx: not connect, disable<br>100:40k<br>101:20k<br>110:10k<br>*For se mode, this res should be connected. If only one channel input ,then set the res the same as AUX input res, the gain is calculated as PREAM_PG.* | RW | 0x0 |
| 15:12 | PREAM1_PG | PREAMP1 OP feedback res select to form different gain with different input res:<br>*For use, first select input mode and input res, then select preamp feedback res, then gain is fixed.*<br>For diff mode (se mode -6db)<br>40k / 20k / 10k<br>0000: -12db / -6db / 0db<br>0001: -9db / -3db / 3db | RW | 0x0 |

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| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | 0010 | -6db | 0db | 6db | | |
| | | 0011 | -3db | 3db | 9db | | |
| | | 0100 | 0db | 6db | 12db | | |
| | | 0101 | 3db | 9db | 15db | | |
| | | 0110 | 6db | 12db | 18db | | |
| | | 0111 | 7.5db | 13.5db | 19.5db | | |
| | | 1000 | 9db | 15db | 21db | | |
| | | 1001 | 10.5db | 16.5db | 22.5db | | |
| | | 1010 | 12db | 18db | 24db | | |
| | | 1011 | 13.5db | 19.5db | 25.5db | | |
| | | 1100 | 15db | 21db | 27db | | |
| | | 1101 | 16.5db | 22.5db | 28.5db | | |
| | | 1110 | 18db | 24db | 30db | | |
| | | 1111 | 19.5db | 25.5db | 31.5db | | |
| 11 | - | R/W forbidden | | | | - | - |
| 10 | FDBUF1_EN | FD BUF OP 1 enable for SE mode<br>0:disable<br>1:enable | | | | RW | 0x0 |
| 9 | PREOP1_EN | PREOP 1 enable<br>0:disable<br>1:enable | | | | RW | 0x0 |
| 8 | ADC1_EN | ADC1 channel SDM enable<br>0:disable<br>1:enable<br>When enable this bit, must enable bit[20] | | | | RW | 0x0 |
| 7:0 | - | R/W forbidden | | | | - | - |

#### 13.2.4.7 ADC_VMIC_CTL

ADC VMIC Register, this register will not be power down or reset when audio be power gated.
Offset = 0x2C

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:4 | - | R/W forbidden | - | - |
| 3:2 | VMIC0_VOL | VMIC0 output voltage control<br>00: 1.57V<br>01: 1.61V<br>10: 1.65V<br>11: 1.7V | R/W | 0x2 |
| 1:0 | VMIC0_EN | VMIC0 enable<br>0x: disable VMIC0<br>10: output voltage fix in VC18;<br>11: enable VMIC0 and voltage set by bit[3:2];<br>**For mems mic, try to use 10 bypass mode to save power.** | R/W | 0x0 |

## 13.3 I2S/PCM

### 13.3.1 Features

**I2S :**

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- TX0/RX0 Support Master / Slave mode; TX1 Support Master mode Only.
- TX0/RX0 Support Slave Mode Sample Rate Detect.
- 5 wire mode：TX0+RX0 Support Master / Slave; TX1+TX0+RX0 Support Master Only.
- TX0/RX0 Formats include I2S /L-J /R-J /TDM4 /TDM8; TX1 NOT Support TDM4/TDM8.
- TX0/RX0/TX1 Support data width 16/20/24 bit.
- TX0/RX0 Support Sample Rate 8k/12k/11.025k/16k/22.05k/24k/32k/44.1k/48k/96k/192k for Dual channel; TX1 NOT Support 192k.
- TX1 Support Volume adjustment.

**PCM:**

- TX0/RX0 Support Master and Slave mode; TX1 Support Master mode Only.
- TX0/RX0 Support Slave Mode Sample Rate Detect.
- TX0/RX0/TX1 Support 13~16/24 bits linear, MSB first.
- TX0/RX0/TX1 Support Short Frame Sync and Long Frame Sync.
- TX0/RX0 Support Sample rate 8k/12k/11.025k/16k/22.05k/24k/32k/44.1k/48k/96k/192k for Dual channel; TX1 NOT Support 192k.
- TX0/RX0 Support 1 /2 /4 /8 channel; TX1 Only Support 2 channel.
- TX0/RX0/TX1 Support 16/20/24 bit data width.
- TX0/RX0/TX1 Per channel can set 16 or 32 pcm_clk cycles.

I2S&PCM:

- TX0 FIFO with 24bit*32 levels, accessed by CPU, DMA, DSP, DSPDMA.
- RX0 FIFO with 24bit*32 levels, accessed by CPU, DMA, DSP, DSPDMA.
- TX1 share 24bit*16 levels DACFIFO, with 3Kbyte PCM RAM.
- I2S_MCLK, I2S_BCLK/PCM_CLK, I2S_LRCLK/PCM_SYNC can independently shut down.
- The max setting frequency of Bclk/PCM_clk/TX0/RX0 is 12.288 MHz, TX1 is 6.144MHz.

## 13.3.1 Module Description

The timing format I2S support are as follows.

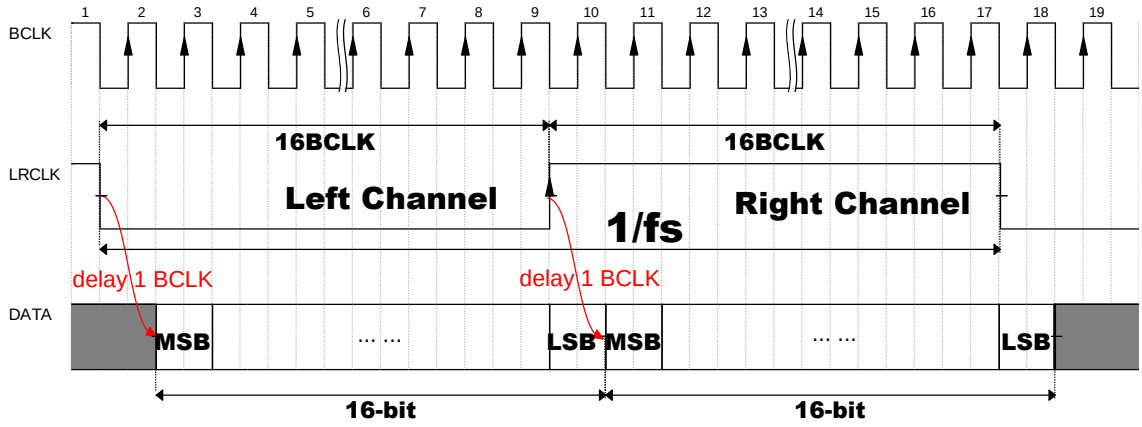



*Figure 13-7 I2S- format -16bit*

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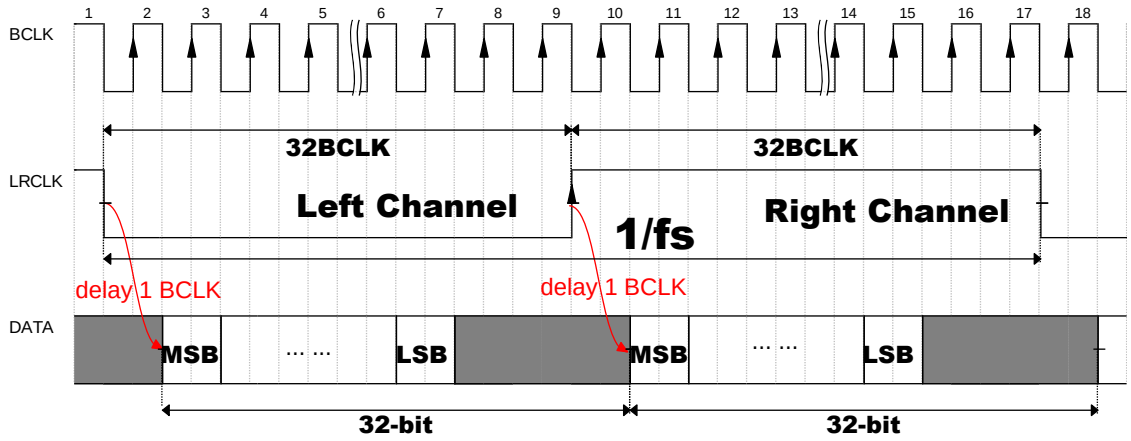



*Figure 13-8 I2S- format -32bit*

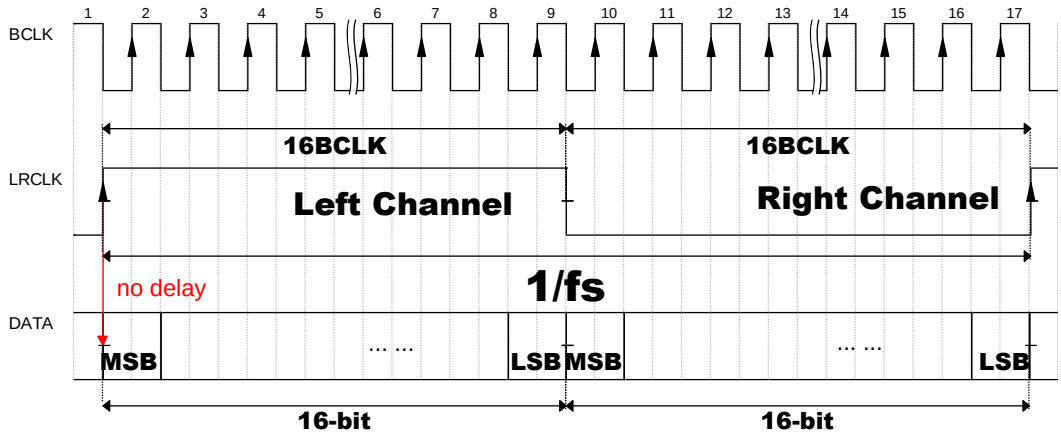



*Figure 13-9 Left justified - format - 16bit*

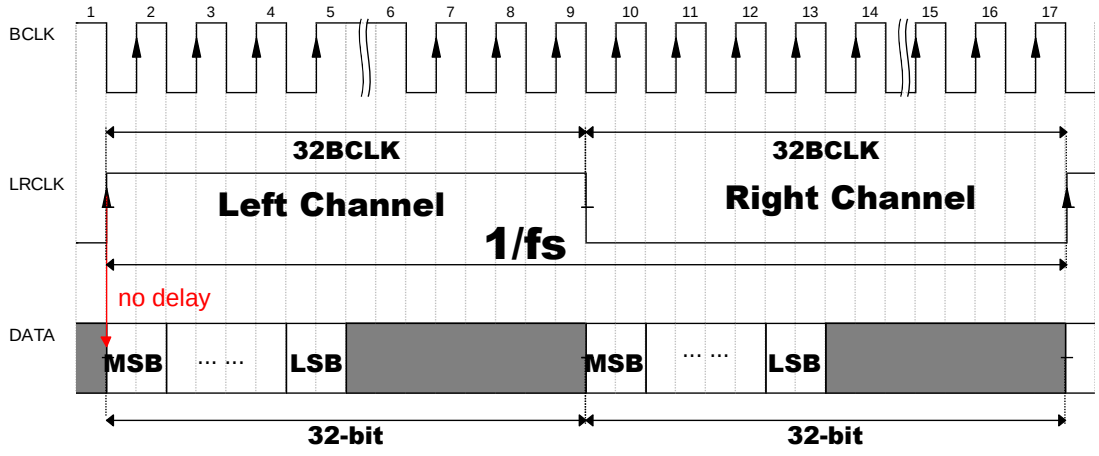



*Figure 13-10 Left justified - format - 32bit*

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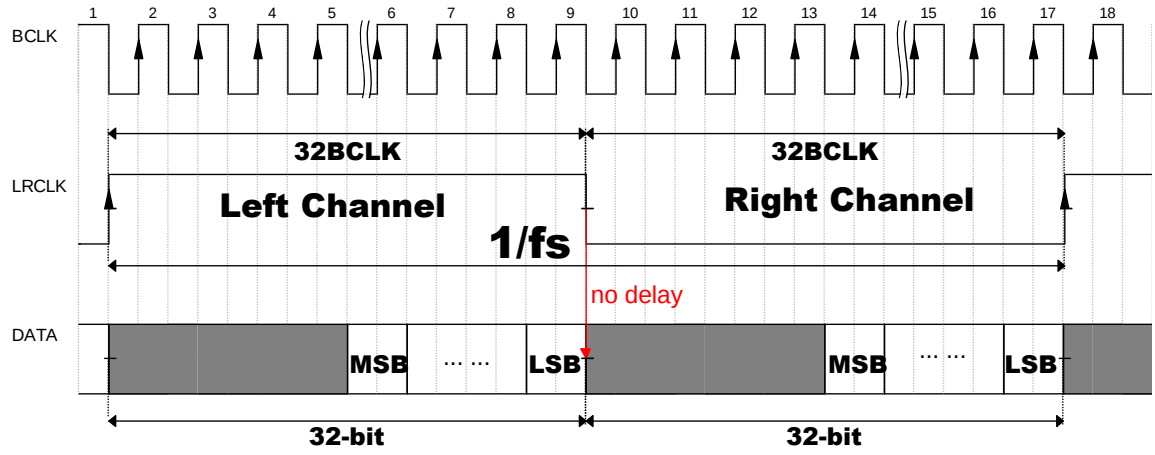



*Figure 13-11 Right justified - format - 32bit*

## 13.3.2 Timing Parameter

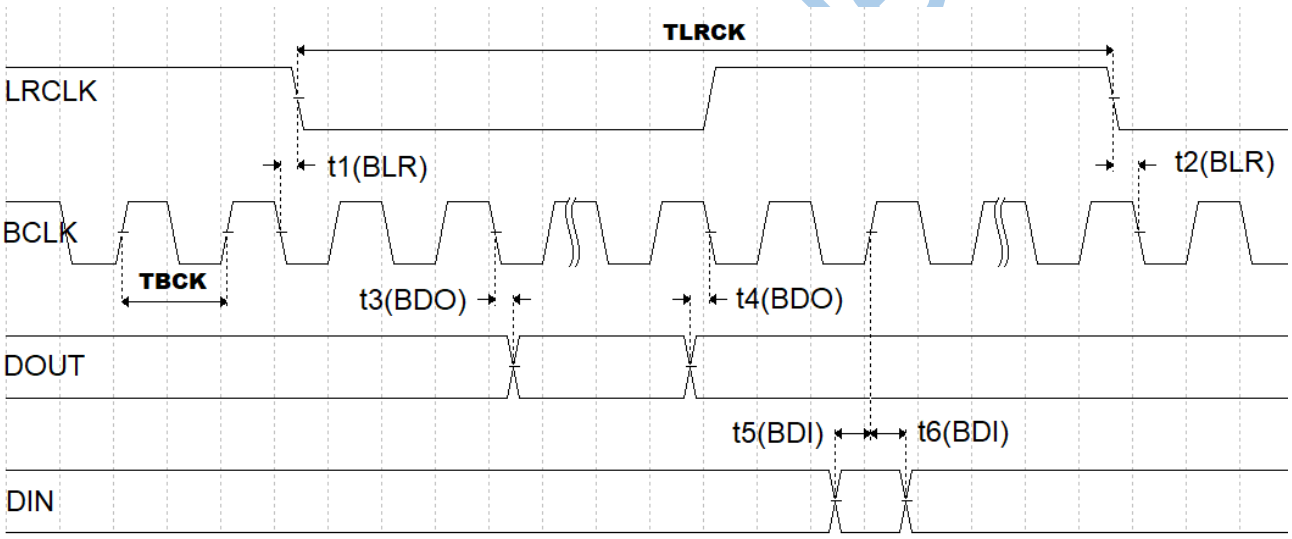



*Figure 13-12 I2S Timing*

*Table 13-8 I2S Timing parameter in Host Mode*

| SYMBOL | PARAMETER | MIN | TYP | MAX | UINT |
|---|---|---|---|---|---|
| TLRCK | LRCLK Period | 1/fs*0.98 | 1/fs | 1/fs*1.02 | ns |
| TBCK | BCLK Period | 1/(64fs)*0.98 | 1/(64fs) | 1/(64fs)*1.02 | ns |
| TBCKH | BCLK High Pulse Duration | | 50% TBCK | | ns |
| TBCKL | BCLK Low Pulse Duration | | 50% TBCK | | ns |
| t1(BLR) | BCLK falling edge ahead LRCLK falling edge time | | | 6 | ns |
| t2(BLR) | BCLK falling edge delay LRCLK falling edge time | | | 6 | ns |
| t3(BDO) | BCLK falling edge ahead data vaild edge time | | | 6 | ns |
| t4(BDO) | BCLK falling edge delay data vaild edge time | | | 6 | ns |
| t5(BDI) | DIN setup time | 16 | | | |
| t6(BDI) | DIN hold time | 12 | | | |

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Note: TA = 25° C, vdd=1.1V

*Table 13-9 I2S Timing parameter in* Slave Mode

| SYMBOL | PARAMETER | MIN | TYP | MAX | UINT |
|---|---|---|---|---|---|
| TLRCK | LRCLK Period | 1/fs*0.98 | | 1/fs*1.02 | ns |
| TBCK | BCLK Period | 1/(64fs)*0.98 | | 1/(64fs)*1.02 | ns |
| TBCKH | BCLK High Pulse Duration | | 50% TBCK | | ns |
| TBCKL | BCLK Low Pulse Duration | | 50% TBCK | | ns |
| t1(BLR) | BCLK falling edge ahead LRCLK falling edge time | | | 15 | ns |
| t2(BLR) | BCLK falling edge delay LRCLK falling edge time | | | 15 | ns |
| t3(BDO) | BCLK falling edge ahead data vaild edge time | 12 | | | ns |
| t5(BDI) | DIN setup time | 12 | | | |
| t6(BDI) | DIN hold time | 12 | | | |

Note: TA = 25° C, vdd=1.1V

## 13.3.3 I2STX Register List

*Table 13-10 I2STX Controller Registers Address*

| Name | Physical Base Address | KSEG1 Base Address |
|---|---|---|
| I2STX_Register | 0x4005C400 | 0x4005C400 |

*Table 13-11 I2STX Controller Registers*

| Offset | Register Name | Description |
|---|---|---|
| 0x0000 | I2STX0_CTL | I2STX0 Control Register |
| 0x0004 | I2STX0_FIFOCTL | I2STX0 FIFO control register |
| 0x0008 | I2STX0_FIFOSTAT | I2STX0 FIFO status register |
| 0x000C | I2STX0_DAT | I2STX0 FIFO data register |
| 0x0010 | I2STX0_SRDCTL | I2STX0 Slave Mode TX Sample Rate Detect Register |
| 0x0014 | I2STX0_SRDSTA | I2STX0 Slave Mode TX Sample Rate Detect State Register |
| 0x0018 | I2STX0_FIFO_CNT | I2STX0 Out FIFO Counter Register |
| 0x0020 | I2STX1_CTL | I2STX1 Control Register |

## 13.3.4 I2STX Register Description

### 13.3.4.1 I2STX0_CTL

I2STX0 Control Register
Offset = 0x00

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31 | TX_ER_PD | The TX0 Channel alignment error pending:<br>0: no irq pending<br>1: irq pending<br>Write '1' to clean this bit. | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 30 | TX_ER_IE | Transmitting Error interrupt enable:<br>0：disable<br>1：enable | R/W | 0x0 |
| 29:27 | - | Reserved | R | 0x0 |
| 26 | PCM_SYNC | PCM Frame Mode Select<br>0: Short Frame Sync Mode<br>1: Long Frame Sync Mode | R/W | 0x0 |
| 25:24 | PCM_CHAN | How many Channels be transmitted per frame<br>0x0: l Slot.<br>0x1: 2 Slot.<br>0x2: 4 Slot.<br>0x3: 8 Slot. | R/W | 0x0 |
| 23 | - | Reserved for [22:21]TXD_DELAY | R | 0x0 |
| 22:21 | TXD_DELAY | **tx data output at:**<br>0x0: 0.5 bclk cycles ahead of the bclk active edge.<br>0x1: 1.0 bclk cycles ahead of the bclk active edge.<br>0x2: 1.5 bclk cycles ahead of the bclk active edge.<br>0x3: 2.0 bclk cycles ahead of the bclk active edge. | R/W | 0x0 |
| 20 | TDMTX_CHAN | 0:4 channel<br>1:8 channel | R/W | 0x0 |
| 19 | TDMTX_MODE | 0:TDM-A (I2S data format)<br>1:TDM-B (left-justified data format) | R/W | 0x0 |
| 18:17 | TDMTX_SYNC | **TDM Frame Start at:**<br>00: the rising edge of LRCLK with a pulse (duration of one BCLK);<br>01: the rising edge of LRCLK with a 50% duty cycle;<br>10: the falling edge of LRCLK with a 50% duty cycle;<br>11: reserved; | R/W | 0x0 |
| 16 | - | Reserved for MULT_DEVICE. | R/W | 0x0 |
| 15:9 | - | Reserved | R | 0x0 |
| 8 | LPEN0 | I2STX0 and I2SRX0 data loopback enable<br>0：disable<br>1：enable<br>When enable,TX0 send clock and data to RX0. | R/W | 0x0 |
| 7 | - | Reserved | R/W | 0x0 |
| 6 | PCMTX0_EN | PCM TX Enable:<br>0: Disable<br>1: Enable | R/W | 0x0 |
| 5:4 | TXWIDTH | Effective width<br>00: data are 16 bit effective<br>01: data are 20 bit effective<br>10: data are 24 bit effective *****<br>11: reserved | R/W | 0x2 |
| 3 | TXBCLKSET | 0:32bit<br>1:16bit | R/W | 0x0 |
| 2:1 | TXMODELSEL | I2S transfer format select:<br>00: S format<br>01: left-justified format<br>10: right-justified format<br>11: TDM | R/W | 0x0 |
| 0 | TXEN | I2S TX Enable.<br>0: Disable | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 1: Enable | | |

### 13.3.4.2 I2STX0_FIFOCTL

I2STX0 FIFO Control Register
Offset = 0x04

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:8 | - | Reserved | R/W | 0x3 |
| 7 | TXFIFO_DMAWIDTH | I2STXFIFO DMA/DSPDMA transfer width configured<br>0 : 32bit<br>1 : 16bit<br>Match with DMA/DSPDMA width which configured at register DMAx_CTL /AUDDSP_DMAx_CTL. I2STXFIFO need not to support 8bit and 64bit DMA data width. | R/W | 0x0 |
| 6:4 | FIFO_IN_SEL | I2STX0_FIFO Input Select:<br>0x00: CPU<br>0x01: DMA<br>0x02: Reserved<br>0x03: DSP<br>0x04: DSPDMA<br>Others: Reserved | R/W | 0x0 |
| 3 | FIFO_DSPDMA_DEN | I2STX0_FIFO DSPDMA DRQ Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 2 | FIFO_IEN | I2STX0_FIFO Half-Empty IRQ Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 1 | FIFO_DEN | I2STX0_FIFO Half-Empty DRQ Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 0 | FIFO_RST | I2STX0_FIFO Reset.<br>0x0: Reset FIFO<br>0x1: Enable FIFO | R/W | 0x0 |

### 13.3.4.3 I2STX0_FIFOSTA

I2STX0 FIFO State Register
Offset = 0x08

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:9 | - | Reserved | R | 0x0 |
| 8 | TFEP | TX0 FIFO Error Pending Bit.<br>0: No Error<br>1: Error<br>Writing 1 to the bit will clear it | RW | 0x0 |
| 7 | IP | I2STX0_FIFO Half-Empty IRQ Pending Bit.<br>0x0: No IRQ<br>0x1: IRQ | RW | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | Writing 1 to the bit is clear it. | | |
| 6 | TFFU | I2STX0_FIFO Full Flag.<br>0x0: Not Full<br>0x1: Full | R | 0x0 |
| 5:0 | STA | I2STX0_FIFO Status<br>These 6 bits shows how many fifo level not filled (can be written into fifo).<br>If no I2STX0_CLK register would display zero.<br>When read as 4, mean can write 4 samples to fifo. | R | 0x0 |

### 13.3.4.4 I2STX0_DAT

I2STX0 FIFO Data Register
Offset = 0x0c

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:8 | DAT | I2STX_FIFO Data<br>FIFO is 24bit x 32 levels. | W | x |
| 7:0 | - | Reserved | R | 0x0 |

### 13.3.4.5 I2STX0_SRDCTL

I2STX0 Slave Mode Sample Rate Detect Control Register
Offset = 0x10

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:13 | - | Reserved | R | 0x0 |
| 12 | MUTE_EN | If detect sample rate or channel width changing, mute the TX output as 0.<br>0:not mute<br>1:mute<br>**Mute would continue until SRC_PD was clear.** | R/W | 0x0 |
| 11 | TO_IE | sample rate detection **timeout** interrupt enable<br>0:disable<br>1:enable | R/W | 0x0 |
| 10 | SRC_IE | sample rate change detection interrupt enable<br>0:disable<br>1:enable | R/W | 0x0 |
| 9 | - | Reserved | R/W | 0x0 |
| 8 | CHW_IE | **channel width change** interrupt enable<br>0:disable<br>1:enable | R/W | 0x0 |
| 7:6 | - | reserved | R/W | 0x0 |
| 5:4 | CNT_TIM | Slave mode sample rate detect counter period select<br>0:2 LRCLK cycle<br>1:4 LRCLK cycle<br>Other: reserved | R/W | 0x0 |
| 3:1 | SRD_TH | The threshold of sample rate detection.<br>0: 8 | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 1: 16<br>…<br>6: 56<br>7: 64<br>Value=(SRD_TH+1)*8 | | |
| 0 | SRD_EN | Slave mode sample rate detect enable:<br>0:disable<br>1:enable | R/W | 0x0 |

### 13.3.4.6 I2STX0_SRDSTA

I2STX0 Slave Mode Sample Rate Detect State Register
Offset = 0x14

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:25 | - | Reserved | R | 0x0 |
| 24:12 | CNT | CNT of LRCLK which sampling by SRDCLK.<br>CNT= SRDCLK / LRCLK.<br>It would clear when module was enabling at first time. | R | 0x0 |
| 11 | TO_PD | sample rate change detection **timeout** interrupt pending<br>0:no irq pending<br>1:irq pending<br>Write '1' to clean this bit.<br>If CNT overflow (0x9c00), if would cause TIMEOUT irq. | RW | 0x0 |
| 10 | SRC_PD | **sample rate change** detection interrupt pending<br>0:no irq pending<br>1:irq pending<br>Write '1' to clean this bit. | RW | 0x0 |
| 9 | - | Reserved for future use | R | 0x0 |
| 8 | CHW_PD | **channel width change** interrupt pending<br>0:no irq pending<br>1:irq pending<br>Write '1' to clean this bit. | RW | 0x0 |
| 7:3 | - | Reserved | R | 0x0 |
| 2:0 | WL | Channel width length ( the rate of BCLK to LRCLK )：<br>000： 32 rate(I2S: 16*2bit)<br>001： 64 rate(I2S: 32*2bit; TDM4: 16*4bit)<br>010：128 rate(TDM8: 16*8bit; TDM4: 32*4bit)<br>011：256 rate(TDM8: 32*8bit)<br>1xx：Reserved | R | 0x0 |

### 13.3.4.7 I2STX0_FIFO_CNT

I2STX0 FIFO Out Counter Register
Offset = 0x20

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:19 | - | Reserved | | |
| 18 | IP | I2STX FIFO counter overflow irq pending<br>0: no pending<br>1: pending | RW | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | Write '1' to clear this bit | | |
| 17 | IE | I2STX FIFO counter overflow irq enable:<br>0:disable<br>1:enable<br>If CNT overflow, it would cause an interrupt. | RW | 0x0 |
| 16 | EN | I2STX FIFO counter enable<br>0:disable<br>1:enable<br>Disable this function could reset the whole counter. | RW | 0x0 |
| 15:0 | CNT | I2STX FIFO counter<br>If overflow count would be clear to zero and cause an interrupt<br>This counter count the valid data output by I2STX FIFO, it means that if FIFO is empty, this counter would not add till FIFO had been written data in again. | R | 0x0 |

### 13.3.4.8 I2STX1_CTL

I2STx1 Control Register
Offset = 0x20

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:27 | - | Reserved | R/W | 0x0 |
| 26 | PCM_SYNC | PCM Frame Mode Select<br>0: Short Frame Sync Mode<br>1: Long Frame Sync Mode | R/W | 0x0 |
| 25:24 | PCM_CHAN | How many Channels be transmitted per frame<br>0x0: l Slot.<br>0x1: 2 Slot.<br>0x2: 4 Slot.<br>0x3: 8 Slot. | R/W | 0x1 |
| 23:21 | - | Reserved for TXD_DELAY | R/W | 0x0 |
| 20 | - | Reserved for TDMTX_CHAN | R/W | 0x0 |
| 19 | - | Reserved for TDMTX_MODE | R/W | 0x0 |
| 18:17 | - | Reserved for TDMTX_SYNC | R/W | 0x0 |
| 16 | - | Reserved for MULT_DEVICE. | R/W | 0x0 |
| 15:10 | - | Reserved | R | 0x0 |
| 9 | TX_MULT | I2STX1 and I2STX0 Multi-Link<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 8 | LPEN1 | I2STX1 and I2SRX0 data loopback enable<br>0：disable<br>1：enable<br>When enable,TX1 send clock and data to RX0. | R/W | 0x0 |
| 7 | - | Reserved | R/W | 0x0 |
| 6 | PCMTX_EN | PCM TX Enable:<br>0: Disable<br>1: Enable | R/W | 0x0 |
| 5:4 | TXWIDTH | Effective width<br>00: data are 16 bit effective | R/W | 0x2 |

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| | | | | |
|---|---|---|---|---|
| | | 01: data are 20 bit effective<br>10: data are 24 bit effective *****<br>11: reserved | | |
| 3 | TXBCLKSET | 0:32bit<br>1:16bit | R/W | 0x0 |
| 2:1 | TXMODELSEL | I2S transfer format select:<br>00: I2S format<br>01: left-justified format<br>10: right-justified format<br>11: TDM | R/W | 0x0 |
| 0 | TXEN | I2S TX Enable.<br>0: Disable<br>1: Enable | R/W | 0x0 |

## 13.3.5 I2SRX Register List

*Table 13-12 I2SRX Controller Registers Address*

| Name | Physical Base Address | KSEG1 Base Address |
|---|---|---|
| I2SRX_Register | 0x4005C500 | 0x4005C500 |

*Table 13-13 I2SRX Controller Registers*

| Offset | Register Name | Description |
|---|---|---|
| 0x0000 | I2SRX_CTL | I2SRX Control Register |
| 0x0004 | I2SRX_FIFOCTL | I2SRX FIFO Control Register |
| 0x0008 | I2SRX_FIFOSTA | I2SRX FIFO State Register |
| 0x000C | I2SRX_DAT | I2SRX FIFO Data Register |
| 0x0010 | I2SRX_SRDCTL | I2SRX Slave Mode Sample Rate Detect Control Register |
| 0x0014 | I2SRX_SRDSTA | I2SRX Slave Mode Sample Rate Detect State Register |

## 13.3.6 I2SRX Register Description

### 13.3.6.1 I2SRX_CTL

I2SRX Control Register
Offset = 0x00

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31 | RX_ER_PD | The RX0 Channel alignment error pending:<br>0: no irq pending<br>1: irq pending<br>Write '1' to clean this bit. | R/W | 0x0 |
| 30 | RX_ER_IE | Receive Error interrupt enable:<br>0：disable<br>1：enable | R/W | 0x0 |
| 29:27 | - | Reserved | R | 0x0 |
| 26 | PCM_SYNC | PCM Frame Mode Select<br>0: Short Frame Sync Mode<br>1: Long Frame Sync Mode | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 25:24 | PCM_CHAN | How many Channels be transmitted per frame<br>0x0: l slot.<br>0x1: 2 slot.<br>0x2: 4 slot.<br>0x3: 8 slot. | R/W | 0x0 |
| 23:17 | - | Reserved | R | 0x0 |
| 16 | TDMRX_CHAN | 0:4 channel<br>1:8 channel | R/W | 0x0 |
| 15 | TDMRX_MODE | 0:TDM-A (I2S format)<br>1:TDM-B (left-justified format) | R/W | 0x0 |
| 14:13 | TDMRX_SYNC | at the beginning of the data frame, the type of LRCLK set:<br>00: at the rising edge of LRCLK with a pulse(duration of one BCLK)<br>01: at the rising edge of LRCLK with a 50% duty cycle;<br>10: at the falling edge of LRCLK with a 50% duty cycle;<br>11: Reserved | R/W | 0x0 |
| 12:7 | - | Reserved | R | 0x0 |
| 6 | PCMRX_EN | PCM RX Enable:<br>0: Disable<br>1: Enable | R/W | 0x0 |
| 5:4 | RXWIDTH | Effective width<br>00: data are 16 bit effective<br>01: data are 20 bit effective<br>10: data are 24 bit effective<br>11: reserved | R/W | 0x2 |
| 3 | RXBCLKSET | 0: 32bit<br>1: 16bit | R/W | 0x0 |
| 2:1 | RXMODELSEL | I2S transfer format select:<br>00 : S format<br>01: left-justified format<br>10: right-justified format<br>11: TDM | R/W | 0x0 |
| 0 | RXEN | I2S RX Enable.<br>0: Disable<br>1: Enable | R/W | 0x0 |

### 13.3.6.2 I2SRX_FIFOCTL

I2SRX FIFO Control Register
Offset = 0x04

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:8 | - | Reserved | R | 0x0 |
| 7 | RXFIFO_<br>DMAWIDTH | I2SRXFIFO DMA transfer width configured<br>0 : 32bit<br>1 : 16bit<br>Set this configure match width DMA width, I2SRXFIFO not need to support 8bit and 64bit DMA data width. | RW | 0x0 |
| 6:4 | RXFOS | I2SRX FIFO Output Select:<br>0x00: CPU<br>0x01: DMA | RW | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0x03: DSP<br>0x04: DSPDMA<br>Others: Reserved | | |
| 3 | RXFF_DSPDMA_DEN | I2SRX0_FIFO DSPDMA DRQ Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 2 | RXFFIE | I2SRX FIFO Half-Filled IRQ Enable.<br>0: Disable<br>1: Enable | RW | 0x0 |
| 1 | RXFFDE | I2SRX FIFO Half-Filled DRQ Enable.<br>0x0: Disable<br>0x1: Enable | RW | 0x0 |
| 0 | RXFRT | I2SRX FIFO Reset.<br>0x0: Reset FIFO<br>0x1: Enable FIFO | RW | 0x0 |

### 13.3.6.3 I2SRX_FIFOSTA

I2SRX FIFO State Register
Offset = 0x08

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:9 | - | Reserved | R | 0x0 |
| 8 | RFEP | RX FIFO Error Pending Bit.<br>0: No Error<br>1: Error<br>Writing 1 to the bit will clear it | RW | 0x0 |
| 7 | RXFEF | I2SRX FIFO Empty Flag.<br>0x0: Not Empty<br>0x1: Empty<br>Can be accessed by DSP. | R | 0x1 |
| 6 | RXFIP | I2SRX FIFO Half-Filled IRQ Pending Bit.<br>0x0: No IRQ<br>0x1: IRQ<br>Writing 1 to the bit is clear it.<br>Can be accessed by DSP. | RW | 0x0 |
| 5:0 | RXFS | I2SRX FIFO Status.<br>This 6 bits shows how many samples fifo filled.<br>when read as 6, means can read 6 samples from fifo.<br>If no I2SRX_CLK register would display zero. | R | 0x0 |

### 13.3.6.4 I2SRX_DAT

I2SRX FIFO Data Register
Offset = 0x0C

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:8 | RXDAT | RX Data<br>FIFO is 24bit x 32 levels. | R | x |
| 7:0 | - | Reserved | R | 0x0 |

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### 13.3.6.5 I2SRX_SRDCTL

I2SRX Slave Mode Sample Rate Detect Control Register
Offset = 0x10

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:13 | - | Reserved | R | 0x0 |
| 12 | MUTE_EN | If detect sample rate or channel width changing, mute the TX output as 0.<br>0:not mute<br>1:mute<br>**Mute would continue until SRC_PD was clear.** | R/W | 0x0 |
| 11 | TO_IE | sample rate detection **timeout** interrupt enable<br>0:disable<br>1:enable | R/W | 0x0 |
| 10 | SRD_IE | **sample rate change** detection interrupt enable<br>0:disable<br>1:enable | R/W | 0x0 |
| 9 | - | Reserved | R/W | 0x0 |
| 8 | CHW_IE | **channel width change** interrupt enable<br>0:disable<br>1:enable | R/W | 0x0 |
| 7:6 | - | Reserved | R/W | 0x0 |
| 5:4 | CNT_TIM | Slave mode sample rate detect counter period select<br>0:2 LRCLK cycle<br>1:4 LRCLK cycle<br>Other: reserved | R/W | 0x0 |
| 3:1 | SRD_TH | The threshold of sample rate detection.<br>0: 8<br>1: 16<br>…<br>6: 56<br>7: 64<br>Value=(SRD_TH+1)*8 | R/W | 0x0 |
| 0 | SRD_EN | Slave mode sample rate detect enable:<br>0:disable<br>1:enable | R/W | 0x0 |

### 13.3.6.6 I2SRX_SRDSTA

I2SRX Slave Mode Sample Rate Detect State Register
Offset = 0x14

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:25 | - | Reserved | R | 0x0 |
| 24:12 | CNT | CNT of LRCLK which sampling by SRDCLK.<br>CNT= SRDCLK / LRCLK.<br>It would clear when module was enabling at first time. | R | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 11 | TO_PD | sample rate changing detection timeout interrupt pending<br>0:no irq pending<br>1:irq pending<br>Write '1' to clean this bit.<br>If CNT overflow (0x9c00), if would cause TIMEOUT irq. | RW | 0x0 |
| 10 | SRC_PD | sample rate changing detection interrupt pending<br>0:no irq pending<br>1:irq pending<br>Write '1' to clean this bit. | RW | 0x0 |
| 9 | - | Reserved for future use | R | 0x0 |
| 8 | CHW_PD | channel width change interrupt pending<br>0:no irq pending<br>1:irq pending<br>Write '1' to clean this bit. | RW | 0x0 |
| 7:3 | - | Reserved | R | 0x0 |
| 2:0 | WL | Channel width length ( the rate of BCLK to LRCLK )：<br>000： 32 rate(I2S: 16*2bit)<br>001： 64 rate(I2S: 32*2bit; TDM4: 16*4bit)<br>010：128 rate(TDM8: 16*8bit; TDM4: 32*4bit)<br>011：256 rate(TDM8: 32*8bit)<br>1xx：Reserved | R | 0x0 |

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# 14 Display

## 14.1 LCD Controller (LCDC)

### 14.1.1 Features

- Support RGB888、RGB666、RGB565、ARGB Source Data Format
- 64*32 level FIFO with DRQ and IRQ
- Support MCU 8bit 8080/6800 interface
- Support (TFT) LCD panels with 3-line 1/2 data lane serial interface optionI/II interface
- Support (TFT) LCD panels with 4-line 1/2 data lane serial interface optionI/II interface
- Support (TFT) LCD panels with quad serial interface(SDR/DDR/SYNC)
- Support TR-J/TR-S parallel HSYNC/VSYNC LCD interface
- Support DMA for Display

### 14.1.2 MCU Interface

- MCU IF support Moto 6800 and Intel 8080

#### 14.1.2.1 MCU Interface signal

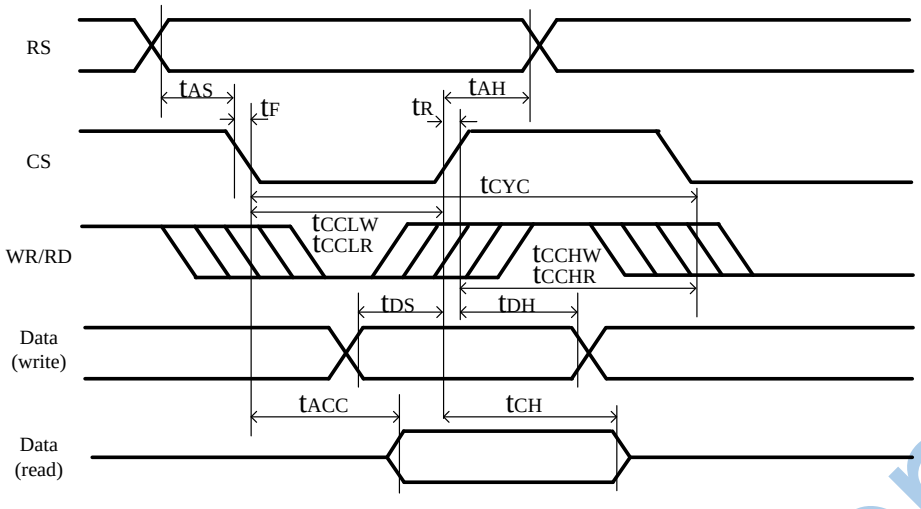
***Table 14-1 MCU Interface signal***

| Signal | Type | Description |
|---|---|---|
| D7~D0 | I/O | Data bus |
| E/RD# | O | M6800: Strobe signal<br>I8080：Reads strobe signal |
| WR# | O | M6800: 1. Read，0. Write.<br>I8080：Writes strobe signal |
| RS | O | Command/Data selection，0：Command，1：Data |
| RST# | O | Reset |
| CS1~0# | O | Chip select |
| TE | I | Tearing effect output pin to synchronize MCU to frame writing |

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### 14.1.2.2 MCU Interface timing




| RS | RD | WR | Function |
|---|---|---|---|
| 0 | 0 | 1 | Read Status |
| 0 | 1 | 0 | Set Index Register |
| 1 | 0 | 1 | Read Display Data |
| 1 | 1 | 0 | Write Display Data |

*Figure 14-1 I8080 interface control signal Timing*

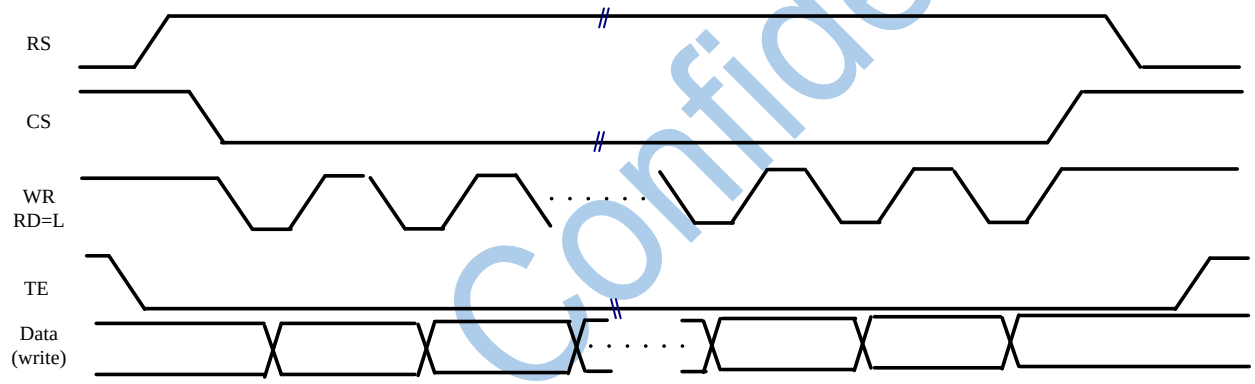
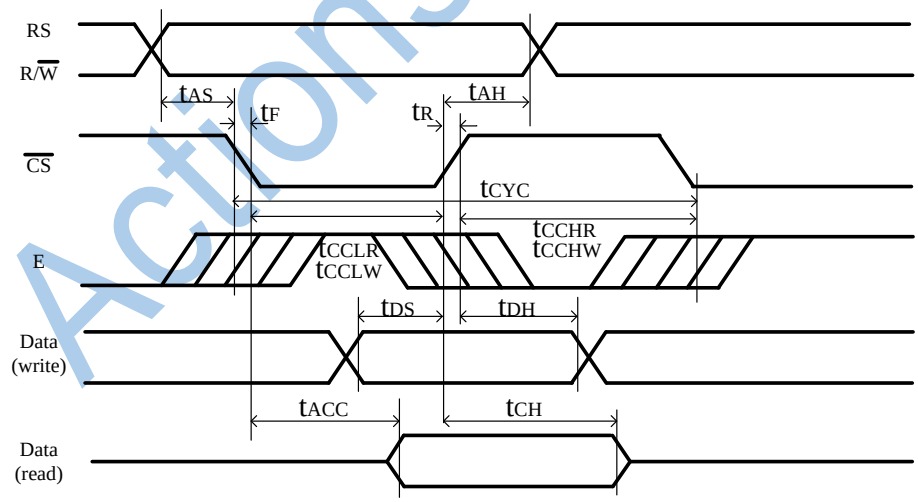



*Figure 14-2 I8080 interface image write Timing*




| RS | R/$\overline{W}$ | Function |
|---|---|---|
| 0 | 1 | Read Status |
| 0 | 0 | Set Index Register |
| 1 | 0 | Write Display Data |
| 1 | 1 | Read Display Data |

*Figure 14-3 M6800 interface control signal Timing*

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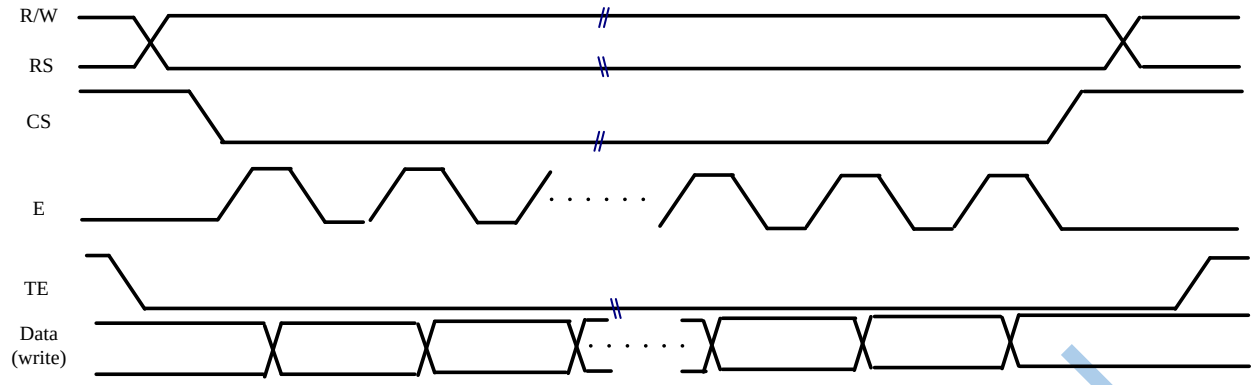



*Figure 14-4 M6800 interface image write Timing*

## 14.1.3 SPI Interface

3-lines and 4-line and Quad serial interfaces for communication between host and Lcd Driver are supported.
The Quad serial mode consists of the chip enable input (CSX), the serial clock input (SCL) and serial data Input/Output (SDI_RDX), input(DCX,DB[0],DB[1]).
The 3-line serial mode consists of the chip enable input (CSX), the serial clock input (SCL) and serial data Input (SDI_RDX) and serial data output (SDO).
The 4-line serial mode consists of the Data/ Command selection input (D/CX), chip enable input (CSX), the serial clock input (SCL) and serial data Input(SDI_RDX) and Output (SDO) for data transmission.

### 14.1.3.1 SPI Interface signal

*Table 14-2 3-line serial interface option I*

| Signal | Type | Description |
|---|---|---|
| CSX | O | Chip Select Signal |
| DCX | O | Clock Signal |
| SDA | I/O | Data in/out |
| TE | I | Tearing effort |

*Table 14-3 3-line serial interface option II*

| Signal | Type | Description |
|---|---|---|
| CSX | O | Chip Select Signal |
| DCX | O | Clock Signal |
| SDA | O | Data Output |
| SDO | I | Data Input |
| TE | I | Tearing effort |

*Table 14-4 4-line serial interface option I*

| Signal | Type | Description |
|---|---|---|
| CSX | O | Chip Select Signal |
| WRX | O | Data is regarded as a command when WRX is low<br>Data is regarded as a parameter or data when WRX is high |
| DCX | O | Clock Signal |
| SDA | I/O | Data In/Out |
| TE | I | Tearing effort |

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*Table 14-5 4-line serial interface option II*

| Signal | Type | Description |
|---|---|---|
| CSX | O | Chip Select Signal |
| WRX | O | Data is regarded as a command when WRX is low<br>Data is regarded as a parameter or data when WRX is high |
| DCX | O | Clock Signal |
| SDA | O | Data Output |
| SDO | I | Data Input |

*Table 14-6 Quad serial interface*

| Signal | Type | Description |
|---|---|---|
| CSX | O | Chip Select Signal |
| WRX_SCL | O | CLK |
| SDI_RDX | I/O | Data Input/Output |
| DCX | O | Data Output |
| DB[0] | O | Data Output |
| DB[1] | O | Data Output |

### 14.1.3.2 SPI Interface Command Write timing

#### 3-Line Serial Interface

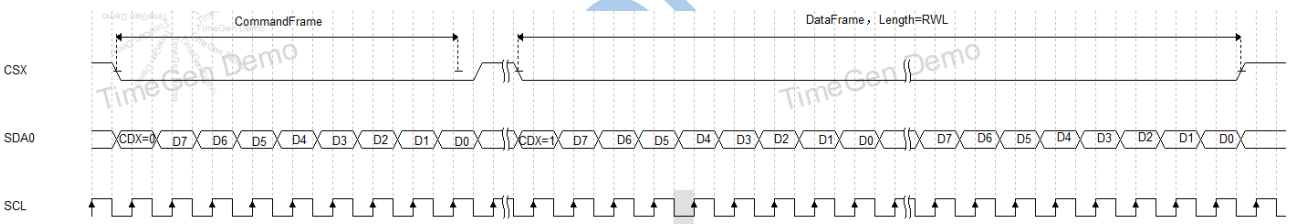



*Figure 14-5 3-line serial interface command write*

#### 4-Line Serial Interface

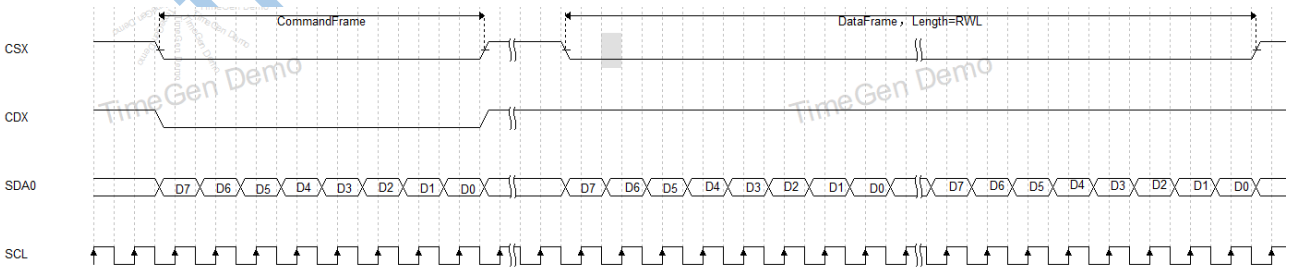



*Figure 14-6 4-line serial interface command write*

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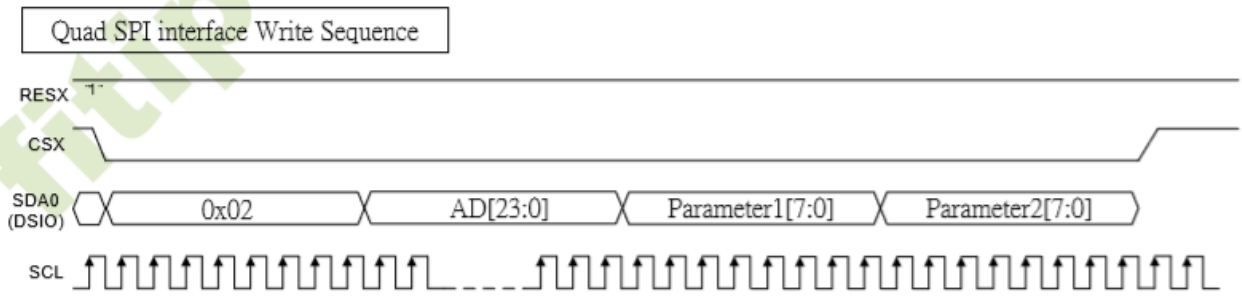



*Figure 14-7 Quad serial interface command write*

#### 14.1.3.3 SPI Interface Read Timing

**3-Line Serial Interface I/II**

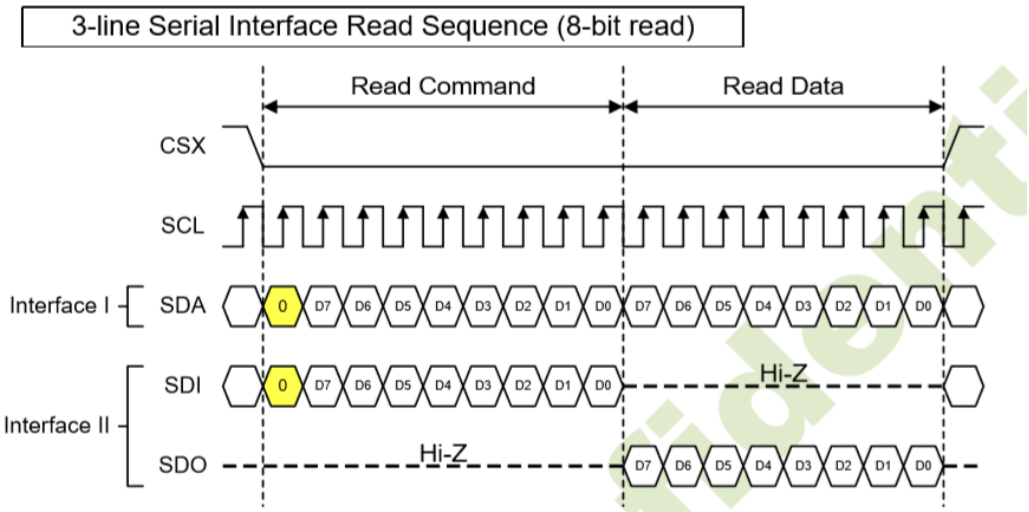



*Figure 14-8 3-line serial read timing (8-bit read)*

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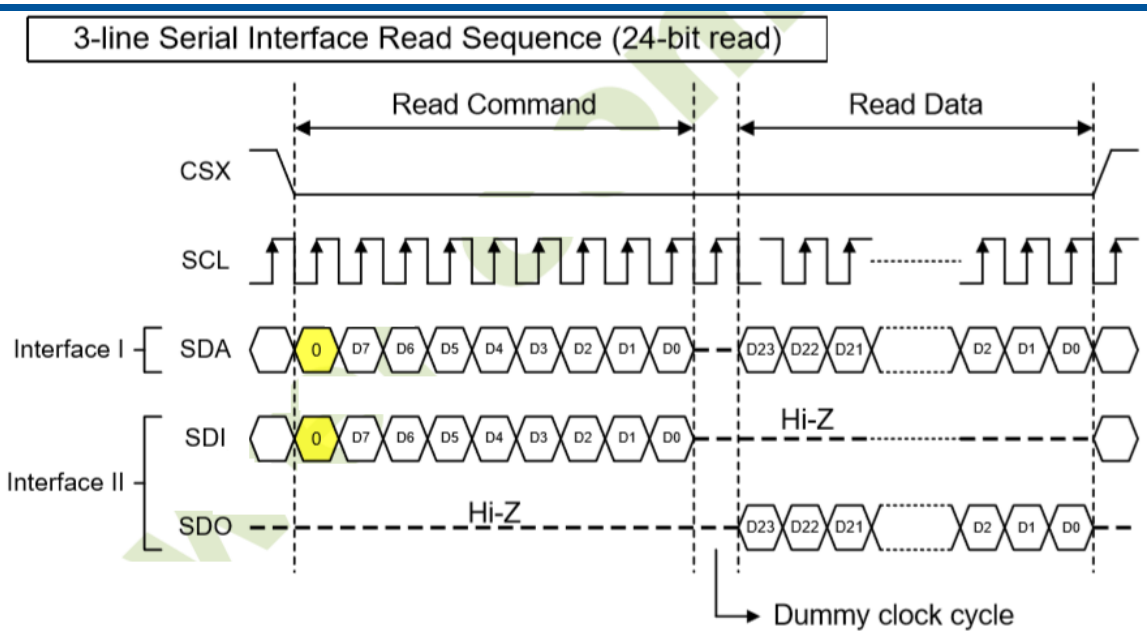



*Figure 14-9 3-line serial read timing (24-bit read)*

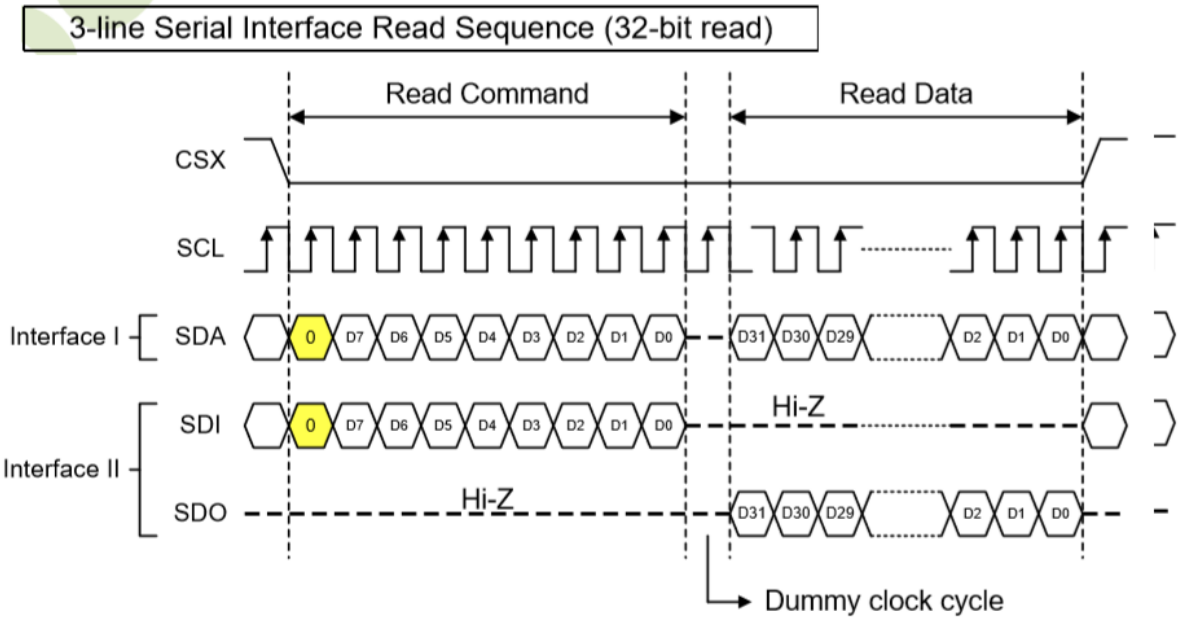



*Figure 14-10 3-line serial read timing (32-bit read)*

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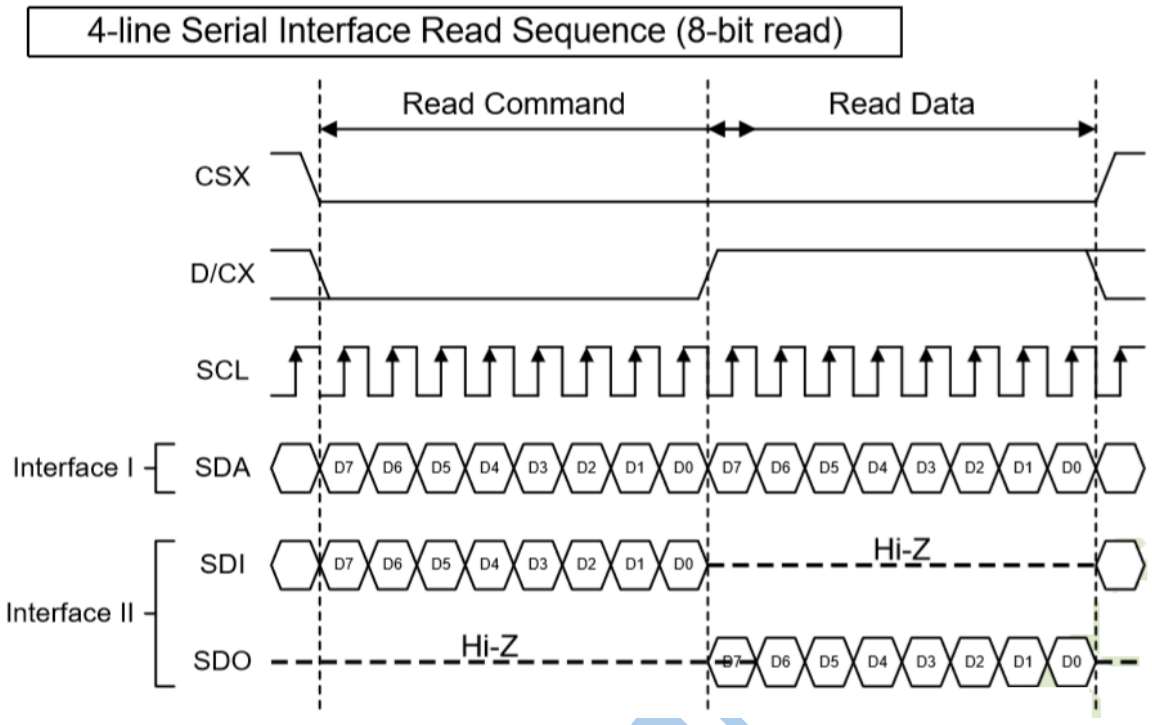



*Figure 14-11 4-line serial read timing (8-bit read)*

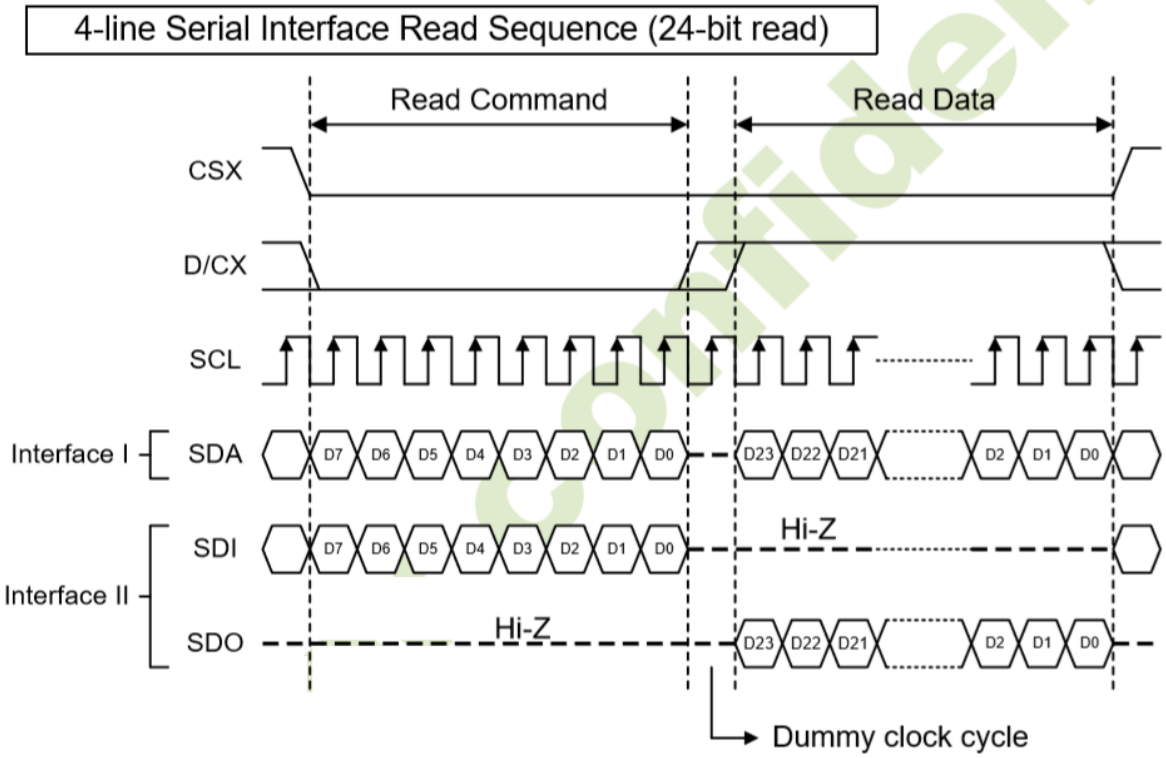



*Figure 14-12 4-line serial read timing (24-bit read)*

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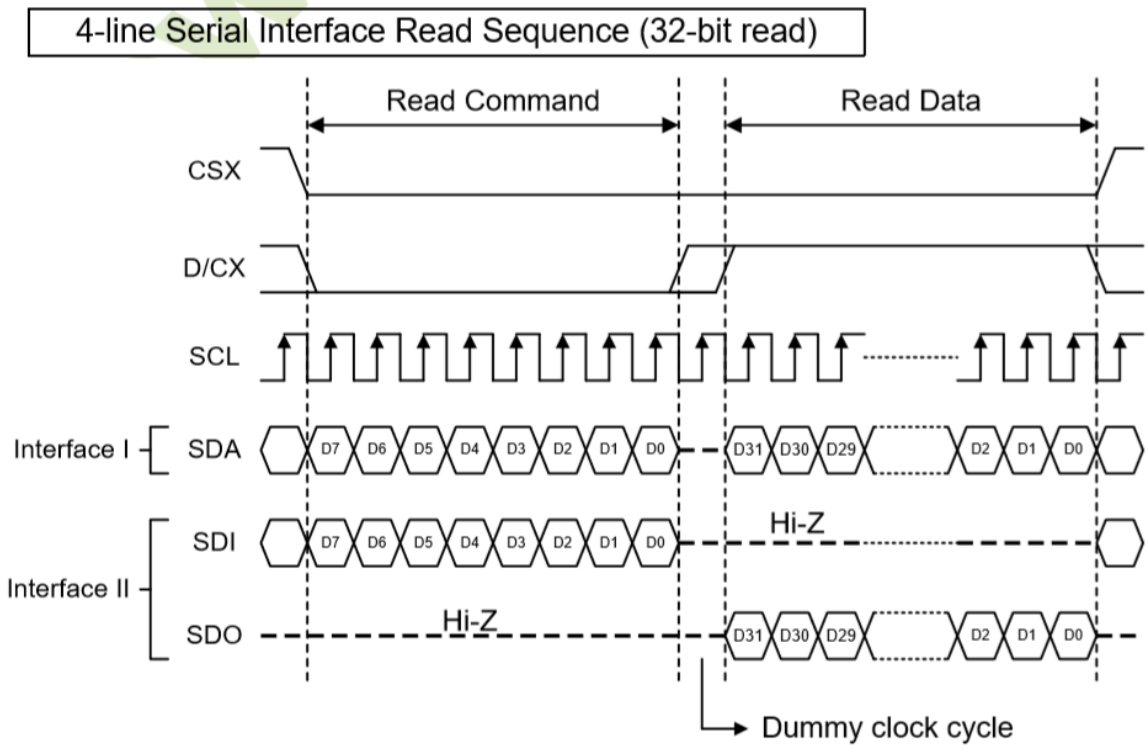



*Figure 14-13 4-line serial read timing (for RDDST command: 32-bit read)*

### Quad Serial Interface

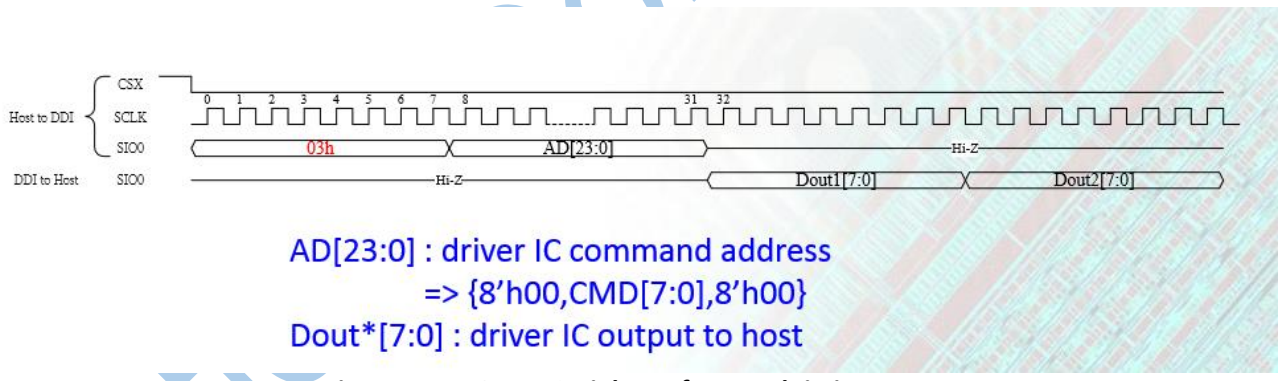


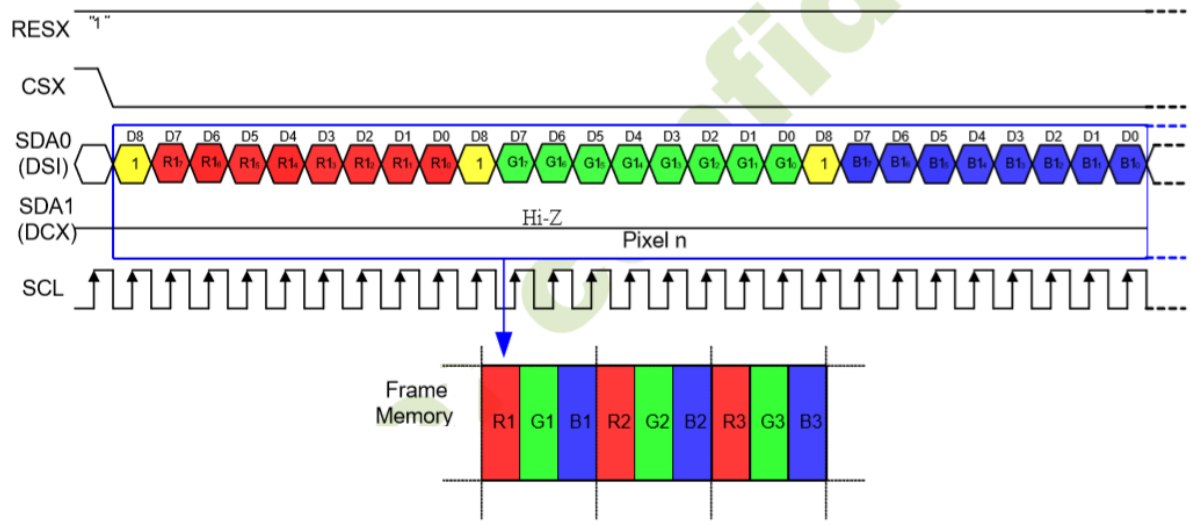

*Figure 14-14 QUAD Serial Interface read timing*

## 14.1.3.4 SPI Interface Image Write timing

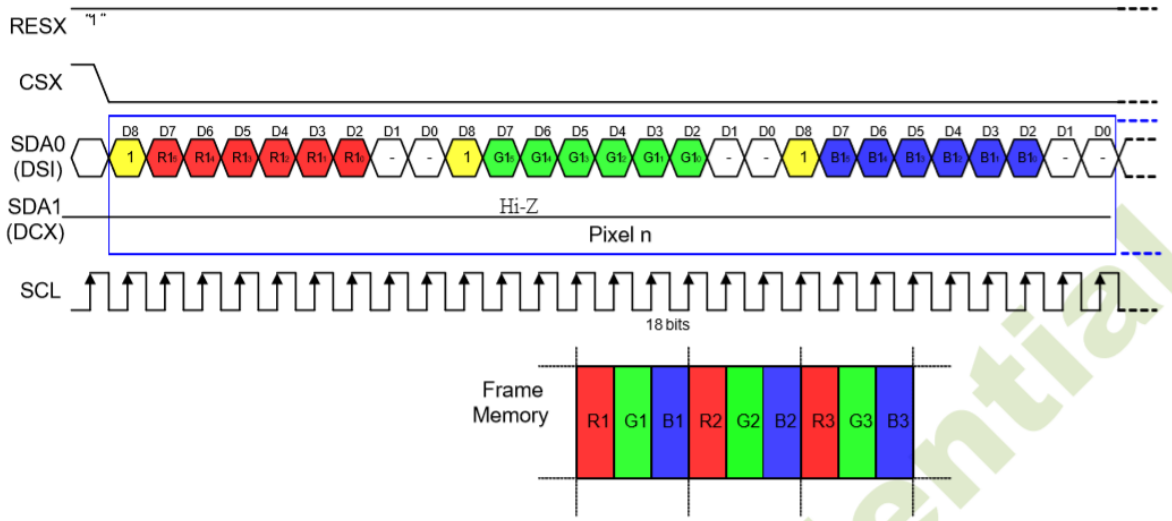
### 3-line Serial Interface

Different display data formats are available for three colors depth supported by the LCM listed below:
(1) RGB888； CDX=1； AHB_CFG=0； DUAL_LANE=0； DCP_MODE=0

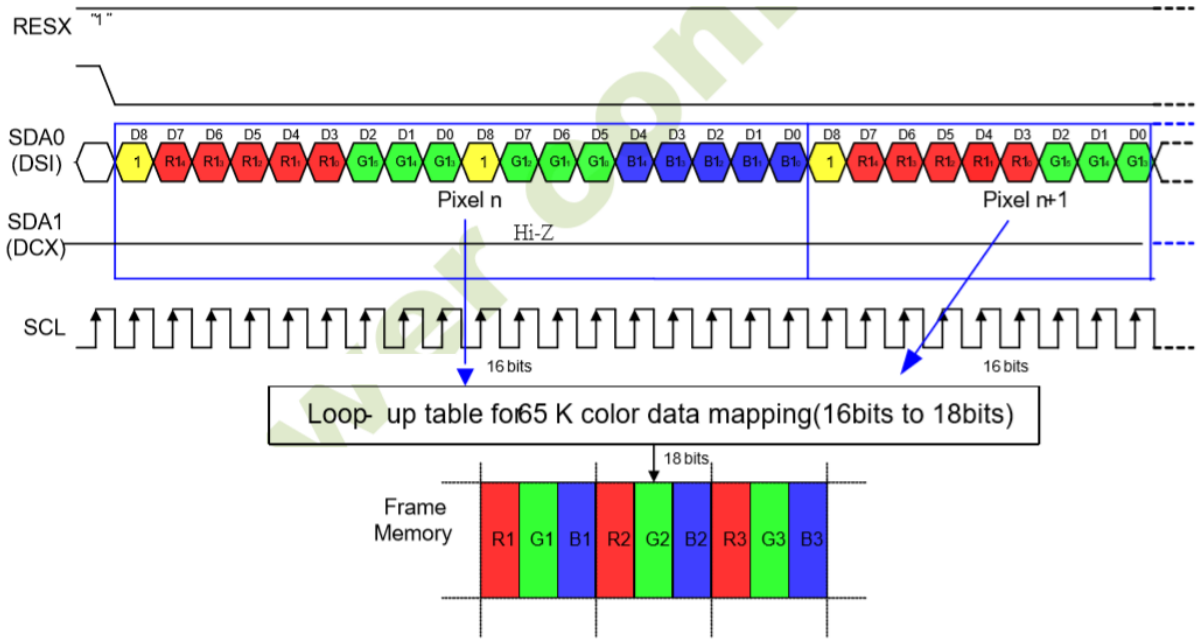
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(2)RGB666；CDX=1；AHB_CFG=0；DUAL_LANE=0；DCP_MODE=0

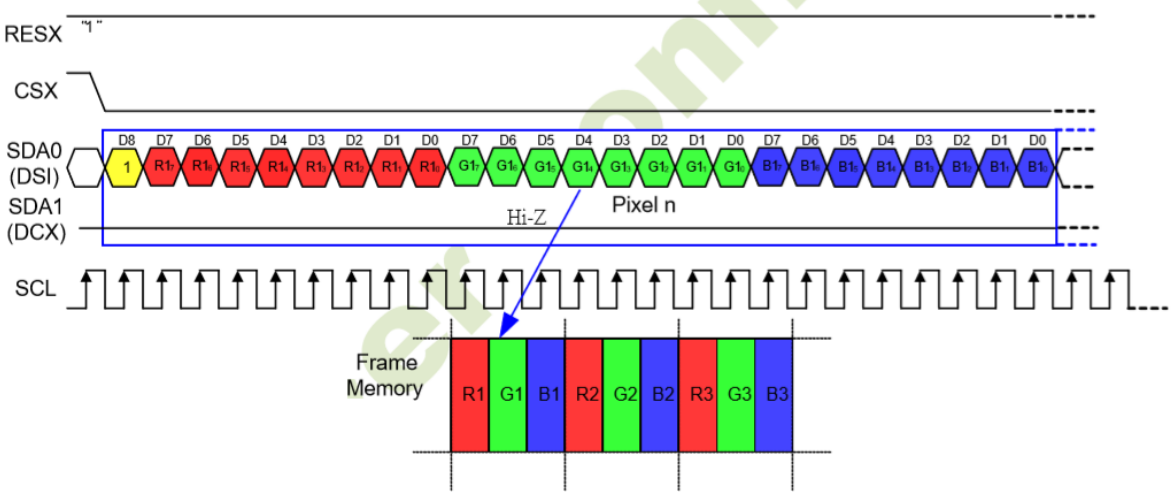



(3) RGB565；CDX=1；AHB_CFG=0；DUAL_LANE=0；DCP_MODE=0

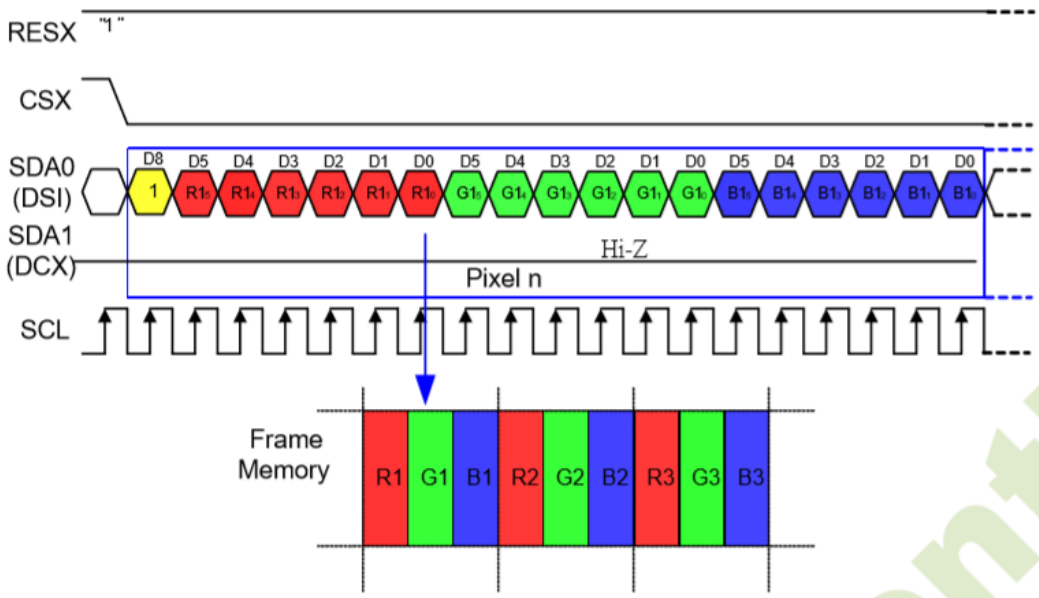



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(4) RGB888；CDX=1；AHB_CFG=0；DUAL_LANE=0；DCP_MODE=1

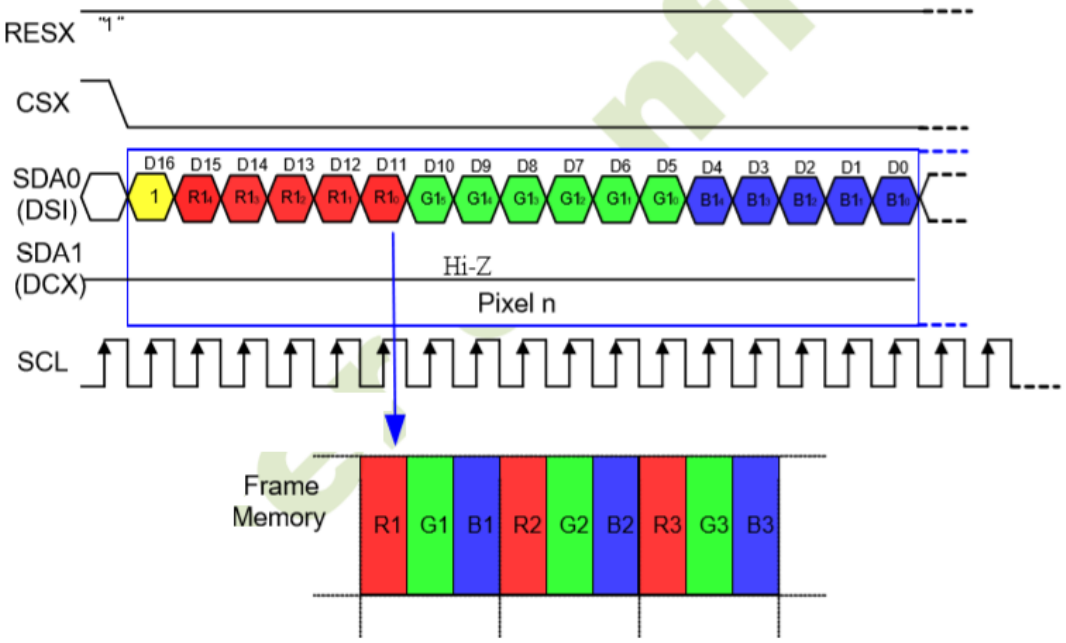



(5) RGB666；CDX=1；AHB_CFG=0；DUAL_LANE=0；DCP_MODE=1



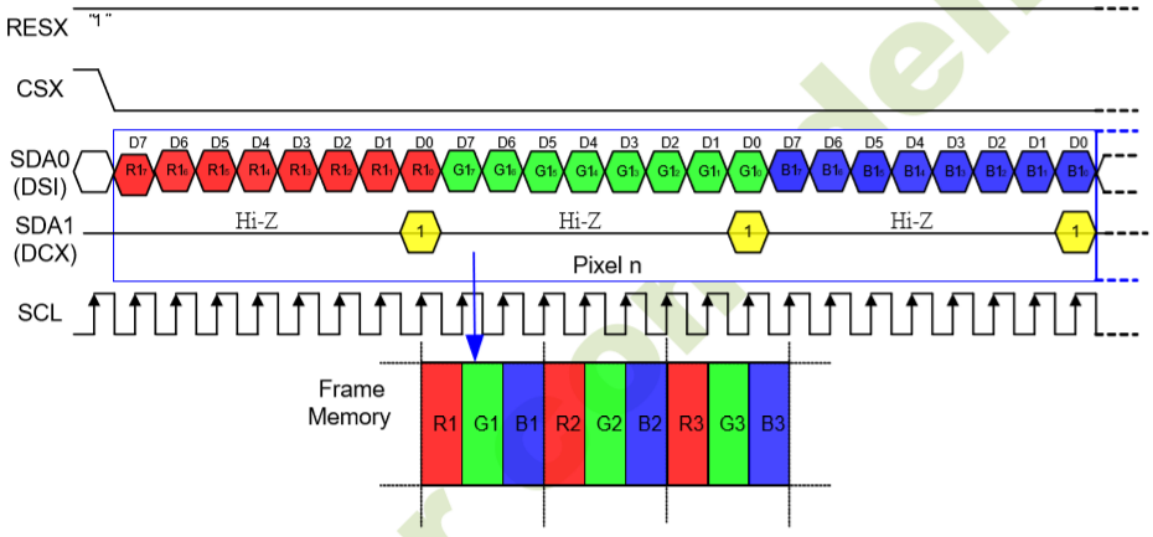

(6) RGB565；CDX=1；AHB_CFG=0；DUAL_LANE=0；DCP_MODE=1

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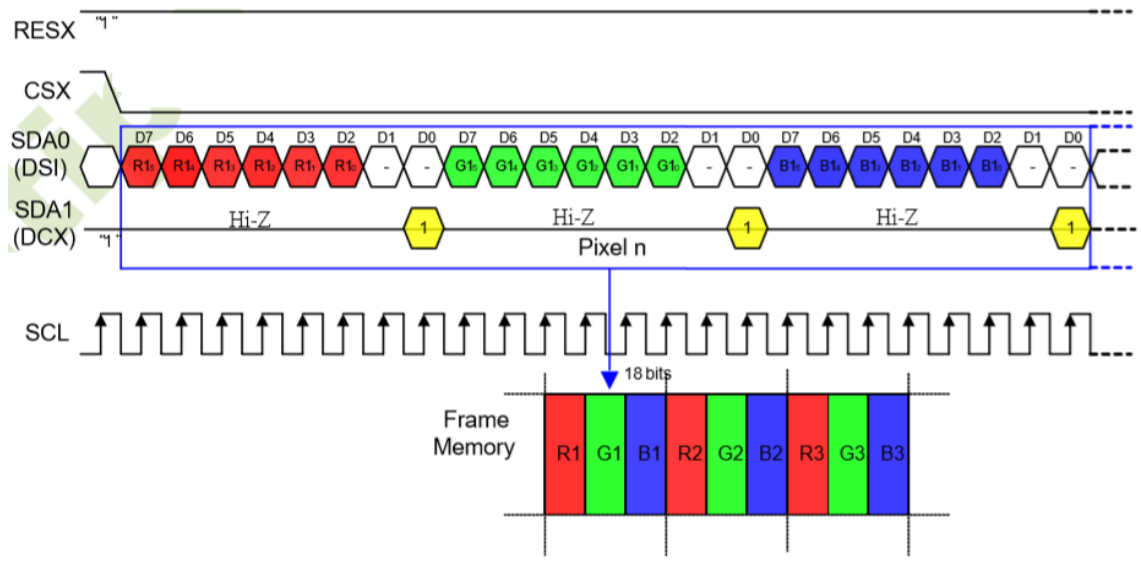

## 4-line Serial Interface

Different display data formats are available for three colors depth supported by the LCM listed below:
(1)RGB888； CDX=1； AHB_CFG=0； DUAL_LANE=0； DCP_MODE=0

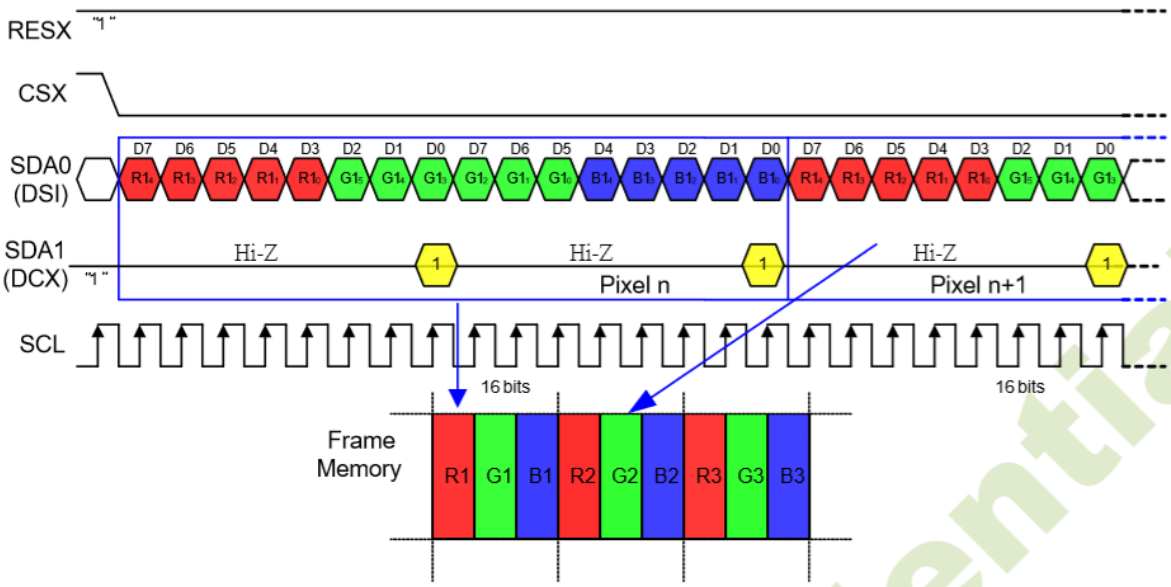



(2) RGB666； CDX=1； AHB_CFG=0； DUAL_LANE=0； DCP_MODE=0

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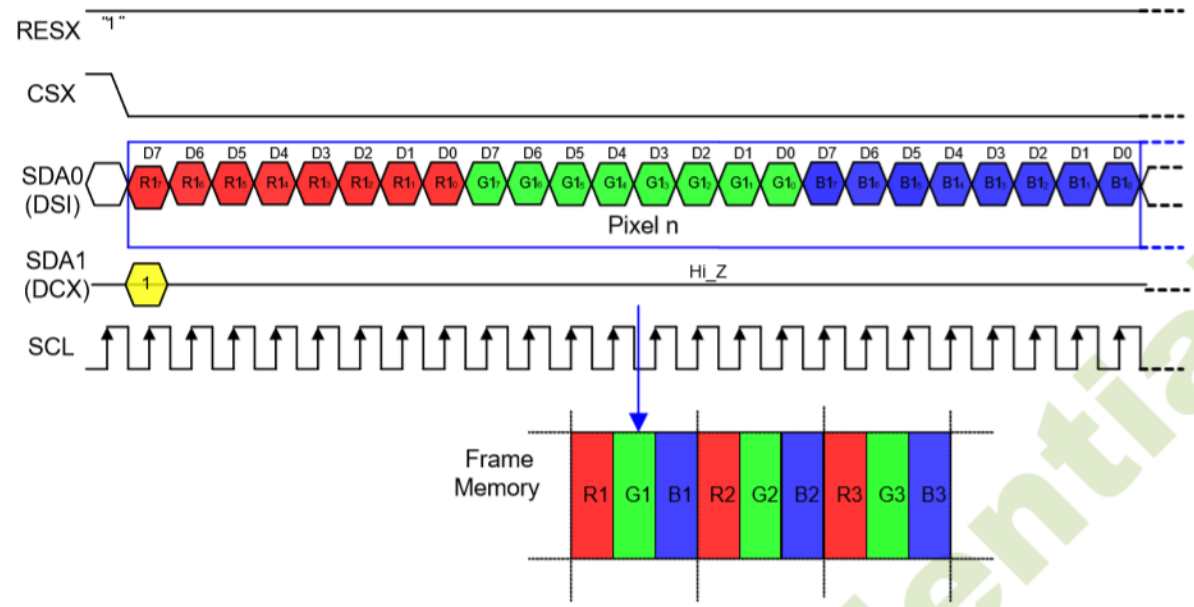

(3) RGB565；CDX=1；AHB_CFG=0；DUAL_LANE=0；DCP_MODE=0



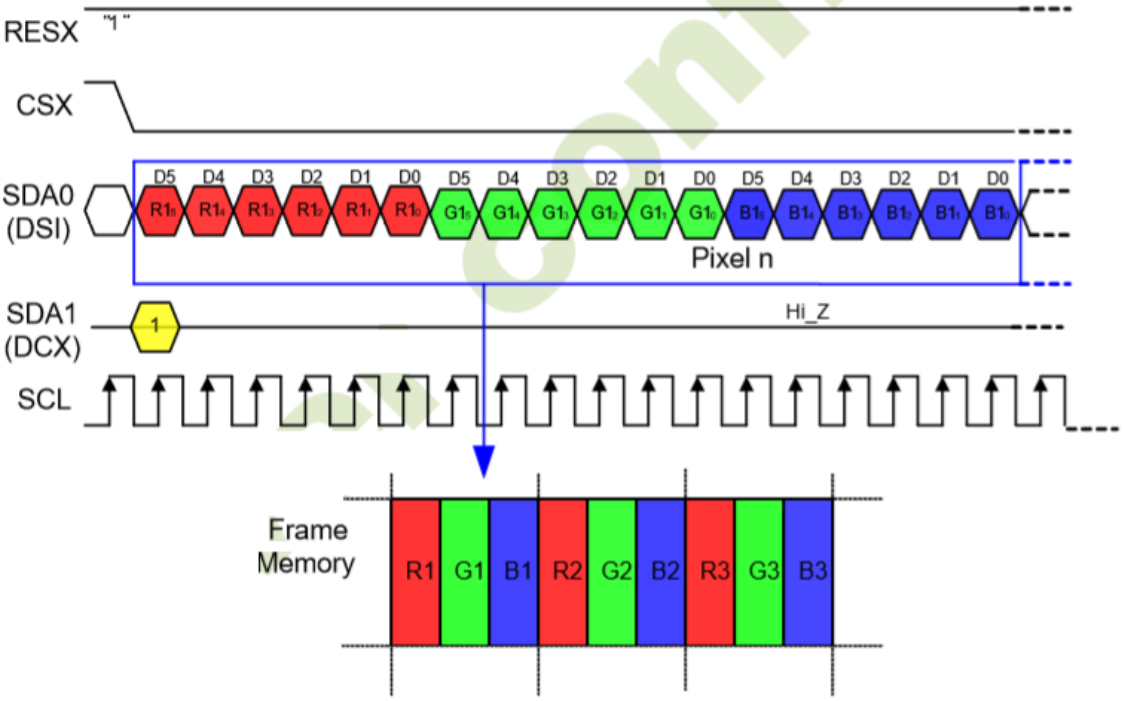

(4) RGB888；CDX=1；AHB_CFG=0；DUAL_LANE=0；DCP_MODE=1

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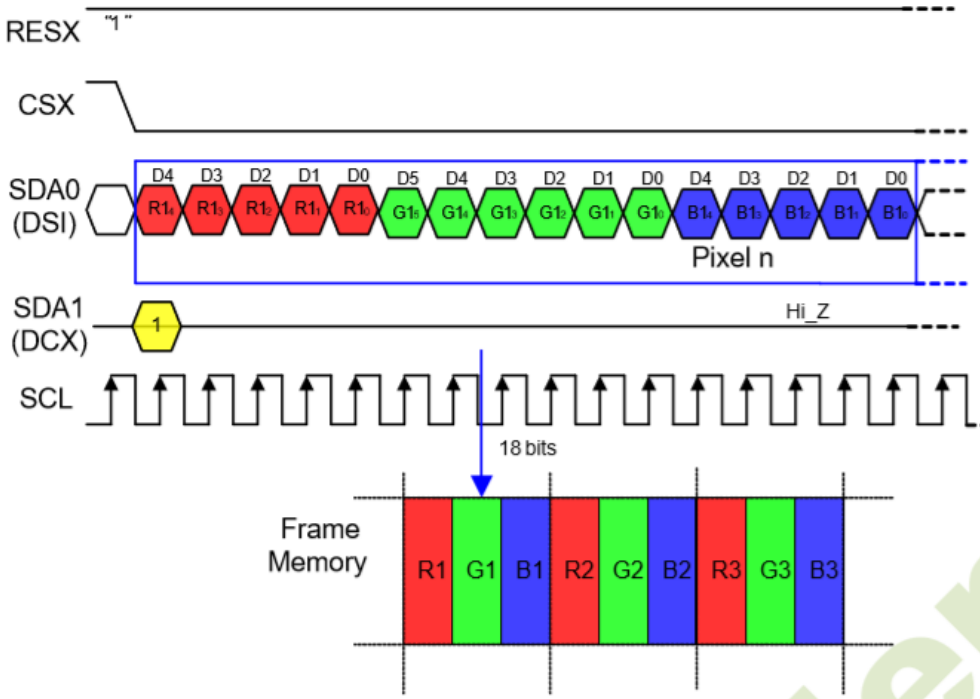



(5) RGB666；CDX=1；AHB_CFG=0；DUAL_LANE=0；DCP_MODE=1



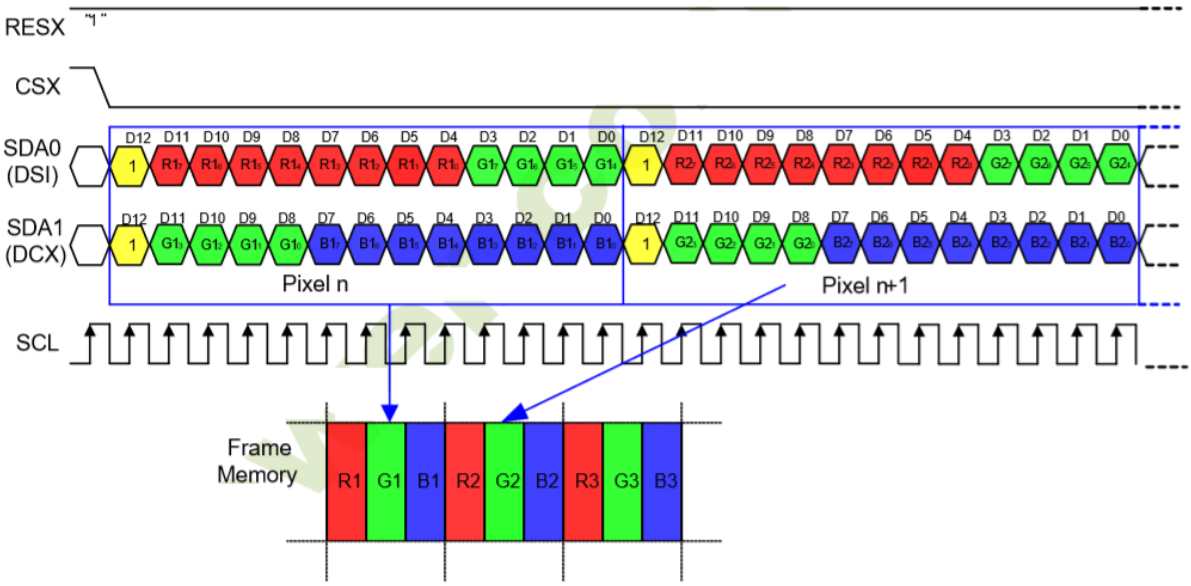

(6) RGB565；CDX=1；AHB_CFG=0；DUAL_LANE=0；DCP_MODE=1

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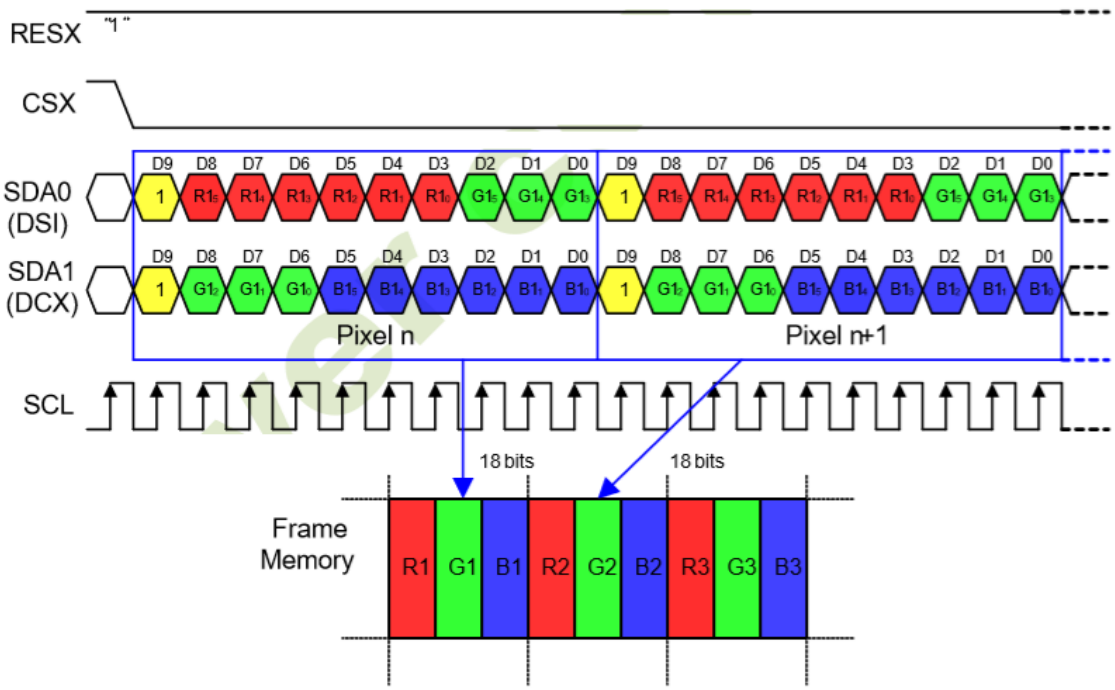

### 3/4-line Dual Lane Serial Interface

(1)RGB888；CDX=1；AHB_CFG=0；DUAL_LANE=1；DCP_MODE= don’t care

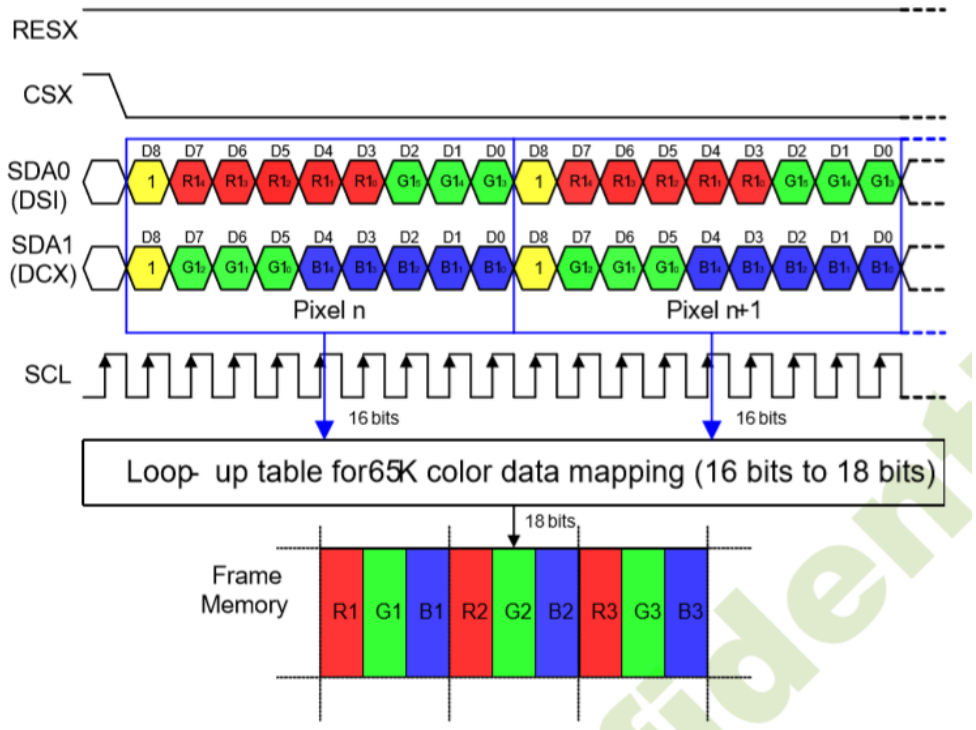



(2) RGB666；CDX=1；AHB_CFG=0；DUAL_LANE=1；DCP_MODE=don’t care

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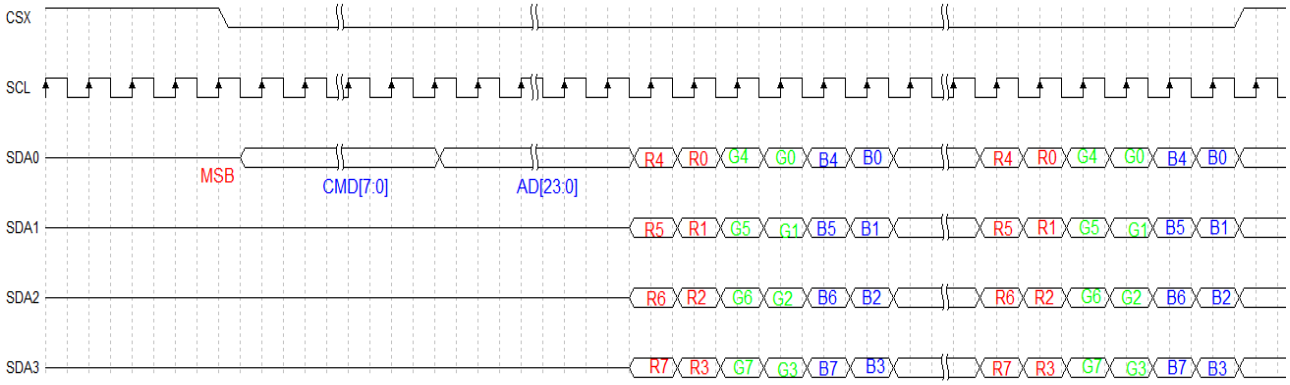

(3) RGB565；CDX=1；AHB_CFG=0；DUAL_LANE=1；DCP_MODE=don't care



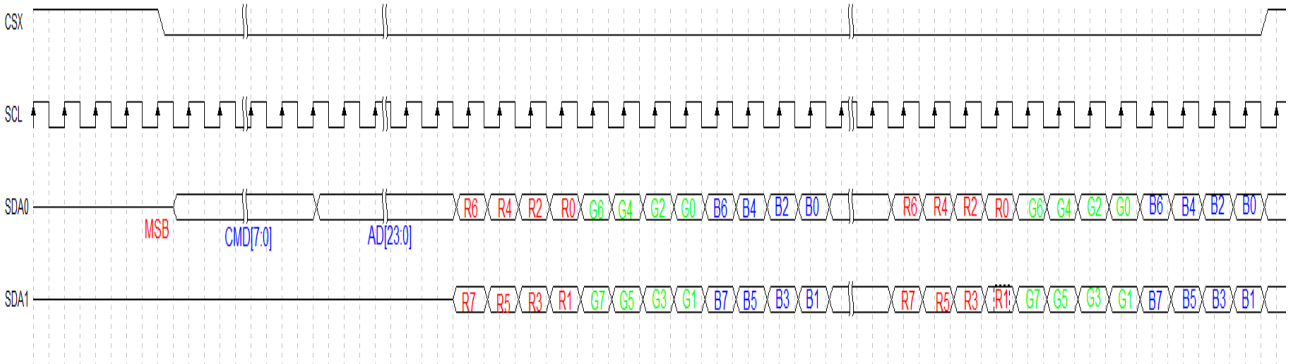

## Quad-line Serial Interface

Different display data formats are available for three colors depth supported by the LCM listed below:
(1) RGB888；CDX=1；AHB_CFG=0；DUAL_LANE=1；DCP_MODE=0；DDR_EN=0

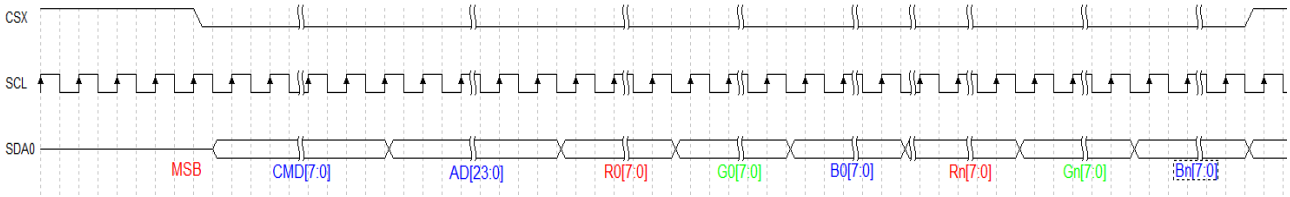
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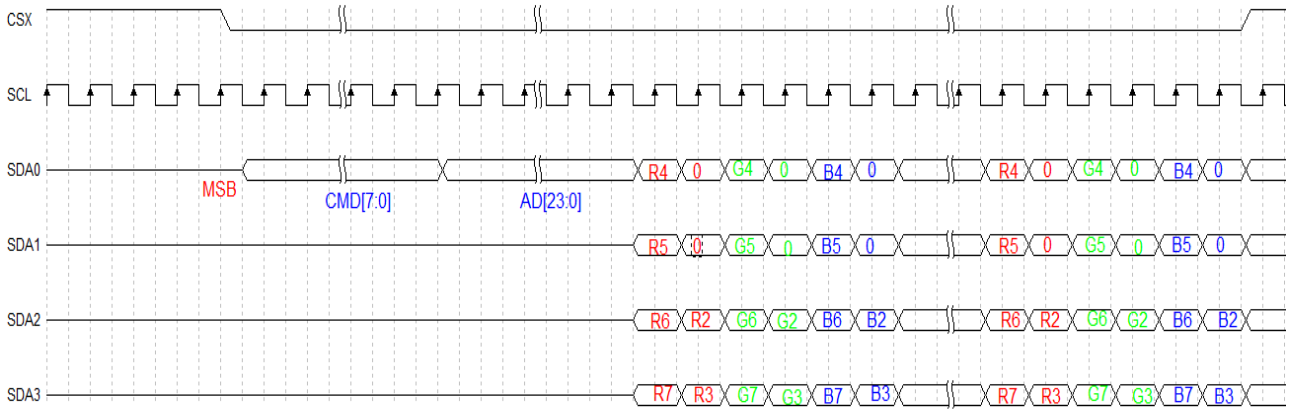

(2) RGB888；CDX=1；AHB_CFG=0；DUAL_LANE=1；DCP_MODE=1；DDR_EN=0




(3) RGB888；CDX=1；AHB_CFG=0；DUAL_LANE=0；DCP_MODE=don’t care；DDR_EN=0

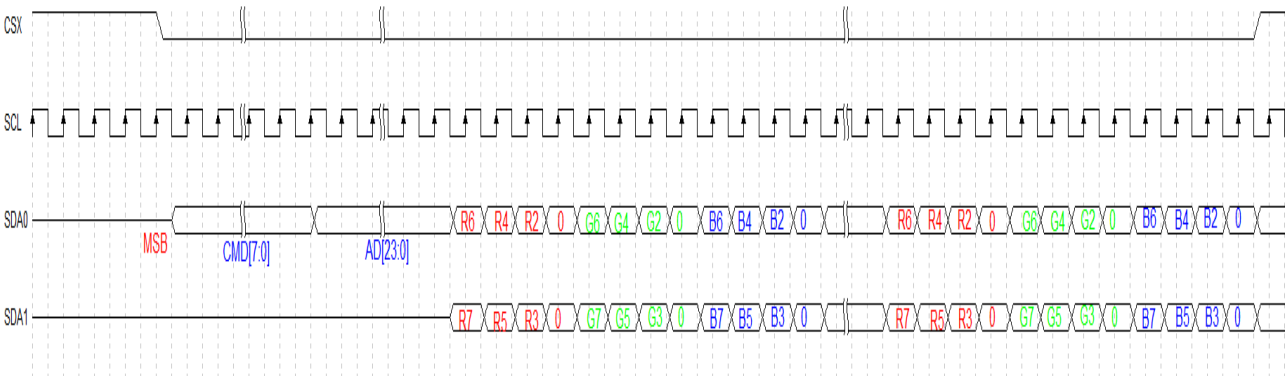



(4) RGB666；CDX=1；AHB_CFG=0；DUAL_LANE=1；DCP_MODE=0；DDR_EN=0

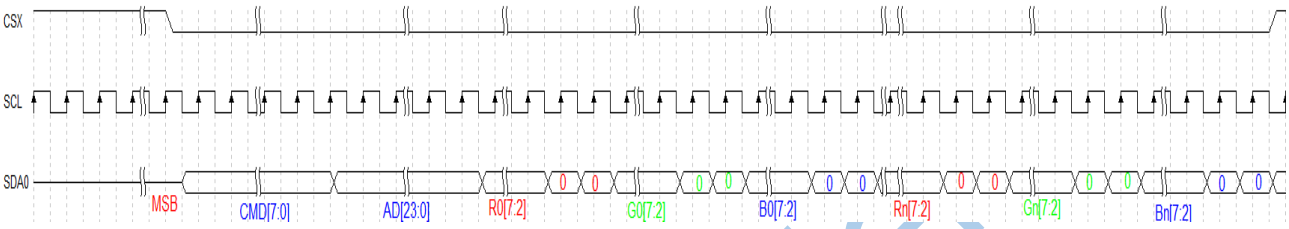



(5) RGB666；CDX=1；AHB_CFG=0；DUAL_LANE=1；DCP_MODE=1；DDR_EN=0

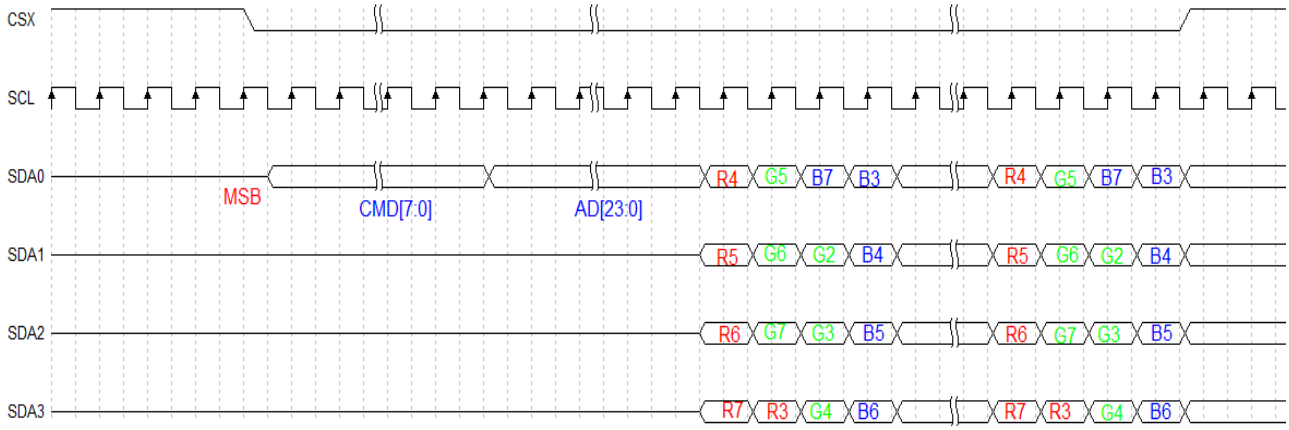
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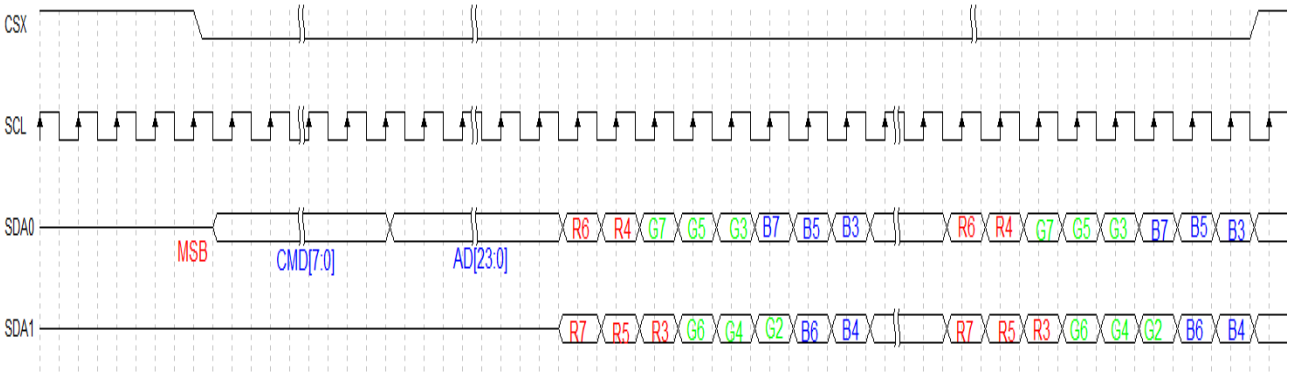

(6) RGB666；CDX=1；AHB_CFG=0；DUAL_LANE=0；DCP_MODE=don't care；DDR_EN=0



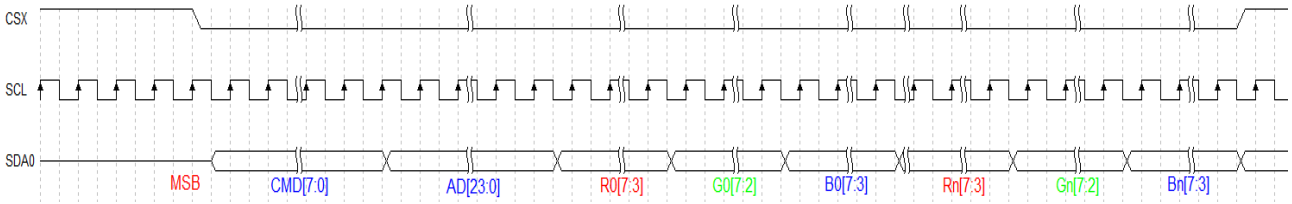

(7) RGB565；CDX=1；AHB_CFG=0；DUAL_LANE=1；DCP_MODE=0；DDR_EN=0




(8) RGB565；CDX=1；AHB_CFG=0；DUAL_LANE=1；DCP_MODE=1；DDR_EN=0



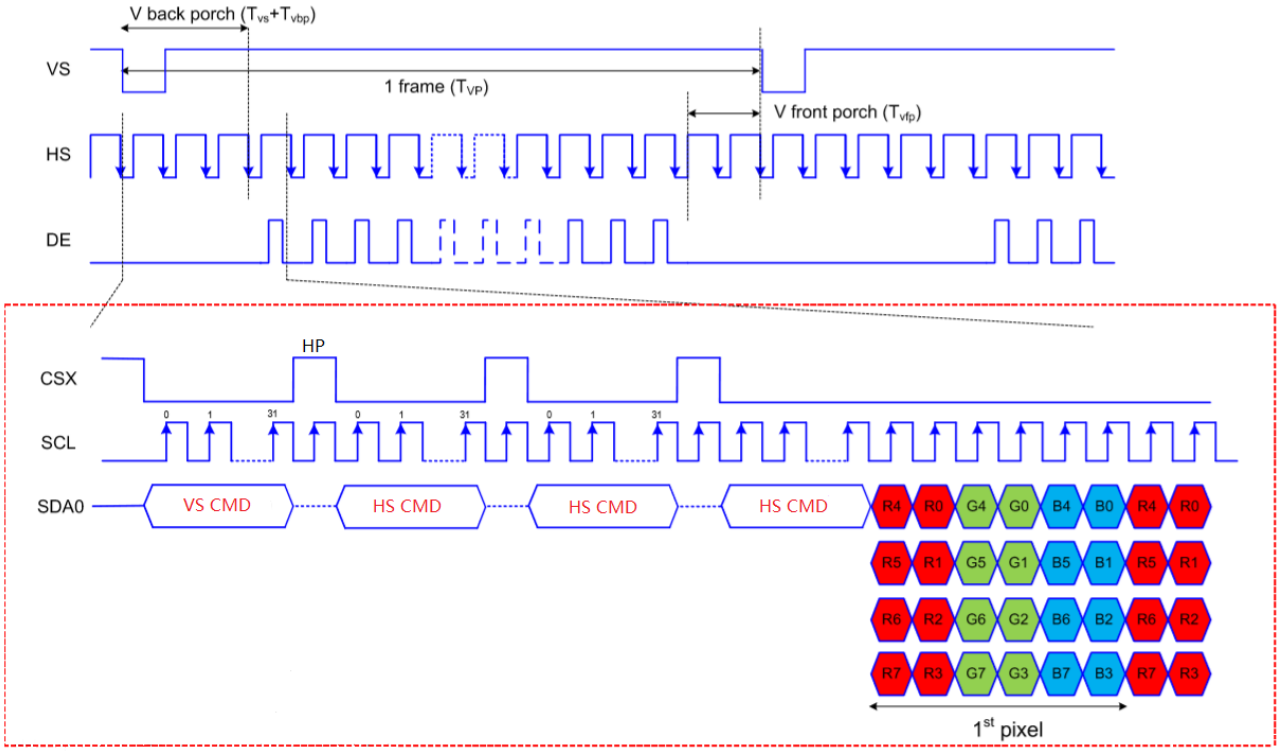

(9) RGB565；CDX=1；AHB_CFG=0；DUAL_LANE=0；DCP_MODE=don't care；DDR_EN=0




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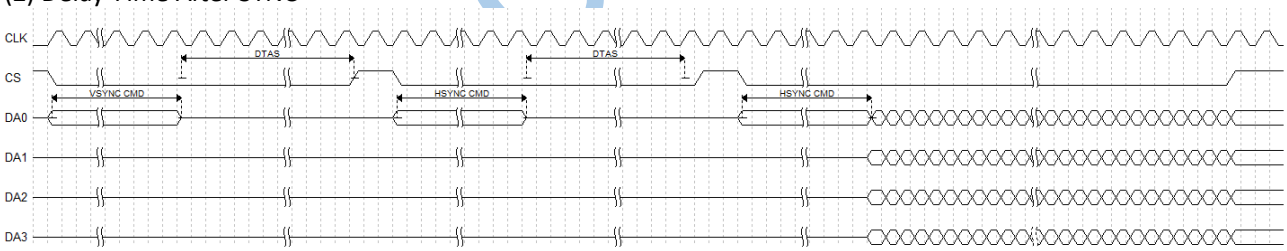
## Sync Mode

(1)Sync Timing



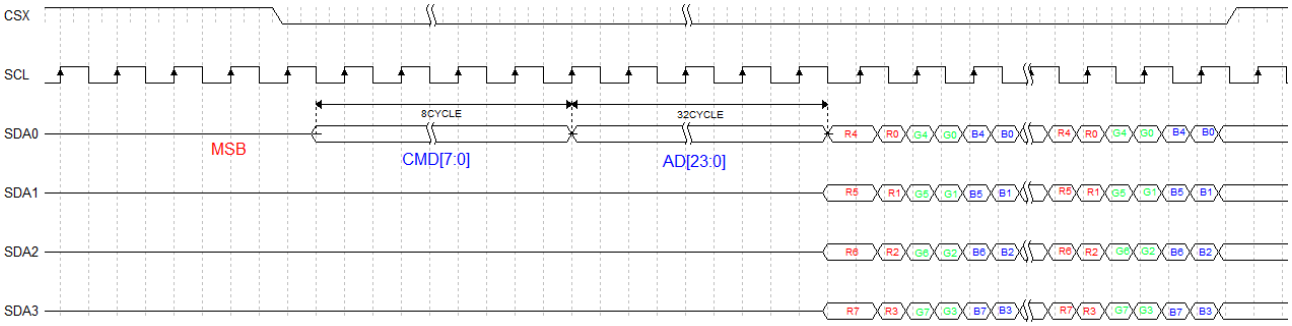

NOTE: Sample for the RGB888,and the same timing with 565/666 format. And Sync Mode support all the quad-line data mode.

(2) Delay Time After SYNC



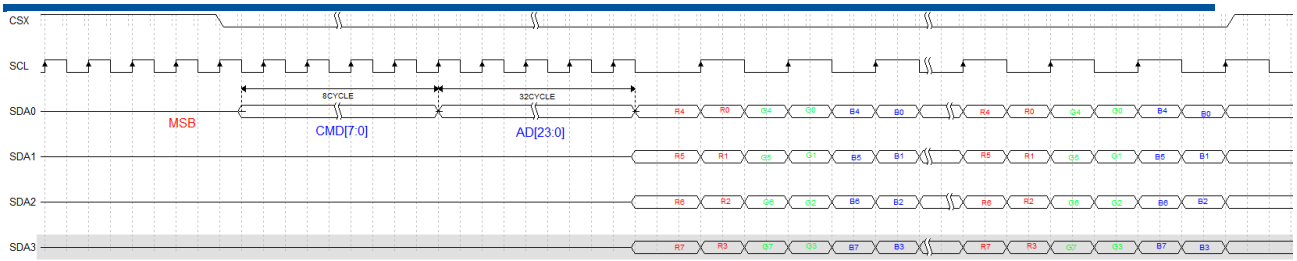

## QSPI DDR Mode

(1) RGB888； CDX=1； AHB_CFG=0； DDR_EN=1； DDR_TYPE=0

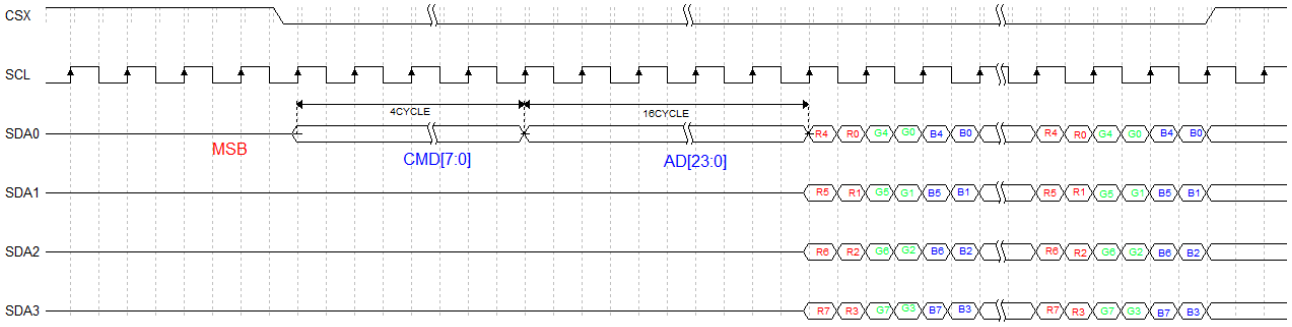



(2)RGB888； CDX=1； AHB_CFG=0； DDR_EN=1； DDR_ TYPE =1 (not support)

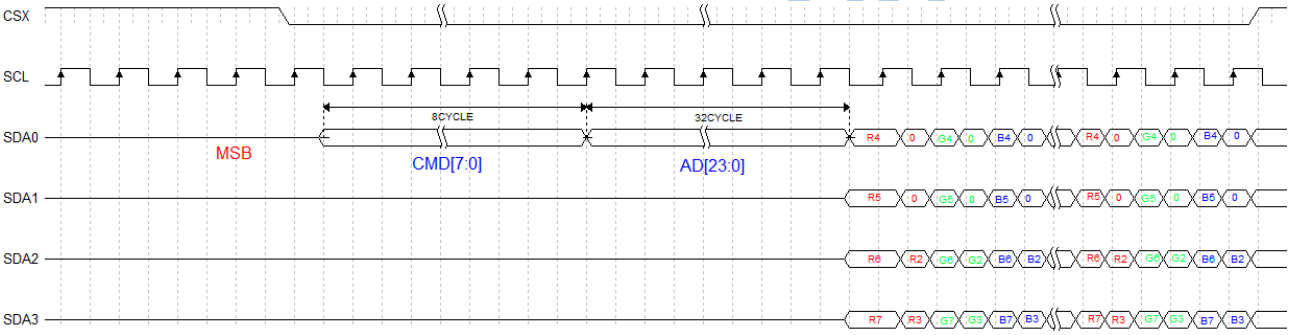
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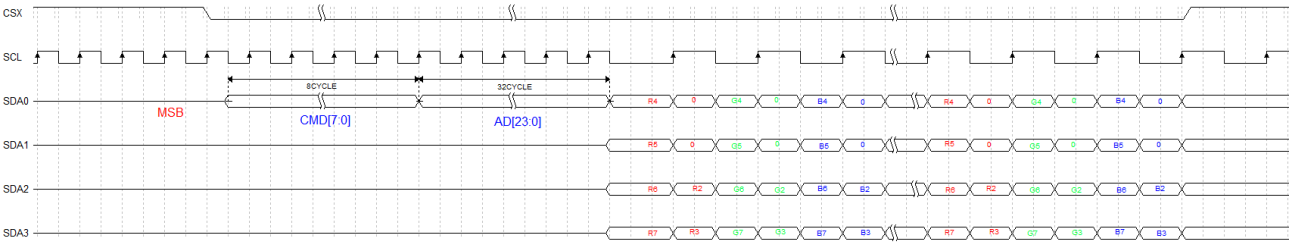

(3)RGB888；CDX=1；AHB_CFG=0；DDR_EN=1；DDR_ TYPE =2



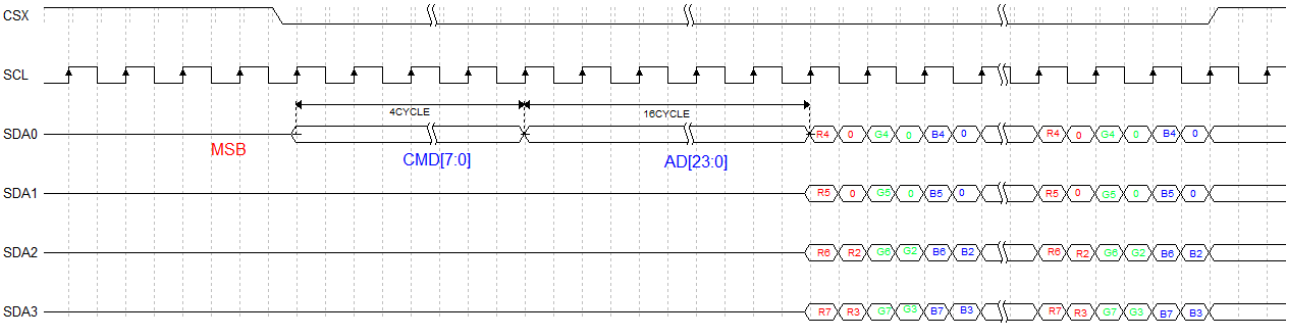

(4)RGB666；CDX=1；AHB_CFG=0；DDR_EN=1；DDR_ TYPE =0




(5)RGB666；CDX=1；AHB_CFG=0；DDR_EN=1；DDR_ TYPE =1

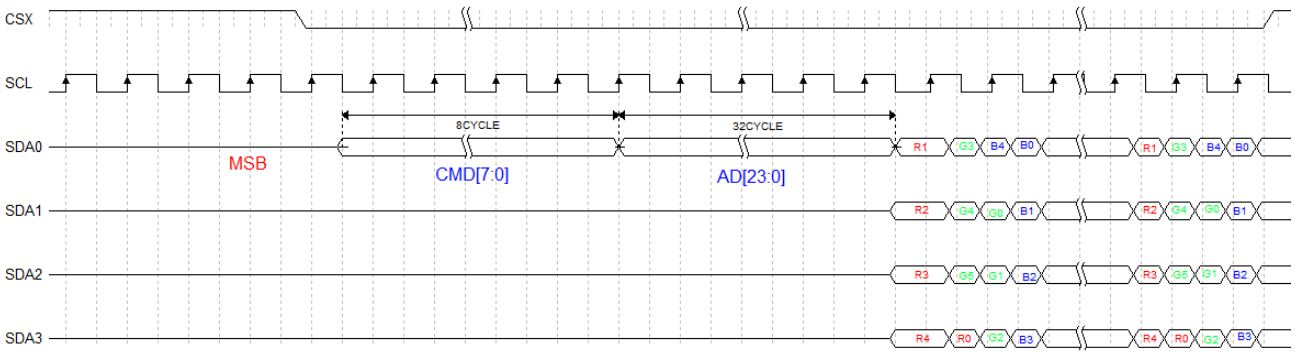



(6)RGB666；CDX=1；AHB_CFG=0；DDR_EN=1；DDR_ TYPE =2

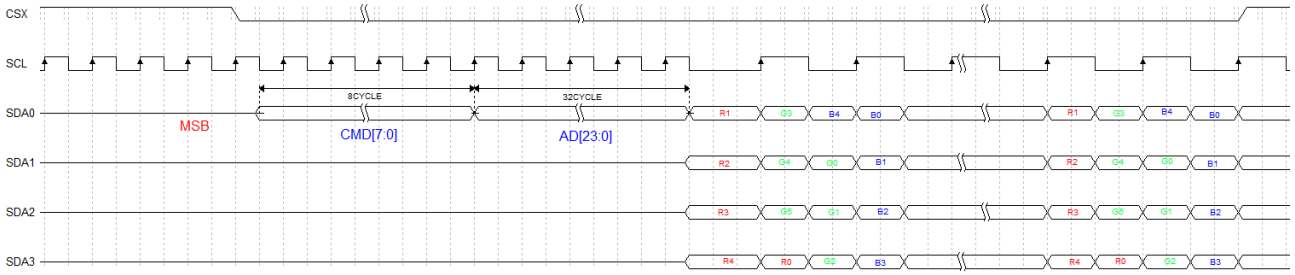



(7)RGB565；CDX=1；AHB_CFG=0；DDR_EN=1；DDR_ TYPE =0

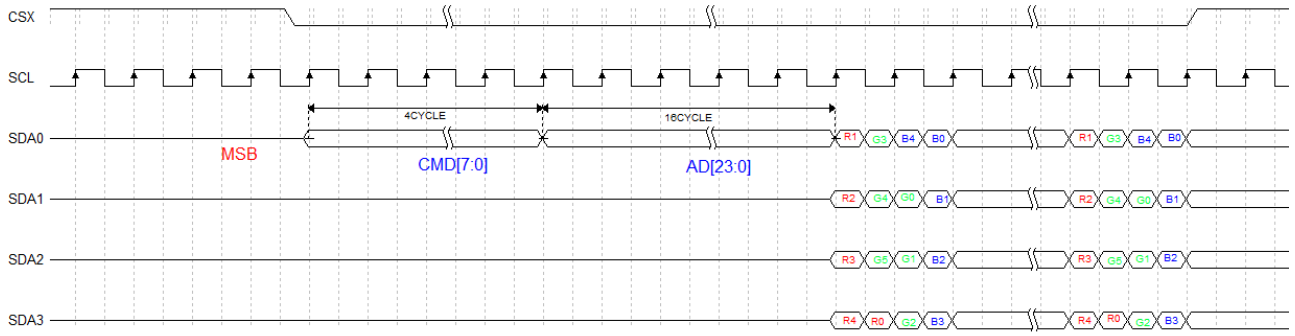
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(8)RGB565；CDX=1；AHB_CFG=0；DDR_EN=1；DDR_ TYPE =1



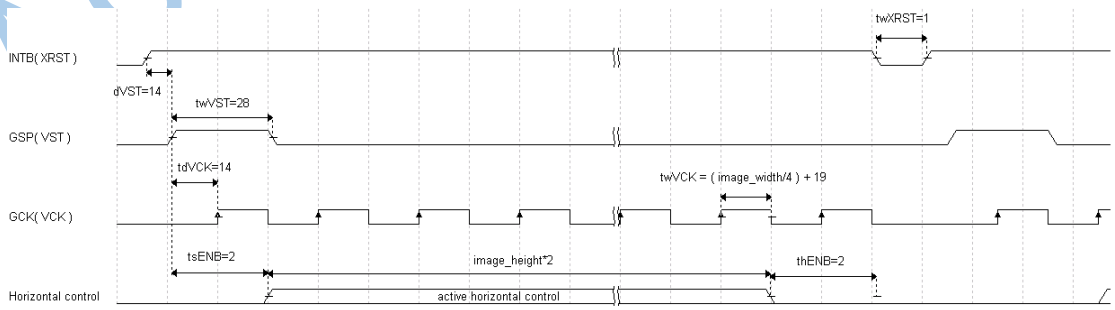

(9)RGB565；CDX=1；AHB_CFG=0；DDR_EN=1；DDR_ TYPE =2




## 14.1.4 TR-LCD Parallel Interface

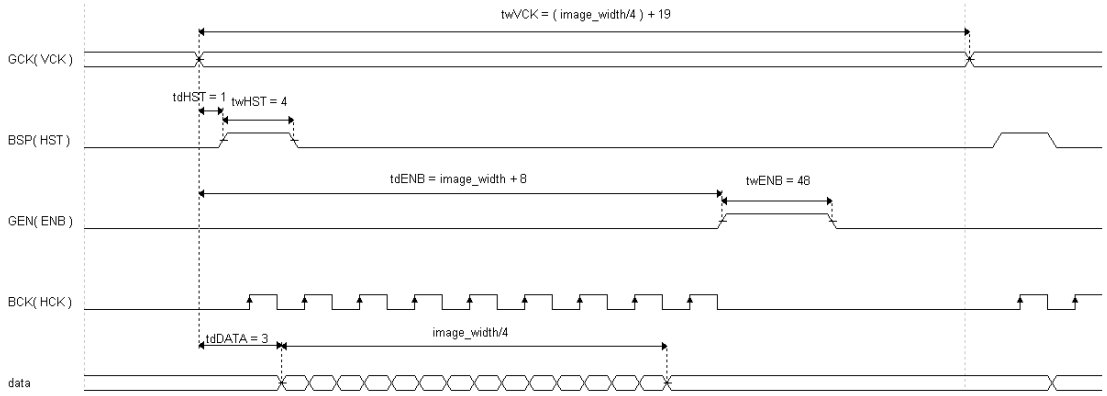
Support TR-J/TR-S Parallel HSYNC/VSYNC LCD interface.

### 14.1.4.1 Veritical




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#### 14.1.4.2 Horizontal




## 14.1.5 LCD Pin Mapping

The LCD Pin Mapping of different interface is as follows.

***Table 14-7 LCD Pin Mapping***

| GPIO | CPU8080 /6800 | 3-LineI | 3-LineII | 4-LineI | 4-LineII | QSPI | TR-S | TR-J |
|---|---|---|---|---|---|---|---|---|
| GPIO33 | E/RD# | | SDO | | SDO | | GEN | ENB |
| GPIO34 | WR# | SCLK | SCLK | SCLK | SCLK | SCLK | BCK | HCK |
| GPIO32 | RS | | | | | | GCK | VCK |
| GPIO30 | CS0 | CS0 | CS0 | CS0 | CS0 | CS0 | BSP | HST |
| GPIO35 | TE | TE | TE | TE | TE | TE | GSP | VST |
| GPIO14 | D0 | SDA0/SDO | SDA0 | SDA0/SDO | SDA0 | SDA0 | R0 | R0 |
| GPIO15 | D1 | | | SDA1/DCX | SDA1/DCX | SDA1 | R1 | R1 |
| GPIO16 | D2 | | | | | SDA2 | G0 | G0 |
| GPIO17 | D3 | | | | | SDA3 | G1 | G1 |
| GPIO18 /GPIO24 | D4 | | | | | | B0 | B0 |
| GPIO19 /GPIO25 | D5 | | | | | | B1 | B1 |
| GPIO20 /GPIO26 | D6 | | | | | | VCOM_VB | VCOM_FRP |
| GPIO21 /GPIO27 | D7 | | | | | | VA | XFRP |
| | | | | | | | INTB | XRST |

## 14.1.6 LCD Register List

***Table 14-8 LCD Controller Registers base address***

| Name | Physical Base Address | KSEG1 Base Address |
|---|---|---|
| LCDC_REGISTER | 0x40064000 | 0x40064000 |

***Table 14-9 LCD Controller Registers***

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| Offset | Register Name | Description |
|---|---|---|
| 0x0000 | LCD_CTL | LCD Control Register |
| 0x0008 | LCD_DISP_SIZE | LCD DMA Source Display Size Register |
| 0x001C | LCD_CPU_CTL | LCD CPU Mode Control Register |
| 0x0020 | LCD_DATA | LCD Data Register |
| 0x0024 | LCD_CPU_CLK | LCD CPU Mode Clock Control Register |
| 0x0028 | LCD_TPL | LCD Transfer Pixel Length Register |
| 0x002C | LCD_SPI_CTL | LCD Serial Interface Control Register |
| 0x0034 | LCD_SPI_CMD | LCD Quad Serial Interface Command and Address register |
| 0x0038 | LCD_QSPI_CMD1 | LCD Quad Serial Interface Command and Address register |
| 0x003C | LCD_QSPI_SYNC_TIM | LCD QSPI SYNC Mode Time Register |
| 0x0040 | LCD_QSPI_IMG_SIZE | LCD QSPI SYNC Mode Display image size Register |
| 0x0044 | 2DGPU_INTERFACE_CTL | 2DGPU Interface Control Register |
| 0x0048 | LCD_PENDING | LCD Quad Serial Interface Command and Address register |
| 0x004c | LCD_QSPI_DTAS | LCD QUAD Serial Interface Delay Time after sync |
| 0x0050 | LCD_DATA1 | LCD Data1 Register |
| 0x0054 | LCD_DATA2 | LCD Data2 Register |
| 0x0058 | LCD_DATA3 | LCD Data3 Register |
| 0x005C | LCD_DATA4 | LCD Data4 Register |
| 0x0060 | LCD_DATA5 | LCD Data5 Register |
| 0x0064 | LCD_DATA6 | LCD Data6 Register |
| 0x0068 | LCD_DATA7 | LCD Data7 Register |
| 0x0070 | LCD_SPI_CTL1 | LCD SPI Control 1 Register |
| 0x0074 | LCD_SPI_CTL2 | LCD SPI Control 2 Register |
| 0x0078 | LCD_QSPI_DDR_CTL | LCD QSPI DDR Control register |
| 0x007c | LCD_SD_CNT | LCD Send Data Counter Register |
| 0x0080 | LCD_SD_CHECKSUM | LCD Send Data CheckSum Register |
| 0x0090 | LCD_TR_CTL | LCD TR-LCD Display Control Register |
| 0x0094 | LCD_TR_IMG_SIZE | LCD TR-LCD Display Image Size Register |
| 0x0098 | LCD_TR_TIM0 | LCD TR-LCD Display Timing0 Register |
| 0x009C | LCD_TR_TIM1 | LCD TR-LCD Display Timing1 Register |
| 0x00A0 | LCD_TR_TIM2 | LCD TR-LCD Display Timing2 Register |
| 0x00A4 | LCD_TR_TIM3 | LCD TR-LCD Display Timing3 Register |
| 0x00A8 | LCD_TR_TIM4 | LCD TR-LCD Display Timing4 Register |
| 0x00AC | LCD_TR_TIM5 | LCD TR-LCD Display Timing5 Register |

## 14.1.7 LCD Register Description

### 14.1.7.1 LCD_CTL

LCD controller control register
Offset=0x0000

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31 | CLK_GATING_EN | CLK Gating Enable<br>0:module clock gating<br>1: mode clock enable | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 30 | CHECK_CLK_EN | Pixel counter, checksum clock enable<br>0: disable clock<br>1: enable clock | R/W | 0x0 |
| 29 | - | Reserved | R | 0x0 |
| 28 | RGB565_OUT_OPT | RGB565 Out Options<br>0:RGB565<br>1:RGB555<br>When RGB565_OUT_OPT =1 the output Green color G[0] force to 0. | R/W | 0x0 |
| 27:26 | RGB666_IN_OPT | RGB666 source options<br>0:RGB666_C0<br>1:RGB666_CLB<br>2:RGB666_CHB<br>3:RGB666_C0 | R/W | 0x0 |
| 25:24 | RGB565_IN_OPT | RGB565 Source Options<br>0:RGB565_C0<br>1:RGB565_CLB<br>2:RGB565_CHB<br>3:RGB565_F555<br>look at data source and converting for details | R/W | 0x0 |
| 23:17 | - | Reserved | R | 0x0 |
| 16 | LCD_HOLD_OPT | LCD Hold Opt<br>0:not hold when FIFO empty<br>1:hold when FIFO empty<br>invalid in QUAD-SYNC or RGB mode | R/W | 0x0 |
| 15 | - | Reserved | R | 0x0 |
| 14 | ARST_FIFO_EN | Auto Reset FIFO per frame start<br>0:disable<br>1:enable | R/W | 0x0 |
| 13 | FIFO_EN | LCDC FIFO Enable<br>0:disable(reset FIFO)<br>1:enable | R/W | 0x0 |
| 12 | TE_EN | Tearing Effort Enable<br>0 : disable<br>1 : enable<br>Invalid in RGB/QUAD-SYNC mode | R/W | 0x0 |
| 11:10 | TE_MOD_SEL | Tearing Effort Mode Select<br>0 : High level<br>1 : Low level<br>2 : Rising edge<br>3: Falling edge<br>Invalid in RGB/QUAD-SYNC mode | R/W | 0x0 |
| 9 | CESEL | Chip Select<br>0: CE0<br>1: CE1<br>for CPU mode and SPI mode | R/W | 0x0 |
| 8 | MLS | Output MSB/LSB Swap<br>0:MSB Frist<br>1:LSB Frist<br>Only for CPU mode and IF_OFS=RGB565 | R/W | 0x0 |
| 7:6 | - | Reserved | R | 0x0 |
| 5:4 | IF_OFS | IF RGB Out Format Select | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0: RGB 565<br>1: RGB 666<br>2: RGB 888<br>Others: Reserved | | |
| 3:1 | IFSEL | Mode select<br>0: I8080 Interface<br>1: M6800 Interface<br>2: Reserved<br>3: SPI Interface<br>4: TR Interface<br>Others: Reserved | R/W | 0x0 |
| 0 | LCD_EN | LCD Module Enable<br>0:Disable<br>1:Enable | R/W | 0x0 |

### 14.1.7.2 LCD_DISP_SIZE

LCD Pixel Size Register
Offset=0x0008

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25:16 | HEIGHT | Screen Height (in lines) for Screen | R/W | 0x0 |
| 15:10 | - | Reserved | R | 0x0 |
| 9:0 | WIDTH | Screen Width (in pixels) for Screen<br>NOTE: include the half word pixel | R/W | 0x0 |

**NOTE:** Only for DMA Source.

### 14.1.7.3 LCD_CPU_CTL

LCD CPU Control Register
Offset=0x001C

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31 | START | Start CPU Mode Transfer<br>0: normal<br>1: start transfer, clean when complete | R/W | 0x0 |
| 30 | FEND_IRQ_EN | Frame End IRQ Enable<br>0 : disable<br>1 : enable | R/W | 0x0 |
| 29:18 | - | Reserved | R | 0x0 |
| 17:15 | SDT | Source Data Type Select:<br>0: RGB565<br>1: RGB666/BGR666<br>2: RGB888/BGR888<br>3: ARGB888<br>4: BGRA888<br>Others: Reserved<br>For AHB & DMA mode | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 13 | RB_SWAP | Input color R/B Swap<br>0:color R/B not swap<br>1:color R/B swap<br>For CPU AHB & DMA Mode | R/W | 0x0 |
| 12:9 | - | Reserved | R | 0x0 |
| 8 | RD_OPT | Read option<br>0:read data when RD rising<br>1:read data when RD rising delay 1 ahb_clk | R/W | 0x0 |
| 7:6 | - | Reserved | R | 0x0 |
| 5 | F565_SEL | RGB565 Format Select<br>0: R[4:0]->LCD_CPU_DATA[23:19]<br>G[5:0]->LCD_CPU_DATA[15:10]<br>B[4:0]->LCD_CPU_DATA[7:3]<br>1: R[4:0]->LCD_CPU_DATA[15:11]<br>G[5:0]->LCD_CPU_DATA[10:5]<br>B[4:0]->LCD_CPU_DATA[4:0]<br>Note: Only CPU AHB MODE | R/W | 0x0 |
| 4 | RS | RS select<br>0: RS output low voltage level<br>1: RS output high voltage level | R/W | 0x0 |
| 3:2 | SRC_SEL | Input Source Select<br>0: AHB<br>1: 2DGPU<br>2: DMA<br>Others: Reserved | R/W | 0x0 |
| 1 | AHB_CFG_DATA | AHB CFG/DATA Select<br>0: Image Data<br>1: Config Data | R/W | 0x0 |
| 0 | CSX | Chip Select pin Level<br>0: CSX=Low Level<br>1: CSX=High Level<br>Only for AHB mode | R/W | 0x1 |

### 14.1.7.4 LCD_DATA

LCD Data Register
Offset=0x0020
**Data for AHB mode:**
ARGB888:

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:24 | A | panel's color A | R/W | 0x0 |
| 23:16 | R | panel's color R | R/W | 0x0 |
| 15:8 | G | panel's color G | R/W | 0x0 |
| 7:0 | B | panel's color B | R/W | 0x0 |

BGRA888:

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:24 | B | panel's color B | R/W | 0x0 |
| 23:16 | G | panel's color G | R/W | 0x0 |
| 15:8 | R | panel's color R | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 7:0 | A | panel's color A | R/W | 0x0 |

RGB888:

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:24 | - | Reserved | R/W | 0x0 |
| 23:16 | R | panel's color R | R/W | 0x0 |
| 15:8 | G | panel's color G | R/W | 0x0 |
| 7:0 | B | panel's color B | R/W | 0x0 |

RGB666:

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:24 | - | Reserved | R/W | 0x0 |
| 23:18 | R | panel's color R | R/W | 0x0 |
| 17:16 | - | Reserved | R/W | 0x0 |
| 15:10 | G | panel's color G | R/W | 0x0 |
| 9:8 | - | Reserved | R/W | 0x0 |
| 7:2 | B | panel's color B | R/W | 0x0 |
| 1:0 | - | Reserved | R/W | 0x0 |

RGB565, F565_SEL=0

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:24 | - | Reserved | R/W | 0x0 |
| 23:19 | R | panel's color R | R/W | 0x0 |
| 18:16 | - | Reserved | R/W | 0x0 |
| 15:10 | G | panel's color G | R/W | 0x0 |
| 9:8 | - | Reserved | R/W | 0x0 |
| 7:3 | B | panel's color B | R/W | 0x0 |
| 2:0 | - | Reserved | R/W | 0x0 |

RGB565, F565_SEL=1

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | - | Reserved | R/W | 0x0 |
| 15:11 | R | panel's color R | R/W | 0x0 |
| 10:5 | G | panel's color G | R/W | 0x0 |
| 4:0 | B | panel's color B | R/W | 0x0 |

Command for AHB mode:

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:8 | - | Reserved | R/W | 0x0 |
| 7:0 | CMD | Transmitted as a command | R/W | 0x0 |

CMD[7:0] transmitted as CMD when 3/4-Line Serial Interface and CPU mode

Data for AHB SPI mode

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | DATA | DATA | R/W | 0x0 |

Transmitted Data when 3/4-Line Serial and QSPI Interface

Note: set LCD_DATA1 ~ LCD_DATA7 and RWL bits before write data to LCD_DATA.

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**Data for DMA mode：**

**RGB565 F565_SEL=0**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| G1[2:0] | B1[4:0] | R1[4:0] | G1[5:3] | G0[2:0] | B0[4:0] | R0[4:0] | G0[5:3] |

31bit 24 23 16 15 8 7 0bit

**RGB565 F565_SEL=1**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| R1[4:0] | G1[5:3] | G1[2:0] | B1[4:0] | R0[4:0] | G0[5:3] | G0[2:0] | B0[4:0] |

31bit 24 23 16 15 8 7 0bit

**RGB666**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| R1[5:0] | 2b00 | B0[5:0] | 2b00 | G0[5:0] | 2b00 | R0[5:0] | 2b00 |
| G2[5:0] | 2b00 | R2[5:0] | 2b00 | B1[5:0] | 2b00 | G1[5:0] | 2b00 |
| B3[5:0] | 2b00 | G3[5:0] | 2b00 | R3[5:0] | 2b00 | B2[5:0] | 2b00 |

31bit 24 23 16 15 8 7 0bit

**RGB888**

| | | | |
|---|---|---|---|
| R1[7:0] | B0[7:0] | G0[7:0] | R0[7:0] |
| G2[7:0] | R2[7:0] | B1[7:0] | G1[7:0] |
| B3[7:0] | G3[7:0] | R3[7:0] | B2[7:0] |

31bit 24 23 16 15 8 7 0bit

**ARGB888**

| | | | |
|---|---|---|---|
| A0[7:0] | R0[7:0] | G0[7:0] | B0[7:0] |
| A1[7:0] | R1[7:0] | G1[7:0] | B1[7:0] |
| A2[7:0] | R2[7:0] | G2[7:0] | B2[7:0] |

31bit 24 23 16 15 8 7 0bit

**BGRA888:**

| | | | |
|---|---|---|---|
| B0[7:0] | G0[7:0] | R0[7:0] | A0[7:0] |
| B1[7:0] | G1[7:0] | R1[7:0] | A1[7:0] |
| B2[7:0] | G2[7:0] | R2[7:0] | A2[7:0] |

31bit 24 23 16 15 8 7 0bit

### 14.1.7.5 LCD_CPU_CLK

LCD CPU Clock Register
Offset=0x0024

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:22 | - | Reserved | R | 0x0 |
| 21:16 | CLKLDU2 | Clock Low Level Duration (from AHB_CLK)<br>from 1 to 64 (EXCL2KL +1) | R/W | 0x0 |
| 15:14 | - | Reserved | R | 0x0 |
| 13:8 | CLKLDU | Clock Low Level Duration (from AHB_CLK)<br>from 1 to 64 (EXCLKL +1) | R/W | 0xF |
| 7:6 | - | Reserved | R | 0x0 |
| 5:0 | CLKHDU | Clock High Level Duration (from AHB_CLK).<br>from 1 to 64 (EXCLKH +1) | R/W | 0xF |

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**NOTE:**
For 2DGPU&DMA source write freq = CMU_LCDCLK/(( clkldu+1)+( Clkhdu+1))
For AHB source write freq = AHB_CLK/(( clkldu+1)+( Clkhdu+1))

### 14.1.7.6 LCD_TPL

LCD Transfer Pixel Length
Offset=0x0028

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | TPL | Transfer Pixel Length(only for SI/CPU in 2DGPU Transfer mode ) | R/W | 0x0 |

**NOTE:** TPL=WIDTH*HEIGHT-1

### 14.1.7.7 LCD_SPI_CTL

LCD Serial Interface Control Register
Offset=0x002C

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31 | START | START Transfer<br>0:normal<br>1:start transfer, auto clean when complete | R/W | 0x0 |
| 30 | FTC_IRQ_EN | Frame Transfer Complete IRQ Enable<br>1'b0 : disable<br>1'b1 : enable<br>Not for sync mode | R/W | 0x0 |
| 29 | FEND_PD_EN | Frame End Pending Enable<br>0:disable<br>1:enable<br>Only for sync mode | R/W | 0x0 |
| 28 | AHB_CFG_DATA | AHB CFG/DATA Select<br>0：Image Data<br>1：Config Data<br>Quad mode | R/W | 0x0 |
| 27 | - | Reserved | R/W | 0x0 |
| 26:24 | SDT | Source Data Type Select:<br>0:RGB565<br>1:RGB666/BGR666<br>2:RGB888/BGR888<br>3:ARGB888<br>4: BGRA888<br>Others: Reserved<br>for AHB&DMA Mode | R/W | 0x0 |
| 23:18 | RWL | SPI Read/Write Data Length<br>0~32Byte(only in quad mode)<br>1~4Byte(in SPI 3/4 wire mode)<br>For read:0~4Byte,others 1Byte<br>Write LCD_DAT when send CMD and RWL=0 | R/W | 0x0 |
| 17 | F565_SEL | RGB565 Format Select<br>0: R[4:0]->LCD_CPU_DATA[23:19]<br>G[5:0]->LCD_CPU_DATA[15:10]<br>B[4:0]->LCD_CPU_DATA[7:3]<br>1: R[4:0]->LCD_CPU_DATA[15:11] | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | G[5:0]->LCD_CPU_DATA[10:5]<br>B[4:0]->LCD_CPU_DATA[4:0]<br>Note: for AHB & DMA Mode | | |
| 16 | DUAL_LANE | Dual Data Lane Sel<br>0:signle data lane<br>1:dual/quad data lane | R/W | 0x0 |
| 15:12 | DELAY_CHAIN | Delay chain<br>0~15ns Delay<br>Valid in read operation | R/W | 0x0 |
| 11:10 | SRC_SEL | Input Source Select<br>0: AHB<br>1: 2DGPU<br>2: DMA<br>Others: Reserved | R/W | 0x0 |
| 9:8 | QSPI_RDP_SEL | QUAD SPI Read Data Port Select<br>0:D0<br>1:D1<br>2:D2<br>3:D3<br>The SPI Command and Dummy will go out by D0, and Read Data Port Decision by QSPI_RDP_SEL.<br>Only in Quad Mode | R/W | 0x0 |
| 7 | DCP_MODE | Data Compact Mode SEL<br>0:normal mode<br>1:compact mode | R/W | 0x0 |
| 6 | - | Reserved | R/W | 0x0 |
| 5 | CSX | ChipSel<br>0:CSX=0<br>1:CSX=1<br>Only for AHB mode | R/W | 0x1 |
| 4 | RB_SWAP | Input color R/B Swap<br>0:color R/B not swap<br>1:color R/B swap<br>For SI AHB or DMA Mode | R/W | 0x0 |
| 3 | CDX | Command/Data Control<br>0: Command<br>1: Display Data or Parameter | R/W | 0x0 |
| 2:0 | TYPE_SEL | SPI type select<br>0: 3-line serial interface type I<br>1: 3-line serial interface type II<br>2: 4-line serial interface type I<br>3 : 4-line serial interface type II<br>4: quad serial interface<br>5: quad serial interface sync mode<br>Others: Reserved | R/W | 0x0 |

### 14.1.7.8 LCD_SPI_CMD

LCD Quad Serial Interface Command and Address Register
Offset=0x0034
For 3/4 line Read

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 31:24 | CMD | Quad Serial Interface Command | R/W | 0x0 |
| 23:0 | - | Reserved | R | 0x0 |

For Quad Mode

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:24 | CMD | Quad Serial Interface Command | R/W | 0x0 |
| 23:0 | AD | Quad Serial Interface Address | R/W | 0x0 |

**NOTE:** QSI Transfer start CMD&AD or VS start command for sync mode

### 14.1.7.9 LCD_QSPI_CMD1

LCD Quad Serial Interface Command and Address Register
Offset=0x0038

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:24 | CMD | Quad Serial Interface Command | R/W | 0x0 |
| 23:0 | AD | Quad Serial Interface Address | R/W | 0x0 |

Note: HS Start Command for sync mode

### 14.1.7.10 LCD_QSPI_SYNC_TIM

LCD Quad Serial Interface sync timing control Register
Offset=0x003C

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | HP | Horizontal Porch Delay(in SCL, Min 200ns) | R/W | 0x0 |
| 15:8 | VFP | Vertical Front Porch Delay(in lines)=VFP+1 | R/W | 0x0 |
| 7:0 | VBP | Vertical Back Porch Delay(in lines)=VBP+1 | R/W | 0x0 |

**NOTE:** only for sync mode

### 14.1.7.11 LCD_QSPI_IMG_SIZE

LCD Quad Serial Interface Display Image Size Register
Offset=0x0040

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25:16 | Y | Screen Height (in lines) for QSI IF | R/W | 0x0 |
| 15:10 | - | Reserved | R | 0x0 |
| 9:0 | X | Screen Width (in pixels) for QSI IF | R/W | 0x0 |

**NOTE:** only for sync mode

### 14.1.7.12 2DGPU_INTERFACE_CTL

2DGPU Interface Control Register
Offset=0x0044

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 31:30 | - | Reserved | R | 0x0 |
| 29:20 | UNIT | Unit<br>Note: Unit can't < 4 | R/W | 0x0F |
| 19:10 | VSFP | Vertical Front Porch Delay | R/W | 0x0 |
| 9:0 | VSPW | Vertical Sync Pulse Width | R/W | 0x0 |

### 14.1.7.13 LCD_PENDING

LCD Pending Register
Offset=0x0048

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:8 | - | Reserved | R | 0x0 |
| 7 | TR_TF_PD | 2DGPU/DMA Mode Transfer Finish Pending<br>0 : busy<br>1 : finish<br>Write 1auto clean<br>Only for TR mode | R/W | 0x0 |
| 6 | FIFO_UDF | FIFO UnderFlow<br>0:normal<br>1:flow | R/W | 0x0 |
| 5 | FEND_PD | QSPI Sync Frame End Pending<br>0:normal<br>1:Frame End<br>Write 1auto clean, Only for QSPI sync mode | R/W | 0x0 |
| 4 | TES | TE signal status | R | 0x0 |
| 3 | SPI_TF_PD | 2DGPU/DMA Mode Transfer Finish Pending<br>0 : busy<br>1 : finish<br>Write 1auto clean<br>For normal SPI mode not include QSPI Sync mode | R/W | 0x0 |
| 2 | CPU_TF_PD | 2DGPU/DMA Mode Transfer Finish Pending<br>0 : busy<br>1 : finish<br>Write 1 auto clean<br>For CPU mode | R/W | 0x0 |
| 1:0 | - | Reserved | R | 0x0 |

### 14.1.7.14 LCD_QSPI_DTAS

QSPI QUAD Serial Interface Delay Time after sync Register
Offset=0x004C

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:17 | - | Reserved | R | 0x0 |
| 16 | DCKEN | SCK output enable when delay time after sync cmd<br>0:disable, the sck output low level<br>1:enable | R/W | 0x0 |
| 15:0 | DTAS | Delay time after sync (in sclk) | R/W | 0x0 |

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### 14.1.7.15 LCD_DATA1

LCD Data1 register for AHB QSPI mode
Offset=0x0050

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | DATA | DATA | R/W | 0x0 |

Note: transmitted Data when QSPI Interface

### 14.1.7.16 LCD_DATA2

LCD Data2 register for AHB QSPI mode
Offset=0x0054

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | DATA | DATA | R/W | 0x0 |

Note: transmitted Data when QSPI Interface

### 14.1.7.17 LCD_DATA3

LCD Data3 register for AHB QSPI mode
Offset=0x0058

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | DATA | DATA | R/W | 0x0 |

Note: transmitted Data when QSPI Interface

### 14.1.7.18 LCD_DATA4

LCD Data4 register for AHB QSPI mode
Offset=0x005C

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | DATA | DATA | R/W | 0x0 |

Note: transmitted Data when QSPI Interface

### 14.1.7.19 LCD_DATA5

LCD Data5 register for AHB QSPI mode
Offset=0x0060

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | DATA | DATA | R/W | 0x0 |

Note: transmitted Data when QSPI Interface

### 14.1.7.20 LCD_DATA6

LCD Data6 register for AHB QSPI mode
Offset=0x0064

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 31:0 | DATA | DATA | R/W | 0x0 |

Note: transmitted Data when QSPI Interface

### 14.1.7.21 LCD_DATA7

LCD Data7 register for AHB QSPI mode
Offset=0x0068

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | DATA | DATA | R/W | 0x0 |

Note: transmitted Data when QSPI Interface

### 14.1.7.22 LCD_SPI_CTL1

LCD SPI Control 1 register
Offset=0x0070

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:18 | - | reserved | R | 0x0 |
| 17:16 | AHB_CLK_DIV | AHB clk division<br>0:1<br>1:1/2<br>2:1/4<br>3:1/8<br>Note: Valid only in AHB mode | R/W | 0x0 |
| 15:9 | - | Reserved | R | 0x0 |
| 8:4 | RDLC | Read Dummy Delay Cycle<br>Valid in 3/4wire & quad spi mode<br>Note:0~31 | R/W | 0x0 |
| 3:0 | - | Reserved | R | 0x0 |

### 14.1.7.23 LCD_SPI_CTL2

LCD SPI Control 2 register
Offset=0x0074

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:30 | - | Reserved | R | 0x0 |
| 29:25 | DELAY_CSX | Delay control for CSX (0.5ns/step)<br>Valid for input or output operation | R/W | 0x0 |
| 24:20 | DELAY_SCL | Delay control for SCL (0.5ns/step)<br>Valid for input or output operation | R/W | 0x0 |
| 19:15 | DELAY_D3 | Out Delay control for D3 (0.5ns/step)<br>Valid for output operation | R/W | 0x0 |
| 14:10 | DELAY_D2 | Out Delay control for D2 (0.5ns/step)<br>Valid for output operation | R/W | 0x0 |
| 9:5 | DELAY_D1 | Out Delay control for D1 (0.5ns/step)<br>Valid for output operation | R/W | 0x0 |
| 4:0 | DELAY_D0 | Out Delay control for D0 (0.5ns/step)<br>Valid for output operation | R/W | 0x0 |

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### 14.1.7.24 LCD_QSPI_DDR_CTL

LCD QSPI DDR Control register
Offset=0x0078

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:30 | SPI_MODE | SPI protocol mode( { cpol, cpha } )<br>0: mode 0<br>1: mode 1<br>2: mode 2<br>3: mode 3<br>Valid when DDR_EN=1 & DDR_TYPE=0. | R/W | 0x3 |
| 29 | SPI_CLK_OPT | SPI CLK Option<br>0: Keep<br>1: Recovery | R/W | 0x0 |
| 28:3 | - | reserved | R | 0x0 |
| 2:1 | DDR_TYPE | DDR Type<br>0:sdr mode<br>1:ddr type0<br>2:ddr type1<br>3:ddr type2<br>Invalid QUAD-SYNC mode | R/W | 0x0 |
| 0 | DDR_EN | DDR mode enable<br>0:disable<br>1:enable<br>Invalid QUAD-SYNC mode | R/W | 0x0 |

### 14.1.7.25 LCD_SD_CNT

LCD Send Counter register
Offset=0x007C

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31 | EN | Check Sum enable<br>0:disable<br>1:enable | R/W | 0x0 |
| 30:0 | CNT | LCDC send data counter(units: pixel)<br>invalid in sync mode and valid in DMA or 2DGPU souce | R | 0x0 |

### 14.1.7.26 LCD_SD_CHECKSUM

LCD Send Data Checksum register
Offset=0x0080

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31 | EN | Check Sum enable<br>0:disable<br>1:enable | R/W | 0x0 |
| 30:0 | SUM | LCDC send data CheckSum(Checksum for data in FIFO)<br>valid in DMA or 2DGPU mode | R | 0x0 |

Note:
If RGB565 out, SUM+=color_h<<8|color;

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Else SUM+=(color_r<<16)| (color_g<<8)|(color_b<<0);

### 14.1.7.27 LCD_TR_CTL

LCD TR-LCD Control register
Offset=0x0090

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31 | START | Start Transfer<br>0:normal<br>1:start Transfer<br>Note: Auto clean when Transfer complete. | R/W | 0x0 |
| 30 | FTC_IRQ_EN | Frame Transfer Complete IRQ Enable<br>0 : IRQ disable<br>1 : IRQ enable | R/W | 0x0 |
| 29 | FRP_END_MOD | FRP End mode<br>0:Frp end immediately<br>1:Frp end when current cycle | R | 0x0 |
| 28:27 | - | Reserved | R | 0x0 |
| 26:24 | SDT | Source Data Type Select:<br>0: RGB565<br>1: RGB666/BGR666<br>2: RGB888/BGR888<br>3: ARGB888<br>4: BGRA888<br>Others: Reserved<br>for DMA Mode | R/W | 0x0 |
| 23:22 | LB_SEL | Low Bit Select<br>0: bit3<br>1: bit4<br>2: bit5<br>3: bit6<br>LSB select. | R/W | 0x0 |
| 21:19 | - | Reserved | R | 0x0 |
| 18 | HCK_TAIL_OPT | HCK Tail Option<br>0: Tail disable<br>1: Tail enable | R/W | 0x1 |
| 17 | F565_SEL | RGB565 Format Select<br>0: R[4:0]->LCD_CPU_DATA[23:19]<br>G[5:0]->LCD_CPU_DATA[15:10]<br>B[4:0]->LCD_CPU_DATA[7:3]<br>1: R[4:0]->LCD_CPU_DATA[15:11]<br>G[5:0]->LCD_CPU_DATA[10:5]<br>B[4:0]->LCD_CPU_DATA[4:0]<br>Note: for DMA Mode | R/W | 0x0 |
| 16 | RB_SWAP | Input color R/B Swap<br>0:color R/B not swap<br>1:color R/B swap<br>For DMA Mode | R/W | 0x0 |
| 15 | VCK_CONT | VCK continue flips during XRST low. | R/W | 0x0 |
| 14 | VCK_MODE | VCK flips during idle period. | R/W | 0x0 |
| 13 | HCK_MODE | VCK continue flips outside active date period. | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 12 | PU_MODE | Partition update mode | R/W | 0x0 |
| 11:10 | SRC_SEL | Input Source Select<br>0: Reserved<br>1: 2DGPU<br>2: DMA<br>Others: Reserved | R/W | 0x0 |
| 9 | FRP_EN | Frame refresh pulse enable | R/W | 0x0 |
| 8 | VCOM_INV | 0: normal<br>1: inverse | R/W | 0x0 |
| 7 | FRP_INV | 0: normal<br>1: inverse | R/W | 0x0 |
| 6 | XFRP_INV | 0: normal<br>1: inverse | R/W | 0x0 |
| 5 | XRST_INV | 0: normal<br>1: inverse | R/W | 0x0 |
| 4 | VST_INV | 0: normal<br>1: inverse | R/W | 0x0 |
| 3 | HST_INV | 0: normal<br>1: inverse | R/W | 0x0 |
| 2 | VCK_INV | 0: normal<br>1: inverse | R/W | 0x0 |
| 1 | HCK_INV | 0: normal<br>1: inverse | R/W | 0x0 |
| 0 | ENB_INV | 0: normal<br>1: inverse | R/W | 0x0 |

### 14.1.7.28 LCD_TR_IMG_SIZE

LCD TR-LCD Display Size register0
Offset=0x0094

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25:16 | IMG_HEIGHT | Screen Height (in lines) | R/W | 0x0 |
| 15:10 | - | Reserved | R | 0x0 |
| 9:0 | IMG_WIDTH | Screen Width (in pixels) | R/W | 0x0 |

Note: Valid in 2DGPU&DMA mode

### 14.1.7.29 LCD_TR_TIM0

LCD TR-LCD Display Timing register0
Offset=0x0098

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:24 | TWXRST | Low level width of XRST | R/W | 0x0 |
| 23 | - | Reserved | R | 0x0 |
| 22:0 | TWVCOM | High/Low level width of VCOM | R/W | 0x0 |

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### 14.1.7.30 LCD_TR_TIM1

LCD TR-LCD Display Timing register1
Offset=0x009C

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:24 | TDVST | Time interval from the rising edge of XRST to the rising edge of VST | R/W | 0x0 |
| 23:16 | TWVST | High level width of VST | R/W | 0x0 |
| 15:8 | TDHST | Time interval from the rising/falling edge of VCK to the rising edge of HST | R/W | 0x0 |
| 7:0 | TWHST | High level width of HST | R/W | 0x0 |

### 14.1.7.31 LCD_TR_TIM2

LCD TR-LCD Display Timing register2
Offset=0x00A0

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:24 | TDVCK | Time interval from the rising edge of VST to the rising edge of VCK | R/W | 0x0 |
| 23:13 | TWVCK | High/Low level width of VCK | R/W | 0x0 |
| 12:11 | TPHCK | HCK phase selection | R/W | 0x2 |
| 10:8 | - | Reserved | R | 0x0 |
| 7:0 | TDHCK | Div coefficient of CMU_LCD_CLK, Must be a multiple of 4 | R/W | 0x4 |

### 14.1.7.32 LCD_TR_TIM3

LCD TR-LCD Display Timing register3
Offset=0x00A4

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:21 | TSENB | Time interval from the rising edge of VST to the first line valid data | R/W | 0x0 |
| 20:10 | THENB | Time interval from the falling edge of XRST to the last valid data | R/W | 0x0 |
| 9:8 | - | Reserved | R | 0x0 |
| 7:0 | TDDATA | Time interval from the rising/falling edge of VCK to the first valid data | R/W | 0x0 |

### 14.1.7.33 LCD_TR_TIM4

LCD TR-LCD Display Timing register4
Offset=0x00A8

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:29 | - | Reserved | R | 0x0 |
| 28:16 | TDENB | Time interval from the rising/falling edge of VCK to the rising | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | edge of ENB | | |
| 15:13 | - | Reserved | R | 0x0 |
| 12:0 | TWENB | High level width of ENB | R/W | 0x0 |

#### 14.1.7.34 LCD_TR_TIM5

LCD TR-LCD Display Timing register5
Offset=0x00AC

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:21 | TSMENB | Time interval from the rising edge of VST to the first minimum VCK(units: VCK/4) | R/W | 0x3 |
| 20:10 | THMENB | Time interval from the last minimum VCK to the falling edge of XRST(units: VCK/4) | R/W | 0x0 |
| 9:0 | TWMVCK | Minimum VCK width(units: HCK) | R/W | 0x1 |

Note: Active only when pu_mode =1.

## 14.2 2.5D GPU

### 14.2.1 Features

- 192MHz 2.5D GPU
- Support any curves and any font rendering by HW tessellation, NO CPU overhead.
- 2.5D perspective transformation.
- Any angle primitive rotation, flip and mirror.
- Anti-aliasing: 16x Anti-aliasing without extra performance drop and memory overhead.
- Solid, linear gradient paint and pattern paint.
- Texture mapping; Point sampling, linear and bilinear filtering.
- Scissoring and viewport clipping.
- Porter Duff Alpha Blending plus additional blend modes.
- 8/16/24/32 bits source image and frame color formats.
- Index 1/2/4/8 format source image supported.
- Command buffer based DMA to minimize CPU overhead.
- Lossy compression to save system memory.

## 14.3 2D GPU

### 14.3.1 Features

- RGB565\666\888\ARGB\ARGB8565\ARGB6666\ARGB1555\Ax\ABMP\XRGB\CLUT source data format
- 256*32 deep FIFO with DRQ and IRQ pre Layer
- 512*32 deep FIFO for Scaling function
- Support 256 Level Alpha
- Support three Image Layer
- Support Master DMA Transfer
- Support any angle rotate with RGB565/ARGB/BGRA source

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- The Layer1/2 Support Round Rotation Function
- The Layer1/2 Support Rotation Function
- The Layer1/2 Support 8Phase 2-TAP (8x~1/15x)Up/down scaling
- All layers support Horizontal Anti-aliasing
- The Layer1/2 Support Super-Sampling Anti-Aliasing(SSAA)

## 14.4 JPEG decoder

### 14.4.1 Features

- Input mode: JPEG Baseline
- Resolution: 48x48 ~ 480x480
- Input data format: YUV 420
- Input data format: RGB888/RGB565

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# 15 GPIO and I/O Multiplexer

## 15.1 Features

**GPIO (General Purpose Input /Output) MFP:**

GPIO can output 0 or 1 and detect the signal level of the external circuit. Each GPIO has its own enable control bit and data registers. But the PADs are limited, so MFP module is designed for multiplexing these PADs. Features of GPIO are listed below:

- 45 Programmable GPIOs, and 4 analog IOs can also config as GPIOs.
- All PADs have internal pull down resistors (100KOhm) or pull up resistors (50 KOhm/10KOhm)
- Driving strength adjustable
- Automatically switching PAD function
- All PADs have external Interrupts function

WIO (Wake up I/O) MFP:

- 4 wake up I/O can wake up SOC externally, and these I/O can be config as Special I/O. The Maximum work frequency should no more than 10KHz.
- WIO2/3 also can be config as LOSCI/LOSCO for connect the low frequency crystal.

Settings in actual practice please consult our engineers. The multiplexing relationship can be found in *Pin Description list*.

## 15.2 Operation Manual

### 15.2.1 Multi-function Switch Operation

1. Some pin can be multiplexed as three kinds of functions: module function (MFP), GPIO function, Analog function, which can be configured by setting AD_Select bit and MFP bit of GPIOx_CTL registers.
2. GPIO and MFP are digital functions. Once the Analog function is selected, the digital functions will fail. The function priority of some multiplexed pin are Analog function > GPIO function > MFP function.
3. Some pin can be multiplexed as analog function and digital function. If the pin is used as digital function, it must be disabled analog function firstly by setting AD_Select register.
4. Some MFP modules have itself pull-up and pull-down resistors, referring the chapter Pad PU control register and Pad PD control register; when the pin is multiplexed as MFP module function, the modules pull-up/pull-down resistors will be enabled automatically and the pull-up/pull-down resistors of GPIOs must be DISABLED, or the voltage level and functions will be abnormal.

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## 15.2.2 GPIO Output

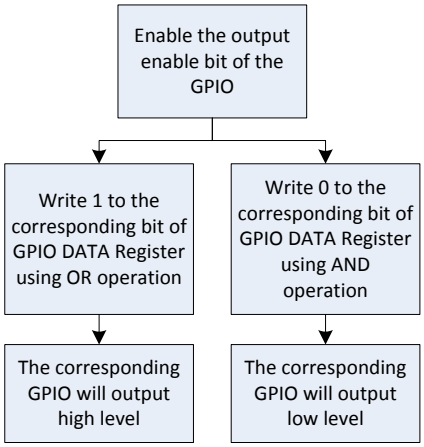



***Figure 15-1 GPIO Output Configuration***

## 15.2.3 GPIO Input

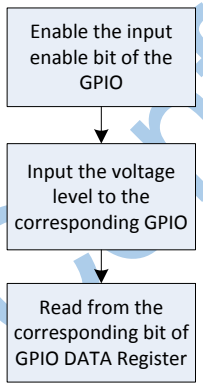



***Figure 15-2 GPIO Input Configuration***

## 15.2.4 GPIO Output/Input Loop Test

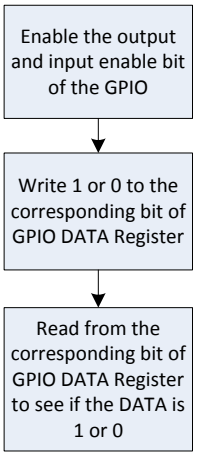
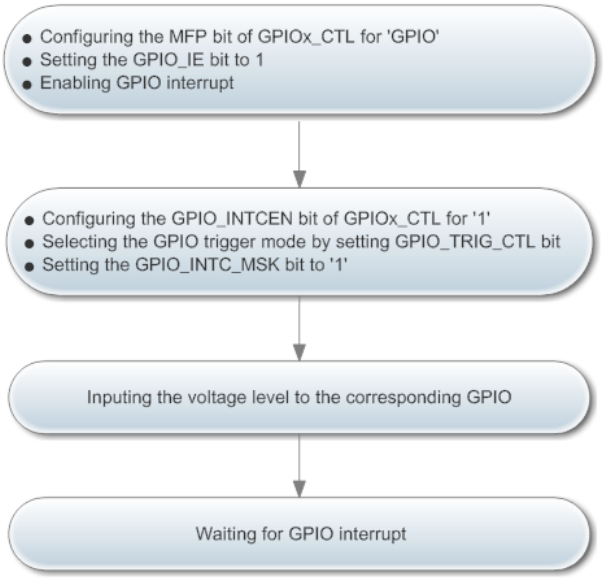



***Figure 15-3 GPIO In/Out Loop Test***

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### 15.2.5 GPIO INTC




## 15.3 GPIO Register List

*Table 15-1 GPIO_MFP Controller Registers Address*

| Name | Physical Base Address | KSEG1 Base Address |
|---|---|---|
| GPIO_MFP_REGISTER | 0x40068000 | 0x40068000 |

*Table 15-2 GPIO&MFP Controller Registers*

| Offset | Register Name | Description | Register Voltage | IO power domain |
|---|---|---|---|---|
| 0x0000 | GPIO0_CTL | GPIO0 control Register | VDD | IOVCC5 |
| 0x0004 | GPIO1_CTL | GPIO1 control Register | VDD | IOVCC5 |
| 0x0008 | GPIO2_CTL | GPIO2 control Register | VDD | IOVCC5 |
| 0x000C | GPIO3_CTL | GPIO3 control Register | VDD | IOVCC5 |
| 0x0018 | GPIO6_CTL | GPIO6 control Register | VDD | IOVCC5 |
| 0x001C | GPIO7_CTL | GPIO7 control Register | VDD | IOVCC5 |
| 0x0038 | GPIO14_CTL | GPIO14 control Register | VDD | IOVCC2 |
| 0x003C | GPIO15_CTL | GPIO15 control Register | VDD | IOVCC2 |
| 0x0040 | GPIO16_CTL | GPIO16 control Register | VDD | IOVCC2 |
| 0x0044 | GPIO17_CTL | GPIO17 control Register | VDD | IOVCC2 |
| 0x0048 | GPIO18_CTL | GPIO18 control Register | VDD | VCC |
| 0x004C | GPIO19_CTL | GPIO19 control Register | VDD | VCC |
| 0x0050 | GPIO20_CTL | GPIO20 control Register | VDD | VCC |
| 0x0054 | GPIO21_CTL | GPIO21 control Register | VDD | VCC |
| 0x0058 | GPIO22_CTL | GPIO22 control Register | VDD | VCC |
| 0x0060 | GPIO24_CTL | GPIO24 control Register | VDD | VCC |
| 0x0064 | GPIO25_CTL | GPIO25 control Register | VDD | VCC |
| 0x0068 | GPIO26_CTL | GPIO26 control Register | VDD | VCC |
| 0x006C | GPIO27_CTL | GPIO27 control Register | VDD | VCC |
| 0x0070 | GPIO28_CTL | GPIO28 control Register | VDD | VCC |
| 0x0074 | GPIO29_CTL | GPIO29 control Register | VDD | VCC |
| 0x0078 | GPIO30_CTL | GPIO30 control Register | VDD | IOVCC2 |
| 0x0080 | GPIO32_CTL | GPIO32 control Register | VDD | IOVCC2 |

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| | | | | |
|---|---|---|---|---|
| 0x0084 | GPIO33_CTL | GPIO33 control Register | VDD | IOVCC2 |
| 0x0088 | GPIO34_CTL | GPIO34 control Register | VDD | IOVCC2 |
| 0x008C | GPIO35_CTL | GPIO35 control Register | VDD | IOVCC2 |
| 0x00C4 | GPIO49_CTL | GPIO49 control Register | VDD | IOVCC4 |
| 0x00C8 | GPIO50_CTL | GPIO50 control Register | VDD | IOVCC4 |
| 0x00CC | GPIO51_CTL | GPIO51 control Register | VDD | IOVCC4 |
| 0x00D0 | GPIO52_CTL | GPIO52 control Register | VDD | IOVCC4 |
| 0x00D4 | GPIO53_CTL | GPIO53 control Register | VDD | IOVCC4 |
| 0x00D8 | GPIO54_CTL | GPIO54 control Register | VDD | IOVCC4 |
| 0x00DC | GPIO55_CTL | GPIO55 control Register | VDD | IOVCC4 |
| 0x00E0 | GPIO56_CTL | GPIO56 control Register | VDD | IOVCC4 |
| 0x00E4 | GPIO57_CTL | GPIO57control Register | VDD | IOVCC4 |
| 0x00E8 | GPIO58_CTL | GPIO58 control Register | VDD | IOVCC4 |
| 0x00EC | GPIO59_CTL | GPIO59 control Register | VDD | IOVCC4 |
| 0x00F0 | GPIO60_CTL | GPIO60 control Register | VDD | IOVCC4 |
| 0x00F4 | GPIO61_CTL | GPIO61 control Register | VDD | IOVCC4 |
| 0x00F8 | GPIO62_CTL | GPIO62 control Register | VDD | IOVCC4 |
| 0x00FC | GPIO63_CTL | GPIO63 control Register | VDD | IOVCC4 |
| 0x0100 | GPIO64_CTL | GPIO64 control Register | VDD | IOVCC1 |
| 0x0128 | GPIO74_CTL | GPIO74 control Register | VDD | IOVCC1 |
| 0x012C | GPIO75_CTL | GPIO75 control Register | VDD | VCC |
| 0x0130 | GPIO76_CTL | GPIO76 control Register | VDD | VCC |
| 0x0168 | GPIO90_CTL | GPIO90 control Register | VDD | AVCC |
| 0x016C | GPIO91_CTL | GPIO91 control Register | VDD | AVCC |
| 0x0178 | GPIO94_CTL | GPIO94 control Register | VDD | AVCC |
| 0x017C | GPIO95_CTL | GPIO95 control Register | VDD | AVCC |
| 0x0200 | GPIO_ODAT0 | GPIO Output Data register0 | VDD | |
| 0x0204 | GPIO_ODAT1 | GPIO Output Data register1 | VDD | |
| 0x0208 | GPIO_ODAT2 | GPIO Output Data register2 | VDD | |
| 0x0210 | GPIO_BSR0 | GPIO Output Data bit set register0 | VDD | |
| 0x0214 | GPIO_BSR1 | GPIO Output Data bit set register1 | VDD | |
| 0x0218 | GPIO_BSR2 | GPIO Output Data bit set register2 | VDD | |
| 0x0220 | GPIO_BRR0 | GPIO Output Data bit set register0 | VDD | |
| 0x0224 | GPIO_BRR1 | GPIO Output Data bit set register1 | VDD | |
| 0x0228 | GPIO_BRR2 | GPIO Output Data bit set register2 | VDD | |
| 0x0230 | GPIO_IDAT0 | GPIO Output Data register0 | VDD | |
| 0x0234 | GPIO_IDAT1 | GPIO Output Data register1 | VDD | |
| 0x0238 | GPIO_IDAT2 | GPIO Output Data register2 | VDD | |
| 0x0240 | GPIO_PD0 | GPIO IRQ Pending register0 | VDD | |
| 0x0244 | GPIO_PD1 | GPIO IRQ Pending register1 | VDD | |
| 0x0248 | GPIO_PD2 | GPIO IRQ Pending register2 | VDD | |
| 0x0300 | WIO0_CTL | Wake up IO0 control Register | SVCC | SVCC |
| 0x0304 | WIO1_CTL | Wake up IO1 control Register | SVCC | SVCC |
| 0x0308 | WIO2_CTL | Wake up IO2 control Register | SVCC | SVCC |
| 0x030C | WIO3_CTL | Wake up IO3 control Register | SVCC | SVCC |
| 0x0320 | DC5V_CTL | DC5V control Register | SVCC | SVCC |

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## 15.4 GPIO Register Description

### 15.4.1 GPIO0_CTL

GPIO0 control Register
Offset=0x0000

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0x1: Enable | | |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO0<br>0x01: SPI0_SS<br>0x0a: SD0_CMD<br>0x16: SPI0_MOSI<br>Others: Reserved | R/W | 0x0 |

## 15.4.2 GPIO1_CTL

GPIO1 control Register
Offset=0x0004

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0x1: Enable | | |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO1<br>0x01: SPI0_MISO<br>0x0A: SD0_CLK0<br>0x16: SPI0_CLK<br>Others: Reserved | R/W | 0x0 |

## 15.4.3 GPIO2_CTL

GPIO2 control Register
Offset=0x0008

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO2<br>0x01: SPI0_CLK<br>0x0A: SD0_DAT0<br>0x16: SPI0_SS<br>Others: Reserved | R/W | 0x0 |

## 15.4.4 GPIO3_CTL

GPIO3 control Register
Offset=0x000c

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | | |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00 : GPIO3<br>0x01 : SPI0_MOSI<br>0x0A: SD0_DAT1<br>0x12: PWM0<br>0x16: SPI0_MISO<br>Others: Reserved | R/W | 0x0 |

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## 15.4.5 GPIO6_CTL

GPIO6 control Register
Offset=0x0018

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO6<br>0x01: SPI0_D2<br>0x06: UART1_CTS<br>0x09: I2C1_CLK<br>0x0A: SD0_DAT2<br>0x11: DMIC01_CLK<br>0x12: PWM2<br>Others: Reserved | R/W | 0x0 |

## 15.4.6 GPIO7_CTL

GPIO7 control Register
Offset=0x001C

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable. | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0x0: Disable<br>0x1: Enable | | |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO7<br>0x01: SPI0_D3<br>0x06: UART1_RTS<br>0x09: I2C1_DAT<br>0x0A: SD0_DAT3<br>0x11: DMIC01_DAT<br>0x12: PWM3<br>Others: Reserved | R/W | 0x0 |

## 15.4.7 GPIO14_CTL

GPIO14 control Register
Offset=0x0038

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO14<br>0x05: UART0_TX<br>0x08: I2C0_CLK<br>0x0B: PPI_TRIGGER0<br>0x0C: IISTX0_DOUT<br>0x12: PWM0<br>0x13: LCD_D0<br>0x17: BT_UART_TX<br>0x18: SPIMT0_SS3<br>0x19: SPIMT1_MISO<br>0x1A: I2CMT0_CLK<br>Others: Reserved | R/W | 0x0 |

## 15.4.8 GPIO15_CTL

GPIO15 control Register
Offset=0x003C

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO15<br>0x05: UART0_RX | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0x08: I2C0_DAT<br>0x0B: PPI_TRIGGER1<br>0x0D: IISRX0_DIN<br>0x12: PWM1<br>0x13: LCD_D1<br>0x17: BT_UART_RX<br>0x18: SPIMT0_SS2<br>0x19: SPIMT1_MOSI<br>0x1A: I2CMT0_DAT<br>Others: Reserved | | |

## 15.4.9 GPIO16_CTL

GPIO16 control Register
Offset=0x0040

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO16<br>0x03: SPI2_SS<br>0x06: UART1_TX<br>0x09: I2C1_DAT<br>0x0B: PPI_TRIGGER2<br>0x0C: IISG0_MCLK<br>0x0D: IISG1_MCLK<br>0x12: PWM2<br>0x13: LCD_D2<br>0x17: BT_REQ<br>0x19: SPIMT1_SS1<br>0x1B: I2CMT1_DAT<br>Others: Reserved | R/W | 0x0 |

## 15.4.10 GPIO17_CTL

GPIO17 control Register
Offset=0x0044

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC. | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | | |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO17<br>0x03: SPI2_CLK<br>0x06: UART1_RX<br>0x09: I2C1_CLK<br>0x0B: PPI_TRIGGER3<br>0x0C: IISG0_BCLK<br>0x0D: IISG1_BCLK<br>0x0F: I2C3_CLK<br>0x12: PWM3<br>0x13: LCD_D3<br>0x17: BT_ACCESS<br>0x19: SPIMT1_SS0<br>0x1B: I2CMT1_CLK<br>OTHER: Reserved | R/W | 0x0 |

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## 15.4.11 GPIO18_CTL

GPIO18 control Register
Offset=0x0048

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO18<br>0x03: SPI2_MISO<br>0x04: SPI3_MISO<br>0x06: UART1_CTS<br>0x08: I2C0_CLK<br>0x0A: SD0_DAT0<br>0x0B: PPI_TRIGGER4<br>0x0C: IISTX0_DOUT<br>0x0D: IISRX0_DIN<br>0x12: PWM1<br>0x13: LCD_D4<br>0x14: SD1_DAT2<br>0x15: IISTX1_DOUT<br>0x17: PTA_GRANT<br>0x18: SPIMT0_MISO<br>0x19: SPIMT1_MISO<br>0x1A: I2CMT0_CLK<br>0x1B: I2CMT1_CLK<br>OTHER: Reserved | R/W | 0x0 |

## 15.4.12 GPIO19_CTL

GPIO19 control Register
Offset=0x004C

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0x1: Slow slew rate | | |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO19<br>0x03: SPI2_MOSI<br>0x04: SPI3_MOSI<br>0x06: UART1_RTS<br>0x08: I2C0_DAT<br>0x0A: SD0_DAT1<br>0x0B: PPI_TRIGGER5<br>0x0C: IISG0_MCLK<br>0x0D: IISG1_MCLK<br>0x12: PWM2<br>0x13: LCD_D5<br>0x14: SD1_DAT3<br>0x18: SPIMT0_MOSI<br>0x19: SPIMT1_MOSI<br>0x1A: I2CMT0_DAT<br>0x1B: I2CMT1_DAT<br>0x1D: HOSCOUT<br>0x1E: LOSCOUT<br>OTHER: Reserved | R/W | 0x0 |

## 15.4.13 GPIO20_CTL

GPIO20 control Register

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Offset=0x0050

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO20<br>0x03: SPI2_SS<br>0x04: SPI3_SS<br>0x06: UART1_TX<br>0x07: UART2_TX<br>0x09: I2C1_CLK<br>0x0A: SD0_DAT2<br>0x0B: PPI_TRIGGER6<br>0x0C: IISG0_BCLK<br>0x0D: IISG1_BCLK<br>0x12: PWM3<br>0x13: LCD_D6<br>0x18: SPIMT0_SS0<br>0x19: SPIMT1_SS0<br>0x1B: I2CMT1_CLK<br>0x1C: LRADC2<br>OTHER: Reserved | R/W | 0x0 |

### 15.4.14 GPIO21_CTL

GPIO21 control Register
Offset=0x0054

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2 | R/W | 0x1 |

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| | | | | |
|---|---|---|---|---|
| | | 0x2: Level 3<br>0x3: Level 4 | | |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO21<br>0x03: SPI2_CLK<br>0x04: SPI3_CLK<br>0x06: UART1_RX<br>0x07: UART2_RX<br>0x08: I2C0_CLK<br>0x09: I2C1_DAT<br>0x0A: SD0_DAT3<br>0x0B: PPI_TRIGGER7<br>0x0C: IISG0_LRCLK<br>0x0D: IISG1_LRCLK<br>0x12: PWM0<br>0x13: LCD_D7<br>0x18: SPIMT0_CLK<br>0x19: SPIMT1_CLK<br>0x1B: I2CMT1_DAT<br>0x1C: LRADC3<br>OTHER: Reserved | R/W | 0x0 |

## 15.4.15 GPIO22_CTL

GPIO22 control Register
Offset=0x0058

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module; | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | | |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO22<br>0x04: SPI3_MISO<br>0x07: UART2_CTS<br>0x08: I2C0_DAT<br>0x0A: SD0_DAT6 | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0x0B: PPI_TRIGGER8<br>0x0E: I2C2_CLK<br>0x12: PWM1<br>0x14: SD1_DAT0<br>0x15: IISTX1_DOUT<br>0x18: SPIMT0_SS3<br>0x19: SPIMT1_MISO<br>0x1A: I2CMT0_CLK<br>0x1D: HOSCOUT<br>0x1E: LOSCOUT<br>OTHER: Reserved | | |

## 15.4.16 GPIO24_CTL

GPIO24 control Register
Offset=0x0060

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0x1: Enable | | |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO24<br>0x04: SPI3_CLK<br>0x06: UART1_TX<br>0x08: I2C0_CLK<br>0x09: I2C1_DAT<br>0x0A: SD0_CLK0<br>0x0B: PPI_TRIGGER10<br>0x0C: IISG0_MCLK<br>0x0D: IISG1_MCLK<br>0x0F: I2C3_CLK<br>0x13: LCD_D4<br>0x18: SPIMT0_SS1<br>0x19: SPIMT1_SS0<br>0x1A: I2CMT0_CLK<br>0x1F: WCI_RX<br>OTHER: Reserved | R/W | 0x0 |

## 15.4.17 GPIO25_CTL

GPIO25 control Register
Offset=0x0064

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | Others: Reserved | | |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO25<br>0x04: SPI3_IO2<br>0x06: UART1_RX<br>0x08: I2C0_DAT<br>0x0A: SD0_CMD<br>0x0B: PPI_TRIGGER11<br>0x0C: IISG0_BCLK<br>0x0D: IISG1_BCLK<br>0x0E: I2C2_CLK<br>0x0F: I2C3_DAT<br>0x13: LCD_D5<br>0x19: SPIMT1_CLK<br>0x1A: I2CMT0_DAT<br>0x1C: LRADC5 | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0x1D: HOSCOUT<br>0x1E : LOSCOUT<br>0x1F: WCI_TX<br>OTHER: Reserved | | |

## 15.4.18 GPIO26_CTL

GPIO26 control Register
Offset=0x0068

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0x1: Enable | | |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO26<br>0x04: SPI3_SS<br>0x07: UART2_TX<br>0x09: I2C1_CLK<br>0x0A: SD0_DAT4<br>0x0B: PPI_TRIGGER0<br>0x0C: IISG0_LRCLK<br>0x0D: IISG1_LRCLK<br>0x0E: I2C2_DAT<br>0x13: LCD_D6<br>0x14: SD1_CMD<br>0x19: SPIMT1_MISO<br>0x1B: I2CMT1_CLK<br>0x1F: VCC_OUT(3.3v)<br>OTHER: Reserved | R/W | 0x0 |

## 15.4.19 GPIO27_CTL

GPIO27 control Register
Offset=0x006C

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO27<br>0x04: SPI3_IO3<br>0x07: UART2_RX<br>0x09: I2C1_DAT<br>0x0A: SD0_DAT5<br>0x0B: PPI_TRIGGER1<br>0x0C: IISTX0_DOUT<br>0x0F: I2C3_CLK<br>0x13: LCD_D7<br>0x14: SD1_CLK<br>0x19: SPIMT1_MOSI<br>0x1B: I2CMT1_DAT<br>0x1C: LRADC6<br>0x1F: VCC_OUT(3.3v)<br>OTHER: Reserved | R/W | 0x0 |

## 15.4.20 GPIO28_CTL

GPIO28 control Register
Offset=0x0070

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| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0x00: GPIO28<br>0x04: SPI3_IO2<br>0x05: UART0_TX<br>0x08: I2C0_CLK<br>0x0A: SD0_DAT6<br>0x0B: PPI_TRIGGER2<br>0x0C: IISTX0_DOUT<br>0x0E: I2C2_CLK<br>0x14: SD1_DAT1<br>0x17: BT_UART_TX<br>0x19: SPIMT1_SS0<br>0x1A: I2CMT0_CLK<br>OTHER: Reserved | | |

## 15.4.21 GPIO29_CTL

GPIO29 control Register
Offset=0x0074

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO29<br>0x03: SPI2_SS<br>0x04: SPI3_IO3<br>0x05: UART0_RX<br>0x08: I2C0_DAT<br>0x0A: SD0_DAT7<br>0x0B: PPI_TRIGGER3<br>0x0D: IISRX0_DIN<br>0x0E: I2C2_DAT<br>0x14: SD1_DAT0<br>0x15: IISTX1_DOUT<br>0x17: BT_UART_RX<br>0x19: SPIMT1_SS1<br>0x1A: I2CMT0_DAT<br>OTHER: Reserved | R/W | 0x0 |

## 15.4.22 GPIO30_CTL

GPIO30 control Register
Offset=0x0078

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0x4: low level<br>Others: Reserved | | |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO30<br>0x03: SPI2_SS<br>0x0B: PPI_TRIGGER0<br>0x0C: IISG0_LRCLK<br>0x0D: IISG1_LRCLK<br>0x0E: I2C2_DAT<br>0x0F: I2C3_DAT<br>0x11: DMIC01_CLK<br>0x13: LCD_CS0<br>0x19: SPIMT1_CLK<br>OTHER: Reserved | R/W | 0x0 |

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## 15.4.23 GPIO32_CTL

GPIO32 control Register
Offset=0x0080

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO32<br>0x03: SPI2_MISO<br>0x08: I2C0_CLK<br>0x0B: PPI_TRIGGER1<br>0x0E: I2C2_CLK<br>0x13: LCD_RS_CD<br>0x15: IISTX1_DOUT<br>0x18: SPIMT0_MISO<br>0x1A: I2CMT0_CLK<br>0x1D: HOSCOUT<br>0x1E: LOSCOUT<br>OTHER: Reserved | R/W | 0x0 |

## 15.4.24 GPIO33_CTL

GPIO33 control Register
Offset=0x0084

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |

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| | | | | |
|---|---|---|---|---|
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO33<br>0x03: SPI2_MOSI<br>0x08: I2C0_DAT<br>0x0B: PPI_TRIGGER2<br>0x13: LCD_RD#<br>0x18: SPIMT0_MOSI<br>0x1A: I2CMT0_DAT<br>0x1F: IOVCC2_OUT(1.8/3.3v)<br>OTHER: Reserved | R/W | 0x0 |

### 15.4.25 GPIO34_CTL

GPIO34 control Register
Offset=0x0088

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable; | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | | |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO34<br>0x03: SPI2_CLK<br>0x09: I2C1_CLK<br>0x0B: PPI_TRIGGER3<br>0x11: DMIC01_CLK<br>0x13: LCD_WR#<br>0x17: BT_UART_TX<br>0x18: SPIMT0_CLK<br>0x1B: I2CMT1_CLK<br>OTHER: Reserved | R/W | 0x0 |

## 15.4.26 GPIO35_CTL

GPIO35 control Register
Offset=0x008C

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO35 | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0x03: SPI2_MISO<br>0x09: I2C1_DAT<br>0x0B: PPI_TRIGGER4<br>0x0F: I2C3_CLK<br>0x11: DMIC01_DAT<br>0x13: LCD_TE<br>0x17: BT_UART_RX<br>0x18: SPIMT0_SS1<br>0x1A: I2CMT0_CLK<br>0x1B: I2CMT1_DAT<br>0x1C: LRADC4<br>OTHER: Reserved | | |

## 15.4.27 GPIO49_CTL

GPIO49 control Register
Offset=0x00C4

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable. | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0x0: Disable<br>0x1: Enable | | |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO49<br>0x03: SPI2_SS<br>0x06: UART1_TX<br>0x08: I2C0_CLK<br>0x0B: PPI_TRIGGER0<br>0x0E: I2C2_CLK<br>0x12: PWM0<br>0x14: SD1_CMD<br>0x17: BT_REQ<br>0x18: SPIMT0_SS0<br>0x1A: I2CMT0_CLK<br>OTHER: Reserved | R/W | 0x0 |

## 15.4.28 GPIO50_CTL

GPIO50 control Register
Offset=0x00C8

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable; | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | | |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO50<br>0x03: SPI2_CLK<br>0x06: UART1_RX<br>0x08: I2C0_DAT<br>0x0B: PPI_TRIGGER1<br>0x0E: I2C2_DAT<br>0x12: PWM1<br>0x14: SD1_CLK<br>0x17: BT_ACCESS<br>0x18: SPIMT0_CLK<br>0x1A: I2CMT0_DAT<br>OTHER: Reserved | R/W | 0x0 |

## 15.4.29 GPIO51_CTL

GPIO51 control Register

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Offset=0x00CC

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO51<br>0x03: SPI2_MISO<br>0x06: UART1_CTS<br>0x09: I2C1_CLK<br>0x0B: PPI_TRIGGER2<br>0x0F: I2C3_CLK<br>0x12: PWM2<br>0x14: SD1_DAT1<br>0x17: PTA_GRANT<br>0x18: SPIMT0_MISO<br>0x1B: I2CMT1_CLK<br>OTHER: Reserved | R/W | 0x0 |

## 15.4.30 GPIO52_CTL

GPIO52 control Register
Offset=0x00D0

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO52<br>0x03: SPI2_MOSI<br>0x06: UART1_RTS<br>0x08: I2C0_CLK<br>0x09: I2C1_DAT<br>0x0B: PPI_TRIGGER3<br>0x0C: IISTX0_DOUT<br>0x0F: I2C3_DAT<br>0x12: PWM3<br>0x14: SD1_DAT0<br>0x15: IISTX1_DOUT<br>0x18: SPIMT0_MOSI<br>0x1B: I2CMT1_DAT<br>OTHER: Reserved | R/W | 0x0 |

## 15.4.31 GPIO53_CTL

GPIO53 control Register
Offset=0x00D4

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | Others: Reserved | | |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO53<br>0x03: SPI2_SS<br>0x04: SPI3_SS<br>0x07: UART2_TX<br>0x08: I2C0_DAT<br>0x0A: SD0_DAT4<br>0x0B: PPI_TRIGGER4<br>0x0D: IISG1_MCLK<br>0x0E: I2C2_CLK<br>0x12: PWM0<br>0x14: SD1_DAT2<br>0x19: SPIMT1_SS0<br>0x1A: I2CMT0_CLK<br>0x1D: HOSCOUT | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0x1E: LOSCOUT<br>OTHER: Reserved | | |

## 15.4.32 GPIO54_CTL

GPIO54 control Register
Offset=0x00D8

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable. | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0x0: Disable<br>0x1: Enable | | |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO54<br>0x03: SPI2_CLK<br>0x04: SPI3_CLK<br>0x07: UART2_RX<br>0x09: I2C1_CLK<br>0x0A: SD0_DAT5<br>0x0B: PPI_TRIGGER5<br>0x0D: IISG1_BCLK<br>0x0E: I2C2_DAT<br>0x12: PWM1<br>0x14: SD1_DAT3<br>0x19: SPIMT1_CLK<br>0x1A: I2CMT0_DAT<br>OTHER: Reserved | R/W | 0x0 |

### 15.4.33 GPIO55_CTL

GPIO55 control Register
Offset=0x00DC

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0x1: Slow slew rate | | |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO55<br>0x03: SPI2_MISO<br>0x04: SPI3_MISO<br>0x07: UART2_CTS<br>0x09: I2C1_DAT<br>0x0A: SD0_DAT6<br>0x0B: PPI_TRIGGER6<br>0x0D: IISG1_LRCLK<br>0x0F: I2C3_CLK<br>0x12: PWM2<br>0x14: SD1_CLK<br>0x19: SPIMT1_MISO<br>0x1B: I2CMT1_CLK<br>OTHER: Reserved | R/W | 0x0 |

## 15.4.34 GPIO56_CTL

GPIO56 control Register
Offset=0x00E0

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | | |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO56<br>0x03: SPI2_MOSI<br>0x04: SPI3_MOSI<br>0x07: UART2_RTS | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0x0A: SD0_DAT7<br>0x0B: PPI_TRIGGER7<br>0x0C: IISTX0_DOUT<br>0x0D: IISRX0_DIN<br>0x0E: I2C2_CLK<br>0x0F: I2C3_DAT<br>0x12: PWM3<br>0x19: SPIMT1_MOSI<br>0x1B: I2CMT1_DAT<br>OTHER: Reserved | | |

## 15.4.35 GPIO57_CTL

GPIO57 control Register
Offset=0x00E4

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO57<br>0x03: SPI2_SS<br>0x08: I2C0_CLK<br>0x0A: SD0_DAT2<br>0x0B: PPI_TRIGGER8<br>0x0E: I2C2_DAT<br>0x12: PWM1<br>0x15: IISTX1_DOUT<br>0x18: SPIMT0_SS2<br>0x1A: I2CMT0_CLK<br>0x1D: HOSCOUT<br>0x1E: LOSCOUT<br>OTHER: Reserved | R/W | 0x0 |

## 15.4.36 GPIO58_CTL

GPIO58 control Register
Offset=0x00E8

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC. | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | | |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO58<br>0x03: SPI2_CLK<br>0x08: I2C0_DAT<br>0x0A: SD0_DAT3<br>0x0B: PPI_TRIGGER9<br>0x0C: IISTX0_DOUT<br>0x0F: I2C3_CLK<br>0x18: SPIMT0_SS3<br>0x1A: I2CMT0_DAT<br>0x1F: WCI_TX<br>OTHER: Reserved | R/W | 0x0 |

## 15.4.37 GPIO59_CTL

GPIO59 control Register
Offset=0x00EC

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO59 | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0x03: SPI2_MISO<br>0x09: I2C1_CLK<br>0x0A: SD0_CLK1<br>0x0B: PPI_TRIGGER10<br>0x0C: IISG0_MCLK<br>0x0D: IISG1_MCLK<br>0x0F: I2C3_DAT<br>0x18: SPIMT0_SS1<br>0x1B: I2CMT1_CLK<br>0x1F: WCI_RX<br>OTHER: Reserved | | |

## 15.4.38 GPIO60_CTL

GPIO60 control Register
Offset=0x00F0

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0x1: Enable | | |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO60<br>0x03: SPI2_MOSI<br>0x09: I2C1_DAT<br>0x0A: SD0_CLK0<br>0x0B: PPI_TRIGGER11<br>0x0C: IISG0_BCLK<br>0x0D: IISG1_BCLK<br>0x0F: I2C3_CLK<br>0x1B: I2CMT1_DAT<br>0x1D: HOSCOUT<br>0x1E: LOSCOUT<br>OTHER: Reserved | R/W | 0x0 |

## 15.4.39 GPIO61_CTL

GPIO61 control Register
Offset=0x00F4

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC. | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | | |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO61<br>0x04: SPI3_IO2<br>0x0A: SD0_DAT0<br>0x0B: PPI_TRIGGER4<br>0x0C: IISG0_LRCLK<br>0x0D: IISG1_LRCLK<br>0x0E: I2C2_CLK<br>0x19: SPIMT1_SS3<br>0x1C: LRADC5<br>0x1D: HOSCOUT<br>0x1E: LOSCOUT<br>OTHER: Reserved | R/W | 0x0 |

## 15.4.40 GPIO62_CTL

GPIO62 control Register
Offset=0x00F8

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| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0x00: GPIO62<br>0x04: SPI3_IO3<br>0x05: UART0_RX<br>0x06: UART1_RX<br>0x07: UART2_RX<br>0x0A: SD0_DAT1<br>0x0B: PPI_TRIGGER5<br>0x0C: IISTX0_DOUT<br>0x0D: IISRX0_DIN<br>0x0E: I2C2_DAT<br>0x19: SPIMT1_SS1<br>0x1C: LRADC4<br>OTHER: Reserved | | |

## 15.4.41 GPIO63_CTL

GPIO63 control Register
Offset=0x00FC

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO63<br>0x03: SPI2_MISO<br>0x04: SPI3_MISO<br>0x05: UART0_TX<br>0x06: UART1_TX<br>0x07: UART2_TX<br>0x0A: SD0_CMD<br>0x0B: PPI_TRIGGER6<br>0x0C: IISG0_MCLK<br>0x0D: IISG1_MCLK<br>0x0F: I2C3_DAT<br>0x15: IISTX1_DOUT<br>0x19: SPIMT1_SS2<br>0x1C: LRADC3<br>0x1D: HOSCOUT<br>0x1E: LOSCOUT<br>0x1F: IOVCC4_OUT(1.8v)<br>OTHER: Reserved | R/W | 0x0 |

## 15.4.42 GPIO64_CTL

GPIO64 control Register
Offset=0x0100

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | | |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO64<br>0x03: SPI2_MISO<br>0x04: SPI3_MISO<br>0x05: UART0_TX<br>0x06: UART1_TX<br>0x07: UART2_TX<br>0x0A: SD0_DAT7<br>0x0B: PPI_TRIGGER11<br>0x0C: IISG0_MCLK<br>0x0D: IISG1_MCLK | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0x0E: I2C2_CLK<br>0x0F: I2C3_DAT<br>0x12: PWM0<br>0x14: SD1_DAT1<br>0x15: IISTX1_DOUT<br>0x18: SPIMT0_SS0<br>0x1B: I2CMT1_DAT<br>0x1D: HOSCOUT<br>0x1E: LOSCOUT<br>0x1F: IOVCC1_OUT(3.3v)<br>OTHER: Reserved | | |

## 15.4.43 GPIO74_CTL

GPIO74 control Register
Offset=0x0128

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0x1: Enable | | |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO74<br>0x03: SPI2_MOSI<br>0x04: SPI3_MOSI<br>0x05: UART0_RX<br>0x07: UART2_RX<br>0x08: I2C0_CLK<br>0x09: I2C1_DAT<br>0x0A: SD0_DAT6<br>0x0B: PPI_TRIGGER4<br>0x0C: IISG0_BCLK<br>0x0D: IISG1_BCLK<br>0x0E: I2C2_DAT<br>0x0F: I2C3_CLK<br>0x11: DMIC01_CLK<br>0x12: PWM1<br>0x14: SD1_DAT0<br>0x18: SPIMT0_CLK<br>0x1B: I2CMT1_CLK<br>0x1C: LRADC6<br>0x1D: HOSCOUT<br>0x1E: LOSCOUT<br>0x1F: WCI_RX<br>OTHER: Reserved | R/W | 0x0 |

## 15.4.44 GPIO75_CTL

GPIO75 control Register
Offset=0x012C

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO75<br>0x03: SPI2_MISO<br>0x04: SPI3_MISO<br>0x05: UART0_TX<br>0x07: UART2_RX<br>0x0C: IISG0_BCLK<br>0x0D: IISG1_BCLK | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0x0E: I2C2_CLK<br>0x10: USBDP<br>0x11: DMIC01_CLK<br>0x12: PWM2<br>0x14: SD1_DAT3<br>0x17: BT_UART_TX<br>0x19: SPIMT1_SS3<br>0x1A: I2CMT0_CLK<br>0x1B: I2CMT1_CLK<br>0x1F: WCI_TX<br>OTHER: Reserved | | |

## 15.4.45 GPIO76_CTL

GPIO76 control Register
Offset=0x0130

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | GPIO_INTC_MSK | GPIO INTC mask.<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the GPIO trigger event is detect | R/W | 0x0 |
| 24 | - | Reserved | R | 0x0 |
| 23:21 | GPIO_TRIG_CTL | GPIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | GPIO_INTCEN | GPIO INTC Enable.<br>0x0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>0x1: Enable;<br>Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'. | R/W | 0x0 |
| 19:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | GPIO4K7PUEN | GPIO 4.7K PU Enable.<br>0x0: Disable | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0x1: Enable | | |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4:0 | MFP | Multi-Function of GPIO<br>0x00: GPIO76<br>0x03: SPI2_MOSI<br>0x04: SPI3_MOSI<br>0x05: UART0_RX<br>0x07: UART2_TX<br>0x0C: IISG0_LRCLK<br>0x0D: IISG1_LRCLK<br>0x0E: I2C2_DAT<br>0x10: USBDM<br>0x11: DMIC01_DAT<br>0x12: PWM3<br>0x14: SD1_CLK<br>0x17: BT_UART_RX<br>0x19: SPIMT1_SS2<br>0x1A: I2CMT0_DAT<br>0x1B: I2CMT1_DAT<br>0x1F: WCI_RX<br>OTHER: Reserved | R/W | 0x0 |

## 15.4.46 GPIO90_CTL

GPIO90 control Register
Offset=0x0168

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable. | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0x0: Disable<br>0x1: Enable | | |
| 10 | - | Reserved | R | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4 | AD_SELECT | GPIO Analog/Digital Select Register<br>0: MFP<br>1: Analog Function(INPUT0P) | R/W | 0x1 |
| 3:0 | MFP | 0:GPIO90<br>Other: Reserved | R/W | 0x0 |

Note:
If the user want to configure INPUT0P PIN to be GPIO90, should do as flow:
1. Enable the power of ADC digital through set the bit26 of PWRGATE_DIG to 1;
2. Wait until the bit26 of PWRGATE_DIG_ACK become to 1;
3. Enable the clock of ADC digital through set the bit3 of CMU_DEVCLKEN1 to 1;
4. Release the reset of ADC digital through set the bit3 of MRCR1 to 1;
5. Release the Isolation of ADC IO through set the bit23&bit22 of ADC_VMIC_CTL to 1;
6. Disable the clock of ADC digital through set the bit3 of CMU_DEVCLKEN1 to 0;
7. Set the INPUT0P PIN to be GPIO90 through GPIO90_CTL.

## 15.4.47 GPIO91_CTL

GPIO91 control Register
Offset=0x016C

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | - | Reserved | R | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4 | AD_SELECT | GPIO Analog/Digital Select Register<br>0: MFP<br>1: Analog Function(INPUT0N) | R/W | 0x1 |
| 3:0 | MFP | 0:GPIO91<br>Other: Reserved | R/W | 0x0 |

Note:
If the user want to configure INPUT0N PIN to be GPIO91, should do as flow:
1. Enable the power of ADC digital through set the bit26 of PWRGATE_DIG to 1;
2. Wait until the bit26 of PWRGATE_DIG_ACK become to 1;
3. Enable the clock of ADC digital through set the bit3 of CMU_DEVCLKEN1 to 1;
4. Release the reset of ADC digital through set the bit3 of MRCR1 to 1;
5. Release the Isolation of ADC IO through set the bit23&bit22 of ADC_VMIC_CTL to 1;
6. Disable the clock of ADC digital through set the bit3 of CMU_DEVCLKEN1 to 0;
7. Set the INPUT0N PIN to be GPIO91 through GPIO91_CTL.

## 15.4.48 GPIO94_CTL

GPIO94 control Register
Offset=0x0178

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:15 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | - | Reserved | R | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |

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|---|---|---|---|---|
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable<br>0x1: enable | R/W | 0x0 |
| 4 | AD_SELECT | GPIO Analog/Digital Select Register<br>0: MFP<br>1: Analog Function(AOUT/AOUT_P) | R/W | 0x1 |
| 3:0 | MFP | 0:GPIO94<br>Other: Reserved | R/W | 0x0 |

## 15.4.49 GPIO95_CTL

GPIO95 control Register
Offset=0x017C

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:16 | - | Reserved | R | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | GPIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | GPIO50KPUEN | GPIO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | - | Reserved | R | 0x0 |
| 9 | GPIO100KPDEN | GPIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | GPIO10KPUEN | GPIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | GPIOINEN | GPIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | GPIOOUTEN | GPIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | SMIT | PAD Schmitt enable bit of GPIO:<br>0x0: disable | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0x1: enable | | |
| 4 | AD_SELECT | GPIO Analog/Digital Select Register<br>0: MFP<br>1: Analog Function(AOUT_N) | R/W | 0x1 |
| 3:0 | MFP | 0:GPIO95<br>Other: Reserved | R/W | 0x0 |

## 15.4.50 GPIO_ODAT0

GPIO Output Data register0
Offset = 0x200

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | GPIO_ODAT | GPIO[31:0] Output Data. | R/W | 0x0 |

## 15.4.51 GPIO_ODAT1

GPIO Output Data register1
Offset = 0x204

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | GPIO_ODAT | GPIO[63:32] Output Data. | R/W | 0x0 |

## 15.4.52 GPIO_ODAT2

GPIO Output Data register2
Offset = 0x208

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | GPIO_ODAT2_1 | GPIO[95:90] Output Data. | R/W | 0x0 |
| 25:17 | - | Reserved | R | 0x0 |
| 16:0 | GPIO_ODAT2_0 | GPIO[80:64] Output Data. | R/W | 0x0 |

## 15.4.53 GPIO_BSR0

GPIO Output Data bit set register0
Offset = 0x210

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | GPIO_BSR | GPIO[31:0] Output Data bit set register<br>0: Not care;<br>1: Set the opposite bit of GPIO_ODAT0 as '1' and Set the GPIO to high level;<br>Write this bit to '1', then will be clear automatically | R/W | 0x0 |

## 15.4.54 GPIO_BSR1

GPIO Output Data bit set register1
Offset = 0x214

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | GPIO_BSR | GPIO[63:32] Output Data bit set register | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0: Not care;<br>1: Set the opposite bit of GPIO_ODAT1 as '1' and Set the GPIO to high level;<br>Write this bit to '1', then will be clear automatically | | |

## 15.4.55 GPIO_BSR2

GPIO Output Data bit set register2
Offset = 0x218

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | GPIO_BSR2_1 | GPIO[95:90] Output Data bit set register<br>0: Not care;<br>1: Set the opposite bit of GPIO_ODAT2 as '1' and Set the GPIO to high level;<br>Write this bit to '1', then will be clear automatically | R/W | 0x0 |
| 25:17 | - | - | R | 0x0 |
| 16:0 | GPIO_BSR2_0 | GPIO[80:64] Output Data bit set register<br>0: Not care;<br>1: Set the opposite bit of GPIO_ODAT2 as '1' and Set the GPIO to high level;<br>Write this bit to '1', then will be clear automatically | R/W | 0x0 |

## 15.4.56 GPIO_BRR0

GPIO Output Data bit reset register0
Offset = 0x220

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | GPIO_BRR | GPIO[31:0] Output Data bit reset register<br>0: Not care;<br>1: Set the opposite bit of GPIO_ODAT0 as '0' and Set the GPIO to low level;<br>Write this bit to '1', then will be clear automatically | R/W | 0x0 |

## 15.4.57 GPIO_BRR1

GPIO Output Data bit reset register1
Offset = 0x224

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | GPIO_BRR | GPIO[63:32] Output Data bit reset register<br>0: Not care;<br>1: Set the opposite bit of GPIO_ODAT1 as '0' and Set the GPIO to low level;<br>Write this bit to '1', then will be clear automatically | R/W | 0x0 |

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## 15.4.58 GPIO_BRR2

GPIO Output Data bit reset register2
Offset = 0x228

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | GPIO_BRR2_1 | GPIO[95:90] Output Data bit reset register<br>0: Not care;<br>1: Set the opposite bit of GPIO_ODAT1 as '0' and Set the GPIO to low level;<br>Write this bit to '1', then will be clear automatically | R/W | 0x0 |
| 25:17 | - | - | R | 0x0 |
| 16:0 | GPIO_BRR2_0 | GPIO[80:64] Output Data bit reset register<br>0: Not care;<br>1: Set the opposite bit of GPIO_ODAT1 as '0' and Set the GPIO to low level;<br>Write this bit to '1', then will be clear automatically | R/W | 0x0 |

## 15.4.59 GPIO_IDAT0

GPIO Input Data register0
Offset = 0x230

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | GPIO_IDAT | GPIO[31:0] Input Data. | R | 0x0 |

## 15.4.60 GPIO_IDAT1

GPIO Input Data register1
Offset = 0x234

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | GPIO_IDAT | GPIO[63:32] Input Data. | R | 0x0 |

## 15.4.61 GPIO_IDAT2

GPIO Input Data register2
Offset = 0x238

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | GPIO_IDAT2_1 | GPIO[95:90] Input Data | R | 0x0 |
| 25:17 | - | Reserved | R | 0x0 |
| 16:0 | GPIO_IDAT2_0 | GPIO[80:64] Input Data. | R | 0x0 |

## 15.4.62 GPIO_PD0

GPIO IRQ Pending register0
Offset = 0x240

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | GPIO_PD | GPIO[31:0] IRQ Pending register.<br>'0': No IRQ | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | '1': IRQ<br>Writing 1 to the bit is clear it. | | |

## 15.4.63 GPIO_PD1

GPIO IRQ Pending register1
Offset = 0x244

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:0 | GPIO_PD | GPIO[63:32] IRQ Pending register.<br>'0': No IRQ<br>'1': IRQ<br>Writing 1 to the bit is clear it. | R/W | 0x0 |

## 15.4.64 GPIO_PD2

GPIO IRQ Pending register2
Offset = 0x248

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:17 | - | - | R | 0x0 |
| 16:0 | GPIO_PD | GPIO[80:64] IRQ Pending register.<br>'0': No IRQ<br>'1': IRQ<br>Writing 1 to the bit is clear it. | R/W | 0x0 |

## 15.4.65 WIO0_CTL

WIO0 control Register (SVCC)
Offset=0x300

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | Reserved | Reserved | R | 0x0 |
| 25 | WIO_INTC_MSK | WIO INTC mask.<br>0: Mask the interrupt, do not send the interrupt to the INTC module;<br>1: Send interrupt to the INTC, when the WIO trigger event is detect | R/W | 0x0 |
| 24 | WIO_INTC_PD | WIO INTC Pending.<br>0: no pending;<br>1: pending<br>Write 1 to clear | R/W | 0x0 |
| 23:21 | WIO_TRIG_CTL | WIO trigger mode<br>000: rising edge<br>001: falling edge<br>010:dual edge<br>011: high level<br>100: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | WIO_INTCEN | WIO INTC Enable.<br>0: Disable;<br>Do not generate IRQ pending and do not send IRQ to INTC.<br>1: Enable; | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when WIO_INTC_MSK is '1'. | | |
| 19:17 | Reserved | Reserved | R | 0x0 |
| 16 | WIODAT | WIO Input/Output Data. | R/W | 0x0 |
| 15 | SR | WIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | WIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | WIO100KPUEN | WIO 100K PU Enable.<br>0: Disable<br>1: Enable | R/W | 0x0 |
| 10 | WIO4K7PUEN | WIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | WIO100KPDEN | WIO 100K PD Enable.<br>0: Disable<br>1: Enable | R/W | 0x0 |
| 8 | 10KPUEN | WIO 10K PU Enable.<br>0: Disable<br>1: Enable | R/W | 0x0 |
| 7 | WIOINEN | WIO Input Enable.<br>0: Disable<br>1: Enable | R/W | 0x0 |
| 6 | WIOOUTEN | WIO Output Enable.<br>0: Disable<br>1: Enable | R/W | 0x0 |
| 5:4 | Reserved | Reserved | R | 0x0 |
| 3:0 | MFP | Multi-Function of WIO<br>0:WIO0<br>1:ONOFF<br>2:I2C1_CLK<br>3:LRADC1<br>OTHER: Reserved | R/W | 0x3 |

Note:
WIO and its' control registers are work in SVCC domain, which will not be power off in all PMU state.

## 15.4.66 WIO1_CTL

Wake up IO1 control Register (SVCC)
Offset=0x0304

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | WIO_INTC_MASK | WIO INTC mask<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module; | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0x1: Send interrupt to the INTC, when the WIO trigger event is detect. | | |
| 24 | WIO_INTC_PD | WIO INTC Pending<br>0x0: no pending<br>0x1: pending<br>Write 1 to clear | R/W | 0x0 |
| 23:21 | WIO_TRIG_CTL | WIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | WIO_INTCEN | WIO INTC Enable<br>0x0: Disable (Don't generate IRQ pending and do not send IRQ to INTC)<br>0x1: Enable (Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when WIO_INTC_MSK is "1") | R/W | 0x0 |
| 19:17 | - | Reserved | R | 0x0 |
| 16 | WIODAT | WIO Input/Output Data. | R/W | 0x0 |
| 15 | SR | WIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | WIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | WIO100KPUEN | WIO 100K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | WIO4K7PUEN | WIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | WIO100KPDEN | WIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | WIO10KPUEN | WIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | WIOINEN | WIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | WIOOUTEN | WIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5:4 | - | Reserved | R | 0x0 |
| 3:0 | MFP | Multi-Function of WIO<br>0x0: WIO1<br>0x2: I2C1_DAT<br>0x3: LRADC1 | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0x6: LOSCOUT<br>OTHER: Reserved | | |

Note:
WIO and its' control registers are work in SVCC domain, which will not be power off in all PMU state.

## 15.4.67 WIO2_CTL

Wake up IO2 control Register (SVCC)
Offset=0x0308

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | WIO_INTC_MASK | WIO INTC mask<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the WIO trigger event is detect. | R/W | 0x0 |
| 24 | WIO_INTC_PD | WIO INTC Pending<br>0x0: no pending<br>0x1: pending<br>Write 1 to clear | R/W | 0x0 |
| 23:21 | WIO_TRIG_CTL | WIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | WIO_INTCEN | WIO INTC Enable<br>0x0: Disable (Don't generate IRQ pending and do not send IRQ to INTC)<br>0x1: Enable (Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when WIO_INTC_MSK is "1") | R/W | 0x0 |
| 19:17 | - | Reserved | R | 0x0 |
| 16 | WIODAT | WIO Input/Output Data. | R/W | 0x0 |
| 15 | SR | GPIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | WIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | WIO100KPUEN | WIO 100K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | WIO4K7PUEN | WIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | WIO100KPDEN | WIO 100K PD Enable.<br>0x0: Disable | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| | | 0x1: Enable | | |
| 8 | WIO10KPUEN | WIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | WIOINEN | WIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | WIOOUTEN | WIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | - | Reserved | R | 0x0 |
| 4 | AD_SELECT | WIO Analog/Digital Select Register<br>0: MFP<br>1: Analog Function(LOSCI) | R/W | 0x1 |
| 3:0 | MFP | Multi-Function of WIO<br>0x0: WIO2<br>0x2: I2C0_CLK<br>OTHER: Reserved | R/W | 0x0 |

Note:
WIO and its' control registers are work in SVCC domain, which will not be power off in all PMU state.

## 15.4.68 WIO3_CTL

Wake up IO3 control Register (SVCC)
Offset=0x030C

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:26 | - | Reserved | R | 0x0 |
| 25 | WIO_INTC_MASK | WIO INTC mask<br>0x0: Mask the interrupt, do not send the interrupt to the INTC module;<br>0x1: Send interrupt to the INTC, when the WIO trigger event is detect. | R/W | 0x0 |
| 24 | WIO_INTC_PD | WIO INTC Pending<br>0x0: no pending<br>0x1: pending<br>Write 1 to clear | R/W | 0x0 |
| 23:21 | WIO_TRIG_CTL | WIO trigger mode<br>0x0: rising edge<br>0x1: falling edge<br>0x2: dual edge<br>0x3: high level<br>0x4: low level<br>Others: Reserved | R/W | 0x0 |
| 20 | WIO_INTCEN | WIO INTC Enable<br>0x0: Disable (Don't generate IRQ pending and do not send IRQ to INTC)<br>0x1: Enable (Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when WIO_INTC_MSK is "1") | R/W | 0x0 |
| 19:17 | - | Reserved | R | 0x0 |
| 16 | WIODAT | WIO Input/Output Data. | R/W | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 15 | SR | WIO Slew Rate<br>0x0: Fast slew rate<br>0x1: Slow slew rate | R/W | 0x0 |
| 14 | - | Reserved | R | 0x0 |
| 13:12 | PADDRV | WIO PAD Drive Control<br>0x0: Level 1<br>0x1: Level 2<br>0x2: Level 3<br>0x3: Level 4 | R/W | 0x1 |
| 11 | WIO100KPUEN | WIO 100K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | WIO4K7PUEN | WIO 4.7K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 9 | WIO100KPDEN | WIO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | WIO10KPUEN | WIO 10K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | WIOINEN | WIO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | WIOOUTEN | WIO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | - | Reserved | R | 0x0 |
| 4 | AD_SELECT | WIO Analog/Digital Select Register<br>0: MFP<br>1: Analog Function(LOSCO) | R/W | 0x1 |
| 3:0 | MFP | Multi-Function of WIO<br>0x0: WIO3<br>0x2: I2C0_DAT<br>0x6: LOSCOUT<br>OTHER: Reserved | R/W | 0x0 |

Note:
WIO and its' control registers are work in SVCC domain, which will not be power off in all PMU state.

## 15.4.69 DC5V_CTL

DC5V control Register (VDD)
Offset=0x320

| Bit(s) | Name | Description | R/W | Reset |
|---|---|---|---|---|
| 31:17 | - | Reserved | R | 0x0 |
| 16 | IODAT | Input/Output Data. | R/W | 0x0 |
| 15:12 | - | Reserved | R | 0x0 |
| 11 | IO50KPUEN | IO 50K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 10 | - | Reserved | R | 0x0 |

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| | | | | |
|---|---|---|---|---|
| 9 | IO100KPDEN | IO 100K PD Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 8 | IO5K6PUEN | IO 5.6K PU Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 7 | IOINEN | IO Input Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 6 | IOOUTEN | IO Output Enable.<br>0x0: Disable<br>0x1: Enable | R/W | 0x0 |
| 5 | - | Reserved | R | 0x0 |
| 4 | AD_SELECT | GPIO Analog/Digital Select Register<br>0x0: Auto<br>0x1: Analog Function(DC5V) | R/W | 0x1 |
| 3:0 | MFP | Multi-Function of GPIO<br>0x0: GPIO<br>0x1: UART1_TX<br>0x2: UART1_RX<br>0x3: UART2_TX<br>0x4: UART2_RX<br>0x5: BT_UART_TX<br>0x6: BT_UART_RX<br>OTHER: Reserved | R/W | 0x0 |

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# 16 Electrical Characteristics

## 16.1 Absolute Maximum Ratings

*Table 16-1 Absolute Maximum Ratings*

| Parameter | Symbol | Min | Max | Unit |
|---|---|---|---|---|
| Ambient Temperature | Tamb | -40 | 85 | °C |
| Storage temperature | Tstg | -55 | +150 | °C |
| ESD Stress voltage | Vesd (Human body model) | 4000 | - | V |
| Supply Voltage | DC5V | -0.3 | 7 | V |
| | BAT | -0.3 | 5 | V |
| Input Voltage | 3.3V IO | -0.3 | IOVCC+0.3 | V |
| | 1.8V IO | -0.3 | IOVCC+0.3 | V |

*Note:*
*Even if one of the above parameters exceeds the absolute maximum ratings momentarily, the quality of the product may be degraded. The absolute maximum ratings, therefore, specify the value exceeding, which the product may be physically damaged. Use the product well within these ratings.*

## 16.2 Recommended PWR Supply

*Table 16-2 Recommended PWR Supply*

| Supply Voltage | Min | Typ | Max | Unit |
|---|---|---|---|---|
| BAT (Li) | 3.2 | 3.8 | 4.5 | V |
| DC5V | 4.5 | 5.0 | 7.0 | V |
| VCC | 2.9 | 3.3 | 3.5 | V |
| VCC2 | 2.9 | 3.1 | 3.5 | V |
| SVCC | -- | 2.5 | -- | V |
| VDD | 0.6 | 1.1 | 1.3 | V |
| VD12 | 0.7 | 1.2 | 1.45 | V |
| VC18 | 1.74 | 1.8 | 2.1 | V |
| VREF | -- | 1.5 | -- | V |

## 16.3 DC Characteristics

*Table 16-3 DC Parameters for +3.3V IO Pin with Schmitt Trigger Off*

| Parameter | Symbol | Min. | Max. | Unit | Condition |
|---|---|---|---|---|---|
| Low-level input voltage | VIL | - | 0.8 | V | VCC = 3.1V<br>Tamb = -10 to 70 °C |
| High-level input voltage | VIH | 2.0 | - | V | |
| Low-level output voltage | VOL | - | 0.4 | V | |
| High-level output voltage | VOH | 2.4 | - | V | |

*Table 16-4 DC Parameter for +3.3V IO Pin with Schmitt Trigger On*

| Parameter | Symbol | Min. | Max. | Unit | Condition |
|---|---|---|---|---|---|
| Schmitt trigger positive-going threshold | VT+ | - | 1.9 | V | VCC=3.1V<br>Tamb = -10 to 70 °C |
| Schmitt trigger negative-going threshold | VT- | 1.2 | - | V | |

*Table 16-5 DC Parameters for +1.8V IO Pin with Schmitt Trigger Off*

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| Parameter | Symbol | Min. | Max. | Unit | Condition |
|---|---|---|---|---|---|
| Low-level input voltage | VIL | - | 0.4 | V | VCC = 1.8V Tamb = -10 to 70 °C |
| High-level input voltage | VIH | 1.2 | - | V | |
| Low-level output voltage | VOL | - | 0.36 | V | |
| High-level output voltage | VOH | 1.4 | - | V | |

*Table 16-6 DC Parameter for +1.8V IO Pin with Schmitt Trigger On*

| Parameter | Symbol | Min. | Max. | Unit | Condition |
|---|---|---|---|---|---|
| Schmitt trigger positive-going threshold | VT+ | - | 1.1 | V | VCC=1.8V Tamb = -10 to 70 °C |
| Schmitt trigger negative-going threshold | VT- | 0.7 | - | V | |

## 16.4 PWR Consumption

*Table 16-7 PWR Consumption Table*

VDD = 1.2V @ 25°C unless otherwise specified

| Parameter | Symbol | Condition | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|---|
| BR Sniff Mode | Cs | 500ms | | 90 | | µA |
| BLE Sniff Mode | Cs | 500ms | | 65 | | µA |
| Standby | Cd | Vbat = 3.8V | | 20 | | µA |

## 16.5 Bluetooth Characteristics

### 16.5.1 Transmitter

*Table 16-8 Basic Data Rate of Transmitter*

| Parameter | Condition | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|
| Maximum RF Transmit PWR | - | - | | 13 | dBm |
| RF PWR Control Step | - | 2 | 4 | 8 | dB |
| 20dB Bandwidth for Modulated Carrier | - | - | 914 | 1500 | KHz |
| Adjacent Channel Transmit | +2 MHz | - | | -20 | dBm |
| | -2 MHz | - | | -20 | dBm |
| | +3 MHz | - | | -40 | dBm |
| | -3 MHz | - | | -40 | dBm |
| Frequency Deviation | Δf1avg Maximum Modulation | 140 | 166 | 175 | KHz |
| | Δf2max Maximum Modulation | 115 | 162 | | KHz |
| | Δf1avg/Δf2avg | 0.8 | 1 | | |
| Initial Carrier Frequency Tolerance | - | -75 | <+/-8 | 75 | KHz |
| Frequency Drift | DH1 Packet | -25 | <+/-10 | 25 | KHz |
| | DH3 Packet | -40 | <+/-10 | 40 | KHz |
| | DH5 Packet | -40 | <+/-10 | 40 | KHz |
| Frequency Drift Rate | - | -20 | 3 | 20 | KHz/50us |

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*Table 16-9 Enhanced Data Rate of Transmitter*

| Description | | Min | Typ. | Max. | Unit |
|---|---|---|---|---|---|
| Relative Transmit PWR(EDR) | | -4 | -2.5 | 1 | dB |
| π/4 DQPSK max carrier frequency stability\|ω0\| | | -10 | +/-3 | 10 | KHz |
| π/4 DQPSK max carrier frequency stability\|ωi\| | | -75 | +/-5 | 75 | KHz |
| π/4 DQPSK max carrier frequency stability\|ω0+ωi\| | | -75 | +/-4 | 75 | KHz |
| 8DPSK max carrier frequency stability \|ω0\| | | -10 | +/-3 | 10 | KHz |
| 8DPSK max carrier frequency stability \|ωi\| | | -75 | +/-5 | 75 | KHz |
| 8DPSK max carrier frequency stability\|ω0+ωi\| | | -75 | +/-5 | 75 | KHz |
| π/4 DQPSK Modulation Accuracy | RMS DEVM | | <7 | 20 | % |
| | 99% DEVM | 99 | 100 | | % |
| | Peak DEVM | | <16 | 35 | % |
| In-band spurious emissions | F >F0 + 3MHz | | | -40 | dBm |
| | F < F0 - 3MHz | | | -40 | dBm |
| | F = F0 + 3MHz | | | -40 | dBm |
| | F = F0 - 3MHz | | | -40 | dBm |
| | F = F0 + 2MHz | | | -20 | dBm |
| | F = F0 - 2MHz | | | -20 | dBm |
| | F = F0 + 1MHz | | | -26 | dBm |
| | F = F0 - 1MHz | | | -26 | dBm |
| EDR Differential Phase Encoding | | 99 | 100 | | % |

*Table 16-10 BLE 1Mbps of Transmitter*

| Description | | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|
| Maximum RF Transmit PWR | | | | 14 | dBm |
| In-band emissions | +2 MHz | | -44 | -20 | dBm |
| | -2 MHz | | -44 | -20 | dBm |
| | +3 MHz | | -41 | -30 | dBm |
| | -3 MHz | | -41 | -30 | dBm |
| Modulation Characteristics | Δf1avg Maximum Modulation | 225 | 255 | 275 | KHz |
| | Δf2max Maximum Modulation | 185 | 235 | | KHz |
| | Δf1avg/Δf2avg | 0.8 | 0.96 | | |
| Carrier Frequency Offset | | -150 | <+/-10 | 150 | KHz |
| Frequency Drift | | -50 | <+/-5 | 50 | KHz |

*Table 16-11 BLE 2Mbps of Transmitter*

| Description | | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|
| Maximum RF Transmit PWR | | | | 14 | dBm |
| Adjacent Channel Transmit | +2 MHz | | -28 | -20 | dBm |
| | -2 MHz | | -28 | -20 | dBm |
| | +3 MHz | | -42 | -30 | dBm |
| | -3 MHz | | -42 | -30 | dBm |
| Frequency Deviation | Δf1avg Maximum Modulation | 450 | 495 | 550 | KHz |
| | Δf2max Maximum Modulation | 370 | 425 | | KHz |
| | Δf1avg/Δf2avg | 0.8 | 0.89 | | |
| Carrier Frequency Offset | | -150 | <+/-10 | 150 | KHz |
| Frequency Drift | | -50 | <+/-5 | 50 | KHz |
| Frequency Drift Rate | | -20 | <+/-5 | 20 | KHz/50us |

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## 16.5.2 Receiver

*Table 16-12 Basic Data Rate of Receiver*

| RF Characteristics | | Min. | Typ. | Max. | Specification |
|---|---|---|---|---|---|
| DH1/DH5 Rx sensitivity with dirty transmit on@0.1% BER | | | -96.5 | | <-70dBm |
| Maximum Input PWR at 0.1% BER | | -20 | 0 | | <-20dBm |
| C/I (dB, Margin relative to Spec.) | Co-channel | | -67 | | <11dB |
| | Adjacent 1Mhz | | -51 | | <0dB |
| | Adjacent 2MHz | | -18 | | <-30dB |
| | Adjacent >=3Mhz | | -18 | | <-40dB |
| | Image interference | | -38 | | <-9dB |
| | Image+/-1Mhz | | -36 | | <-20dB |
| Blocking(dBm) | 30Mhz~2000Mhz | -30 | | | -30 |
| | 2000Mhz~2400MHz | -35 | | | -35 |
| | 2500MHz~3000MHz | -35 | | | -35 |
| | 3000MHz~12.75Ghz | -30 | | | -30 |
| Inter-Modulation | Payload length: 37, n: 3,4,5<br>Packets: 1500<br>RF Level: -64 dBm | | -30 | | PER≦30.8%<br>≧-50 |

*Table 16-13 Enhanced Data Rate of Receiver(2M)*

| RF Characteristics | | Min. | Typ. | Max. | Specification |
|---|---|---|---|---|---|
| 2DH5 Rx sensitivity with dirty transmit on@0.1% BER | | | -95 | | <-70dBm |
| Maximum Input PWR at 0.1% BER | | | 0 | | >-20dBm |
| C/I (dB, Margin relative to Spec.) | Co-channel | | -71 | | <13dB |
| | Adjacent 1Mhz | | -51 | | <0dB |
| | Adjacent 2MHz | | -19 | | <-30dB |
| | Adjacent >=3Mhz | | -19 | | <-40dB |
| | Image interference | | -36 | | <-7dB |
| | Image+/-1Mhz | | -34 | | <-20dB |
| Blocking(dBm) | 30Mhz~2000Mhz | -30 | | | -30 |
| | 2000Mhz~2400MHz | -35 | | | -35 |
| | 2500MHz~3000MHz | -35 | | | -35 |
| | 3000MHz~12.75Ghz | -30 | | | -30 |
| Inter-Modulation | Payload length: 37, n: 3,4,5<br>Packets: 1500<br>RF Level: -64 dBm | | -30 | | PER≦30.8%<br>≧-50 |

*Table 16-14 Enhanced Data Rate of Receiver(3M)*

| RF Characteristics | | Min. | Typ. | Max. | Specification |
|---|---|---|---|---|---|
| 3DH5 Rx sensitivity (8DPSK) | | | -89.5 | | <-65dBm |
| Maximum Input PWR at 0.1% BER (8DPSK) | | | 0 | | <-20dBm |
| C/I (dB) | Co-channel | | -77 | | <21dB |
| | Adjacent 1Mhz | | -57 | | <5dB |
| | Adjacent 2MHz | | -23 | | <-25dB |
| | Adjacent >=3Mhz | | -23 | | <-33dB |
| | Image interference | | -43 | | <0dB |
| | Image+/-1Mhz | | -41 | | <-13dB |
| Blocking(dBm) | 30Mhz~2000Mhz | -30 | | | -30 |

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| | | | | | |
|---|---|---|---|---|---|
| | 2000Mhz~2400MHz | -35 | | | -35 |
| | 2500MHz~3000MHz | -35 | | | -35 |
| | 3000MHz~12.75Ghz | -30 | | | -30 |
| Inter-Modulation | Payload length: 37, n: 3,4,5<br>Packets: 1500<br>RF Level: -64 dBm | | -30 | | PER≦30.8%<br>≧-50 |

*Table 16-15 BLE 1Mbps of Receiver*

| RF Characteristics | | Min. | Typ. | Max. | Specification |
|---|---|---|---|---|---|
| Sensitivity LE 1M (dBm) | | | -99.5 | | <-70 |
| Maximum Input PWR at 0.1% BER | | -10 | -2 | | -10 |
| C/I<br>(dB, Margin relative to Spec.) | Co-channel | | -75 | | <21dB |
| | Adjacent 1Mhz | | -60 | | <15dB |
| | Adjacent 2MHz | | -28 | | <-17dB |
| | Adjacent >=3Mhz | | -23 | | <-27dB |
| | Image interference | | -34 | | <-9dB |
| | Image+/-1Mhz | | -32 | | <-15dB |
| Blocking(dBm) | 30Mhz~2000Mhz | -30 | | | -30 |
| | 2000Mhz~2400MHz | -35 | | | -35 |
| | 2500MHz~3000MHz | -35 | | | -35 |
| | 3000MHz~12.75Ghz | -30 | | | -30 |
| Inter-Modulation | Payload length: 37, n: 3, 4, 5<br>Packets: 1500<br>RF Level: -64 dBm | | -30 | | PER≦30.8%<br>≧-50 |

*Table 16-16 BLE 2Mbps of Receiver*

| RF Characteristics | | Min. | Typ. | Max. | Specification |
|---|---|---|---|---|---|
| Sensitivity LE 2M(dBm) | | | -96 | | <-70 |
| Maximum Input PWR at 0.1% BER | | -10 | 0 | | -10 |
| C/I<br>(dB, Margin relative to Spec.) | Co-channel | | -75 | | <21dB |
| | Adjacent 2Mhz | | -60 | | <15dB |
| | Adjacent 4MHz | | -40 | | <-17dB |
| | Adjacent >=6Mhz | | -33 | | <-27dB |
| | Image interference | | -29 | | <-9dB |
| | Image+/-2Mhz | | -33 | | <-15dB |
| Blocking(dBm) | 30Mhz~2000Mhz | -30 | | | -30 |
| | 2000Mhz~2400MHz | -35 | | | -35 |
| | 2500MHz~3000MHz | -35 | | | -35 |
| | 3000MHz~12.75Ghz | -30 | | | -30 |
| Inter-Modulation | Payload length: 37, n: 3,4,5<br>Packets: 1500<br>RF Level: -64 dBm | | -30 | | PER≦30.8%<br>≧-50 |

## 16.6 Audio ADC

*Table 16-17 Audio ADC Parameters*

| Analogue to Digital Converter | | | | | |
|---|---|---|---|---|---|
| **Parameter** | **Conditions** | **Min** | **Typ** | **Max** | **Unit** |
| Full Scale Input Voltage | THD+N < 1%<br>(single-ended input aux input) | - | - | 550m | Vrms |
| | THD+N < 1% | - | - | 1.10 | Vrms |

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| | | | | | | |
|---|---|---|---|---|---|---|
| | (diff mode input aux input) | | | | | |
| Analogue gain | OP | Single Ended | -18 | - | 25.5 | dB |
| | | Full Differential | -12 | - | 31.5 | |
| **Analogue to Digital Converter** | | | | | | |
| Resolution | - | | - | - | 24 | Bits |
| Input Sample Rate | - | | 8 | - | 96 | kHz |
| SNR | **Single end input:**<br>fin = 1kHz@Full Scale Input Voltage<br>B/W = 22Hz~1/2FS<br>Fs=16kHz(a-weighting) | | - | 97 | - | dB |
| | **Differential input:**<br>fin = 1kHz@Full Scale Input Voltage<br>B/W = 22Hz~1/2FS<br>Fs=16kHz(a-weighting) | | - | 98 | - | dB |
| Dynamic Range | **Single end input:**<br>fin = 1kHz@-40dBFS Input Voltage<br>B/W = 22Hz~1/2FS<br>Fs=16kHz(a-weighting) | | - | 97 | - | dB |
| | **Differential input:**<br>fin = 1kHz@-40dBFS Input Voltage<br>B/W = 22Hz~1/2FS<br>Fs=16kHz(a-weighting) | | - | 98 | - | dB |
| THD+N | **Single end input:**<br>fin = 1kHz<br>B/W = 22Hz~1/2FS<br>Fs=16kHz(a-weighting) | | - | - | -81 | dB |
| | **Differential input:**<br>fin = 1kHz<br>B/W = 22Hz~1/2FS<br>Fs=16kHz(a-weighting) | | - | - | -88 | dB |
| Digital gain | - | | 0 | - | 52.5 | dB |

## 16.7 Audio DAC

***Table 16-18 Audio DAC Parameters***

| **Digital to Analogue Converter(Differential Output)** | | | | | |
|---|---|---|---|---|---|
| **Parameter** | **Conditions** | **Min** | **Typ** | **Max** | **Unit** |
| Resolution | - | - | - | 24 | Bits |
| Output Sample Rate | - | 8 | - | 96 | kHz |
| SNR | fin = 1kHz@0dBFS input<br>B/W = 22Hz~22kHz<br>Fs=48kHz,Load=32Ω<br>(A-Weighting) | - | 113 | - | dB |
| Dynamic Range | fin = 1kHz@-48dBFS input<br>B/W = 22Hz~22kHz<br>Fs=48kHz,Load=32 Ω<br>(A-Weighting) | - | 111 | - | dB |
| THD+N | fin = 1kHz@0dBFS input<br>B/W = 22Hz~22kHz<br>Fs=48kHz,Load=10KΩ(A-Weighting) | | | -98 | dB |

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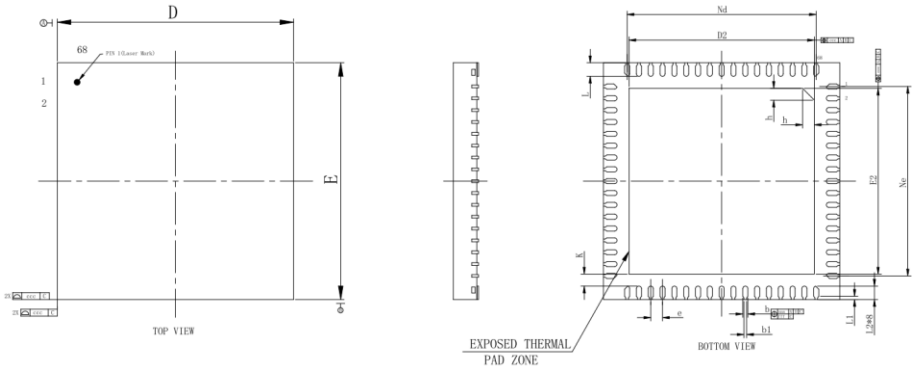
| | | | | | | |
|---|---|---|---|---|---|---|
| | fin = 1kHz@0dBFS input<br>B/W = 22Hz~22kHz<br>Fs=48kHz,Load=32Ω<br>(A-Weighting) | | - | - | -88 | dB |
| Digital gain | 0.375db/step<br>Can set to mute | | -98 | - | 24 | dB |
| Max Amplitude/PWR | fin = 1kHz@0dBFS input<br>Fs=48kHz,Load=10KΩ | Differential Output | - | - | 0.95 | Vrms |

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# 17 Package information

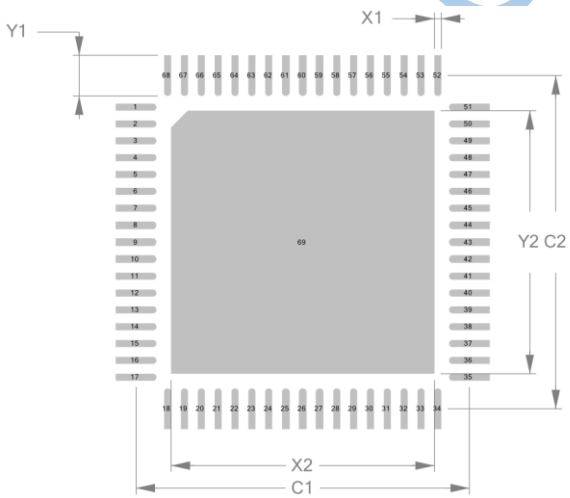
## 17.1 Package Dimensions




| SYMBOL | MILLIMETER | | |
|---|---|---|---|
| | MIN | NOM | MAX |
| A | 0.70 | 0.75 | 0.80 |
| A1 | — | 0.02 | 0.05 |
| b | 0.10 | 0.15 | 0.20 |
| b1 | 0.08REF | | |
| c | 0.18 | 0.20 | 0.25 |
| D | 6.90 | 7.00 | 7.10 |
| D2 | 5.39 | 5.49 | 5.59 |
| e | 0.35BSC | | |
| Nd | 5.60BSC | | |
| E | 6.90 | 7.00 | 7.10 |
| E2 | 5.39 | 5.49 | 5.59 |
| Ne | 5.60BSC | | |
| L | 0.35 | 0.40 | 0.45 |
| L1 | 0.10REF | | |
| L2 | 0.30 | 0.40 | 0.50 |
| K | 0.20 | — | — |
| h | 0.30 | 0.35 | 0.40 |
| aaa | 0.07 | | |
| bbb | 0.08 | | |
| ccc | 0.10 | | |
| ddd | 0.10 | | |
| eee | 0.10 | | |
| fff | 0.05 | | |

*Figure 17-1 QFN68(7*7*0.75mm, Pitch 0.35mm) Dimensions*

## 17.2 Recommended Footprint




| SYMBOL | MILLIMETER |
|---|---|
| C1 | 6.90 |
| Y1 | 0.85 |
| X1 | 0.15 |
| C2 | 6.90 |
| Y2 | 5.45 |
| X2 | 5.45 |

*Figure 17-2 QFN68(7*7*0.75mm, Pitch 0.35mm) recommended footprint*

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# 18 Reflow Solder Information

The ATS3085Sx is constructed with all RoHS compliant material and should be reflowed accordingly. This chip is Moisture Sensitivity Level MSL3 and must be stored and handled accordingly.

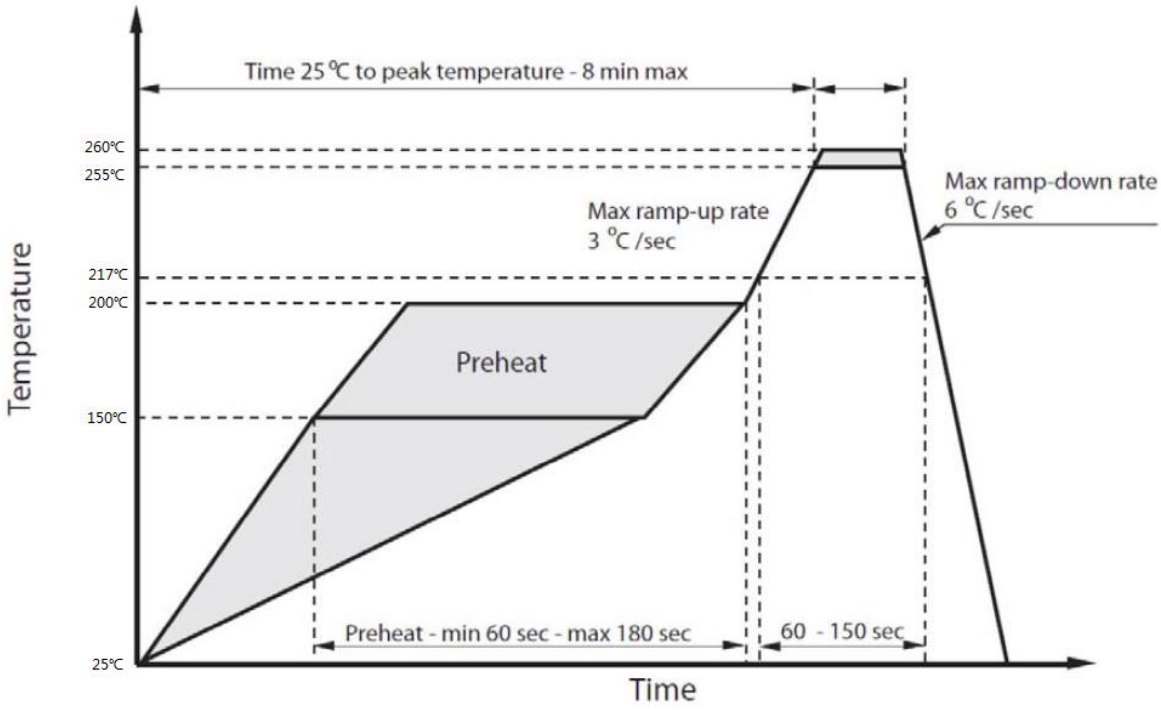



***Figure 18-1 Typical Reflow Solder Profile***

# 19 Ordering Information

| Order Number | | | Package | Package Size | MOQ (Tray) | MOQ (Tape & Reel Packaging) |
|---|---|---|---|---|---|---|
| Device Name | PSRAM Density | Feature Code | | | | |
| ATS3085S | 64Mbit | MPC (1) | SQFN68 | 7mm x 7mm x 0.75mm | 4160 | 3000 |
| ATS3085S4 | 32Mbit | MPW (1) | SQFN68 | 7mm x 7mm x 0.75mm | 4160 | 3000 |

(1) The difference between feature codes lies in the density of built-in PSRAM devices. For further information, please contact sales team.

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# Acronyms and Abbreviations

| Abbreviations | Descriptions |
|---|---|
| AEC | acoustic echo cancellers |
| ADC | Analog-to-Digital-Converter |
| ALU | Arithmetic Logic Unit |
| CC | Constant Current |
| DAC | Digital-to-Analog-Converter |
| DMA | Direct Memory Access |
| GPIO | General Purpose Input Output |
| HOSC | High Frequency OSC |
| HCL | High Frequency Calibration Low Frequency |
| INTC | Interrupt Controller |
| IRQ | Interrupt Request |
| LED | Light Emitting Diode |
| LCD | Liquid Crystal Display |
| LOSC | Internal Low Frequency OSC |
| MIC | Microphone |
| MMU | Memory Management Unit |
| MFP | Multiple Function PAD |
| NMI | Nonmaskable Interrupt |
| OSC | Oscillator |
| PA | Power Amplifier |
| UTMI | USB Transceiver Macro Interface |

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